Observers Are All You Need

B. Müller Peter Nguyen

March 23, 2026

Paper release: `r1347` **Released:** March 23, 2026

Abstract

Observer-Patch Holography (OPH) is a mathematical reconstruction program that starts from a simple requirement: overlapping local observer descriptions on a holographic screen must agree where they overlap. From that starting point, we develop conditional routes toward general relativity, toward reconstruction of the Standard Model gauge structure, toward a first-principles particle-mass program, and toward string/worldsheet descriptions emerging from the same underlying framework. We also report early IBM Quantum Cloud benchmarks of OPH-motivated reduced sectors that match the predicted recoverability ordering and exact-ratio patterns on real hardware, providing preliminary real-device evidence for those local reduced-sector signatures. In the interpretive epilogue, we examine the strange-loop hypothesis that, under OPH, the universe can exist as a self-referential timeless causal structure in which observer reconstruction and information constraints close into a single consistency loop. Throughout, theorem-level results, conditional branches, quantitative continuations, experimental benchmarks, and interpretive material are kept explicitly distinct.

Contents

I Core Framework and Axioms **10**

1 Observer-Patch Holography **10**
1.1 Overview . . . 10
1.2 Model and Axioms . . . 10
1.2.1 Observers and access model . . . 10
1.2.2 Screen, patches, and algebra net . . . 10
1.2.3 Five OPH axioms . . . 11
1.2.4 External inputs and theorem-local technical premises . . . 11
1.2.5 Notation . . . 13
1.2.6 Summary: premise ledger and dependency DAG . . . 13
1.3 Information-Theoretic Tools . . . 16
1.3.1 Strong subadditivity and Markov states . . . 16
1.3.2 Approximate recovery . . . 16
1.3.3 Collar refinement and sufficient mechanisms . . . 16
1.3.4 Concrete UV realization: quantum link models . . . 18
1.3.5 Conformal-modular fixed point microphysics (CMFP) . . . 20
1.4 Overlap Consistency and Gluing . . . 22
1.4.1 Constructive gluing on tree covers . . . 23

1.4.2 Gauge-as-gluing and loops . . . 23
1.4.3 Loop obstruction class (central defect) . . . 24
1.4.4 Non-central obstruction (2-group cocycle) . . . 24
1.5 Modular Flow and Lorentz Kinematics . . . 25
1.5.1 Modular additivity in the Markov collar limit . . . 25
1.5.2 Theorem: BW_{S^2} on the OPH geometric branch . . . 26
1.5.3 Theorem: BW_{S^2} implies Lorentz kinematics . . . 27
1.6 Gravity from Entanglement Equilibrium . . . 27
1.6.1 Cap first law . . . 27
1.6.2 Null-surface modular bridge . . . 28
1.6.3 Modular energy as stress-tensor charge (UV CFT) . . . 34
1.6.4 Localized generalized entropy from Markov + MaxEnt . . . 34
1.6.5 Entanglement equilibrium from MaxEnt . . . 35
1.6.6 Einstein equation from cap equilibrium . . . 35
1.6.7 Overlaps supply all timelike directions . . . 36
1.6.8 Non-tunable numerical constants . . . 36
1.6.9 Quantitative Markov error and controlled corrections . . . 36
1.6.10 Focusing/QNEC internalization via relative entropy . . . 38
1.6.11 Discrete horizon area spectrum and Hawking emission (established) . . . 40
1.6.12 Classical mechanics from emergent GR . . . 44
1.6.13 Precision gravity predictions and experimental bounds . . . 45
1.6.14 Theory-side error propagation from Markov bounds . . . 46
1.6.15 Dark matter as modular anomaly (program-level) . . . 47
1.6.16 De Sitter holography: static patch vs boundary-at-infinity . . . 49
1.7 Standard Model from Gluing Consistency . . . 50
1.7.1 Edge sector category and gauge group reconstruction . . . 50
1.7.2 Selecting the SM factors (derived from MAR) . . . 51
1.7.3 Refinement stability and unprotected relevant operators . . . 53
1.7.4 Generation number from CP violation and refinement stability (derived from MAR) . . . 54
1.7.5 Hilbert-space formulation of gluing data . . . 55
1.7.6 Loop obstruction class (central defect) . . . 56
1.7.7 EFT reduction to anomaly cancellation . . . 56
1.7.8 Hypercharge from anomaly freedom and Yukawas . . . 57
1.7.9 Witten anomaly and the number of colors . . . 58
1.7.10 Bond-dimension gatekeeping . . . 58
1.7.11 Inevitability of photon and graviton . . . 59
1.7.12 Mass predictions from symmetry . . . 59
1.7.13 Coupling extraction from edge-sector probabilities . . . 61
1.7.14 Numerical validation of the heat-kernel law . . . 63
1.7.15 IBM Quantum Cloud hardware benchmarks . . . 67
1.7.16 Particle mass extraction from spectroscopy . . . 68
1.7.17 Composite masses and the path to predictions . . . 69
1.7.18 Gauge unification and spectrum constraints . . . 69
1.7.19 The Z_6 quotient: edge-sector selection rules and center-label entropy . . . 79
1.7.20 Electroweak scale as a calibration-sector output . . . 80
1.7.21 Heavy-top heuristic and its status . . . 81
1.7.22 Yukawa hierarchy from Z_6 defect suppression . . . 81

1.7.23 Higgs/top critical-surface branch . . . 83
1.7.24 Rigorous derivation chain: axioms to predictions . . . 83
1.8 Open Questions . . . 86
1.9 Critical Evaluation . . . 87
1.9.1 Classification of results and remaining dependence . . . 87
1.9.2 Structural assessment . . . 89
1.9.3 Testable items and continuation templates . . . 89
1.9.4 What is not predicted (gaps) . . . 91
1.9.5 Comparison with other unification approaches . . . 92
1.10 References . . . 94
1.10.1 Foundational results used in this work . . . 94
1.10.2 Standard Model and unification (borrowed results) . . . 95

II Gauge Group Derivation **96**

2 Gauge Group Derivation: Standard Model Closure under Extended MAR **96**
2.1 Premise Summary . . . 96
2.1.1 The Extended Theory . . . 96
2.1.2 What the Core Proves . . . 97
2.1.3 The Transportability Premise . . . 97
2.2 Admissibility Conditions . . . 97
2.2.1 (i) Loop-coherent / transportable . . . 98
2.2.2 (ii) Anomaly-free . . . 98
2.2.3 (iii) Refinement-stable with light chiral matter . . . 98
2.2.4 (iv) Single-Higgs Yukawa-completable . . . 98
2.2.5 (v) Intrinsically CP-capable . . . 98
2.2.6 (vi) Weak-sector UV-completable . . . 99
2.3 Selection Axiom MAR . . . 99
2.4 Main Theorem . . . 100
2.5 Proof: Gauge Group . . . 100
2.5.1 Step 1: Some compact group exists . . . 100
2.5.2 Step 2: The gauge group must be a product . . . 100
2.5.3 Step 3: The factors are SU(3), SU(2), and U(1) . . . 101
2.5.4 Step 4: Nothing else , the commutant argument . . . 102
2.5.5 Step 5: The global quotient is $\mathbb{Z}_6$ . . . 102
2.5.6 Summary of Step 1–5 . . . 103
2.6 Proof: Number of Colors . . . 103
2.7 Proof: Number of Generations . . . 103
2.8 Corollaries . . . 104
2.8.1 Corollary 1: No gauge-mediated proton decay . . . 104
2.8.2 Corollary 2: Uniqueness . . . 104
2.8.3 Corollary 3: Hypercharge quantization . . . 105
2.9 Complete Derivation Chain . . . 105
2.10 Future Work . . . 105

III Spectrum Derivation **106**

3 From Pixel Area to Particle Masses: The Complete OPH Spectrum Derivation 106
3.1 Inputs and Contract . . . 106
3.1.1 Physical Inputs . . . 106
3.1.2 What Counts as "Derived" . . . 106
3.1.3 No-Leakage Guarantee . . . 107
3.2 Complete Audit of All Constants . . . 107
3.2.1 The Physical Input . . . 107
3.2.2 Structural Integers Derived from Axioms . . . 108
3.2.3 Standard QFT Constants (Fixed by the Gauge Group + Matter Content) . . . 110
3.2.4 Dimensional/Definitional Constants . . . 111
3.2.5 Numerical-Method Parameters . . . 111
3.2.6 Summary: What Is and Isn't Assumed . . . 112
3.3 Calibration vs. Prediction: Epistemic Classification of Outputs . . . 113
3.3.1 The Calibration Structure . . . 113
3.3.2 Why "Downstream of P" Does Not Mean "Recovered Core" . . . 113
3.3.3 Classification . . . 113
3.3.4 What This Part Actually Contributes . . . 114
3.4 Stage 1: Fundamental Scales . . . 114
3.4.1 Unification Scale . . . 114
3.4.2 Cell Energy Scale . . . 115
3.5 Stage 2: Gauge Closure and the Pixel Constraint . . . 115
3.5.1 The Heat-Kernel Entropy . . . 115
3.5.2 The Pixel Constraint . . . 116
3.5.3 Gauge Running and the α_U Solution . . . 116
3.5.4 Self-Consistent Fixed Point . . . 116
3.6 Stage 3: Electroweak Observables . . . 116
3.6.1 Z Boson Pole Mass . . . 117
3.6.2 W Boson Mass . . . 117
3.7 Stage 4: Critical Surface , Higgs and Top . . . 117
3.7.1 The Critical Surface Constraint . . . 117
3.7.2 RG Evolution to Low Scales . . . 118
3.7.3 Top Mass . . . 118
3.7.4 Higgs Mass . . . 119
3.8 Stage 5: Discrete Spectrum , Quarks and Leptons . . . 119
3.8.1 The $\mathbb{Z}_6$ Defect Parameter . . . 119
3.8.2 Integer Exponents (Algorithmically Derived) . . . 119
3.8.3 The Koide Phase . . . 119
3.8.4 Diagonal Quark Masses . . . 119
3.8.5 CKM Mixing . . . 120
3.8.6 Charged Lepton Masses . . . 120
3.9 Stage 6: QCD Scale and Hadron Masses . . . 120
3.9.1 $\Lambda_{\overline{\mathrm{MS}}}$ from $\alpha_s(m_{Z,\mathrm{run}})$ . . . 120
3.9.2 Hadron Masses . . . 120
3.10 Stage 7: Neutrino Masses . . . 121
3.10.1 The Capacity Model . . . 121
3.11 Massless Particles . . . 121
3.12 Complete Spectrum: OPH Outputs vs PDG . . . 122
3.12.1 Gauge Bosons and Higgs . . . 122

3.12.2 Charged Leptons . . . 122
3.12.3 Neutrinos . . . 123
3.12.4 Quarks . . . 123
3.12.5 Gauge Couplings at $m_{Z,\mathrm{run}}$ . . . 125
3.12.6 Derived Scales . . . 125
3.12.7 Hadron Masses . . . 125
3.13 Analysis of Discrepancies . . . 126
3.13.1 Sub-Permille Outputs ($< 0.1\%$) . . . 126
3.13.2 Percent-Level Outputs ($0.1\% - 2\%$) . . . 126
3.13.3 Qualitative Agreement Only: Quark Masses ($16\% - 73\%$) . . . 126
3.13.4 Hadron Masses . . . 127
3.13.5 Neutrinos: No Direct Comparison Available . . . 127
3.14 Reproduction Instructions . . . 127
3.14.1 Prerequisites . . . 127
3.14.2 Code Files . . . 128
3.14.3 Running the Full Prediction . . . 128
3.14.4 No-Leakage Verification . . . 128
3.14.5 Individual Stages . . . 129
3.14.6 Fetching PDG Reference Data . . . 129
3.15 Summary . . . 129

IV String-Theory Derivation **130**

4 String Theory from Observer Patch Holography **130**
4.1 A Continuation-Level Route . . . 130

5 Overview: The Claimed Route **130**

6 OPH Inputs Used **130**
6.1 OPH Axioms . . . 131
6.2 Technical Premises . . . 131
6.3 Key Derived Results . . . 131

7 Einstein Equation from Entanglement Equilibrium **131**
7.1 Setup: Null Deformation Problem . . . 131
7.2 Generalized Entropy Variation . . . 132
7.3 Area Variation via Raychaudhuri . . . 132
7.4 Bulk Entropy Variation via Modular Theory . . . 132
7.5 Entanglement Equilibrium Implies Einstein . . . 132
7.6 Upgrade to Full Einstein Equation . . . 133

8 Edge Sectors and 2D Yang-Mills **133**
8.1 Edge-Sector Structure . . . 133
8.2 Heat-Kernel Weights from MaxEnt . . . 133
8.3 Edge Partition Function . . . 133
8.4 Identification with Group Heat Kernel . . . 134
8.5 Gluing = Markov = Area Additivity . . . 134
8.6 Exact Match with 2D Yang-Mills . . . 134

9 2D Yang-Mills as String Theory **134**
9.1 The Gross-Taylor Expansion . . . 134
9.2 Worldsheet Interpretation . . . 135
9.3 The Large-N Regime . . . 135
9.4 Genus Expansion . . . 135

10 Worldsheet Kinematics from Modular Flow **135**
10.1 Conformal Structure . . . 135
10.2 Sewing Structure . . . 135
10.3 The Analogy with Standard String Theory . . . 136

11 The B-Field Slot: Higher Gauge Structure **136**
11.1 Nonabelian Gluing Cocycles . . . 136
11.2 Strictification and Trivialization . . . 136
11.3 D-Brane Interpretation . . . 136

12 OPH String Scale and Parameters **136**
12.1 String Scale from Pixel Area . . . 136
12.2 Worldsheet YM Coupling . . . 137
12.3 Current Algebra Central Charge (WZW) . . . 137

13 From 2D Yang-Mills to Critical Superstrings **137**
13.1 Established Results . . . 138
13.2 Extensions to Critical Superstrings . . . 138

14 Summary **138**

15 Appendix A: Entanglement First Law Numerical Check **139**
15.1 A.1 Setup . . . 139
15.2 A.2 Expected Result . . . 139
15.3 A.3 Code . . . 139

16 Appendix B: Key Equations Summary **139**

17 References **140**

V Technical Supplement **141**

18 Technical Supplement: Mathematical Foundations of OPH **141**
18.1 For Theoretical Physicists . . . 141

19 Core Axioms and Definitions **141**
19.1 Five OPH Axioms . . . 141
19.2 Technical Premises and Regulator Names . . . 142
19.3 Edge-Center Completion (EC) . . . 143
19.4 External Continuous Inputs . . . 144

20 Emergence of Gravity and Quantum Mechanics **144**
20.1 Local Lorentz Structure . . . 144
20.2 Geometric Modular Flow . . . 144
20.3 Null Modular Bridge (EFT Conditions N1-N3) . . . 145
20.4 Stress Tensor Reconstruction . . . 148
20.5 Entanglement Equilibrium and Einstein's Equation . . . 149
20.6 Newton's Constant from Edge Entropy . . . 149

21 The Measurement Problem **149**
21.1 Records as Central Variables . . . 149
21.2 Markov Structure Theorem . . . 150
21.3 Approximate Markov and Recovery . . . 150
21.4 Definiteness of Outcomes . . . 151
21.5 Born Rule as Unique Probability Measure . . . 151
21.6 Collapse as Conditioning . . . 152

22 The Cosmological Principle **152**
22.1 MaxEnt Symmetry Inheritance . . . 152
22.2 Isotropy from SO(3)-Invariant Constraints . . . 152
22.3 Homogeneity from Isotropy Everywhere . . . 152
22.4 FLRW Emergence . . . 153

23 Supersymmetry and Gauge Coupling Unification **153**
23.1 One-Loop Unification Algebra . . . 153
23.2 SM vs MSSM Coefficients . . . 153
23.3 Heat-Kernel Edge Sector Weights . . . 154
23.4 Peter-Weyl and Effective Multiplicity . . . 154
23.5 Beta Function Shifts . . . 154

24 The Horizon Problem **155**
24.1 Anisotropy Bound from Markov Control . . . 155
24.2 CMB Anisotropy Requirement . . . 155
24.3 Homogeneity as MaxEnt Default . . . 155

25 Black Hole Information Paradox **155**
25.1 The Information Trilemma . . . 155
25.2 Edge-Center Resolution . . . 156
25.3 Recoverability of Interior . . . 156
25.4 Hawking Temperature from Modular Theory . . . 156
25.5 Discrete Area Spectrum . . . 156

26 The Cosmological Constant Problem **156**
26.1 Vacuum Energy Blindness . . . 156
26.2 Structural Separation . . . 157
26.3 Λ from Capacity . . . 157
26.4 No-Go for Local Λ Prediction . . . 157

27 Dark Matter as Modular Anomaly **157**
27.1 The Modular Additivity Defect . . . 157
27.2 Anomalous Stress-Energy . . . 158
27.3 Properties of the Anomaly Sector . . . 158
27.4 The MOND Acceleration Scale . . . 158
27.5 Polarization Response (MOND-like Scaling) . . . 158

28 Proton Stability and Spin **159**
28.1 Sector Factorization and Product Group . . . 159
28.2 No Leptoquarks, No Gauge Proton Decay . . . 159
28.3 Proton Spin Fraction . . . 159

29 Baryogenesis **160**
29.1 The Z_6 Defect Suppression . . . 160
29.2 The Baryon-Violating Channel . . . 160
29.3 Defect-Order Identification . . . 160
29.4 CP Bias from Overlap Holonomy . . . 160

30 Generation Structure and Yukawa Hierarchy **160**
30.1 Why Three Generations . . . 160
30.2 Yukawa as Defect-Mediated Overlap . . . 161
30.3 The Yukawa Spectrum . . . 161
30.4 Deterministic Completion (RK Hamiltonian) . . . 161

31 The Koide Formula **161**
31.1 Numerical Verification . . . 161
31.2 Z_3 Holonomy Parameterization . . . 162
31.3 Why Q = 2/3 . . . 162
31.4 The Koide Phase from OPH . . . 162

32 Cosmological Parameters **163**
32.1 The de Sitter Scale . . . 163
32.2 Emergent Age of the Universe . . . 163
32.3 Screen Capacity . . . 163
32.4 Pixel Parameters . . . 163

33 Appendix A: Summary of Derived Results **163**
33.1 A.1 Theorem-Level Results and Dependency Status . . . 163
33.2 A.2 Secondary Quantitative and Continuation Benchmarks . . . 164
33.3 A.3 Future Directions . . . 165

34 Appendix B: Mixed-Status Numerical Benchmarks **165**

35 References **166**

VI Interpretive Appendix: Strange-Loop Proposal **167**

36 Interpretive Status, the Habitat Theorem, and the Remaining Strange-Loop Gap 167

37 Timeless Causal Chain 169

38 Why Reality Exists and Why It Has This Form 169

39 Additional Problem Closures 170

40 Standalone Status 171

VII Speculative Appendix: Observer Continuation and Backup 172

41 Observer as Algebraic Pattern 172

42 Markov Collar Factorization 172

43 Checkpoint and Restoration Map 172

44 Physical Meaning 173

Part I
Core Framework and Axioms

1 Observer-Patch Holography

OPH is an observer-centric reconstruction program for fundamental physics. Physical data live on a horizon screen S^2, and reality is encoded by the mutual consistency of overlapping patch descriptions. This paper works with five OPH axioms, two explicit external inputs used only in the quantitative branches, and theorem-local technical premises stated where needed for the scaling, gauge, and transport arguments below.

1.1 Overview

The recovered core of the paper is the D1–D5 and D7–D9 chain: overlap-consistency normal form, collar and edge-center structure, the conditional Lorentz and Einstein branches, compact gauge reconstruction, and the realized Standard Model branch with exact hypercharge structure, $N_g = 3$, $N_c = 3$, and symmetry-protected massless gauge bosons. D6 is a separate input-dependent corollary tied to total screen capacity. D10 and D11 are quantitative branches tied to the pixel-area input and the running and matching conventions stated later in the manuscript. D12 collects phenomenological continuations.

The Lorentz/null-modular/Einstein chain is a scaling-limit statement with explicit fixed-cutoff control. The gauge chain reconstructs a compact group from a refinement-stable sector colimit and then specializes to the realized MAR-admissible branch. The worldsheet/string material and the flavor, neutrino, hadron, dark-sector, and spectroscopy branches are kept separate from the recovered core.

Two external continuous inputs appear only where declared. The pixel area $P \equiv a_{\text{cell}}/\ell_P^2$ sets the particle-physics scale in the quantitative branch. The screen capacity $N_{\text{scr}} \equiv \log \dim \mathcal{H}_{\text{tot}}$ enters the cosmological-capacity branch. They are implementation inputs, not additional axioms.

1.2 Model and Axioms

1.2.1 Observers and access model

An observer O is a tuple $(P_O, \mathcal{A}(P_O), \rho_O, R_O)$ where:

- $P_O \subset S^2$ is a connected screen patch (the observer's access region).
- $\mathcal{A}(P_O)$ is the von Neumann algebra associated to P_O.
- ρ_O is the local state, obtained by restricting the global state to $\mathcal{A}(P_O)$.
- R_O is a set of records: stable internal correlations within P_O.

Observers are internal patterns in the global state. Different observers correspond to different patches and their compatible marginals.

1.2.2 Screen, patches, and algebra net

We work in a single static patch with a horizon screen S^2. Each connected subregion $P \subset S^2$ is assigned a von Neumann algebra $\mathcal{A}(P)$. The net satisfies isotony:

$$P \subset Q \implies \mathcal{A}(P) \subset \mathcal{A}(Q).$$

A global state ω is a positive linear functional on the inductive-limit algebra. Overlap consistency is imposed algebraically: for overlaps $P_1 \cap P_2$, ω restricted to $\mathcal{A}(P_1 \cap P_2)$ is the same from either side.

1.2.3 Five OPH axioms

This paper uses five OPH axioms. Later theorem statements sometimes use theorem-local shorthand such as premise ME (MaxEnt), premise ROT (rotational symmetry), premise RS (refinement stability), premise MX (mixing), and regulator premises R0–R1; Section 1.4 defines them.

1. Screen Net. A horizon screen S^2 carries a net of algebras $P \mapsto \mathcal{A}(P)$.

2. Overlap Consistency. Local states agree on shared observables for any overlap.

3. Local MaxEnt and Refinement Stability. At the UV scale the realized state maximizes entropy subject to a finite family of gauge-invariant local constraints, and the resulting family of states is stable under coarse-graining so symmetry-allowed relevant operators are not held at zero by unexplained fine tuning.

4. Recoverable Generalized Entropy. A generalized entropy functional exists on caps, obeys quantum focusing on null generators, and comes with the recoverability structure used throughout the manuscript: collar tripartitions have small conditional mutual information with controlled recovery maps, with stronger collar/null-strip hypotheses stated explicitly where needed.

5. Minimal Admissible Realization (MAR). On the admissible low-energy branch, the realized sector package is the lexicographically minimal one under the complexity vector $C(\mathfrak{S})$. MAR is an explicit structural-economy axiom on admissible realized branches, not a theorem derived from the preceding four axioms.

1.2.4 External inputs and theorem-local technical premises

The quantitative branches discussed here use exactly two external continuous inputs:

$$P \equiv a_{\text{cell}}/\ell_P^2 = 1.63094, \qquad N_{\text{scr}} \equiv \log \dim \mathcal{H}_{\text{tot}} \sim 10^{122}.$$

Here P is calibrated from the gauge sector and N_{scr} is inferred from the observed cosmological constant. The first sets the particle-physics scale; the second enters the cosmological-capacity branch and a separate order-of-magnitude neutrino estimate. These are implementation inputs. They are not extra axioms.

Later theorem statements may abbreviate the named local premises by ME, ROT, GG, CD, COL, GB, RS, LR, MX, R0, R1, and $[z] = 0$. These shorthands do not introduce additional axioms.

Premise ME (MaxEnt selection with local constraints): At the regulator scale ℓ_{UV}, the global state ω maximizes von Neumann entropy subject to:

1. A finite set $\{O_a\}$ of gauge-invariant local operators, each supported on a ball of radius $\leq r_0 = O(\ell_{\text{UV}})$.
2. Constraint equations $\langle O_a(x) \rangle = c_a$ for each cell x in the UV lattice.
3. Optionally, a finite number of global constraints (total energy, charge).

This is the minimal specification that turns MaxEnt into a theorem-engine for deriving the local Gibbs form (Lemma 2.6).

Clarification (MaxEnt $\neq$ thermal equilibrium). MaxEnt here is **local state selection** given constraints, not "the universe is in thermal equilibrium." The Lagrange multipliers (inverse temperatures) may vary slowly in space and time. Non-equilibrium physics appears as gradients

in these multipliers and as controlled violations of exact Markov additivity (bounded by premise MX). Equilibrium is an approximation regime with explicit error terms.

Premise ROT (Rotationally invariant constraints): Constraint sets are SO(3)-invariant on S^2.

Premise GG (Gauge-as-gluing): Overlap identifications are not unique; the freedom that leaves overlap observables invariant forms a local groupoid.

Premise CD (Central defect): On triple overlaps, the only failure of strict coherence is central, so

$$\varphi_{ij}\varphi_{jk}\varphi_{ki} = \mathrm{Ad}(z_{ijk}), \qquad z_{ijk} \in Z(\mathcal{A}_{ijk}).$$

Premise COL (Collar refinement, double scaling): There exists a UV length ℓ_{UV} such that for any cap C and collar width δ, in the refinement limit $\delta \to 0$ and $\ell_{\mathrm{UV}} \to 0$ with $\delta/\ell_{\mathrm{UV}} \to \infty$, the Markov error satisfies

$$I(A_\delta : D_\delta \mid B_\delta)_\omega \le \varepsilon(\delta/\ell_{\mathrm{UV}}), \qquad \varepsilon(x) \to 0 \text{ as } x \to \infty.$$

See Section 2.3 for the collar tripartition definitions and the regulated EC proof that yields this limit under R0/R1.

Premise GB (cap-modular geometric branch): The refinement-stable OPH branch for caps lies in the geometric modular class, together with the controlled collar/refinement limit that carries the Markov and recovery remainders explicitly.

Derived LR / modular-locality control: At scale ℓ_{UV}, the dynamics generated by the effective Hamiltonian H_{eff} (from Lemma 2.6) has a finite Lieb-Robinson velocity v_{LR}: for local operators A, B supported on regions separated by distance d,

$$\|[A(t), B]\| \le c\|A\|\|B\| \min(|R_A|, |R_B|) e^{-(d - v_{\mathrm{LR}}|t|)/\xi}$$

for $d > v_{\mathrm{LR}}|t|$, where $\xi = O(\ell_{\mathrm{UV}})$.

This is the branch-internal control statement that turns the quasi-local structure of LG into explicit support control for time-evolved operators; it is not an extra premise beyond the third OPH axiom.

Approximate modular covariance on the geometric branch. Under R0 + Lemma 2.6 (local Gibbs) + the derived LR control + collar refinement (COL/MX), the modular flow $\sigma_t^{\omega,C}$ maps $\mathcal{A}(R)$ into a slightly thickened region algebra:

$$\sigma_t^{\omega,C}(\mathcal{A}(R)) \subseteq \mathcal{A}(R^{+v_{\mathrm{mod}}|t|}) \quad \text{up to error } \eta(d - v_{\mathrm{mod}}|t|),$$

where R^{+s} denotes the s-neighborhood thickening, v_{mod} is a "modular propagation velocity" controlled by v_{LR} and local norm bounds, and d is the distance from R to ∂C.

Heuristic motivation. The modular Hamiltonian $K_C = -\log \rho_C$ is quasi-local by Lemma 4.1a-b (modular additivity localizes it to the collar). The derived Lieb-Robinson control then bounds support spreading under $e^{iK_C t}$. In the double-scaling collar limit ($\delta/\ell_{\mathrm{UV}} \to \infty$), the thickening vanishes in macroscopic units. This is heuristic support for the tangent-limit package used later.

Continuum-limit heuristic. Define the induced region flow

$$f_t^C(R) := \lim_{\ell_{\mathrm{UV}} \to 0} R^{+v_{\mathrm{mod}}|t|}.$$

Then $\sigma_t^{\omega,C}(\mathcal{A}(R)) = \mathcal{A}(f_t^C(R))$ becomes exact in the continuum limit, with error controlled by

$$\eta(\delta) \lesssim 2\sqrt{\ln 2 \cdot c \cdot |\partial C|_{\mathrm{UV}}}\, e^{-\delta/(2\xi)}.$$

The discussion above is heuristic support for the tangent-limit package used in Theorem 4.2.

Premise RS (Refinement stability / RG consistency): There exists a family of coarse-graining channels $\Phi_{\ell\to L}$ between UV scale ℓ and IR scale L such that the MaxEnt-selected states are self-similar under refinement,

$$\Phi_{\ell\to L}(\omega_\ell) = \omega_L,$$

with the constraint set fixed and finite. Equivalently, the MaxEnt family is an RG fixed point or a low-dimensional stable manifold determined only by the constraints.

Regulator premises (R0, R1): At a UV scale ℓ_{UV}, local patch algebras are type-I with finite-dimensional Hilbert spaces, and gauge-as-gluing is realized as a boundary group action whose fixed-point algebra defines physical observables. These premises are used in Section 2.3 to derive EC.

External mathematical inputs: SSA and recovery theorems (Petz 1986, 1988; Fawzi and Renner 2015), Jacobson's entanglement-equilibrium derivation (Jacobson 1995, 2016), and one of the following EFT bridges: (i) the null-surface modular route (Section 5.2), or (ii) a UV CFT regime on sufficiently small caps (Section 5.3). For SM contact we also use the Doplicher-Roberts reconstruction (Doplicher and Roberts 1989, 1990) once localized transportable sectors are assumed in the small-region limit. Full citations appear in the References.

Label convention. Later theorem statements may abbreviate premise ME (MaxEnt), premise RS (refinement stability), premise ROT (rotational symmetry), premises COL and MX (collar control and mixing), and the regulator or gauge premises R0/R1, LR, GG, CD, $[z] = 0$, and GB. These labels do not introduce additional axioms.

1.2.5 Notation

- ρ_C: reduced state on cap C.
- $K_C := -\log \rho_C^\omega$: modular Hamiltonian of the reference state.
- B_C: geometric generator of the cap-preserving conformal dilation.
- $S_{\mathrm{gen}}(C)$: generalized entropy on a cap.
- ℓ_{UV}: UV length scale of the refined screen net.
- δ: collar width around a cap boundary.

1.2.6 Summary: premise ledger and dependency DAG

For reference, the paper uses the five OPH axioms above, the two external inputs P and N_{scr} only in the branches that require them, and the theorem-local technical premises declared in Section 1.4.

Item	Meaning, typical use, and status
Axiom 1	Screen Net; use: global net structure; status: axiom.
Axiom 2	Overlap Consistency; use: compatibility on overlaps; status: axiom.
Axiom 3	Local MaxEnt and Refinement Stability; use: state selection and RG stability; status: axiom.
Axiom 4	Recoverable Generalized Entropy; use: focusing plus recoverability structure; status: axiom.

Item	Meaning, typical use, and status
Axiom 5	Minimal Admissible Realization (MAR); use: selection on admissible gauge branches; status: axiom.
P	Pixel area $a_{\mathrm{cell}}/\ell_P^2$; use: quantitative particle-physics scale; status: external input.
N_{scr}	Total screen capacity; use: cosmological-capacity branch; status: external input.
ME (MaxEnt), **RS (refinement stability)**	Regulator-scale MaxEnt clause and refinement-stability clause of Axiom 3; use: Sections 2, 4, and 6; status: theorem-local shorthand.
ROT (rotational symmetry)	Rotationally invariant constraint set on S^2; use: the rotational-symmetry branch in Theorem 4.2 and the cosmological-principle continuation in the Technical Supplement; status: theorem-local technical premise.
COL (collar refinement), **MX (mixing)**	Recoverability and the theorem-local collar/null-strip hypotheses used under Axiom 4; use: Sections 2 and 5; status: theorem-local shorthand for explicit local hypotheses, not a separate axiom package.
R0, **R1**	Type-I regulator and boundary gauge invariants; use: EC and gauge reconstruction; status: technical premise.
LR	Derived Lieb-Robinson locality at the UV scale; use: modular-locality control; status: theorem-local shorthand from Axiom 3.
GG (gauge-as-gluing), **CD (central defect)**, $[z] = 0$	Gluing, central-defect, and transportability hypotheses; use: gauge branch; status: technical premise.
GB (geometric branch)	Cap-modular tangent-limit package used in Theorem 4.2; use: BW/KMS normalization and geometric modular action; status: theorem-local scaling-limit package.

Theorem-local shorthands used later include the cap-modular tangent-limit package GB and the EFT-bridge labels N1–N3. These are packaged hypotheses or alternative routes inside the technical program; none of them replace the five OPH axioms.

Master dependency map. The completion program is organized into the following nodes. The point of the table is to make explicit which results use the OPH axioms alone, which import standard mathematics, and which require extra physical premises or external inputs.

Node	Main-manuscript output	Immediate OPH ingredients	Standard mathematics used	Extra physical premises / external inputs	Claim tier
D1	Theorem 3.1: schedule-independent normal-form statement	overlap-consistency setup on finite patch nets	Newman's lemma	finite patch net, termination, and local confluence	structural theorem
D2	Section 2.3 and §5.4: EC/Markov collar and generalized-entropy split	Axioms 1–4 plus R0/R1 when invoked	HJPW; Petz; Fawzi–Renner	exact-Markov or idealized recoverability hypotheses when exact identities are used; coarse-grained $A/(4G)$ dictionary	structural lemma/proposition layer
D3	Theorems 4.2–4.3: geometric modular flow and Lorentz branch	Axioms 1–4 plus premise ROT	Bisognano–Wichmann modular geometry; conformal-group classification	controlled refinement limit together with the OPH Bisognano–Wichmann geometric-branch hypothesis used in Theorem 4.2	conditional scaling-limit theorem

Node	Main-manuscript output	Immediate OPH ingredients	Standard mathematics used	Extra physical premises / external inputs	Claim tier
D4	Section 5.2: null modular bridge	D2+D3	Borchers–Wiesbrock and endpoint differentiation	the theorem-local inherited-strip decomposition and exact-or-controlled Markov hypotheses of Section 5.2, plus the downstream density-upgrade hypotheses and the relativistic null-stress premise	conditional bridge theorem
D5	Theorem 5.1: Einstein branch	D3+D4	Jacobson small-ball mathematics; null-to-tensor reconstruction modulo a metric term	fixed-cap stationarity and the locally Lorentzian $d = 4$ scaling regime	conditional scaling-limit theorem
D6	capacity/Λ reduction statements	D5 leaves $+\Lambda g_{ab}$ undetermined	de Sitter entropy relation and dimensional analysis	external input N_{scr} and the screen-capacity identification	input-dependent corollary / estimate
D7	Theorem 6.1: compact gauge reconstruction	Axioms 1–4	Doplicher–Roberts / Tannaka reconstruction	rigid symmetric bosonic colimit, transportability, and fiber-functor premises	conditional structural theorem
D8	Section 6.2: product gauge structure up to finite quotient	D7 + Axiom 5 (MAR)	compact Lie representation classification; Schur's lemma	connected admissible class, one connected abelian factor, and $[z] = 0$	realized-branch theorem
D9	Theorem 6.13, Proposition 6.9, Theorem 6.14, Proposition 6.6: hypercharge lattice, the counting chain $N_g = 3$ then $N_c = 3$, and the $\mathbb{Z}_6$ quotient	D8	anomaly algebra; Witten anomaly; CKM CP counting	realized one-generation/one-Higgs package and MAR admissibility premises	realized-branch theorem/corollary chain
D10	gauge/electroweak calibration closure documented in Part III and the Technical Supplement	D9 + P	RG running and matching conventions recorded in the supplement	pixel closure and the printed threshold conventions	calibration sector
D11	Section 6.22 Higgs/top critical-surface branch	D10	RG transport recorded in Part III and the Technical Supplement	critical-surface condition $\lambda = \beta_\lambda = 0$ at M_U together with the UV synchronization scale μ_{sync} used in the transport layer	quantitative branch
D12	downstream flavor, dark-sector, baryogenesis, spectroscopy, and string branches	various subsets of D6, D9, D10, and D11	branch-specific EFT and phenomenological manipulations	explicit additional ansätze	phenomenological continuation / program branch

In particular, D3–D5 are scaling-limit branches rather than fixed-cutoff matrix-algebra theorems; D7–D9 are realized-branch structural outputs in the bosonic internal-gauge sector; D10–D11 are quantitative branches; and D12 remains phenomenological.

1.3 Information-Theoretic Tools

1.3.1 Strong subadditivity and Markov states

For any tripartite state ρ_{ABC},

$$I(A:C \mid B) := S(AB) + S(BC) - S(B) - S(ABC) \geq 0.$$

Exact Markov states satisfy $I(A:C \mid B) = 0$ and admit a recovery map:

$$\rho_{ABC} = (\mathrm{id}_A \otimes \mathcal{R}_{B\to BC})(\rho_{AB}).$$

1.3.2 Approximate recovery

If $I(A:C \mid B) \leq \varepsilon$ (bits), there exists a CPTP recovery map $\mathcal{R}$ with

$$\|\rho_{ABC} - (\mathrm{id}_A \otimes \mathcal{R})(\rho_{AB})\|_1 \leq 2\sqrt{\ln 2\, \varepsilon}.$$

1.3.3 Collar refinement and sufficient mechanisms

Fix a cap $C \subset S^2$ with boundary circle. For collar width δ define

$$B_\delta := \{x \in S^2 : \mathrm{dist}(x, \partial C) \leq \delta\}, \qquad A_\delta := C \setminus B_\delta, \qquad D_\delta := (S^2 \setminus C) \setminus B_\delta.$$

Then $S^2 = A_\delta \cup B_\delta \cup D_\delta$ with A_δ and D_δ interacting only through B_δ. Premise COL is the requirement that $I(A_\delta : D_\delta \mid B_\delta) \to 0$ in the collar double-scaling limit. At fixed regulator scale, R0/R1 provide the finite-dimensional realization of the same patch-net and overlap-gluing data used below. We therefore isolate that realized presentation first, then separate the exact-Markov and quantitative routes.

Derived regulator realization (read-along for R0/R1). Choose a finite UV cellulation at scale ℓ_{UV} and let R be a finite union of cells. Hilbertize each cell's finite local data to a finite-dimensional space $\tilde{\mathcal{H}}_i \cong \mathbb{C}^{n_i}$. The extended algebra before quotienting by overlap redundancy is

$$\widetilde{\mathcal{A}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R), \qquad \tilde{\mathcal{H}}_R = \bigotimes_{i \subset R} \tilde{\mathcal{H}}_i.$$

Overlap-preserving changes of local trivialization act only on cut data. On each finite-dimensional chart, their unitary image has compact closure, so one may represent the boundary gluing redundancy by a compact group $G_{\partial R} \subset U(\tilde{\mathcal{H}}_R)$. The physical algebra is the fixed-point algebra

$$A(R) = \widetilde{\mathcal{A}}(R)^{G_{\partial R}} = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

Historical shorthand: $R0$ means the finite type-I extended presentation above; $R1$ means the boundary fixed-point realization of overlap gluing.

Theorem 2.3 (EC from regulated overlap gluing). For a collar B_δ around a cap boundary Σ, there is a canonical decomposition

$$H_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_\Sigma} = \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

with

$$Z(A(B_\delta)) = \bigoplus_\alpha \mathbb{C}\,\mathbf{1}_\alpha,$$

such that $\mathcal{A}(A_\delta B_\delta)$ acts only on $H_{b_L^\alpha}$ and $\mathcal{A}(B_\delta D_\delta)$ acts only on $H_{b_R^\alpha}$ within each block.

Proof. Split the collar into half-collars B_L and B_R meeting on $\Sigma = \partial C$. By the realized regulator presentation above, the physical collar Hilbert space is the diagonal invariant subspace $(\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{G_\Sigma}$. Decompose each side into irreps:

$$\tilde{\mathcal{H}}_{B_L} = \bigoplus_\alpha (V_\alpha \otimes H_{b_L^\alpha}), \qquad \tilde{\mathcal{H}}_{B_R} = \bigoplus_\beta (V_\beta^* \otimes H_{b_R^\beta}).$$

Then

$$\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R} = \bigoplus_{\alpha,\beta} (V_\alpha \otimes V_\beta^*) \otimes (H_{b_L^\alpha} \otimes H_{b_R^\beta}).$$

By Schur's lemma,

$$(V_\alpha \otimes V_\beta^*)^{G_\Sigma} \cong \begin{cases} \mathbb{C}, & \alpha = \beta, \\ 0, & \alpha \neq \beta. \end{cases}$$

Therefore the invariant subspace is

$$H_{B_\delta} = \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

as claimed. The invariant algebra is

$$A(B_\delta) = \bigoplus_\alpha (B(H_{b_L^\alpha}) \otimes B(H_{b_R^\alpha})),$$

so the center is generated by the block projectors. Adjacent region algebras act on the left or right factor only because the gluing action is supported on Σ. QED.

Remark. If premise CD holds, replace G_Σ by its central extension. The sector label α then ranges over irreps of the extension; the decomposition is unchanged.

We refer to the decomposition in Theorem 2.3 as **edge-center completion (EC)**.

Exact Markov route. If the reference state on $A_\delta B_\delta D_\delta$ is exact Markov, or in an explicitly stated idealized limit that reduces to exact Markovity, then

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_\alpha p_\alpha \big(\rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta}\big), \qquad I_\omega(A_\delta : D_\delta \mid B_\delta) = 0.$$

This is the HJPW normal form applied to the EC blocks. It is the exact identity used when the manuscript writes literal Markov-modular equalities.

Interpreting collar refinement as the inductive limit of these regulators with $\delta/\ell_{\mathrm{UV}} \to \infty$, Theorem 2.3 supplies the kinematic edge-center decomposition. Exact Markovity is one idealized route. The following lemma and axiom provide the quantitative decay route used when the manuscript keeps the approximation explicit.

Lemma 2.6 (MaxEnt with local constraints implies local Gibbs form). Under premise ME (MaxEnt with local constraints) and the finite-dimensional regulator realization above, the MaxEnt state has the Gibbs form

$$\omega = \frac{e^{-H_{\text{eff}}}}{\operatorname{Tr} e^{-H_{\text{eff}}}}, \qquad H_{\text{eff}} = \sum_x \sum_a \lambda_a O_a(x) + (\text{global terms}),$$

where the sum runs over UV cells x and constraint operators O_a. The effective Hamiltonian H_{eff} is quasi-local with range $O(\ell_{\text{UV}})$.

Proof. On a finite-dimensional algebra, the unique state maximizing $S(\rho) = -\operatorname{Tr}(\rho \log \rho)$ subject to linear constraints $\operatorname{Tr}(\rho O_i) = c_i$ is given by Lagrange multipliers:

$$\rho = \frac{e^{-\sum_i \lambda_i O_i}}{\operatorname{Tr} e^{-\sum_i \lambda_i O_i}}.$$

Strict concavity of von Neumann entropy ensures uniqueness. When the constraints are "translated local" (the same O_a at each cell x), the exponent is a sum of local terms. QED.

Premise MX (Exponential mixing): There exist constants c and correlation length $\xi = O(\ell_{\text{UV}})$ such that

$$I_\omega(A_\delta : D_\delta \mid B_\delta) \leq c\, |\partial C|_{\text{UV}}\, e^{-\delta/\xi}, \qquad |\partial C|_{\text{UV}} \sim \frac{\text{length}(\partial C)}{\ell_{\text{UV}}}.$$

This is the standard clustering/mixing condition for local Gibbs states, equivalent to assuming the MaxEnt state lies in a Dobrushin uniqueness regime or has a uniform spectral gap. It is a physically natural condition on the UV state and is not derived from premise ME.

Theorem 2.5 (Local Gibbs + mixing implies collar refinement). Under Lemma 2.6 (local Gibbs form from premise ME) and premise MX (exponential mixing), premise COL holds in the collar double-scaling limit $\delta \to 0$, $\ell_{\text{UV}} \to 0$ with $\delta/\ell_{\text{UV}} \to \infty$.

Proof. The bound in MX has polynomial growth in $|\partial C|_{\text{UV}}$ and exponential decay in δ/ℓ_{UV}. In the double-scaling limit the exponential dominates, so $I_\omega(A_\delta : D_\delta \mid B_\delta) \to 0$. QED.

This bound is the quantitative hinge for constructive gluing.

1.3.4 Concrete UV realization: quantum link models

Section 2.3 internalizes the regulator package: the fixed-cutoff type-I algebra and boundary fixed-point structure are the realized form of the screen net plus overlap gluing. Quantum link models are included here only as an explicit microscopic example of that structure, not as a separate axiom layer.

UV regulator. Triangulate S^2 at scale ℓ_{UV}, giving vertices v, oriented links ℓ, and plaquettes p. Refinement corresponds to $\ell_{\text{UV}} \to 0$ with increasing lattice size.

Degrees of freedom. Attach to every oriented link ℓ a **finite-dimensional** Hilbert space $\mathcal{H}_\ell$. In ordinary Wilson lattice gauge theory, the continuum/refinement-limit edge description is modeled by $L^2(G)$ (infinite-dimensional for continuous G), but the microscopic OPH regulator is instead a **quantum link model** with finite-dimensional link Hilbert spaces that preserve gauge symmetry in operator form (see Chandrasekharan and Wiese, hep-lat/9609042). Optionally attach matter Hilbert spaces $\mathcal{H}_v$ at vertices. Then:

$$\tilde{\mathcal{H}}_{\text{total}} = \bigotimes_{\ell} \mathcal{H}_\ell \otimes \bigotimes_{v} \mathcal{H}_v,$$

finite-dimensional on any finite lattice. This is a concrete realization of the extended type-I presentation used in Section 2.3.

Boundary gluing as Gauss-law invariants. Define a local gauge transformation group G_v at each vertex v acting on incident links (and matter at v). Physical states satisfy:

$$|\psi\rangle \in \mathcal{H}_{\text{phys}} \quad \Longleftrightarrow \quad U(g_v)|\psi\rangle = |\psi\rangle \;\; \forall\, v,\, g_v \in G_v.$$

Equivalently: $\mathcal{H}_{\text{phys}} = \tilde{\mathcal{H}}_{\text{total}}^{\prod_v G_v}$.

Region algebras. For any region $R \subset S^2$, define an extended Hilbert space $\tilde{\mathcal{H}}_R$ from the links/vertices in R. The **boundary gauge group** $G_{\partial R}$ acts on the cut degrees of freedom (the "half-links" ending on ∂R). Define:

$$\mathcal{A}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

This is the concrete quantum-link instance of the fixed-point algebra used in Section 2.3. The same definition gives isotony, overlap consistency, and the edge-center structure on collars.

Why EC follows immediately. Take a cap C and a collar B_δ around ∂C. Because the *only* coupling between inside and outside is through the boundary gauge constraint, the collar Hilbert space decomposes into superselection blocks labeled by boundary irreps:

$$\mathcal{H}_{B_\delta} \cong \bigoplus_{\alpha} (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

with center generated by the projectors P_α. This is precisely the Schur-lemma mechanism of Theorem 2.3. The labels α are the familiar edge-mode / electric-flux labels appearing whenever one factorizes gauge theories across an entangling cut (see Donnelly and Wall, PRL 114 (2015)). Exact Markovity depends on the state and is not forced by the decomposition alone.

Dynamics and MaxEnt. The natural Hamiltonian is a 2+1D lattice gauge Hamiltonian on the screen worldvolume: plaquette ("magnetic") terms, electric terms on links, vertex Gauss terms as constraints, plus local matter couplings. In quantum link form this remains finite-dimensional per link while behaving like gauge theory in the continuum limit. Then the MaxEnt assumption becomes concrete: the MaxEnt state is a Gibbs state $\rho \propto e^{-\sum_i \lambda_i O_i}$ with quasi-local O_i, precisely the LG (local Gibbs) regime.

Geometry and G. This microphysics naturally supplies the emergent geometric objects:

- **Edge entropy / area operator:** $L_C = \sum_\alpha (\log d_\alpha) P_\alpha$ becomes "log of boundary irrep dimension" in the gauge link model.
- **Newton constant G:** the conversion factor between edge entropy density per boundary UV cell and macroscopic geometric area.

Thus area is an operator living in the center of the boundary algebra, because in gauge systems the center is where the cut labels live.

What this example does and does not close. The quantum-link realization makes the fixed-cutoff matrix/fixed-point bookkeeping concrete. What it does **not** automatically guarantee is that modular flow on caps becomes geometric conformal dilation with the 2π KMS normalization, that is, the modular-covariance condition and the GB package feeding Theorem 4.2. That requires the state to sit in a regime that is effectively relativistic/QFT-like in the continuum limit. Viable

architectures for this include holographic quantum error-correcting codes (e.g., Pastawski et al., JHEP 2015) and quantum double / string-net Hamiltonians (Levin and Wen, PRB 71 (2005)).

1.3.5 Conformal-modular fixed point microphysics (CMFP)

After Section 2.3, the remaining gap is narrower. On the local finite-constraint MaxEnt branch of the third OPH axiom, the logarithm of the selected state is a quasi-local UV generator, and the refinement-stable branch lies inside one common finite-dimensional multiplier family. The open task is to land in a phase whose modular flow is geometric on caps and whose null limits support the later relativistic bridge.

CMFP-1 (Local finite-constraint MaxEnt branch). The constraint family $\mathcal{C}$ is generated by finitely many gauge-invariant local densities $\{O_a(x)\}$ of UV range $O(\ell_{\mathrm{UV}})$, with the same finite label set retained under refinement. This is not an extra postulate beyond the third OPH axiom; it is that axiom unpacked at regulator level.

Theorem 2.6 (Local constraints imply LG). If the MaxEnt constraints are expectations of finitely many quasi-local operators $\{O_a\}$ with bounded support size at scale ℓ_{UV}, then the entropy maximizer is

$$\omega \propto \exp\left(-\sum_a \lambda_a O_a\right),$$

so the MaxEnt generator $H_{\mathrm{MaxEnt}} = -\log\omega$ is a UV-range quasi-local sum. This is exactly LG.

Proof. Standard exponential-family result: maximum entropy subject to linear constraints $\langle O_a\rangle = c_a$ yields the Gibbs state with Lagrange multipliers λ_a. QED.

This turns "LG is an assumption" into "LG is a corollary of what constraints we allow."

Derived propagation control on the same branch. Because H_{MaxEnt} is a finite-range or quasi-local sum on a finite type-I regulator net, standard Lieb–Robinson estimates apply to the automorphism group it generates, or to any branch generator lying in the same bounded-support algebraic closure. Thus there is a finite propagation velocity v_{LR} and constants C, ξ such that for local observables A_X, B_Y,

$$\|[\tau_t(A_X), B_Y]\| \le C\,\|A_X\|\,\|B_Y\|\,\min(|X|,|Y|)\,e^{-(d(X,Y)-v_{\mathrm{LR}}|t|)/\xi}.$$

This is the later T2 control. Because changing a bounded interval endpoint only adds or removes an $O(|\Delta v|)$ collar of local terms, the same local branch gives bounded-interval endpoint-Lipschitz matrix elements,

$$\left|\langle\psi, (K[I'] - K[I])\phi\rangle\right| \le C_{\psi,\phi,I_{\max}}\,|I'\Delta I|,$$

which is the later T3 control. So propagation control is not an external regularity selector; it is the local-constraint MaxEnt branch written in dynamical form.

CMFP-2 (Refinement-stable multiplier branch). Because the same finite constraint family is preserved under coarse-graining, the regulator states lie in one common finite-dimensional multiplier family rather than in unrelated state spaces at different cutoffs. The "refinement-stable" language used later therefore means persistence along the stable or fixed branch of this multiplier family. In particular, whenever the gauge derivation speaks of a refinement-stable directed colimit of sectors, the refinement-stable qualifier is supplied by the third OPH axiom; only the transportability, symmetry, bosonic, and fiber-functor clauses remain extra.

CMFP-3 (Scaling limit with geometric modular action). In the refinement limit, the net $\mathcal{A}(P)$ with cyclic and separating Ω satisfies the **geometric modular action** property for caps and their conformal images: the modular group of a cap algebra acts as the unique conformal transformation preserving that cap.

This is precisely the Bisognano–Wichmann / geometric modular action package known to hold in conformal AQFT; see Brunetti et al., Rev. Math. Phys. 5 (1993).

Proposition 2.6 (CMFP-3 implies modular covariance and G). Under CMFP-3:

- **The derived modular-covariance property** on the cap net holds because modular flow *is* the geometric conformal flow.
- **The cap-modular tangent-limit package G** is realized, with its 2π KMS normalization fixed by the modular-geometric identification via the same rigidity that fixes Unruh/Hawking temperature.

Proof. In conformal AQFT, geometric modular-action results identify the modular group with the corresponding geometric symmetry for wedges and double cones. Once modular flow is geometric, the 2π normalization follows from the KMS condition. QED.

Alternative derivation via net regularity. The same modular-covariance property can also be derived directly from a standard AQFT regularity condition, without invoking the full CMFP-3 package.

(NR) Outer regularity / minimal support. For any operator O, the intersection of all connected regions P with $O \in \mathcal{A}(P)$ is again a connected region, denoted $\mathrm{supp}(O)$.

Proposition 2.7 (Modular covariance from net regularity). Under (NR), define for any region $R \subset C$

$$f_t^C(R) := \bigcup_{O \in \mathcal{A}(R)} \mathrm{supp}\left(\sigma_t^{\omega,C}(O)\right).$$

Then $\sigma_t^{\omega,C}(\mathcal{A}(R)) = \mathcal{A}(f_t^C(R))$, which is exactly the desired modular-covariance property.

Proof. Since $\sigma_t^{\omega,C}$ is an automorphism of $\mathcal{A}(C)$, and (NR) allows one to read support from the net labeling, the map $R \mapsto f_t^C(R)$ is well-defined and consistent. QED.

This shows the modular-covariance condition serves as a regularity condition on how the geometric labeling $P \subset S^2$ matches the algebra net. It does not add extra physics and is required if locality is to be meaningful.

Null-surface modular structure. Under CMFP-3, the null-surface modular machinery narrows as follows:

- **N1 (fixed-cutoff null-strip bridge).** The null-strip package proves transferred cut-center data, theorem-local inherited left/right strip structure, exact-or-controlled four-term strip additivity on one inherited strip model, renormalized endpoint control up to the weak tail generator, and the derived half-sided modular pair whose Borchers–Wiesbrock consequence is an explicit positive null-translation generator on its Stone domain. No separate LR, endpoint-control, or half-sided-inclusion premise is imported beyond that theorem-local inherited-strip hypothesis and the geometric scaling branch of Theorem 4.2.

- **N2 (derived half-sided modular inclusion).** Nested null half-line algebras satisfy half-sided modular inclusion on the OPH geometric scaling branch by Corollary 5.2e; Borchers/Wiesbrock then identifies the positive null-translation generator on its Stone domain together with the affine half-line modular relation; see Wiesbrock, CMP 157 (1993).

- **N3 (weak continuity/finite variation).** The bounded-interval and half-line endpoint control follow from the local MaxEnt branch and define the weak tail generator at fixed cutoff. CMFP-3 supplies the continuum null-generator setting in which that weak-tail data can be identified with the geometric null modular Hamiltonian of the relativistic phase.

Constraint set specification. Under CMFP-1, the "correct fixed-cap constraint set" becomes explicit: constraints are the local conserved charges of the symmetries used in the derivation:

1. **Edge/cap label constraints:** fix the distribution of collar-sector labels, equivalently $\langle L_C \rangle$ for each cap size, giving the area term.
2. **Gauge charges:** fix boundary flux or charge operators.
3. **Geometric (conformal) charges:** fix the expectation of the conformal Killing charges that preserve the cap, i.e. the generator B_C or its microscopic lattice approximation.

MaxEnt then selects the unique invariant state compatible with those conserved charges. In the CMFP-3 scaling limit, this is exactly the vacuum or canonical state whose modular group is geometric.

QNEC internalization. The Quantum Null Energy Condition has rigorous QFT proofs in broad settings; see Bousso et al., PRD 93 (2016). Under CMFP-3, the Recoverable Generalized Entropy axiom can be supported internally by:

- "Generalized entropy exists" is derived from EC + MaxEnt as $S_{\text{gen}} = S_{\text{bulk}} + \langle L_C \rangle$ (Section 5.4).
- "Focusing" becomes a semiclassical consequence of QNEC + the derived Einstein equation + Raychaudhuri, in the regime where the EFT bridge holds.

Summary. The CMFP package separates the dependency structure into two layers:

- **Internal to OPH:** LG, quasi-local propagation, bounded-interval endpoint control, and the refinement-stable notion of sector persistence, all from the local finite-constraint MaxEnt branch.
- **Carried by the CMFP phase statement:** derived modular covariance, the 2π normalization, the geometric null-modular phase, and the null half-sided-inclusion bridge.

CMFP-3 is therefore a phase statement: the refinement-stable MaxEnt fixed point must lie in the geometric modular action class. That is a concrete condition on the UV completion rather than an abstract axiom. The fixed-cutoff regulator bookkeeping, quasi-local propagation, and refinement stability belong to the OPH branch itself rather than to CMFP-3.

1.4 Overlap Consistency and Gluing

The constructive part of overlap consistency is the tree-gluing theorem below. The structural part is the origin of the gluing redundancy itself. On the ordinary or central-defect branch, gauge-as-gluing is the finite-regulator overlap redundancy of local chart presentations: once the overlap algebras are realized in finite-dimensional charts, overlap-consistent rechartings of a cut form a compact unitary transition system. The collar theorem of Section 2.3 should therefore be read with its boundary group G_Σ understood as shorthand for this derived compact boundary action, not as a model-specific add-on. The only fixed-cutoff obstruction to that ordinary-group picture is a genuinely noncentral 2-group defect, kept explicit in Section 3.4.

1.4.1 Constructive gluing on tree covers

Theorem 3.1 (tree gluing). Let a rooted tree of patches satisfy a tree-ordered overlap structure and a tripartite factorization (A_k, B_k, C_k) at step k. If a target state ρ^* obeys $I(A_k : C_k \mid B_k) \leq \varepsilon_k$, then there exist recovery maps $\mathcal{R}_k$ such that

$$\|\rho^*_{A_k B_k C_k} - (\mathrm{id}_{A_k} \otimes \mathcal{R}_k)(\rho^*_{A_k B_k})\|_1 \leq \delta_k,$$

with

$$\delta_k = 2\sqrt{\ln 2\, \varepsilon_k}.$$

The iteratively glued state $\hat{\rho}$ satisfies

$$\|\hat{\rho} - \rho^*\|_1 \leq \min\left(2, \sum_{k=2}^{n} \delta_k\right).$$

Proof. Induct on k. The recovery error contracts under CPTP maps, so the errors add. QED.

1.4.2 Gauge-as-gluing and loops

At finite regulator scale, premise GG identifies the overlap-consistency redundancy of local chart presentations. Choose finite-dimensional local presentations of the overlap algebras on a connected cut Σ. Because the overlap algebras are matrix algebras, any overlap-consistent change of chart is inner and is implemented by a unitary on the cut Hilbert space. Fixing a reference chart, the compact closure of the subgroup generated by all such recharting unitaries is the boundary gluing group K_Σ; before fixing the reference chart, the same data form a compact unitary groupoid.

Proposition 3.2a (Derived gauge-as-gluing at finite regulator). Let a finite regulator chart be chosen for the patches meeting along a connected interface Σ. Then overlap consistency determines a compact unitary transition system on the cut data. On the ordinary or central-defect branch, this transition system reduces to a compact boundary group K_Σ, and when the triple-overlap defect is central its projective composition law lifts to an honest action of a compact central extension $\widehat{K}_\Sigma$. Gauge is therefore the overlap redundancy itself, not an extra primitive.

Proof. On a finite-dimensional matrix algebra every *-automorphism is inner, so each overlap-consistent recharting is conjugation by a unitary on the cut Hilbert space. The subgroup generated by those unitaries has compact closure inside a finite-dimensional unitary group. If triple-overlap defects are central, the resulting projective composition law lifts to a central extension. QED.

Lemma 3.2b (trees vs loops). If the patch adjacency graph is a tree, one can choose local charts h_i so that the overlap labels satisfy $g_{ij} = h_i^{-1} h_j$ on all edges. If loops exist, the loop holonomy

$$H(\gamma) = g_{i_1 i_2} g_{i_2 i_3} \cdots g_{i_n i_1}$$

is invariant under local frame changes. Nontrivial holonomy is the obstruction to global trivialization. QED.

Corollary 3.2c (Collar consequence on the ordinary or central-defect branch). Let $B_\delta = B_L \cup B_R$ be a collar around a cap boundary Σ, and set $\widehat{K}_\Sigma = K_\Sigma$ on the ordinary branch. Then the EC theorem of Section 2.3 has the derived interpretation

$$H_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K}_\Sigma} \cong \bigoplus_\alpha (H_{b_L^\alpha} \otimes H_{b_R^\alpha}),$$

with

$$Z(A(B_\delta)) = \bigoplus_\alpha \mathbb{C}\,\mathbf{1}_\alpha.$$

The right half-collar carries the contragredient action because it sees the inverse transport across the same cut. Exact Markov normal forms used later require the additional state hypothesis $I_\omega(A_\delta : D_\delta \mid B_\delta) = 0$ or the explicitly stated idealized recoverability limit; the block decomposition itself is kinematic and follows from the derived boundary action.

Proof sketch. Decompose the left boundary data into irreps $(V_\alpha \otimes H_{b_L^\alpha})$ of $\widehat{K}_\Sigma$ and the right boundary data into the dual modules $(V_\beta^* \otimes H_{b_R^\beta})$. Then Schur's lemma leaves a singlet only when $\alpha = \beta$, producing the displayed direct sum. QED.

1.4.3 Loop obstruction class (central defect)

Under premise CD, define central defects z_{ijk} by

$$\varphi_{ij}\varphi_{jk}\varphi_{ki} = \mathrm{Ad}(z_{ijk}) \quad \text{on } \mathcal{A}_{ijk}.$$

Then $\{z_{ijk}\}$ is a Čech 2-cocycle, and its cohomology class $[z]$ is gauge invariant. Central defects do not obstruct the collar block decomposition: they only replace K_Σ by its central extension $\widehat{K}_\Sigma$. Ordinary loop-coherent gluing exists iff $[z] = 0$. (A full proof appears in Section 6.4 below, in the algebra-net language.)

1.4.4 Non-central obstruction (2-group cocycle)

When defects are not central, the natural coefficient data is a crossed module $(H \to G)$ with an action of G on H by conjugation. Here G is the reconstructed gauge group, and H is the unitary group acting on edge multiplicity spaces, with boundary map $\partial : H \to G$.

A crossed module is a homomorphism $\partial : H \to G$ together with an action of G on H such that

$$\partial(g \triangleright h) = g\,\partial(h)\,g^{-1}, \qquad \partial(h) \triangleright h' = hh'h^{-1}.$$

On a good cover $\{P_i\}$, a weakly coherent gluing is encoded by:

$$g_{ij} : P_{ij} \to G, \qquad h_{ijk} : P_{ijk} \to H,$$

obeying the 2-cocycle conditions

$$g_{ij}g_{jk} = \partial(h_{ijk})g_{ik},$$

and on quadruple overlaps,

$$h_{jkl}h_{ijl} = (g_{ij} \triangleright h_{ikl})h_{ijk}.$$

Gauge changes act by 1- and 2-cochains in the standard way for crossed-module cohomology.

Theorem 3.4 (non-central obstruction). Loop-coherent gluing exists iff the 2-cocycle (g_{ij}, h_{ijk}) is equivalent to the trivial cocycle in nonabelian Čech H^2 with values in the crossed module $(H \to G)$.

Proof sketch. Strict gluing corresponds to $h_{ijk} = 1$ and $g_{ij}g_{jk} = g_{ik}$. Gauge changes are exactly the crossed-module coboundaries, so strictification exists iff the 2-class is trivial. QED.

The central-defect case is the abelian truncation with H central and trivial action, which reduces to Section 3.3. A genuinely noncentral class is the precise blocker to promoting the ordinary-group EC theorem to the fully generic higher-gauge branch.

1.5 Modular Flow and Lorentz Kinematics

1.5.1 Modular additivity in the Markov collar limit

Consider a collar tripartition $A : B : D$ around a cap boundary, with the EC decomposition of Section 2.3 understood. Define, for a faithful reference state ω,

$$\Delta K(\omega) := K_{ABD}(\omega) - K_{AB}(\omega) - K_{BD}(\omega) + K_B(\omega).$$

Exact modular additivity is not a consequence of EC alone. It is available only on the exact Markov set, or along a controlled fixed-cutoff family that approaches that set on one fixed collar model. Three quantities must therefore be kept separate:

1. the raw collar conditional mutual information $I(A : D \mid B)$;
2. the constructive Fawzi–Renner comparison error
$$r_{\mathrm{FR}}(\varepsilon) := 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon};$$
3. the fixed-collar exact-Markov replacement modulus
$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) := \sup \left\{ \inf_{\sigma \in \mathfrak{M}_{A:B:D}} \|\rho - \sigma\|_1 : I(A : D \mid B)_\rho \le \varepsilon \right\},$$
where
$$\mathfrak{M}_{A:B:D} := \{\sigma_{ABD} : I(A : D \mid B)_\sigma = 0\}.$$

Lemma 4.1a (Exact Markov implies exact additivity up to a central term). If $I(A : D \mid B)_\omega = 0$, then the EC decomposition of Section 2.3 puts the state in HJPW block form

$$\omega_{ABD} = \bigoplus_\alpha p_\alpha \, \omega^{(\alpha)}_{Ab^\alpha_L} \otimes \omega^{(\alpha)}_{b^\alpha_R D},$$

and $\Delta K(\omega)$ is central. On the canonical HJPW block model one may take

$$\Delta K(\omega) = 0.$$

Equivalently, there exists a central operator $K_{\partial,ABD}(\omega) \in Z(\mathcal{A}(B))$ such that

$$K_{ABD}(\omega) = K_{AB}(\omega) + K_{BD}(\omega) - K_B(\omega) + K_{\partial,ABD}(\omega).$$

Proof. On each HJPW block the modular Hamiltonians of ABD, AB, BD, and B are the logarithms of tensor-product states with the same classical block weight p_α. The blockwise logarithms therefore cancel exactly in the combination $K_{ABD} - K_{AB} - K_{BD} + K_B$. If one keeps the blockwise endpoint-label bookkeeping explicit rather than absorbing it into the canonical block identification, the remainder is a direct sum of block constants, hence central. QED.

Proposition 4.1b (Controlled exact-Markov replacement on a fixed collar). Because the state space is compact and conditional mutual information is continuous in finite dimension,

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \longrightarrow 0 \qquad (\varepsilon \downarrow 0).$$

Thus small collar CMI converges to the exact HJPW normal form only in this controlled fixed-cutoff sense; the manuscript does not claim a dimension-free one-shot trace-norm bound from small $I(A:D\mid B)$ directly to an exact Markov state.

Corollary 4.1c (Carried collar defect operator). Let ω_ε satisfy $I(A:D\mid B)_{\omega_\varepsilon}\le\varepsilon$, and choose $\sigma_\varepsilon\in\mathfrak{M}_{A:B:D}$ with

$$\|\omega_\varepsilon-\sigma_\varepsilon\|_1\le\delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Define the carried defect operator

$$\mathfrak{D}_{A:B:D}(\omega_\varepsilon,\sigma_\varepsilon):=\Delta K(\omega_\varepsilon)-\Delta K(\sigma_\varepsilon).$$

Then

$$K_{ABD}(\omega_\varepsilon)=K_{AB}(\omega_\varepsilon)+K_{BD}(\omega_\varepsilon)-K_B(\omega_\varepsilon)+K_{\partial,ABD}(\sigma_\varepsilon)+\mathfrak{D}_{A:B:D}(\omega_\varepsilon,\sigma_\varepsilon),$$

where $K_{\partial,ABD}(\sigma_\varepsilon)=\Delta K(\sigma_\varepsilon)$ is central. For every bounded observable X supported on $A\cup B$,

$$|\mathrm{Tr}[X(\omega_\varepsilon-\sigma_\varepsilon)]|\le\|X\|_\infty\,\delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

If the relevant marginals of ω_ε and σ_ε are uniformly faithful with lower spectral bound $\lambda_*>0$, then

$$\|\mathfrak{D}_{A:B:D}(\omega_\varepsilon,\sigma_\varepsilon)\|_\infty\le 4\lambda_*^{-1}\,\delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Independently, Fawzi–Renner recovery supplies a constructive comparison state $\omega_\varepsilon^{\mathrm{rec}}$ with

$$\|\omega_\varepsilon-\omega_\varepsilon^{\mathrm{rec}}\|_1\le r_{\mathrm{FR}}(\varepsilon),\qquad r_{\mathrm{FR}}(\varepsilon):=2\sqrt{1-e^{-\varepsilon}}\le 2\sqrt{\varepsilon}.$$

This $O(\varepsilon^{1/2})$ term is the finite-stage observable error. The modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$ is instead the fixed-collar quantity that justifies replacing the physical state by an exact Markov normal form in the later geometric modular arguments.

Accordingly, the manuscript's later exact collar formulas should be read in one of two ways:

1. literal exactness when the reference state is exact Markov on the relevant collar; or
2. a controlled collar family for which $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon_\delta)\to 0$, with the constructive Fawzi–Renner remainder and the carried operator defect $\mathfrak{D}_{A:B:D}$ kept explicit at finite stage.

1.5.2 Theorem: BW_{S^2} on the OPH geometric branch

Theorem 4.2 (BW_{S^2} on the OPH Bisognano–Wichmann branch). Let $C\subset S^2$ be a cap and let $(A_\delta,B_\delta,D_\delta)$ be a shrinking collar family around ∂C. Assume: (i) the collar-control/refinement limit used throughout Section 4, so that exact collar formulas are read either at exact Markovity or along a controlled family with

$$\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta)\to 0;$$

(ii) MaxEnt selection with rotationally invariant local constraints, i.e. premise ROT; and (iii) the refinement-stable OPH branch for caps lies in the Bisognano–Wichmann geometric modular class, i.e. for each cap C the limiting modular action is implemented by the standard cap-preserving conformal one-parameter subgroup $\lambda_C(s)$, with geometric parameter s normalized so that the null blow-up near a smooth cut acts by $v\mapsto e^{-s}v$. If B_C denotes the infinitesimal generator of $\lambda_C(s)$, then cap modular flow is implemented by that subgroup, and its normalization is fixed to 2π. Equivalently,

$$K_C=2\pi B_C.$$

More precisely, at collar scale δ,

$$K_C^{(\delta)} = 2\pi B_C + E_C^{(\delta)},$$

where the carried remainder is controlled by the finite-stage errors separated above,

$$r_{\mathrm{FR}}(\varepsilon_\delta) \qquad \text{and} \qquad 4\lambda_*^{-1}\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$

and therefore vanishes on every controlled collar family whose faithfulness bound survives the limit.

Proof. Markov locality does one precise job: it reduces the modular problem to the shrinking collar, with the finite-stage deviation from an exact Markov reference carried explicitly by $r_{\mathrm{FR}}(\varepsilon_\delta)$ and $4\lambda_*^{-1}\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta)$. On the stated OPH Bisognano–Wichmann refinement branch, assumption (iii) provides the standard cap-preserving conformal subgroup $\lambda_C(s)$, so the limiting modular Hamiltonian must be $K_C = \kappa B_C$ for some $\kappa > 0$. The modular group is KMS by definition, and because s is normalized by unit null dilation on the branch, the usual Bisognano–Wichmann relation fixes $\kappa = 2\pi$. Therefore

$$K_C = 2\pi B_C,$$

with the displayed $E_C^{(\delta)}$ as the carried finite-stage remainder. QED.

Scope note. The theorem proves the Bisognano–Wichmann conclusion *on the OPH refinement branch lying in the stated BW geometric modular class.* On that branch, the standard cap-preserving conformal subgroup is part of the branch data, and the theorem fixes its 2π normalization together with the carried collar remainder. Phase selection of the BW branch is encoded in the stated hypotheses.

1.5.3 Theorem: BW_{S^2} implies Lorentz kinematics

Theorem 4.3 (Lorentz kinematics on the screen). Under the branch and controlled-limit hypotheses of Theorem 1.5.2, the induced kinematic group is

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1).$$

Proof. Theorem 1.5.2, together with its BW geometric-branch hypothesis, identifies cap modular flows with the standard cap-preserving conformal dilations on S^2. Together with the rotation subgroup fixed by the screen geometry, these are the standard one-parameter subgroups of the orientation-preserving conformal group of the sphere. But

$$\mathrm{Conf}^+(S^2) = \mathrm{PSL}(2,\mathbb{C}),$$

the Möbius group, and this is isomorphic to the connected Lorentz group $\mathrm{SO}^+(3,1)$. Hence the local kinematics induced by the modular action is Lorentzian. QED.

1.6 Gravity from Entanglement Equilibrium

1.6.1 Cap first law

For a reference state ω and small cap C,

$$K_C := -\log \rho_C^\omega,$$

and for $\rho(\varepsilon)$ with $\rho(0) = \omega$,

$$\delta S_C = \delta\langle K_C\rangle.$$

By Section 4.2, $K_C = 2\pi B_C + \text{const}$, hence

$$\delta S_C = 2\pi\delta\langle B_C\rangle.$$

1.6.2 Null-surface modular bridge

This subsection extends the fixed-cutoff weak-tail-generator boundary to the derived positive null-translation stage. At fixed cutoff one can transfer cut-center data to regulated null strips, obtain exact or controlled strip additivity on one inherited strip model, and derive a weak tail generator for the renormalized half-line family. On the OPH geometric scaling branch of Theorem 4.2, the null half-line blow-up net then inherits geometric dilation and therefore half-sided modular inclusion, so Borchers–Wiesbrock supplies an explicit positive null-translation generator on its Stone domain. What is not proved here is the downstream density upgrade and the relativistic null-stress identification. Accordingly the c.12-local bridge consists of:

- null-cut center transfer and the inherited split condition;
- exact or controlled four-term strip additivity on one fixed inherited strip model;
- endpoint-Lipschitz control for the renormalized half-line family and the resulting weak tail generator;
- derived half-sided modular inclusion on the null half-line blow-up net, and the resulting Borchers positive translation generator.

The later density formula and T_{kk} bridge are kept below only as explicit downstream templates rather than as fixed-cutoff theorems proved in this subsection. Lemma 5.2f makes the positive null-translation generator itself explicit: it records the Borchers unitary group, positivity and self-adjointness on the Stone domain, the corrected affine commutator relation, and the half-line modular-Hamiltonian identity $K_a = K_0 - 2\pi a P_\Omega$. Bounded-interval formulas remain downstream.

Proposition 5.2a (Null-cut center transfer and inherited split). Fix a regulated null tripartition

$$I_- = (v_1, v_2), \qquad J = (v_2, v_3), \qquad I_+ = (v_3, v_4),$$

with cuts $\Gamma_- := \{v = v_2\}$ and $\Gamma_+ := \{v = v_3\}$. Under R0 and R1, assume that the two null cuts inherit the ordinary or central-defect boundary-redundancy data used in the spatial collar branch, so that in a compatible type-I regulator presentation one has

$$\tilde{\mathcal{H}}_{I_-} \cong \bigoplus_{\alpha_-} W_{\alpha_-} \otimes \mathcal{H}_{i_-^{\alpha_-}},$$

$$\tilde{\mathcal{H}}_J \cong \bigoplus_{\alpha_-,\alpha_+} W^*_{\alpha_-} \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes W_{\alpha_+},$$

$$\tilde{\mathcal{H}}_{I_+} \cong \bigoplus_{\alpha_+} W^*_{\alpha_+} \otimes \mathcal{H}_{i_+^{\alpha_+}},$$

where opposite sides of each cut carry inverse transport. If the strip algebra is the commutant of the transported cut actions, then

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-,\alpha_+}}), \qquad Z(\mathcal{A}(J)) = \bigoplus_{\alpha_-,\alpha_+} \mathbb{C}\, P_{\alpha_-,\alpha_+}.$$

If, in addition, each multiplicity space factors as

$$\mathcal{H}_{j^{\alpha_-,\alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}},$$

with $\mathcal{A}(I_- \cup J)$ acting blockwise only on $\mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}}$ and $\mathcal{A}(J \cup I_+)$ acting blockwise only on $\mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}$, then

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j_L^{\alpha_-,\alpha_+}}) \otimes \mathcal{B}(\mathcal{H}_{j_R^{\alpha_-,\alpha_+}}),$$

and

$$\mathcal{H}_{I_- \cup J \cup I_+} \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

Proof. Complete reducibility gives the displayed decompositions. Commuting with the two transported cut actions forces strip operators to preserve the sector pair (α_-, α_+) and act only on the multiplicity space; Schur's lemma kills the off-diagonal intertwiners and leaves the direct-sum algebra above. Taking invariants across both cuts in the glued tripartition again uses Schur's lemma and leaves exactly the matching sector pairs. The left/right split of the multiplicity spaces is additional structure; it is not forced by the center transfer alone. QED.

Corollary 5.2b (Exact or controlled four-term null modular relation on an inherited strip model). On one fixed finite-dimensional strip model satisfying Proposition 5.2a, define

$$\mathfrak{M}_{I_-:J:I_+} := \{\sigma : \ I(I_- : I_+ \mid J)_\sigma = 0\},$$

and

$$\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon) := \sup \left\{ \inf_{\sigma \in \mathfrak{M}_{I_-:J:I_+}} \|\rho - \sigma\|_1 : I(I_- : I_+ \mid J)_\rho \le \varepsilon \right\}.$$

For any strip state η, let

$$\Delta K_J(\eta) := K_{I_- \cup J \cup I_+}(\eta) - K_{I_- \cup J}(\eta) - K_{J \cup I_+}(\eta) + K_J(\eta).$$

If the strip reference state ω is exact Markov, then $\Delta K_J(\omega)$ is central, and on the canonical inherited HJPW model one may take

$$\Delta K_J(\omega) = 0.$$

Equivalently, there exists a central operator $K_{\partial,J}(\omega) \in Z(\mathcal{A}(J))$ such that

$$K_{I_- \cup J \cup I_+}(\omega) = K_{I_- \cup J}(\omega) + K_{J \cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\omega).$$

If instead $I(I_- : I_+ \mid J)_\omega \le \varepsilon$, choose $\widetilde{\omega}_J \in \mathfrak{M}_{I_-:J:I_+}$ with

$$\|\omega - \widetilde{\omega}_J\|_1 \le \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

Define

$$\mathfrak{D}_J(\omega, \widetilde{\omega}_J) := \Delta K_J(\omega) - \Delta K_J(\widetilde{\omega}_J).$$

Then

$$K_{I_- \cup J \cup I_+}(\omega) = K_{I_- \cup J}(\omega) + K_{J \cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\widetilde{\omega}_J) + \mathfrak{D}_J(\omega, \widetilde{\omega}_J),$$

with $K_{\partial,J}(\widetilde{\omega}_J) = \Delta K_J(\widetilde{\omega}_J) \in Z(\mathcal{A}(J))$. Every bounded observable on $I_- \cup J$ or $J \cup I_+$ then differs from the exact-Markov reference by at most

$$\|O\|_\infty \, \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

If the relevant strip marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_J(\omega, \widetilde{\omega}_J)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

Independently, Fawzi–Renner gives a recovered comparison state with trace-norm error

$$r_{\mathrm{FR}}(\varepsilon) = 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Thus the exact four-term strip relation is available only at exact Markovity or in a controlled strip family on one fixed inherited strip model with $\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon_J) \to 0$, while $\mathfrak{D}_J$ is carried explicitly at finite stage.

Proof. On the inherited split of Proposition 5.2a, the exact-Markov case is the same HJPW block calculation as in the spatial collar theorem: the four logarithms cancel blockwise on the canonical model, and any residual bookkeeping term is central. The controlled replacement is the same fixed-model compactness argument used for ordinary collars, after relabeling $A, B, D \mapsto I_-, J, I_+$; the operator-norm bound on $\mathfrak{D}_J$ is the corresponding relabeling of the modular-transport estimate. QED.

Definition (Renormalized null modular functional). For a null interval I on generator Ω, let $K_\partial(I, \Omega)$ denote the central endpoint-label term singled out by Corollary 5.2b on the inherited strip model, or by its controlled exact-Markov replacement when that model is used as reference. Define

$$\widetilde{K}[I, \Omega] := K[I, \Omega] - K_\partial(I, \Omega), \qquad \widetilde{K}_a(\Omega) := \widetilde{K}[(a, \infty), \Omega].$$

Proposition 5.2c (Endpoint-Lipschitz null modular families and weak tail generator). For renormalized modular Hamiltonians on one null generator,

$$\widetilde{K}[I, \Omega] := K[I, \Omega] - K_\partial(I, \Omega), \qquad \widetilde{K}_a(\Omega) := \widetilde{K}[(a, \infty), \Omega],$$

the branch-internal endpoint-control coming from the local finite-constraint MaxEnt branch gives:

$$\big|\langle \psi, (\widetilde{K}[(a', b'), \Omega] - \widetilde{K}[(a, b), \Omega])\phi\rangle\big| \le C_{\psi,\phi,\Omega}\big(|a' - a| + |b' - b|\big)$$

for bounded intervals in a compact endpoint window, and

$$\big|\langle \psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle\big| \le C_{\psi,\phi,\Omega}|a' - a|$$

for half-lines. Hence

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi\rangle$$

is locally Lipschitz and therefore absolutely continuous, and its distributional derivative defines the weak tail generator

$$\langle \psi, q(a, \Omega)\phi\rangle := -\frac{1}{2\pi}\, \partial_a f_{\psi,\phi}(a).$$

For $a < b$,

$$\langle \psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle = -2\pi \int_a^b \langle \psi, q(v, \Omega)\phi\rangle\, dv.$$

At this stage $q(a, \Omega)$ is a weak tail generator, not yet a local operator-valued density; its identification with the positive self-adjoint Borchers generator occurs only after Corollary 5.2e and Lemma 5.2f.

Proof. The derived endpoint-control estimate applies to the renormalized family after removal of the central endpoint term. For bounded intervals, the symmetric-difference length is bounded

by $|a'-a|+|b'-b|$; for half-lines it is exactly $|a'-a|$. Therefore the matrix-element functions are Lipschitz in the endpoints. A Lipschitz function on a compact interval is absolutely continuous, so $f_{\psi,\phi}(a)$ has an L^∞_{loc} derivative and obeys the fundamental theorem of calculus. Defining q as the rescaled negative derivative gives the displayed integral relation. QED.

Downstream c.12–c.13 boundary. The fixed-cutoff bridge established above is the end of this subsection's proved content.

Lemma 5.2d (Downstream density-upgrade template). If, in addition, the weak tail generator $q(a,\Omega)$ is weakly differentiable in a and both $f_{\psi,\phi}(a)$ and $q_{\psi,\phi}(a)$ vanish at $+\infty$, then one may define a density

$$\langle\psi, p(a,\Omega)\phi\rangle := -\partial_a\langle\psi, q(a,\Omega)\phi\rangle = \frac{1}{2\pi}\,\partial_a^2\langle\psi, \widetilde{K}_a(\Omega)\phi\rangle$$

and obtain the half-line formula

$$\langle\psi, \widetilde{K}_a(\Omega)\phi\rangle = 2\pi\int_a^\infty (v-a)\,\langle\psi, p(v,\Omega)\phi\rangle\,dv.$$

This is a downstream density-upgrade step and is not proved from the fixed-cutoff strip arguments above.

Null half-line blow-up net. Fix a smooth cut point and choose affine coordinate v on generator Ω so that the cut sits at $v=0$. For $a\geq 0$, write

$$H_a := (a,\infty), \qquad \mathcal{M}_a(\Omega) := \overline{\bigvee_{a<c<d<\infty} \mathcal{A}((c,d),\Omega)}.$$

By isotony of the null interval net,

$$a\leq b \quad\Longrightarrow\quad \mathcal{M}_b(\Omega)\subseteq\mathcal{M}_a(\Omega).$$

Corollary 5.2e (Derived half-sided modular inclusion on null half-lines). On the OPH geometric scaling branch of Theorem 4.2, the blow-up modular action near a smooth entangling cut acts on the null coordinate by

$$v\mapsto e^{-2\pi t}v.$$

Therefore, for every $a\geq 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) = \mathcal{M}_{e^{-2\pi t}a}(\Omega).$$

Hence, for every $a>0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega))\subseteq\mathcal{M}_a(\Omega) \qquad (t\leq 0),$$

so the inclusion $\mathcal{M}_a(\Omega)\subset\mathcal{M}_0(\Omega)$ is half-sided modular. After the harmless convention change $t\mapsto -t$, this is the standard positive-time half-sided-inclusion form. The old N2 premise is therefore derived from OPH null-interval structure, isotony, and the scaling-limit geometric action rather than imported separately.

Proof. Blow up the cap modular flow of Theorem 4.2 near a smooth cut and restrict to the chosen null generator. In the tangent limit the cap-preserving flow becomes the null dilation $v\mapsto e^{-2\pi t}v$. For every bounded interval $(c,d)\subset H_a$, this sends $\mathcal{A}((c,d),\Omega)$ to $\mathcal{A}((e^{-2\pi t}c, e^{-2\pi t}d),\Omega)$, and taking the von Neumann closure of the interval net yields the displayed identity for $\mathcal{M}_a(\Omega)$. If $t\leq 0$, then $e^{-2\pi t}a\geq a$, so $H_{e^{-2\pi t}a}\subseteq H_a$, and isotony gives $\mathcal{M}_{e^{-2\pi t}a}(\Omega)\subseteq\mathcal{M}_a(\Omega)$. QED.

Lemma 5.2f (Positive null-translation generator). For the derived half-sided modular inclusion

$$\mathcal{M}_a(\Omega)\subset\mathcal{M}_0(\Omega) \qquad (a>0),$$

let $\Delta_0(\Omega)$ be the modular operator of the standard pair $(\mathcal{M}_0(\Omega), \omega)$ and define $K_0(\Omega) := -\log \Delta_0(\Omega)$. Borchers–Wiesbrock then yields a unique strongly continuous one-parameter unitary group

$$U_\Omega(a) = e^{iaP_\Omega}, \qquad a \in \mathbb{R},$$

such that

$$U_\Omega(a)\omega = \omega, \qquad U_\Omega(a)\mathcal{M}_b(\Omega)U_\Omega(a)^* = \mathcal{M}_{a+b}(\Omega) \quad (a, b \geq 0),$$

and whose generator P_Ω is positive and self-adjoint on the Stone domain

$$D(P_\Omega) := \left\{ \psi \in \mathcal{H} : \lim_{a \to 0} \frac{U_\Omega(a)\psi - \psi}{ia} \text{ exists} \right\}.$$

Moreover,

$$\Delta_0(\Omega)^{it} U_\Omega(a) \Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t} a), \qquad \Delta_0(\Omega)^{it} P_\Omega \Delta_0(\Omega)^{-it} = e^{-2\pi t} P_\Omega,$$

and on the common invariant analytic core of $K_0(\Omega)$ and P_Ω,

$$[K_0(\Omega), P_\Omega] = -\, i\, 2\pi P_\Omega.$$

If $\Delta_a(\Omega)$ denotes the modular operator of $(\mathcal{M}_a(\Omega), \omega)$ and $K_a(\Omega) := -\log \Delta_a(\Omega)$, then

$$\Delta_a(\Omega) = U_\Omega(a)\Delta_0(\Omega)U_\Omega(a)^*, \qquad K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*,$$

so

$$K_a(\Omega) = K_0(\Omega) - 2\pi a\, P_\Omega$$

as a quadratic-form identity on $D(K_0(\Omega)) \cap D(P_\Omega)$. Equivalently,

$$\langle \psi, (K_b(\Omega) - K_a(\Omega))\phi \rangle = -2\pi(b - a)\langle \psi, P_\Omega \phi \rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$. In the canonical normalization of Corollary 5.2b, the weak endpoint derivative from Proposition 5.2c is therefore exactly the Borchers generator. Bounded-interval modular-Hamiltonian formulas remain downstream and require the separate interval-preserving branch recorded in Theorem 5.2g.

Proof. Corollary 5.2e gives the required half-sided modular inclusion. Borchers–Wiesbrock then yields the unique strongly continuous unitary group $U_\Omega(a)$, its positive self-adjoint generator P_Ω, and the affine covariance relation with the modular group; Stone's theorem gives the displayed domain formula. Differentiating

$$\Delta_0(\Omega)^{it} U_\Omega(a) \Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t} a)$$

first at $a = 0$ and then at $t = 0$ on the common analytic core gives

$$[K_0(\Omega), P_\Omega] = -\, i\, 2\pi P_\Omega.$$

Because $U_\Omega(a)\omega = \omega$ and $U_\Omega(a)\mathcal{M}_0(\Omega)U_\Omega(a)^* = \mathcal{M}_a(\Omega)$, uniqueness of modular data for the translated standard pair gives the displayed formulas for $\Delta_a(\Omega)$ and $K_a(\Omega)$. Differentiating $K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*$ in a yields $dK_a/da = -2\pi P_\Omega$ as a quadratic-form identity, and integrating from 0 to a gives the stated half-line modular-Hamiltonian relation. QED.

Theorem 5.2g (Downstream null-stress identification template). If one adds the density-upgrade hypotheses of Lemma 5.2d and the relativistic scaling-limit null-stress premise,

then on the half-line blow-up net whose half-sided modular inclusion is derived in Corollary 5.2e and whose positive null-translation generator is identified in Lemma 5.2f, one expects an operator-valued distribution $T_{kk}(v,\Omega)$ with

$$\widetilde{K}[(a,\infty),\Omega] = 2\pi \int_a^\infty (v-a)\, T_{kk}(v,\Omega)\, dv + E^{(\eta)}_{(a,\infty),\Omega},$$

and, for bounded null intervals that inherit geometric modular action,

$$\widetilde{K}[I,\Omega] = 2\pi \int_I g_I(v)\, T_{kk}(v,\Omega)\, dv + E^{(\eta)}_{I,\Omega}.$$

Here, for each relevant half-line or bounded interval I, ε_I denotes the strip-CMI control parameter, δ_I^{M} the corresponding fixed-strip exact-Markov modulus from Corollary 5.2b, and

$$\eta_I := r_{\mathrm{FR}}(\varepsilon_I) + 4\lambda_*^{-1}\delta_I^{\mathrm{M}}(\varepsilon_I) + \eta_I^{\mathrm{EFT}},$$

so that

$$\left|\langle \psi, E^{(\eta)}_{I,\Omega}\phi\rangle\right| \le C_{I,\Omega,\psi,\phi}\, \eta_I$$

for domain vectors ψ,ϕ. The bounded-interval kernel $g_I(v)$ is affine-covariant and vanishes at finite interval endpoints. The full modular Hamiltonian is

$$K[I,\Omega] = \widetilde{K}[I,\Omega] + K_\partial(I,\Omega),$$

with K_∂ central and boundary-supported, and knowledge of T_{kk} for all null directions reconstructs a symmetric tensor T_{ab} up to a metric term. This subsection does not prove this theorem; it records the exact downstream burden beyond the derived half-sided-inclusion bridge.

Boundary of the conditional null bridge. The null bridge is conditional rather than automatic. Its explicit burden is:

- transferred cut-center data alone give the central sector decomposition of the null strip, not yet the left/right HJPW split;
- the extra decomposition-inheritance condition of Proposition 5.2a is exactly what upgrades those sectors to the same collar-type block structure used in the spatial branch;
- N1 is exact for exact Markov strip states on that inherited decomposition, and otherwise is replaced by a controlled limit with the carried defect operator $\mathfrak{D}_J$;
- the local finite-constraint MaxEnt branch gives endpoint-Lipschitz control strong enough to define a weak tail generator for renormalized half-lines, and the OPH geometric scaling branch then derives half-sided modular inclusion on the half-line blow-up net;
- Borchers–Wiesbrock therefore supplies a genuine positive null-translation generator P_Ω inside the bridge itself, together with its Stone domain and the half-line modular-Hamiltonian relation $K_a = K_0 - 2\pi a P_\Omega$;
- the density upgrade and the final identification with a local continuum T_{kk} are the remaining explicit downstream steps.

This is the route from null-strip factorization to the later Jacobson branch without identifying the null modular bridge itself with the downstream density-upgrade and null-stress steps.

1.6.3 Modular energy as stress-tensor charge (UV CFT)

If one assumes a UV CFT regime on sufficiently small caps, the modular Hamiltonian is explicitly local. This serves as an alternative EFT bridge to Section 5.2.

For a CFT vacuum on a ball, the modular Hamiltonian is local:

$$H_\zeta = \int_\Sigma T_{ab}\zeta^b d\Sigma^a,$$

where ζ is the conformal Killing field preserving the diamond. For a small diamond of size ℓ in $d = 4$,

$$\delta\langle H_\zeta\rangle = \frac{4\pi\ell^4}{15}\delta\langle T_{00}\rangle + O(\ell^5),$$

in the diamond rest frame.

1.6.4 Localized generalized entropy from Markov + MaxEnt

Using the collar decomposition and premise COL (double-scaling, established at regulator level via Theorem 2.3, or alternatively via LG+MX), the state takes the Markov normal form. MaxEnt selection maximizes entropy within each edge sector, producing

$$\rho_C = \bigoplus_\alpha p_\alpha \left(\rho^\alpha_{\text{bulk},C} \otimes \frac{\mathbf{1}^\alpha_{\text{edge}}}{d_\alpha}\right).$$

The entropy splits as

$$S(\rho_C) = H(p_\alpha) + \sum_\alpha p_\alpha S(\rho^\alpha_{\text{bulk},C}) + \sum_\alpha p_\alpha \log d_\alpha.$$

Convention: Throughout this paper, "log" denotes the natural logarithm (ln), so entropies are measured in **nats** (1 nat = 1/ln 2 ≈ 1.443 bits). This is standard in thermodynamics and QFT; the Bekenstein-Hawking formula S = A/4G uses nats. When clarity requires it, we write $\log_2$ explicitly for bits.

Define

$$S_{\text{bulk}}(C) := H(p_\alpha) + \sum_\alpha p_\alpha S(\rho^\alpha_{\text{bulk},C}),$$

and the central area operator

$$L_C := \sum_\alpha (\log d_\alpha) P_\alpha.$$

Then

$$S_{\text{gen}}(C) := \text{Tr}(\rho L_C) + S_{\text{bulk}}(C).$$

Deriving Newton's constant from edge entropy density.

Rather than normalize L_C by fiat, we *derive* the relation to G from the UV edge structure. In the collar double-scaling limit, the edge contribution becomes extensive along the entangling surface $\Sigma = \partial C$:

$$\text{Tr}(\rho L_C) \approx N_\Sigma \cdot \bar{\ell}(t), \qquad \bar{\ell}(t) := \sum_\alpha p_\alpha \log d_\alpha,$$

where N_Σ is the number of UV cut elements covering Σ and $\bar{\ell}(t)$ is the **single-cell edge entropy** from the heat-kernel distribution (Theorem 6.20). Similarly, the geometric area is extensive:

$$A(C) \approx N_\Sigma \cdot a_{\text{cell}},$$

where a_{cell} is the area per UV cut element in the emergent metric.
Matching these expressions gives the **derived formula for Newton's constant**:

$$G = \frac{a_{\text{cell}}}{4\,\bar{\ell}(t)}$$

where:

- a_{cell} is fixed operationally from the UV correlation/mixing length ξ via $a_{\text{cell}} \sim \xi^2$ (from premise MX),
- $\bar{\ell}(t) = \sum_R p_R(t) \log d_R$ is computed from the heat-kernel edge distribution with $p_R \propto d_R e^{-t\lambda_R}$.

Explicitly:

$$\bar{\ell}(t) = \frac{\sum_R d_R e^{-t\lambda_R} \log d_R}{\sum_R d_R e^{-t\lambda_R}}.$$

Thus G is the inverse edge-entropy density per geometric area, computable from the UV regulator and the reference-state Gibbs parameter t, rather than a normalization convention.

1.6.5 Entanglement equilibrium from MaxEnt

MaxEnt selection implies that for variations preserving cap labels (fixed size and charges),

$$\delta S_{\text{gen}}(C) = 0.$$

Using the split above and the first law for the bulk term,

$$\delta S_{\text{gen}}(C) = \delta\langle L_C\rangle + \delta\langle K_{\text{bulk}}\rangle.$$

1.6.6 Einstein equation from cap equilibrium

In the EFT regime, combine:

1) Modular energy as stress-tensor charge (Section 5.2 or Section 5.3), and
2) The geometric identity for area variation at fixed volume:

$$\delta A|_{V,\lambda} = -\frac{\Omega_{d-2}\ell^d}{d^2-1}(G_{00} + \lambda g_{00}).$$

The equilibrium condition yields

$$G_{00} + \Lambda g_{00} = 8\pi G\langle T_{00}\rangle,$$

in the diamond rest frame, with Λ fixed by the reference curvature.

1.6.7 Overlaps supply all timelike directions

Different observers through the same bulk point select different local rest frames u. Overlap consistency forces the scalar relation to hold for all timelike u, so

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle.$$

1.6.8 Non-tunable numerical constants

The gravity chain yields specific numerical constants as rigid outputs of the axiom chain.

The 2π KMS normalization. From Theorem 4.2, once the OPH refinement branch is geometric, cap modular flow is fixed by the KMS condition to the standard 2π normalization. This is the same rigidity that fixes Unruh/Hawking temperature normalization. The period is determined by the stated scaling-limit package together with the geometric-branch hypothesis.

The geometric coefficient $\Omega_{d-2}/(d^2-1)$. This coefficient appears in both (a) the CFT-ball modular Hamiltonian weight integral and (b) the geometric area-variation identity. It is an exact integral identity:

$$\int_{B_\ell^{d-1}} \frac{\ell^2 - r^2}{2\ell}\, d^{d-1}x = \frac{\Omega_{d-2}\, \ell^d}{d^2-1}.$$

In $d = 4$:

$$\frac{\Omega_2}{4^2-1} = \frac{4\pi}{15} \approx 0.8377580409572781.$$

This is the reason prefactors cancel cleanly when going from $\delta S_{\text{gen}} = 0$ to the Einstein equation (leaving $8\pi G$ with the 2π fixed by Theorem 4.2).

What is predicted. The framework cleanly separates:

- **Non-tunable constants**: 2π (KMS period), $\Omega_{d-2}/(d^2-1)$ (geometric coefficient), the existence of the Einstein form.
- **Micro-dependent constants**: G (Newton's constant) is the conversion between edge entropy and geometric area (a density of edge degrees of freedom per geometric area), which is model-dependent. Λ is fixed by the reference state/constraints. These require specifying the microscopic model.

1.6.9 Quantitative Markov error and controlled corrections

The role of this subsection is to keep three distinct quantities separate:

1. the raw collar conditional mutual information

$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega;$$

2. the constructive recovery error

$$r_{\text{FR}}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \le 2\sqrt{\varepsilon_\delta};$$

3. the exact-Markov replacement error on one fixed faithful collar model,

$$\eta_\delta^{\text{M}} := 4\lambda_*^{-1}\, \delta^{\text{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$

where $\lambda_* > 0$ is the lower spectral bound needed to compare modular Hamiltonians.

The exact identity for the modular defect expectation is unchanged:

$$\langle \Delta K_\delta \rangle_\omega = -I(A_\delta : D_\delta \mid B_\delta)_\omega = -\varepsilon_\delta,$$

with

$$\Delta K_\delta := K_{A_\delta B_\delta D_\delta} - K_{A_\delta B_\delta} - K_{B_\delta D_\delta} + K_{B_\delta}.$$

What changes is the interpretation of later "exact" collar formulas. For each fixed finite-dimensional collar one may choose an exact Markov reference state σ_δ with

$$\|\omega_\delta - \sigma_\delta\|_1 \le \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$

and a constructive recovered comparison state $\omega_\delta^{\mathrm{rec}}$ with

$$\|\omega_\delta - \omega_\delta^{\mathrm{rec}}\|_1 \le r_{\mathrm{FR}}(\varepsilon_\delta).$$

These two comparisons do different jobs:

Bounded-observable control. For every bounded observable X supported on $A_\delta \cup B_\delta$,

$$|\mathrm{Tr}[X(\omega_\delta - \omega_\delta^{\mathrm{rec}})]| \le \|X\|_\infty \, r_{\mathrm{FR}}(\varepsilon_\delta),$$

and

$$|\mathrm{Tr}[X(\omega_\delta - \sigma_\delta)]| \le \|X\|_\infty \, \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta).$$

The first bound is constructive and dimension-free; the second is the fixed-collar route to the exact Markov set.

Modular-additivity control. If the relevant marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\Delta K_\delta(\omega) - \Delta K_\delta(\sigma_\delta)\|_\infty \le \eta_\delta^{\mathrm{M}}.$$

Since $\Delta K_\delta(\sigma_\delta)$ is central by exact Markovity, the finite-stage modular defect is within η_δ^{M} of the exact splice or additivity value.

Therefore the manuscript's exact collar identities should be read as limits of a carried remainder

$$\eta_\delta := r_{\mathrm{FR}}(\varepsilon_\delta) + \eta_\delta^{\mathrm{M}},$$

together with the separate long-wavelength derivative remainders present in the small-ball expansion. Exact identities at finite collar width require exact Markovity; otherwise one works with η_δ-controlled approximations and then lets $\eta_\delta \to 0$ in the controlled collar limit.

Modified Einstein relation with an explicit carried remainder. In the small-ball rest-frame calculation, write

$$\delta S_C = \frac{8\pi^2 \ell^4}{15}\, \delta\langle T_{00}\rangle + \delta\langle E_C^{(\delta)}\rangle + O(\ell^5 \partial T),$$

where $E_C^{(\delta)}$ packages the carried collar remainder. Then the fixed-cap stationarity argument gives

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle + \mathcal{E}_\delta,$$

with

$$\mathcal{E}_\delta := \frac{15G}{\pi \ell^4}\, \delta\langle E_C^{(\delta)}\rangle + O(\ell\, \partial T),$$

and, by the bounds above,

$$|\delta\langle E_C^{(\delta)}\rangle| \le C_C\,\eta_\delta$$

for some collar-dependent constant C_C on the fixed faithful model.

If one wishes to package this carried remainder as an effective anomalous energy density, the natural definition is

$$\delta\langle T_{00}^{\mathrm{anom}}\rangle := \frac{15}{8\pi^2\ell^4}\,\delta\langle E_C^{(\delta)}\rangle.$$

Then

$$\delta(G_{00}+\Lambda g_{00}) = 8\pi G\,\delta(\langle T_{00}\rangle + \langle T_{00}^{\mathrm{anom}}\rangle) + O(\ell\,\partial T).$$

This is a bookkeeping definition of the finite-stage remainder, not an upgrade of the recovered-core theorem. Its significance is exactly that it is controlled:

$$|\delta\langle T_{00}^{\mathrm{anom}}\rangle| \le \frac{15\,C_C}{8\pi^2\ell^4}\,\eta_\delta.$$

The framework carries the Markov error through three distinct controls:

1. $r_{\mathrm{FR}}(\varepsilon_\delta)$ controls bounded observables at one stage;
2. $\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta)$ controls convergence to the exact Markov normal form on a fixed collar;
3. η_δ^{M} controls modular-Hamiltonian replacements once faithfulness is assumed;
4. the Einstein-branch remainder is a carried $O(\eta_\delta)$ term, not a silent exact identity at finite cutoff.

This is the concrete bridge from "axioms about screens" to "precision GR predictions plus a disciplined carried remainder".

1.6.10 Focusing/QNEC internalization via relative entropy

Once the fixed-cutoff null modular structure of Section 5.2 is combined with the downstream density-upgrade and null-stress premises, focusing constraints follow from information-theoretic principles in the same scaling-limit branch rather than from a separate fixed-cutoff postulate.

Derivation chain. QNEC and focusing are supported conditionally within the same branch:

- Fixed-cutoff null modular bridge (§5.2) $\Longrightarrow$ exact-or-controlled strip additivity and a weak tail generator
- Derived half-sided modular pair + downstream density upgrade $\Longrightarrow$ half-line modular densities and [K, P] = i 2π P
- Relative entropy monotonicity $\Longrightarrow$ QNEC
- Einstein (Thm 5.1) + Raychaudhuri $\Longrightarrow$ QFC for S_gen

Relative entropy monotonicity argument. The key input is the monotonicity of relative entropy under partial trace, which is pure information theory:

$$S(\rho_{AB}\|\sigma_{AB}) \ge S(\rho_A\|\sigma_A).$$

For null deformations parameterized by λ, consider nested null regions $R(\lambda)\subset R(\lambda')$ obtained by varying the entangling cut along v. The modular Hamiltonian K_λ generates the modular flow, and relative entropy satisfies convexity:

$$\frac{d^2}{d\lambda^2}S(\rho_\lambda\|\sigma_\lambda) \ge 0.$$

Proposition 5.10a (Conditional internal QNEC). Under the fixed-cutoff null bridge of §5.2 together with the downstream density-upgrade and resulting null-stress-identification premises, the second null variation of von Neumann entropy satisfies

$$\frac{d^2 S_{\text{bulk}}}{d\lambda^2} \leq 2\pi \langle T_{kk}(\lambda) \rangle,$$

with the 2π normalization fixed by Theorem 4.2.

Proof. Use the derived half-sided modular-inclusion statement of Corollary 5.2e and its Borchers-Wiesbrock consequence from Lemma 5.2f. This gives the translation structure with $[K, P] = -i2\pi P$.

Consider the relative entropy $S(\rho_\lambda \| \omega_\lambda)$ between the state ρ restricted to $R(\lambda)$ and the reference state ω. Monotonicity under restriction to smaller regions ($\lambda' > \lambda$) gives:

$$S(\rho_{R(\lambda)} \| \omega_{R(\lambda)}) \leq S(\rho_{R(\lambda')} \| \omega_{R(\lambda')}).$$

Using the first law $\delta S = \delta \langle K \rangle$ and the geometric half-line modular form supplied by Section 5.2, expand to second order in the deformation. The convexity of relative entropy yields the QNEC inequality. The bound saturates for coherent states in the standard way. QED.

Corollary 5.10b (QFC for generalized entropy). With the central area operator L_C from EC/MaxEnt (Section 5.4), define

$$S_{\text{gen}} = \text{Tr}(\rho L_C) + S_{\text{bulk}}.$$

Given the conditional Einstein branch (Theorem 5.1) and the classical Raychaudhuri identity for null congruences, the Quantum Focusing Conjecture (QFC) follows within the same scaling regime: the generalized expansion Θ_{gen} is non-increasing along null generators.

Proof sketch. The Raychaudhuri equation relates expansion evolution to R_{kk}. Einstein's equation gives $R_{kk} = 8\pi G(T_{kk} - \frac{1}{2} g_{kk} T)$. For null k, this simplifies to $R_{kk} = 8\pi G T_{kk}$. The QNEC (Prop 5.10a) then bounds the bulk entropy production, ensuring $d\Theta_{\text{gen}}/d\lambda \leq 0$. QED.

Significance. This section shows how the Recoverable Generalized Entropy focusing package is internally supported inside the same null-modular branch once the stress-tensor identification and Einstein relation are in place. That package remains part of the stated axiom set.

Theorem 5.1 (Observer-consistency implies a conditional scaling-limit Einstein branch). Under the five OPH axioms, the theorem-local premises ME, ROT, GG, CD, COL, and GB from Section 1.4, the conditional null-stress identification of Section 5.2, and admissible first variations about a maximally symmetric reference state, the cap equilibrium condition implies the rest-frame first-variation relation

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G \, \delta \langle T_{00} \rangle.$$

If this relation holds for all local directions and reference states in the locally Lorentzian scaling regime, overlap consistency upgrades it to the semiclassical Einstein equation modulo the expected Λg_{ab} ambiguity.

Proof sketch. The cap first law identifies δS_C with $\delta\langle K_C\rangle$, while the conditional null bridge relates the bulk modular variation to a null stress-tensor charge. Since the reference state is maximally symmetric, the fixed-volume small-ball area identity enters at first order in the perturbation, so entanglement equilibrium yields the displayed rest-frame relation. Overlap consistency then supplies all local directions, and the null-to-tensor lemma upgrades the relation to the tensor equation modulo Λg_{ab}. QED.

1.6.11 Discrete horizon area spectrum and Hawking emission (established)

The edge-sector structure implies a discrete area spectrum with observable consequences for black hole emission. This section develops a established but sharp prediction.

Area eigenvalues from edge sectors. The central area operator (Section 5.4) is

$$L_C = \sum_\alpha (\log d_\alpha) P_\alpha,$$

where $d_\alpha \in \mathbb{N}$ is the dimension of the edge Hilbert space in sector α. With the normalization $\mathrm{Tr}(\rho L_C) = \langle A\rangle/4G$, the area eigenvalues are

$$A_\alpha = 4G \log d_\alpha = 4\ell_p^2 \ln d_\alpha,$$

where $\ell_p^2 = \hbar G/c^3$ is the Planck area. Since d_α is a positive integer, **areas are discretely spaced** with logarithmic gaps.

Hawking emission energy quantization. For a Schwarzschild black hole with $A(M) = 16\pi G^2 M^2/c^4$, a transition between sectors $d \to d'$ changes the area by

$$\Delta A = 4\ell_p^2 \ln(d'/d).$$

The corresponding energy at infinity is $\Delta E = c^2 \Delta M$, with $\Delta M = \Delta A/(dA/dM)$. This gives

$$\Delta E = \frac{\hbar c^3}{8\pi G M} \ln(d'/d).$$

Using the Hawking temperature $T_H = \hbar c^3/(8\pi G k_B M)$, whose 2π normalization is fixed by the BW_{S^2} tangent-limit normalization of Theorem 4.2:

$$\Delta E = k_B T_H \ln(d'/d).$$

Integer transitions. If dominant transitions multiply the edge dimension by an integer k (i.e., $d'/d = k$), the spectrum becomes a discrete comb:

$$\Delta E_k = k_B T_H \ln k, \qquad \Delta f_k = \frac{c^3}{16\pi^2 G M} \ln k.$$

Structural condition: comb vs. generic discreteness. The log-integer *comb* structure requires the additional dynamical assumption that integer-multiplication transitions (d $\to$ kd) dominate. If generic transitions between arbitrary integers dominate instead, the set of ln(d'/d) values becomes a dense log-rational set that may appear quasi-continuous after folding in linewidths and astrophysical effects. What follows directly from the axioms is the *discrete area spectrum*; the clean comb pattern is conditional on the selection rule.

Continuation-level template (Discrete Hawking spectrum). If the additional integer-transition selection rule is realized, the Hawking emission spectrum consists of discrete lines with

spacing $\Delta E_k = k_B T_H \ln k$, where k is an integer characterizing the dominant sector transitions, instead of a continuous thermal profile.

Mass-independent fractional linewidth. Using Page's semiclassical calculation for emission power $P(M) = p_0 \hbar c^6/(G^2 M^2)$ with $p_0 \approx 2 \times 10^{-4}$, the emission rate is $\dot{N} \approx P/\langle E \rangle$ where $\langle E \rangle = a\, k_B T_H$ with $a \sim \mathcal{O}(1-10)$. The natural linewidth $\Gamma \sim \hbar \dot{N}$ divided by the level spacing gives:

$$\frac{\Gamma}{\Delta E_k} \approx \frac{64\pi^2 p_0}{a \ln k} \approx 3 - 5\%$$

independent of black hole mass. Within this continuation branch, the emission lines are narrow (few-percent fractional width) and the fraction is mass-independent.

Connection to quasinormal modes. The highly-damped Schwarzschild quasinormal modes have asymptotic real part (Motl, 2002):

$$\mathrm{Re}\,\omega \to \frac{c^3}{8\pi G M} \ln 3.$$

This matches exactly the $k = 3$ transition frequency $\Delta E_3/\hbar$. If one adopts a Bohr-type identification between quantum transition frequencies and asymptotic QNM frequencies, this selects

$$\Delta A = 4\ell_p^2 \ln 3 \approx 4.39\, \ell_p^2$$

as the fundamental area quantum.

Conditionality statement. The area quantization follows from the edge-sector structure (derived). The $k = 3$ selection requires the additional interpretive identification with QNM frequencies (not derived from axioms). The linewidth prediction uses standard semiclassical inputs.

Numerical examples. For Δf_k = ($c^3/16\pi^2$GM) ln k:

- **M = 30 M⊙**: k=2 at 29.7 Hz, k=3 at 47.1 Hz
- **M = 1 M⊙**: k=2 at 891 Hz, k=3 at 1412 Hz
- **M = 10^{12} kg (primordial)**: k=2 at 7.3 MeV, k=3 at 11.6 MeV

These frequencies track $k_B T_H \ln k$ exactly and are in principle distinguishable from a continuous thermal spectrum.

Experimental test: PBH burst searches with comb template.

The discrete Hawking comb provides an OPH-unique signature that can be tested against existing gamma-ray data. The smoking gun is **log-integer energy ratios**: if two emission lines are observed at energies E_2 and E_3, their ratio must satisfy

$$\frac{E_3}{E_2} = \frac{\ln 3}{\ln 2} \approx 1.585$$

exactly, independent of black hole mass. This is a parameter-free prediction.

Available instruments and energy coverage. The $k = 2$ line energy $E_2 = k_B T_H \ln 2$ determines which instruments can see a given BH mass:

Instrument	Energy band	BH mass range (k=2 in band)
Fermi GBM (BGO)	0.15–40 MeV	2×10^{11}–5×10^{13} kg
Fermi LAT	0.1–300 GeV	2×10^{7}–7×10^{10} kg
H.E.S.S.	0.1–100 TeV	7×10^{4}–7×10^{7} kg
LHAASO-WCDA	1–15 TeV	5×10^{5}–7×10^{6} kg

Instrument	Energy band	BH mass range (k=2 in band)

Detector resolution vs. intrinsic linewidth. The predicted intrinsic linewidth is 3–5% (mass-independent). Current detector energy resolutions:

- Fermi GBM: $< 10\%$ (0.1–1 MeV), $\sim 4\%$ at 10 MeV (BGO)
- Fermi LAT: $< 10\%$ (1–100 GeV)
- H.E.S.S.: $\sim 15\%$ (TeV)
- LHAASO-WCDA: $\sim 33\%$ (TeV)

The comb is in principle resolvable with GBM/LAT; at TeV energies it would appear as moderately broad bumps rather than sharp lines.

Search protocol. A dedicated OPH-comb search would:

1. Select burst-like candidates (10–120 s time windows, matching existing PBH burst search protocols).
2. Fit each candidate with null model (smooth continuum) vs. OPH comb model (peaks at $E_k = E_0 \ln k$ convolved with detector response).
3. Scan over the single scale parameter $E_0 = k_B T_H$ (equivalently, BH mass).
4. Require at least two lines satisfying log-integer ratio to claim detection.
5. Correct significance for trials (time windows $\times$ sky positions $\times$ E_0 scan).

Observational context. Dedicated PBH burst searches (H.E.S.S., LHAASO) report **no significant bursts**. An OPH-specific comb-template analysis of archival data would:

- Set upper limits on OPH-comb PBH burst rates
- Demonstrate direct testability of the discrete spectrum prediction
- Provide constraints comparable to or stronger than generic PBH burst limits

Data availability. Fermi GBM provides public Time-Tagged Event (TTE) burst data; Fermi LAT provides public photon event lists with documented analysis workflows. H.E.S.S. has a small public test data release.

GW horizon spectroscopy: conditional continuation template for Kerr remnants.

The same continuation-level discrete-horizon branch extends to gravitational wave observables. For Kerr black holes, the thermodynamic first law is $\delta M = T_H \delta S + \Omega_H \delta J$, so the entropy change for absorbing a quantum with frequency ω and azimuthal number m is:

$$\delta S = \frac{\hbar(\omega - m\Omega_H)}{k_B T_H}.$$

In the edge-sector framework, $\delta S = \ln(d'/d)$, so the discreteness condition becomes:

$$\hbar(\omega - m\Omega_H) = k_B T_H \ln k, \qquad k \in \{2, 3, 4, \ldots\}$$

Under those additional discrete-horizon premises, this gives the **GW horizon spectroscopy comb**: discrete resonant frequencies where the horizon can efficiently absorb or emit energy.

Kerr line frequencies. For a remnant with mass M and dimensionless spin $\chi = a_*/M$, define the spin correction factor:

$$g(\chi) = \frac{2\sqrt{1-\chi^2}}{1+\sqrt{1-\chi^2}}, \qquad \Omega_H(M,\chi) = \frac{c^3}{2GM} \cdot \frac{\chi}{1+\sqrt{1-\chi^2}}.$$

The line frequencies are:

$$f_{k,m}(M,\chi) = \frac{m\,\Omega_H(M,\chi)}{2\pi} + \frac{c^3}{16\pi^2 GM}\, g(\chi)\, \ln k$$

This is rigidly constrained: once LIGO/Virgo infers (M,χ) for a remnant, the entire line pattern is fixed with no free parameters.

Line weights from GR envelope + discretization. The line strengths are not arbitrary; they are fixed by matching to the known GR greybody absorption spectrum in the semiclassical limit. The discretization rule gives bin width $\Delta\omega_k \approx \omega_T \ln(1+1/k)$ where $\omega_T = k_B T_H/\hbar$. The net line weight (absorption minus stimulated emission) is:

$$W^{\text{net}}_{k,\ell m} = \Gamma^{\text{GR}}_{\ell m}(\omega_{k,m}) \cdot \Delta\omega_k \cdot \frac{k-1}{k}$$

where $\Gamma^{\text{GR}}_{\ell m}$ is the standard GR greybody factor and the $(k-1)/k$ factor arises from KMS detailed balance with $e^{(\omega - m\Omega_H)/T_H} = k$.

Universal stacking coordinate. Define the dimensionless rescaled frequency:

$$x := \frac{GM}{c^3 g(\chi)}(\omega - m\Omega_H).$$

Then the predicted line locations collapse to universal constants:

$$x_k = \frac{\ln k}{8\pi} \qquad (k = 2, 3, 4, \ldots)$$

Numerically: $x_2 = 0.02758$, $x_3 = 0.04371$, $x_4 = 0.05516$, $x_5 = 0.06404$.

Stacking test. Multiple BBH events can be mapped to this universal x coordinate and stacked. If the comb is real, peaks align across events with different (M,χ); detector noise does not stack coherently.

Comparison to existing work. Prior area-quantization searches (e.g., arXiv:2011.03816) used parameterized models with one free spacing constant. The OPH prediction is more constrained: multiple lines with exact $\ln k$ ratios, plus the $(k-1)/k$ weight hierarchy from detailed balance.

Numerical example (GW170608). Remnant parameters: $M_f \approx 18.0 M_\odot$, $\chi_f \approx 0.69$. For $m = 2$, the horizon rotation frequency is $m\Omega_H/(2\pi) \approx 719$ Hz. The **thermal comb spacing** (the part that encodes the area quantization) is:

k	$\Delta f_k := \frac{c^3 g(\chi)}{16\pi^2 GM} \ln k$ (Hz)	Relative weight $(k-1)/k$
2	41.6	0.500
3	65.9	0.667
4	83.2	0.750
5	96.5	0.800
6	107.5	0.833

The full physical frequencies are $f_{k,2} = 719 + \Delta f_k$ Hz (i.e., 760–827 Hz), outside LIGO's most sensitive band for this remnant. However, the **stacking analysis** uses the rescaled coordinate

$x = GM(\omega - m\Omega_H)/(c^3 g(\chi))$, which maps the thermal spacing to universal constants $x_k = \ln k/8\pi$ regardless of the rotation offset.

Measurement contradiction criterion. The smoking gun is the rigid arithmetic pattern: after rescaling by (M, χ), spectral features must satisfy $f_k/f_2 = \ln k/\ln 2$ exactly, independent of remnant parameters. Absence of coherent stacking at the predicted x_k values identifies a measurement contradiction with the log-integer area spectrum.

1.6.12 Classical mechanics from emergent GR

Once the Einstein equation is established, the framework inherits standard GR consequences. This section makes explicit how classical mechanics emerges.

Stress-energy conservation is automatic. The contracted Bianchi identity is geometric:

$$\nabla^a G_{ab} = 0.$$

Combined with the Einstein equation, this implies:

$$\nabla^a \langle T_{ab} \rangle = 0.$$

Geodesic motion from dust limit. For pressureless classical matter ("dust"), $T^{ab} = \rho\, u^a u^b$. Conservation yields:

$$\nabla_a(\rho u^a u^b) = 0 \quad \Rightarrow \quad u^b \nabla_a(\rho u^a) + \rho\, u^a \nabla_a u^b = 0.$$

Projecting orthogonally to u^b using $h^b{}_c = \delta^b{}_c + u^b u_c$ kills the first term, giving:

$$\rho\, u^a \nabla_a u^b = 0 \quad \Rightarrow \quad u^a \nabla_a u^b = 0.$$

This is the geodesic equation: free classical bodies follow spacetime geodesics.

Newtonian limit from weak-field GR. Take the weak-field, slow-motion limit with metric:

$$g_{00} \approx -(1 + 2\Phi/c^2), \qquad g_{0i} \approx 0, \qquad g_{ij} \approx \delta_{ij}(1 - 2\Phi/c^2),$$

and velocities $|\mathbf{v}| \ll c$. Then $G_{00} \approx 2\nabla^2\Phi/c^2$ (leading order), and $T_{00} \approx \rho c^2$. The Einstein equation reduces to:

$$\nabla^2 \Phi = 4\pi G \rho.$$

Geodesic motion reduces to:

$$\ddot{\mathbf{x}} = -\nabla\Phi.$$

These are Newton's gravitational law and Newton's second law. Classical mechanics is recovered as a controlled limit of the emergent GR dynamics.

Precision classical predictions. Once the field equation is fixed to Einstein form, the framework inherits the standard GR precision toolbox (post-Newtonian expansion, lensing, time delay, etc.), with no free "shape" parameters beyond G and Λ.

Selected precision predictions (in the regime where the GR derivation applies):

Light bending by mass M: For impact parameter b,

$$\Delta\theta = \frac{4GM}{c^2 b}.$$

For the Sun with $b \approx R_\odot$: $\Delta\theta \approx 1.751$ arcsec.

Mercury perihelion advance: Per orbit,

$$\Delta\varpi = \frac{6\pi GM}{a(1-e^2)c^2}.$$

Using Mercury's orbital parameters: $\Delta\varpi \approx 42.98$ arcsec/century.

Gravitational redshift: Between two radii in a static potential,

$$\frac{\Delta\nu}{\nu} \approx \frac{\Delta\Phi}{c^2}.$$

For the Sun (surface to infinity): $z \approx 2.12 \times 10^{-6}$.

These predictions are fixed functions of G and known source parameters, and are confirmed observationally to high precision. The framework inherits them automatically once the Einstein equation is derived.

1.6.13 Precision gravity predictions and experimental bounds

The gravity sector makes symmetry-protected exact-zero predictions that can be confronted with the tightest available experimental bounds. This section translates the theoretical predictions into the specific observables that experiments actually constrain.

Speed of gravitational waves. The derived GR regime implies massless gravitons propagating on the same null cones as photons:

$$\frac{c_{\mathrm{GW}} - c}{c} = 0 \text{ exactly.}$$

Current bound (GW170817 + GRB 170817A multi-messenger):

$$-3 \times 10^{-15} < \frac{c_{\mathrm{GW}} - c}{c} < +7 \times 10^{-16} \quad \text{(90\% credibility).}$$

For a source at $\sim$ 40 Mpc, this fractional difference corresponds to only a few seconds of propagation-time mismatch across $\sim 10^8$ years of travel.

Graviton mass. The gauge redundancy (diffeomorphism invariance) forbids a hard mass term:

$$m_g = 0 \text{ exactly.}$$

Current bound (GW dispersion analysis, PDG 2025):

$$m_g \leq 1.76 \times 10^{-23} \text{ eV}/c^2 \quad \text{(90\% credibility).}$$

This corresponds to a reduced Compton wavelength $\bar{\lambda}_C \gtrsim 1.6 \times 10^{16}$ m, i.e., order ~ 1.6 light-years.

No dipole radiation. Many modified gravity theories predict extra channels (scalar/vector) producing dipolar radiation at (-1)PN order. The derived GR limit predicts no such channel.

Current bound (GW170817 inspiral phasing, PDG 2025):

$$-4 \times 10^{-6} < \delta\hat{p}_{-2} < 2 \times 10^{-5} \quad \text{(90\% credibility).}$$

Only tensor polarizations. The GR outcome means only the two tensor (helicity-2) modes propagate. Pure non-tensor hypotheses are disfavored by observational constraints, and mixed tensor-scalar/vector models are tightly constrained.

Equivalence principle tests. Additional null checks from the derived GR structure:

- Universality of free fall (space tests): precision ~10^{-15}
- Nordtvedt parameter ($\eta = 4\beta$ - γ): $(0.47 \pm 0.55) \times 10^{-4}$
- Binary pulsar radiative damping (PSR J0737-3039): 0.999963 ± 0.000063

1.6.14 Theory-side error propagation from Markov bounds

The framework provides exact-zero predictions and quantitative control over how well those predictions hold. The Markov/recovery machinery can be propagated through the entire GR emergence chain.

The key quantitative hook. From Theorem 3.1, if the target state satisfies

$$I(A_k : C_k \mid B_k) \leq \varepsilon_k,$$

then recovery maps exist with trace-distance error

$$\delta_k = 2\sqrt{\ln 2 \cdot \varepsilon_k}.$$

Trace distance gives immediate bounds on observable errors. Using the standard dual norm inequality:

$$|\langle O \rangle_\rho - \langle O \rangle_\sigma| \leq \|O\|_\infty \|\rho - \sigma\|_1 = 2\|O\|_\infty D(\rho, \sigma),$$

where $D(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$ is the trace distance.

Exponential decay from MX. The mixing assumption (Section 2.3) provides:

$$I_\omega(A_\delta : D_\delta \mid B_\delta) \leq c \cdot |\partial C|_{\mathrm{UV}} \cdot e^{-\delta/\xi}.$$

Combining these gives an explicit precision dial:

$$\delta_{\mathrm{step}} \lesssim 2\sqrt{\ln 2 \cdot c \cdot |\partial C|_{\mathrm{UV}}} \cdot e^{-\delta/(2\xi)}.$$

What precision requires. To match the GW speed bound ($\sim 10^{-15}$ fractional accuracy), the recovery-map error must satisfy:

$$\delta \lesssim 10^{-15} \quad \Rightarrow \quad \varepsilon \lesssim \frac{(\delta/2)^2}{\ln 2} \approx 3.6 \times 10^{-31}.$$

This is extremely small, but achievable: with a macroscopic boundary ($|\partial C|_{\mathrm{UV}} \sim 10^{35}$ for a meter-scale boundary at Planck UV scale), the exponential decay $e^{-\delta/\xi}$ with $\delta/\xi \sim$ a few hundred easily pushes below 10^{-31} once the prefactor is included.

Precision summary. The framework provides:

1. Exact-zero predictions ($m_g = 0$, $c_{\mathrm{GW}} = c$) from symmetry protection.
2. Translation of those zeros into the specific observables experiments constrain.
3. Explicit bounds on how far derived geometric statements can drift, using the conditional mutual information $\rightarrow$ trace distance $\rightarrow$ observable error chain.

This is the concrete path from "axioms about screens" to "precision GR predictions with quantitative error control."

1.6.15 Dark matter as modular anomaly (program-level)

This subsection is a program-level continuation, not a recovered-core or Phase-II theorem. The formulas below record the normalization estimate obtained if the modular-anomaly term dominates the relevant deep-IR phenomenology; the dark-matter identification and RAR-like scaling law remain additional assumptions.

The modular anomaly term T_{ab}^{anom} derived in Section 5.9 provides a natural candidate for what is observationally interpreted as dark matter, without introducing new particle species.

The identification. The anomalous stress-energy contribution

$$\langle T_{00}^{\text{anom}} \rangle = \frac{15}{8\pi^2} \cdot \frac{\delta \langle K_C^{(\text{anom})} \rangle}{\ell^4}$$

is "dark" by construction: it arises from information-theoretic/gravitational structure (modular Markov imperfections), not from Standard Model fields. It gravitates but does not couple electromagnetically. This is precisely what "dark matter" means observationally.

Connection to the cosmological constant. The framework makes Λ a global capacity parameter:

$$\Lambda = \frac{3\pi}{GN_{\text{scr}}}, \quad r_{dS} = \sqrt{\frac{3}{\Lambda}}.$$

This introduces an unavoidable IR length scale r_{dS}. Galaxy "dark matter" phenomenology is an IR phenomenon-it appears when accelerations are small and distances are large.

Emergent acceleration scale. In the Newtonian/weak-field regime, any IR modification from T_{00}^{anom} must:

1. Vanish if $r_{dS} \to \infty$ (infinite capacity, no de Sitter scale)
2. Be controlled by r_{dS} as the only new IR scale
3. Carry non-tunable coefficients from the derivation

The anomaly enters with prefactor $\frac{15}{8\pi^2}$. The natural acceleration scale constructible from (Λ, c) with this coefficient is:

$$\boxed{a_0^{(\text{OPH})} := \frac{15}{8\pi^2} \cdot c^2 \sqrt{\frac{\Lambda}{3}} = \frac{15}{8\pi^2} \cdot \frac{c^2}{r_{dS}}}$$

Continuation-level normalization estimate. Using Planck 2018 ΛCDM parameters ($H_0 \approx 67.4$ km/s/Mpc, $\Omega_\Lambda \approx 0.685$):

- $\Lambda \approx 1.09 \times 10^{-52}$ m^{-2}
- $r_{dS} \approx 1.66 \times 10^{26}$ m
- Therefore:

$$\boxed{a_0^{(\text{OPH})} \approx 1.03 \times 10^{-10} \text{ m/s}^2}$$

For comparison, observational fits to galaxy regularities (RAR/MDAR/MOND phenomenology) quote $a_0 \sim 1.2 \times 10^{-10}$ m/s^2. The prediction lands within 15% without introducing a new free parameter beyond screen capacity/Λ.

Phenomenological consequences. If T_{00}^{anom} sources the inferred dark matter, the Newtonian limit becomes:

$$\nabla^2\Phi = 4\pi G(\rho_b + \rho_{\text{anom}}),$$

i.e., baryons plus an effective extra density. The radial acceleration relation (RAR) takes the form:

$$g_{\text{obs}} \approx g_b + \sqrt{a_0 \cdot g_b}, \quad g_{\text{DM}} := g_{\text{obs}} - g_b \approx \sqrt{a_0 \cdot g_b}.$$

With $a_0 = a_0^{(\text{OPH})}$ fixed, this predicts:
(i) Baryonic Tully-Fisher relation.

$$V^4 \approx G \cdot M_b \cdot a_0^{(\text{OPH})}$$

where V is the asymptotic rotation velocity and M_b is baryonic mass.
(ii) Flat rotation curves. For a point mass M_b:

$$g_{\text{DM}}(r) = \frac{\sqrt{GM_b\, a_0^{(\text{OPH})}}}{r} \quad \Rightarrow \quad M_{\text{DM}}(r) \propto r$$

i.e., inferred dark mass grows linearly with radius, producing flat rotation curves.
(iii) Characteristic surface density.

$$\Sigma_0^{(\text{OPH})} = \frac{a_0^{(\text{OPH})}}{2\pi G} \approx 0.25 \text{ kg/m}^2 \approx 120\, M_\odot/\text{pc}^2.$$

This is in the range of observed central halo surface densities.
Status. What is grounded in the framework developed here:

- The modular anomaly term exists with fixed coefficient $\frac{15}{8\pi^2}$
- Λ and r_{dS} are determined by screen capacity
- The anomaly acts as "effective extra gravity" in the Newtonian limit

What is an additional assumption (derivation target):

- That T_{00}^{anom} is the dominant source of galaxy-scale "dark matter" phenomenology
- That in the deep IR the anomaly organizes into RAR-like scaling with normalization inherited from the $\frac{15}{8\pi^2}$ prefactor

This is best viewed as a **program-level derivation target** (like the θ_{QCD} program in Section 8.4): the framework contains the natural ingredients, and the precise prediction follows if those ingredients dominate the relevant phenomenology.

Falsifiability. The prediction $a_0^{(\text{OPH})} \approx 1.03 \times 10^{-10}$ m/s^2 is sharp. If galaxy data definitively require a different value (say, $a_0 > 1.5 \times 10^{-10}$ m/s^2), or if the RAR normalization varies systematically with environment in ways incompatible with a universal Λ-derived scale, this interpretation would be contradicted.

1.6.16 De Sitter holography: static patch vs boundary-at-infinity

A natural question arises: how does this framework relate to the "unsolved problem" of de Sitter holography?

What the usual dS holography problem is. When people say "dS holography is unsolved," they typically mean: we don't have anything as sharp as AdS/CFT, where the bulk has a timelike asymptotic boundary supporting a well-defined dual CFT with a precise dictionary. For de Sitter, there is no asymptotic timelike boundary in the static patch where you can just "put the field theory." The classic dS/CFT proposal (Euclidean CFT at future infinity) has notorious issues including potential non-unitarity and complex weights in the would-be dual.

What this model does differently. Our framework begins with an observer's static patch and its horizon screen (S^2), building a net of subregion algebras on that screen. This is a fundamental fork away from AdS/CFT-style holography:

AdS/CFT	This framework
Codimension-1 boundary at infinity	Codimension-2 horizon screen (S^2)
Single global boundary theory	Observer-dependent patches that overlap
Dual CFT required	Only algebras + consistency conditions
Negative Λ	Positive Λ natural

This aligns with the **static patch / complementarity** approach in the dS literature, where the fundamental description is patch-based and different static patches are related by consistency rules, not by a single god's-eye boundary theory.

The mechanism: Λ as global capacity, not local physics.

A key structural result (Lemma 5.2) shows that null modular data can reconstruct T_{ab} only up to an additive ϕg_{ab} ambiguity. This is the statement that vacuum energy / cosmological constant shifts are invisible to the null-data route. The Einstein equation derived from entanglement equilibrium is fixed only up to Λg_{ab}.

Therefore Λ cannot be determined by local consistency. It must be fixed by a **global constraint**: the total number of degrees of freedom on the screen. The de Sitter link is:

$$\Lambda = \frac{3\pi}{GN_{\text{scr}}}$$

If the screen Hilbert space has finite dimension $\dim(\mathcal{H}_{\text{tot}}) = \exp(S_{dS})$, then the natural semi-classical interpretation of that finite entropy is a cosmological horizon, and matching to GR via the entropy-area relation gives positive Λ.

What this "solves" vs what it assumes.

The model does **not** solve the classic "give me a unitary CFT on the boundary at infinity for dS" problem. It doesn't aim there. Instead, it provides a coherent route to **patch holography** where de Sitter static patches are natural:

1. The fundamental object is a horizon screen in a static patch description.
2. Λ is a capacity parameter tied to finite Hilbert space dimension, not locally reconstructible "vacuum energy."
3. Einstein-like dynamics emerge up to Λg_{ab}; the numerical value of Λ is inferred from the observed cosmological constant, not predicted.

Many observers, one Λ. The philosophical stance ("no objective reality, only subjective perspectives that must agree on overlaps") maps onto dS static-patch intuition: each timelike observer has a horizon and a patch; there's no operational access to a single global description. The de Sitter parameter Λ is the one thing that **can** be shared across overlaps: a global capacity constraint that all consistent overlapping descriptions inherit.

Summary. The model gets de Sitter by moving the holographic screen from "infinity" to an observer's horizon and by elevating de Sitter entropy (finite screen capacity) to a fundamental input. The usual dS holography obstacles are exactly the ones avoided by refusing a global, boundary-at-infinity viewpoint. This is not a bug-it's the point.

1.7 Standard Model from Gluing Consistency

This section has two logically distinct parts:

Part I (§6.1): The mathematical reconstruction machinery. Given edge-center completion (Theorem 2.3), we get fixed-cutoff sector categories and a refinement-stable colimit category. If that colimit satisfies standard categorical properties (rigid, symmetric, C^*), Tannaka-Krein reconstruction yields *some* compact gauge group G. This part has explicit hypotheses and is unconditional once the refinement-stable category is specified.

Part II (Derived from MAR, §6.2 onward): The Selection Axiom MAR (Minimal Admissible Realization) narrows from "some G" to the Standard Model gauge group once the admissible class includes one connected abelian charge factor. The former Selectors S1–S3 and the separate minimality steps for N_c and N_g are all consequences of MAR applied to the admissible class. The SM derivation follows from the extended gauge-selection package T_{ext}: the five OPH axioms together with the explicit transportability premise $[z] = 0$ and the gauge-reconstruction hypotheses used in Theorem 6.1. See Part II of this manuscript for the complete proof.

1.7.1 Edge sector category and gauge group reconstruction

Edge-center completion (Theorem 2.3) provides, at each regulator scale r, a finite tensor category Sect_r of transportable edge charges. The reconstruction category is the refinement-stable directed colimit

$$\mathsf{Sect}_\infty := \varinjlim_r \mathsf{Sect}_r,$$

where the transition functors are induced by refinement and only sectors/intertwiners that persist under refinement are retained. This gives a tensor category Sect_∞ of edge charges:

- objects: refinement-stable transportable sector labels,
- morphisms: intertwiners between sectors,
- tensor product: fusion by collar concatenation, $\alpha \otimes \beta = \bigoplus_\gamma N_{\alpha\beta}{}^{\gamma}\, \gamma$,
- duals: orientation reversal $\alpha \leftrightarrow \bar{\alpha}$,
- symmetric braiding in the EFT regime (no anyonic statistics in 3+1D).

Let $\mathcal{F} : \mathsf{Sect}_\infty \to \mathsf{Hilb}_{\text{fd}}$ be the fiber functor that sends each sector to its edge multiplicity space. The fibers are finite-dimensional objectwise even though Sect_∞ can have infinitely many simple objects.

Theorem 6.1 (Tannaka/DR reconstruction). If Sect_∞ is a rigid symmetric C^* tensor category with a faithful fiber functor $\mathcal{F}$, then there exists a compact group G, unique up to isomorphism, such that $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$. Moreover,

$$G = \mathrm{Aut}_\otimes(\mathcal{F})$$

is a compact subgroup of a product of unitary groups.

Proof sketch. Define G as the group of monoidal natural automorphisms of $\mathcal{F}$. This is compact because it is closed in a product of unitary groups. By Tannaka-Krein/DR reconstruction, objects and morphisms of Sect_∞ are recovered as finite-dimensional representations and intertwiners of G. QED.

Corollary 6.1 (field algebra reconstruction, conditional). If in the small-region limit the edge sectors are localized and transportable in the DHR sense (i.e., charges can be moved between patches without changing their fusion), then there exists a field algebra $\mathcal{F}$ and a compact group G such that $\mathcal{A} = \mathcal{F}^G$. This is the Doplicher-Roberts reconstruction of local gauge symmetry from sectors. QED.

Proposition 6.1a (Transportability from gluing obstruction). DHR transportability is not an independent assumption. In the gluing framework (Section 3), transportability is precisely the statement that charges can be moved between patches without changing fusion rules. The obstruction to path-independent transport is the Čech-type central cocycle z_{ijk} from premise CD.

Explicitly: the gluing framework gives a gauge-invariant Čech 2-cocycle class $[z]$ built from triple overlaps (Section 6.6). Transportability holds iff this class vanishes:

$$\text{DHR transportable} \iff [z] = 0 \iff \text{loop-coherent gluing.}$$

Proof. Transportability means charges can be moved along any path without affecting the result. In gluing language, this is path-independent parallel transport of edge labels. Lemma 6.12 shows that loop-coherent global gluing exists iff $[z] = 0$. But loop-coherent gluing is exactly path-independent transport, so the equivalence holds. QED.

Corollary. The "DHR transportability" condition in Corollary 6.1 is internal to the gluing framework: it is equivalent to requiring that the central obstruction class vanishes. This is a constraint on the allowed sector structure, not an external physical assumption.

1.7.2 Selecting the SM factors (derived from MAR)

Theorem 6.1 yields *some* compact G. The Selection Axiom MAR (Minimal Admissible Realization) uniquely determines which G is realized.

Selection Axiom MAR (Minimal Admissible Realization). Among all OPH-realizable sector packages $\mathfrak{S}$ consisting of the connected Lie gauge-sector image relevant in the low-energy EFT, its admissible light chiral matter content, and one Higgs doublet, and which are (i) loop-coherent / transportable ($[z] = 0$), (ii) anomaly-free, (iii) refinement-stable with light chiral matter, (iv) single-Higgs Yukawa-completable with one connected abelian charge factor acting nontrivially on the coupled carrier, (v) intrinsically CP-capable, (vi) weak-sector UV-completable, the realized low-energy package is the lexicographically minimal one under

$$C(\mathfrak{S}) = (\chi_{\mathrm{cpl}},\ N_{\mathrm{nonab}},\ N_c,\ N_g).$$

See Part II of this manuscript for the complete formal statement, admissibility definitions, and proof.

Here χ_{cpl} is the **coupled edge capacity**: the dimension of the minimal unitary carrier containing a common irreducible block on which the admissible pseudoreal and complex nonabelian charge types both act nontrivially. This is intentionally stronger than the abstract minimal faithful representation dimension. For $S(U(3) \times U(2))$, the block-diagonal action on $\mathbb{C}^3 \oplus \mathbb{C}^2$ is faithful of dimension 5, but it is not coupled and therefore does not enter MAR. The object minimized by MAR is the sector package $\mathfrak{S}$, not the bare tensor category by itself. MAR is thus an explicit structural-economy selector on the admissible class, not a theorem derived from the earlier axioms.

Note on [z]=0. The loop-coherent gluing condition $[z] = 0$ (Proposition 6.1a) is kept as an explicit premise of the extended theory, not hidden inside MAR. It ensures that the reconstructed compact group acts as a genuine global gauge symmetry. By Proposition 6.1a, this is equivalent to DHR transportability.

What MAR derives. The former Selectors S1 (sector factorization), S2 (minimal sector content), and S (coupled edge-capacity minimality) are all consequences of MAR applied to the admissible class:

- **Product structure** (formerly S1): follows from the minimal coupled carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$, which enforces commuting color and weak actions.
- **Minimal sector content** (formerly S2): the pseudoreal doublet and complex triplet are the minimal nonabelian representations satisfying the admissibility conditions, while the connected abelian charge factor is an explicit admissibility input.
- **Coupled edge-capacity minimality** (formerly S): is the first component of MAR's complexity vector.

With MAR stated, the SM derivation proceeds via standard lemmas:

Lemma 6.2 (S1 implies product group). If $\mathsf{Sect} \simeq \mathrm{Rep}(G)$ and $\mathsf{Sect} \simeq \mathsf{Sect}_1 \boxtimes \mathsf{Sect}_2$, then

$$G \cong G_1 \times G_2, \qquad \mathsf{Sect}_i \simeq \mathrm{Rep}(G_i).$$

QED.

Lemma 6.3 (SU(2) from a pseudoreal doublet). Let H be a positive-dimensional compact connected Lie group with a faithful 2D pseudoreal unitary representation V. Then the semisimple part of H contains an $SU(2)$ factor acting as the fundamental doublet. Finite or disconnected counterexamples are not part of the theorem package, which only applies the lemma to the identity component on the relevant nonabelian image. QED.

Lemma 6.4 (SU(3) from an irreducible triplet). Let H be a positive-dimensional compact connected Lie group with a faithful irreducible complex 3D unitary representation W. Then the semisimple image contains an $SU(3)$ factor acting as the fundamental triplet. Finite or disconnected counterexamples are not part of the theorem package, which only applies the lemma to the identity component on the relevant nonabelian image. QED.

Lemma 6.5 (Connected abelian factor criterion). If the admissible sector package contains a connected abelian charge factor acting nontrivially on the coupled carrier, then the identity component of the abelianized reconstructed group contains a one-torus. Under the single connected abelian-factor admissibility premise, this factor is $U(1)$. QED.

Proposition 6.6 (physical group quotient). If the realized matter spectrum has hypercharges quantized in sixths, then the kernel acting trivially on all realized sectors is Z_6, so

$$G_{\text{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

QED.

Proposition 6.6a (SM from MAR). Under the Selection Axiom MAR:

- Admissibility conditions (iii)–(iv) require both a pseudoreal nonabelian charge type and a complex nonabelian charge type, and include one connected abelian charge factor acting nontrivially on the coupled carrier (Lemma 6.7, Corollary 6.8).
- The minimal faithful pseudoreal representation is the doublet ($\chi = 2$), giving $SU(2)$. No intrinsically complex 2D irrep qualifies in the connected compact Lie class, because the connected derived image of any irreducible 2D unitary representation lies in $SU(2)$, so the nonabelian action is pseudoreal up to abelian twist. The first intrinsically complex case is therefore the triplet ($\chi = 3$), giving $SU(3)$.
- The minimal coupled carrier for both is $\mathbb{C}^3 \otimes \mathbb{C}^2$, giving coupled edge capacity $\chi_{\text{cpl}} = 6$.
- The block-diagonal faithful representation $\mathbb{C}^3 \oplus \mathbb{C}^2$ of $S(U(3) \times U(2))$ has dimension 5, but it is not coupled and therefore is not the MAR minimizer.
- The maximal compact subgroup of $U(6)$ acting on $\mathbb{C}^3 \otimes \mathbb{C}^2$ with commuting actions is $(SU(3) \times SU(2) \times U(1))/(\text{finite center})$.
- The commutant of $SU(3) \times SU(2)$ inside $U(6)$ is exactly $U(1)$, so any connected abelian factor acting nontrivially on the coupled carrier is necessarily a single $U(1)$, and no additional continuous factors appear without increasing χ_{cpl}.
- Product structure (formerly Selector S1) is not separately assumed: it follows from the tensor product structure of the minimal coupled carrier.

Combined with Proposition 6.6 (hypercharges quantized in sixths from the realized spectrum), this yields:

$$G_{\text{phys}} = \frac{SU(3) \times SU(2) \times U(1)}{Z_6}.$$

The full proof is in Part II of this manuscript.

1.7.3 Refinement stability and unprotected relevant operators

The refinement-stability used later in the gauge branch is contained in the third OPH axiom. The UV states lie in one finite-dimensional MaxEnt family built from the same local constraint list at every cutoff. Premise RS names the selected stable branch of this family; it does not add a separate axiom beyond the MaxEnt branch itself.

Concretely, if

$$\omega_\ell(\lambda) = Z_\ell(\lambda)^{-1} \exp\left(-\sum_x \sum_a \lambda_a O_a(x) - \sum_b \mu_b Q_b \right),$$

then coarse-graining acts on the finite set of multipliers rather than on an uncontrolled infinite space of couplings. The only question is whether the realized trajectory stays on a stable manifold of this multiplier map. Relevant operators that are neither symmetry-forbidden nor retained in the selected constraint family are precisely the directions that try to push the branch off that stable manifold.

Lemma 6.7 (refinement-stable MaxEnt branch forbids unprotected relevant operators). Assume the local finite-constraint MaxEnt/refinement branch of the third OPH axiom. Let

$\mathcal{O}$ be a gauge-invariant Lorentz-scalar relevant deformation in the emergent EFT sense ($\Delta < 4$ in $3+1$D), allowed by symmetry and absent from the retained constraint family. Then the selected branch cannot keep the coupling of $\mathcal{O}$ at zero without fine tuning. Generic refinement induces a nonzero component along that direction, the component grows under RG, and the flow leaves the selected critical branch for a gapped IR phase. In the near-vacuum regime at fixed macroscopic charges or energy, such a gapped phase has strictly smaller entropy density than the corresponding critical phase, so MaxEnt favors spectra in which $\mathcal{O}$ is symmetry-forbidden or explicitly retained among the constraints.

Proof sketch. Linearize the renormalization map on the finite-dimensional multiplier space around the selected branch. A relevant operator gives an unstable direction with scaling exponent $y > 0$. If $\mathcal{O}$ is not fixed by symmetry and is not part of the retained constraint family, generic UV mismatch produces a nonzero component along that direction. Because $y > 0$, repeated coarse-graining amplifies the component and pushes the flow off the selected branch unless one fine-tunes it away at every scale. Turning on the relevant coupling generates a mass scale and gaps the IR. At fixed low energy density, the gapped phase has lower entropy density than the corresponding gapless or critical branch, so the MaxEnt selector disfavors it. QED.

Corollary 6.8 (chirality selector). A gauge-invariant Dirac mass term is a relevant scalar. If both chiralities exist in conjugate representations, the mass term is allowed and will be generated under refinement unless symmetry-forbidden or explicitly retained as a protected constraint. Therefore the MaxEnt/refinement-stable branch selects chiral fermion content, or else an explicit protecting mechanism, as the natural way to keep light fermions without fine tuning. QED.

1.7.4 Generation number from CP violation and refinement stability (derived from MAR)

Anomaly cancellation is generation-by-generation, so it does not fix the number of generations. The admissibility conditions constrain the window; MAR selects the minimum.

Proposition 6.9 (The number of generations is N_g = 3). Under (i) intrinsic CP violation in the quark sector, (ii) UV-completability of SU(2)_L (asymptotic freedom at one loop), (iii) MaxEnt/refinement stability selecting minimal viable spectrum, and (iv) the derived N_c = 3 from Theorem 6.14, the generation number is

$$N_g = 3.$$

Inputs.

1. **Intrinsic CP violation exists** in the quark sector (empirical fact; also, the framework treats "intrinsic CP violation" as a selector input).
2. **UV-completability proxy**: SU(2)_L is asymptotically free at one loop in the emergent EFT.
3. **MaxEnt + refinement stability** penalizes unnecessary unfixed flavor structure, selecting the minimal viable spectrum.
4. Use the derived N_c = 3 from Theorem 6.14.

Step 1: CP violation lower bound. The number of physical CP-violating phases in an N_g × N_g CKM matrix is:

$$\#(\text{CP phases}) = \frac{(N_g - 1)(N_g - 2)}{2}.$$

- For N_g = 1, 2: this is 0 → **no intrinsic CP violation possible**.
- For N_g = 3: this is 1 → **intrinsic CP violation possible**.

So intrinsic CP violation requires:

$$N_g \geq 3.$$

Step 2: SU(2) asymptotic freedom upper bound. The one-loop coefficient is:

$$b_{\mathrm{SU}(2)} = \frac{22}{3} - \frac{1}{3}N_g(N_c + 1) - \frac{1}{6},$$

where the final $-1/6$ is the contribution of one complex Higgs doublet. Asymptotic freedom means $b_{\mathrm{SU}(2)} > 0$, i.e.,

$$N_g(N_c + 1) < \frac{43}{2}.$$

With N_c = 3, we have N_c + 1 = 4, so:

$$4N_g < \frac{43}{2} \quad \Rightarrow \quad N_g \leq 5.$$

Combining: 3 ≤ N_g ≤ 5.

Step 3: MAR selection. Given the allowed window {3, 4, 5}, MAR (fourth component of the complexity vector $C(\mathfrak{S})$) selects the smallest viable choice:

$$N_g = 3.$$

QED.

Why this is convincing.

- It predicts a **single integer**.
- It uses **two admissibility conditions** (CP violation exists; weak sector is UV-completable) plus MAR's lexicographic minimality.
- It is not a fit to a continuous number.
- Under T_{ext}, this is a derived result, not conditional.

1.7.5 Hilbert-space formulation of gluing data

Let {P_i} be a good cover of the screen. For each patch, fix a representation

$$\pi_i : \mathcal{A}_i \to \mathcal{B}(\mathcal{H}_i).$$

For each overlap, choose a unitary intertwiner

$$U_{ij} : \mathcal{H}_j \to \mathcal{H}_i$$

such that for all $O \in \mathcal{A}_{ij}$,

$$\pi_i(O) = U_{ij}\pi_j(O)U_{ij}^{\dagger}.$$

Normalize U_ii = 1 and U_ji = U_ij†.

Lemma 6.10 (centrality on triple overlaps). On a triple overlap define

$$\Omega_{ijk} := U_{ij} U_{jk} U_{ki}.$$

For all $O \in \mathcal{A}_{ijk}$,

$$\Omega_{ijk} \pi_i(O) = \pi_i(O) \Omega_{ijk}.$$

Proof. Conjugation by U_ki sends π_i(O) to π_k(O), by U_jk to π_j(O), by U_ij back to π_i(O). Thus conjugation by Ω_ijk fixes π_i(O), so Ω_ijk commutes with π_i(O). QED.

Lemma 6.11 (gauge behavior). If $\tilde{U}$_ij = V_i U_ij V_j† with V_i acting trivially on overlap observables, then

$$\tilde{\Omega}_{ijk} = V_i \Omega_{ijk} V_i^{\dagger}.$$

In particular, if Ω_ijk is central, its class is gauge invariant. QED.

1.7.6 Loop obstruction class (central defect)

Assume the defect is central and write

$$\varphi_{ij} := \mathrm{Ad}(U_{ij}) \quad \text{on } \mathcal{A}_{ij}.$$

This is the abelian truncation of the full 2-group obstruction in Section 3.4.

Then there exist central unitaries z_{ijk} such that

$$\varphi_{ij} \varphi_{jk} \varphi_{ki} = \mathrm{Ad}(z_{ijk}).$$

Theorem 6.12 (loop-coherent gluing iff vanishing obstruction). The family $\{z_{ijk}\}$ is a Čech 2-cocycle, and its class $[z]$ is gauge invariant. On any quadruple overlap P_{ijkl},

$$z_{jkl} z_{ikl}^{-1} z_{ijl} z_{ijk}^{-1} = 1.$$

A loop-coherent global gluing exists iff [z] = 0.

Proof. Compare two parenthesizations of ϕ_ij ϕ_jk ϕ_kl ϕ_li on a quadruple overlap to obtain the cocycle condition above. Gauge changes shift z by a coboundary. If [z]=0, rephase by a 1-cochain to eliminate defects and obtain path-independent transport. Conversely, loop-coherent gluing implies z_ijk = 1. QED.

1.7.7 EFT reduction to anomaly cancellation

Assume ExtEFT: a low-energy 3+1D chiral gauge theory exists with group G. Then the obstruction class $[z]$ is structurally analogous to the EFT 't Hooft anomaly class and is expected to map to it after a separate anomaly-descent construction. That map is not constructed in this manuscript, so $[z] = 0$ is used here as an internal gluing/transportability condition rather than a proved equivalence to EFT anomaly cancellation.

1.7.8 Hypercharge from anomaly freedom and Yukawas

Theorem 6.13 (Hypercharge from anomaly freedom and Yukawas). Assume gauge group SU(N_c) × SU(2) × U(1)_Y and one generation of left-handed Weyl fermions (Q, u^c, d^c, L, e^c), with a Higgs doublet H and Yukawa terms

$$QHu^c, \qquad QH^\dagger d^c, \qquad LH^\dagger e^c.$$

Then anomaly freedom and Yukawa invariance fix the hypercharges up to an overall normalization, yielding the Standard Model pattern for N_c = 3.

Proof. Yukawa invariance gives

$$Y_u = -(Y_Q + Y_H), \quad Y_d = -Y_Q + Y_H, \quad Y_e = -Y_L + Y_H.$$

Anomaly cancellation yields

$$\begin{aligned} SU(2)^2U(1): &\quad N_cY_Q + Y_L = 0, \\ \text{grav}^2U(1): &\quad 2N_cY_Q + N_cY_u + N_cY_d + 2Y_L + Y_e = 0. \end{aligned}$$

Solving gives

$$Y_L = -N_cY_Q, \quad Y_H = N_cY_Q, \quad Y_u = -(N_c+1)Y_Q, \quad Y_d = (N_c-1)Y_Q, \quad Y_e = 2N_cY_Q.$$

With these relations, SU(N_c)^{2}U(1) and U(1)^3 anomalies vanish automatically. Fixing the normalization by Q = T_3 + Y and Q(ν_L)=0 gives

$$Y_Q = \frac{1}{2N_c}.$$

For N_c = 3,

$$Y_Q = \frac{1}{6}, \quad Y_L = -\frac{1}{2}, \quad Y_e = 1, \quad Y_u = -\frac{2}{3}, \quad Y_d = \frac{1}{3}, \quad Y_H = \frac{1}{2}.$$

Without Yukawas, the cubic anomaly leaves two discrete branches (Y_u, Y_d exchange). Yukawa invariance selects the branch with a single Higgs doublet. QED.

Corollary 6.13a (Exact rational hypercharges). With the derived N_c = 3, the hypercharge assignments are uniquely fixed to exact rational values:

$$Y_Q = \tfrac{1}{6}, \quad Y_L = -\tfrac{1}{2}, \quad Y_u = -\tfrac{2}{3}, \quad Y_d = \tfrac{1}{3}, \quad Y_e = 1, \quad Y_H = \tfrac{1}{2}.$$

Why this is convincing.

- These are **exact rationals**, not approximate numbers.
- Their ratios are fixed by anomaly freedom + Yukawa invariance, and the absolute lattice is fixed by the standard normalization, with no continuous parameters to adjust.
- This high-precision set of numbers strongly constrains the particle spectrum and matches observation exactly.

1.7.9 Witten anomaly and the number of colors

Theorem 6.14 (The number of colors is N_c = 3). Under the gauge structure SU(N_c) × SU(2)_L × U(1)_Y with one left-handed quark doublet Q per color and one left-handed lepton doublet L per generation, the global SU(2) anomaly (Witten, 1982) requires N_c to be odd. Under MAR (lexicographic minimization of $C(\mathfrak{S})$, where N_c is the third component), this yields:

$$N_c = 3.$$

Inputs.

1. Low-energy gauge group contains an SU(2)_L factor and an SU(N_c) color factor.
2. The matter content per generation includes:
   - one left-handed quark doublet Q which is an SU(2) doublet and carries color,
   - one left-handed lepton doublet L which is an SU(2) doublet and color singlet.
3. **Witten's global SU(2) anomaly constraint** (Witten, 1982): the number of left-handed SU(2) doublets must be even.
4. **MAR** (Selection Axiom): among allowed values, the third component of the complexity vector $C(\mathfrak{S})$ is minimized.

Proof. Count SU(2) doublets per generation:

- Quark doublets: N_c copies (one per color),
- Lepton doublets: 1 copy.

Total doublets per generation:

$$N_c + 1.$$

Witten anomaly cancellation requires this to be even:

$$N_c + 1 \equiv 0 \pmod 2 \quad \Rightarrow \quad N_c \text{ is odd.}$$

The Witten constraint alone allows N_c ∈ {1, 3, 5, 7, ...}. N_c = 1 fails admissibility (SU(1) is trivial, no complex nonabelian charge type). MAR (input 4) then selects N_c = 3 as the smallest nontrivial value. QED.

Status. The Witten anomaly derives N_c odd. The specific value N_c = 3 follows from MAR applied to the admissible class. Under the extended theory T_{ext}, this is derived, not assumed.

Why this is convincing.

- It predicts a **single integer** given the minimality selector.
- The odd constraint is independent of continuous parameters, RG running, masses, or Yukawa values.
- It cannot be adjusted without changing the basic notion of electroweak doublets and color replication.

1.7.10 Bond-dimension gatekeeping

In tensor-network or code realizations, gauge actions act on edge factors of size χ, so emergent compact gauge groups embed in U(χ). This shows a capacity constraint: accommodating SU(3) color and SU(2) weak factors shows $\chi \geq 6$ in the minimal case, consistent with the MAR-derived gauge group.

1.7.11 Inevitability of photon and graviton

The model requires photons and gravitons.

Photon inevitability chain:

1. Premise GG (gauge-as-gluing) states that overlap identifications have redundancy forming a local groupoid.
2. Theorem 2.3 (edge-center completion) decomposes collar Hilbert spaces into sectors labeled by boundary gauge representations.
3. Theorem 6.1 (Tannaka/DR reconstruction) recovers a compact gauge group G from the fusion rules of these edge sectors.
4. Corollary 6.1 (conditional on DHR transportability) reconstructs a field algebra with G as a local gauge symmetry.
5. For the Standard Model, G includes U(1)_em after electroweak symmetry breaking.
6. A gauge boson is the quantum of the gauge field. Once U(1)_em emerges from overlap redundancy, its gauge field exists, and its quantum (the photon) must exist.

The photon is not postulated. It is forced by the axioms through the chain above. The photon mediates the correlations between charged excitations in different patches; it is how the U(1) redundancy structure propagates through the algebra net.

Graviton inevitability chain:

1. Theorem 4.2 (BW_{S^2}) shows that under collar Markov locality, MaxEnt selection with rotational invariance, and the tangent-half-space Bisognano–Wichmann comparison, modular flow on caps becomes geometric conformal dilation.
2. Theorem 4.3 identifies the induced kinematic group as $\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$, the Lorentz group.
3. Theorem 5.1 shows that, under the stated scaling-limit null-stress and reference-state premises, the condition δS_gen = 0 implies the rest-frame first-variation Einstein relation, with overlap consistency upgrading it to the semiclassical Einstein equations in the EFT regime.
4. The metric tensor emerges as the compression of modular flow data, and its dynamics are fixed by entanglement equilibrium.
5. A dynamical metric in a quantum theory requires a spin-2 quantum field. Its quantum (the graviton) must exist.

The graviton is not postulated. It is forced by the axioms through the chain above. Diffeomorphism invariance emerges because the bulk spacetime description is a compression of screen data; different coordinate descriptions are redundancies in how that compression is presented.

1.7.12 Mass predictions from symmetry

The model makes sharp numerical predictions for certain particle masses where symmetry protection applies.

Theorem 6.17 (Photon mass vanishes exactly). From the chain:

- single Higgs doublet H = (1, 2, 1/2),
- unbroken U(1)_em after EWSB,
- gauge-as-gluing (premise GG) identifying U(1)_em as a genuine redundancy on overlaps,

a hard photon mass term (Proca mass) would break the U(1)_em gauge redundancy. Therefore it is forbidden, and

$$m_\gamma = 0 \text{ exactly.}$$

Experimental status: PDG compiles upper limits of order 10^{-18} eV (approximately 10^{-27} GeV). Existing bounds are consistent with the exact-zero consequence at available experimental sensitivity.

Theorem 6.18 (Graviton mass vanishes exactly). In the semiclassical GR regime derived in Section 5, spacetime diffeomorphism invariance emerges from overlap consistency. A graviton mass term would break this gauge redundancy. Therefore

$$m_g = 0 \text{ exactly.}$$

Experimental status: PDG 2025 lists m_g $\leq$ 1.76 $\times$ 10^{-23} eV/c^2 (90% credibility) from gravitational wave dispersion analysis. The GW speed bound from GW170817 constrains (c_GW - c)/c to ~10^{-15}. Existing bounds are consistent with the exact-zero consequence at available experimental sensitivity.

Theorem 6.19 (Charge quantization and no fractional color singlets). If the global gauge group is

$$G_{\text{phys}} = \frac{\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)}{Z_6},$$

as derived in Proposition 6.6, then

All color-singlet states have integer electric charge.

Equivalently: no stable isolated particles with charges like $\pm 1/3$ can exist as color singlets.

Proof. The Z_6 quotient identifies the center elements (eˆ{2πi/3}, -1, eˆ{iπ/3}) $\in$ SU(3) $\times$ SU(2) $\times$ U(1) with the identity. For a color-singlet state ($\tau = 0$), the SU(3) factor acts trivially. The remaining identification requires the SU(2) $\times$ U(1) quantum numbers to satisfy

$$(-1)^{2j} \cdot e^{i\pi n/3} = 1,$$

where j is the SU(2) spin and n = 6Y is the integer hypercharge label. This gives n $\equiv$ -6j (mod 6), i.e., n $\equiv$ 0 (mod 6) for integer j and n $\equiv$ 3 (mod 6) for half-integer j. Equivalently: Y is integer when j is integer, and Y is half-integer when j is half-integer.

After electroweak breaking, Q = T_3 + Y. For integer j, $T_3 \in \mathbb{Z}$ and Y $\in \mathbb{Z}$, so Q $\in \mathbb{Z}$. For half-integer j, $T_3 \in \mathbb{Z} + 1/2$ and Y $\in \mathbb{Z} + 1/2$, so Q = (half-integer) + (half-integer) $\in \mathbb{Z}$. In both cases, Q $\in \mathbb{Z}$. QED.

Experimental status: No fractionally charged color-singlet particles have been observed. Three independent high-precision bounds confirm this:

1. **Neutrality of matter** (PDG 2024): The proton-electron charge sum satisfies

$$|q_p + q_e|/e < 1 \times 10^{-21},$$

confirming charge quantization to 21 decimal places.

2. **Fractional charge searches in bulk matter**: Silicone oil drop experiments limit fractionally charged particle abundance to

$$\text{(fractionally charged particles)/nucleon} \lesssim 10^{-22}.$$

3. **Collider searches** (CMS, PRL 134, 2025): Exclusions for stable particles with $q \in [e/3, 0.9e]$ up to masses ~ 640 GeV (95% CL).

Existing searches are fully consistent with these structural consequences.

These are exact symmetry- or quotient-protected outputs: the empirical content is that violations remain excluded up to available experimental sensitivity.

1.7.13 Coupling extraction from edge-sector probabilities

The edge-center completion (Theorem 2.3) yields sector probabilities p_α on collar boundaries. These probabilities encode the renormalized gauge coupling through a heat-kernel/Laplacian weighting law.

Abelian case (Z_n). For a Z_n gauge theory, the edge sectors are labeled by charge q $\in$ {0, 1, ..., n-1}. The correct "Casimir" eigenvalue is the Laplacian eigenvalue of the boundary random walk:

$$\lambda_q = 4\sin^2\left(\frac{\pi q}{n}\right).$$

Note: only in the limit n $\to \infty$ and q $\ll$ n does λ_q $\approx (2\pi q/n)^2 \propto q^2$. For finite n, the exact form is essential.

The sector probabilities follow a heat-kernel law:

$$p_q \propto e^{-t(\mu)\lambda_q},$$

where t(μ) is the "modular time" parameter encoding the scale. The extraction formula is:

$$t(\mu) = -\frac{\log(p_q/p_0)}{\lambda_q}, \qquad g^2_{\text{ent}}(\mu) = \frac{t(\mu)}{2\pi}.$$

Consistency requires that t extracted from different charges q agrees; this has been verified numerically (see Section 6.14).

Electric-center measurement. The edge sectors are measured using the *electric-center* prescription. For a region A and boundary vertex v $\in \partial$A, define the restricted star operator:

$$Q_v^{(A)} = \prod_{\ell \in \text{star}(v)\cap A} X_\ell^{\pm 1},$$

where X_ℓ is the shift operator on link ℓ. The sector projectors are:

$$P_{v,q} = \frac{1}{n}\sum_{m=0}^{n-1} \omega^{-mq}\left(Q_v^{(A)}\right)^m, \qquad \omega = e^{2\pi i/n},$$

and the probabilities are p_{v,q} = $\langle$P_{v,q}$\rangle$. This electric-center operator, built from X's rather than Z's, correctly captures the boundary gauge charge/flux that labels entanglement edge sectors.

Non-abelian generalization. For SU(N) gauge theories, the edge sectors are labeled by irreducible representations with probabilities:

$$p_j \propto d_j\, e^{-t(\mu)C_2(j)},$$

where d_j is the dimension and C_2(j) the quadratic Casimir. Extraction:

$$t(\mu) = -\frac{\log(p_j/p_0)}{C_2(j)}, \qquad g^2_{\text{ent}}(\mu) = \frac{t(\mu)}{2\pi}.$$

Theoretical derivation. The heat-kernel law can be derived from the axioms under one additional assumption (LG: local Gibbs generator).

Theorem 6.20 (Heat-kernel law from MaxEnt + gauge structure). Under the OPH axioms, together with premise ME (local MaxEnt), premise GG (gauge-as-gluing), the local Gibbs hypothesis LG, and the regulator premises R0 and R1, the edge-sector probability distribution satisfies:

$$p_R = \frac{d_R\, e^{-t\lambda_R}}{\sum_{R'} d_{R'}\, e^{-t\lambda_{R'}}}$$

where λ_R is the Laplacian eigenvalue on the R-isotypic component and t is determined by the collar Gibbs parameter.

Proof.

Step 1 (Edge Hilbert space). From gauge-as-gluing (GG) and the regulator premises R0 and R1, the edge degrees of freedom at a boundary circle Σ = ∂C live in a Hilbert space transforming under the gauge group G. Microscopically this is a finite-dimensional quantum-link edge space; the effective representation-theoretic description in the refinement limit is modeled by $L^2(G)$. By the Peter-Weyl theorem:

$$L^2(G) \cong \bigoplus_R V_R \otimes V_R^*$$

where V_R is the carrier space of irrep R.

Step 2 (Gauge invariance). The Gauss law constrains physical states. For an entanglement cut at Σ, the physical edge Hilbert space decomposes as $\mathcal{H}$_{edge}^{phys} = ⊕_R W_R where W_R contains states with flux in representation R.

Step 3 (Natural Hamiltonian). From LG, the MaxEnt generator restricted to edge modes takes the form H_{edge} = Σ_R h_R P_R where P_R is the projector onto the R-sector. The key claim is that h_R = λ_R.

Justification: For a compact simple factor, the group Laplacian Δ_G = -Σ_a (T^a)^2 is the unique bi-invariant second-order differential operator up to scale. For a product group $G = \prod_i G_i$, the most general bi-invariant second-order operator is a positive linear combination $\sum_i c_i \Delta_{G_i}$, with one coefficient per factor. Any other gauge-invariant local choice would require higher derivatives, violating locality. For finite groups, the Cayley graph Laplacian plays the same role: λ_R = |S| - (1/d_R) Σ_{s∈S} χ_R(s).

Step 4 (MaxEnt selection). MaxEnt (premise ME) selects the Gibbs state:

$$\rho_{\text{edge}} = \frac{1}{Z} e^{-tH_{\text{edge}}} = \frac{1}{Z} \sum_R e^{-t\lambda_R} P_R.$$

Step 5 (Sector probabilities). The probability of sector R is p_R = Tr(ρ_{edge} P_R). The effective dimension for entanglement is d_R (not d_R^2) because we trace over one side of the cut. This gives:

$$p_R = \frac{d_R\, e^{-t\lambda_R}}{Z}.$$

QED.

Why the entropy rank is d_R (instead of d_R^2). The full edge space has dimension d_R^2 in sector R (from $V_R \otimes V_R^*$). Entanglement entropy, however, measures correlations *across* the cut. After tracing over one side, the reduced density matrix has effective rank d_R. Mathematically: in the Markov normal form, the edge factor on one side contributes $\log d_R$ to the entropy.

Status. The derivation is complete up to the factorwise Laplacian choice just stated. The LG assumption (quasi-local MaxEnt generator) is derived from Theorem 2.6: if MaxEnt constraints are expectations of finitely many quasi-local operators, the entropy maximizer is automatically a Gibbs state with a quasi-local generator. What remains is the specific *Laplacian form* of that generator; this follows from gauge invariance plus the factorwise uniqueness of the bi-invariant second-order operator on each compact simple factor.

Normalization anchor: 2D Yang-Mills. The parameter t can be exactly matched to a conventional coupling in 2D Yang-Mills, where the physical Hamiltonian is literally the group Laplacian:

$$H = \frac{g^2}{2}\Delta_G, \qquad \Delta_G \chi_R = -C_2(R)\chi_R \quad \Rightarrow \quad E_R = \frac{g^2}{2}C_2(R).$$

Euclidean evolution for "time" A (the area of a cylinder in 2D YM) gives weight(R) $\propto$ exp(-A E_R) = exp(-g^2 A C$_2$(R)/2). Comparing with the heat-kernel expansion K_t(U) = Σ_R d_R χ_R(U) e^{-t C$_2$(R)} yields the exact identification:

$$t_{\text{phys}} = \frac{g^2 A}{2} \quad \text{(in 2D YM, no ambiguity).}$$

This shows that the Laplacian + MaxEnt $\to$ heat-kernel structure is not just plausible; it is exactly how continuum Yang-Mills behaves in a solvable case. The coefficient in front of C$_2$ is fixed. In any regime where the edge theory reduces to an effective 2D YM with known "Euclidean thickness" A_eff:

$$g^2(\mu) = \frac{2}{A_{\text{eff}}(\mu)} \cdot \frac{\Delta_R(\mu)}{C_2(R)},$$

and the RHS must be R-independence. This R-independence is an internal precision consistency test; the formula itself is the normalization map that connects t to the conventional gauge coupling.

1.7.14 Numerical validation of the heat-kernel law

The heat-kernel/Laplacian weighting of edge sectors has been validated in explicit 2D Z_n gauge models on closed geometries.

Model. A 2×2 periodic lattice gauge theory (8 links) with Z_n link Hilbert spaces and Hamiltonian:

$$H = -K\sum_p \mathrm{Re}(B_p) - h\sum_\ell \mathrm{Re}(X_\ell) - \Gamma\sum_v \mathrm{Re}(A_v),$$

where X_ℓ is the Z_n shift on link ℓ, B_p is the oriented plaquette operator (product of Z's around plaquette p), and A_v is the oriented star/Gauss operator (outgoing X, incoming X†). With K = 1 and Γ = 5, the ground state satisfies ⟨A_v⟩ = 1 at all vertices to numerical precision.

Region and edge operator. Region A consists of links whose tail has x = 0 ("half-lattice" cut). At each boundary vertex v, the electric-center edge charge is the restricted star Q_v^{(A)} = ∏_{ℓ ∈ star(v) ∩ A} X_ℓ^{±1}.

Results for Z_2. With $\lambda_1 = 4\sin^2(\pi/2) = 4$:

h	p_0	p_1	t	g_ent
0.5	0.8266	0.1734	0.391	0.249
1.0	0.9612	0.0388	0.803	0.357
2.0	0.9917	0.0083	1.194	0.436

Results for Z_3 (overconstrained test). With $\lambda_1 = \lambda_2 = 4\sin^2(\pi/3) = 3$:

h	p_0	p_1	p_2	t(q=1)	t(q=2)	g_ent	m_plaq
0.2	0.4395	0.2803	0.2803	0.1500	0.1500	0.154	2.22
0.5	0.7509	0.1245	0.1245	0.5989	0.5989	0.309	1.75
1.0	0.9606	0.0197	0.0197	1.2956	1.2956	0.454	4.07
1.5	0.9851	0.0074	0.0074	1.6288	1.6288	0.509	7.06
2.0	0.9921	0.0039	0.0039	1.8440	1.8440	0.542	10.10

The equality $p_1 = p_2$ is exact (charge conjugation symmetry in Z_3). The equality t_{q=1} = t_{q=2} is the crucial **overconstrained** check: at h = 1.0, extracting t from q = 1 and q = 2 independently gives t_{q=1} ≈ 1.2956389318579 and t_{q=2} ≈ 1.2956389318521. The agreement to $\sim 10^{-14}$ (machine precision) confirms that the edge distribution genuinely follows the heat-kernel/Laplacian form.

Region-choice robustness. At h = 1, the extracted g_ent is nearly independent of region size:

- 2 links (one vertex's outgoing links): g_ent ≈ 0.453
- 4 links (half-lattice): g_ent ≈ 0.454
- 6 links (three vertices): g_ent ≈ 0.453

This locality confirms that the coupling is dominated by physics near the cut, not global book-keeping, exactly what is expected if this behaves like a local QFT observable.

Results for Z_5 (golden ratio test). The Z_5 case provides a stringent test because the Laplacian eigenvalues have a distinctive ratio involving the golden ratio $\phi = (1+\sqrt{5})/2$:

$$\lambda_q = 4\sin^2\left(\frac{\pi q}{5}\right), \qquad \frac{\lambda_2}{\lambda_1} = \frac{\sin^2(72^\circ)}{\sin^2(36^\circ)} = \phi^2 \approx 2.618.$$

This ratio distinguishes the Laplacian law from naive alternatives: a linear model ($\lambda_q \propto q$) would predict ratio 2, while a quadratic model ($\lambda_q \propto q^2$) would predict ratio 4.

Simulations on a 2×2 torus in the dual/flux basis (125 states in the zero-winding sector) give:

h	Measured ratio $\ln(p_2/p_0)/\ln(p_1/p_0)$	Deviation from ϕ^2
0.5	2.25	14%
1.0	2.51	4%
2.0	2.619	$< 0.1\%$

In the weak-field limit ($h \to 0$, strong magnetic coupling), the simulation converges to the golden ratio squared. This confirms that the vacuum entanglement spectrum encodes the precise geometric structure of the gauge group Laplacian.

Significance. This validates the mathematical law (sector probabilities weighted by Laplacian eigenvalues) in honest 2D gauge-invariant models with non-flat sector distributions. The Z_3 and Z_5 tests are structurally identical to SU(2)/SU(3): multiple irreps overconstrain the slope, and agreement confirms the mechanism works before jumping to nonabelian groups.

Results for S_3 (first nonabelian test). The abelian tests above use charge-sector projectors that reduce to Fourier modes. For nonabelian groups, the edge-sector projector must be generalized to character projectors:

$$P_{v,R} = \frac{d_R}{|G|} \sum_{h \in G} \chi_R(h^{-1}) Q_v^{(A)}(h),$$

where d_R is the dimension of irrep R, χ_R is its character, and Q_v^{(A)}(h) is the restricted gauge action at boundary vertex v acting only on links in region A.

For S_3 (the smallest nonabelian group, order 6), there are three irreps: trivial (d=1), sign (d=1), and standard (d=2). The Cayley-graph Laplacian eigenvalues for the transposition generating set are:

$$\lambda_{\text{triv}} = 0, \qquad \lambda_{\text{sign}} = 6, \qquad \lambda_{\text{std}} = 3.$$

Exact reduction on one plaquette. For the single-plaquette model (4 links), imposing Gauss's law at all vertices means the physical wavefunction depends only on the plaquette holonomy's conjugacy class. Since S_3 has exactly 3 conjugacy classes, the gauge-invariant Hilbert space is 3-dimensional, spanned by the character states $\{|\chi_R\rangle\}$. In this basis, the edge-sector probabilities are exactly p_R = |c_R|^2 where $|\psi_0\rangle = \Sigma$_R c_R $|\chi$_R$\rangle$. This is not an approximation; it is an exact identity for the one-plaquette gauge-invariant sector.

The heat-kernel ansatz predicts p_R $\propto$ d_R exp(-t λ_R). Extracting t independently from the sign and standard irreps provides an overconstrained test: the ratio λ_sign/λ_std = 6/3 = 2 is a parameter-free prediction. Results from a single-plaquette S_3 lattice gauge model (K=1, Γ=5):

h	p_triv	p_sign	p_std	t (sign)	t (std)	Δt/t	log-ratio
0.5	0.909	0.0013	0.089	1.09	1.01	8.4%	2.17
1.0	0.980	7.5×10^{-5}	0.020	1.58	1.54	2.8%	2.06
2.0	0.996	4.3×10^{-6}	0.004	2.06	2.04	1.0%	2.02
5.0	0.9993	1.0×10^{-7}	0.00066	2.68	2.67	0.3%	2.006
12	0.9999	3.0×10^{-9}	0.00011	3.27	3.27	0.1%	2.002
100	1.0000	6.1×10^{-13}	2.0×10^{-6}	4.69	4.69	0.009%	2.0002

The "$\Delta t/t$" column shows the fractional difference (t_sign - t_std) / $\bar{t}$. The "log-ratio" column shows log(p_sign/p_0) / log(p_std/(2 p_0)), which should equal λ_sign/λ_std = 2 if the heat-kernel form holds exactly.

As h increases, both diagnostics converge: $\Delta t/t$ drops below 10^{-4} and the log-ratio approaches 2.000. This is exactly the expected behavior: finite-size corrections are largest at strong coupling; the heat-kernel form becomes exact as the perturbative regime is approached.

This is the first nonabelian validation of the edge-sector extraction mechanism. The structure (character projectors, Laplacian eigenvalues from the group's Cayley graph, overconstrained t extraction) is identical to what will be used for SU(2) and SU(3).

Parameter-free predictions for SU(2) and SU(3). The heat-kernel law yields exact, parameter-free ratio predictions that require no scheme matching. Define the "Casimir log-gap":

$$\Delta_R \equiv \ln\left(\frac{p_0}{d_0}\right) - \ln\left(\frac{p_R}{d_R}\right) = t\, C_2(R).$$

Ratios of Δ_R cancel all unknowns (t, partition function):

$$\frac{\Delta_{R_1}}{\Delta_{R_2}} = \frac{C_2(R_1)}{C_2(R_2)} \quad \text{(exact, parameter-free).}$$

SU(2) predictions. Irreps labeled by spin j have d_j = 2j+1 and C_2(j) = j(j+1). The framework predicts:

- $\Delta_1/\Delta_{1/2} = 2/(3/4) = \mathbf{8/3} \approx \mathbf{2.667}$
- $\Delta_{3/2}/\Delta_{1/2} = (15/4)/(3/4) = \mathbf{5}$
- $\Delta_{3/2}/\Delta_1 = (15/4)/2 = \mathbf{15/8 = 1.875}$

SU(3) predictions. Irreps labeled by Dynkin indices (p,q) have C_2(p,q) = ($p^2 + q^2 + pq + 3p + 3q$)/3. Using the fundamental **3** = (1,0) with $C_2 = 4/3$ as the reference:

- $\Delta_8/\Delta_3 = 3/(4/3) = \mathbf{9/4 = 2.25}$
- $\Delta_6/\Delta_3 = (10/3)/(4/3) = \mathbf{5/2 = 2.5}$
- $\Delta_{10}/\Delta_3 = 6/(4/3) = \mathbf{9/2 = 4.5}$
- $\Delta_{15}/\Delta_3 = (16/3)/(4/3) = \mathbf{4}$
- $\Delta_{27}/\Delta_3 = 8/(4/3) = \mathbf{6}$

These are the SU(2)/SU(3) analogs of the Z_5 golden-ratio test: exact rational numbers fixed entirely by group theory, with no adjustable parameters.

Preliminary SU(3) results. A one-plaquette SU(3) "quantum link" model (finite truncated irrep basis, n_max = 12, $\kappa = 2$) has been used to extract t from 14 different irreps simultaneously. The results show internal consistency at the 1-3% level:

bare g^2	extracted t (mean±std)	g_ent	gap
0.3	0.314±0.0005	0.224	1.92
0.5	0.539±0.0025	0.293	1.83
0.8	0.896±0.012	0.378	1.72
1.0	1.144±0.025	0.427	1.64

The standard deviation across irreps provides a built-in error estimate. This is a QCD proton-physics surrogate rather than a full proton calculation; it lacks dynamical quarks and operates on a single plaquette. It nevertheless demonstrates that the nonabelian extraction machinery produces self-consistent outputs without tuning.

Extracting the normalization factor A_eff. The 2D YM anchor (Section 6.13) gives t = g^2 A / 2, so the "effective Euclidean thickness" is

$$A_{\text{eff}} = \frac{2t}{g^2}.$$

Computing this from the SU(3) table:

bare g^2	extracted t	A_eff
0.3	0.314	2.093
0.5	0.539	2.156
0.8	0.896	2.240
1.0	1.144	2.288

Mean: A_eff ≈ 2.19 with point-to-point scatter ~4%.

Extrapolation to weak coupling. The systematic drift in A_eff shows fitting A_eff(g^2) = A_0 + a · g^2. A weighted linear fit gives:

$$A_0 = 2.004 \pm 0.012$$

with χ^2/dof ≈ 0.09, indicating excellent consistency. This strongly shows that, in this toy UV completion, the "missing normalization" converges to A_eff → 2 as g^2 → 0.

The normalization factor behaves like a quasi-constant and extrapolates to a simple value (≈ 2) in the weak-coupling limit. This provides a concrete path to absolute coupling predictions: once A_eff is determined from microphysics, the conversion g^2 = 2t/A_eff fixes the gauge coupling without additional free parameters.

Internal validation summary. The heat-kernel law has been validated with increasing precision across multiple gauge groups:

- **Z_3**: Overconstrained t extraction (q=1 vs q=2), precision ~10^{-14}
- **Z_5**: Golden ratio squared ($\lambda_2/\lambda_1 = \phi^2$), precision 0.04%
- **S_3**: Casimir log-ratio (λ_sign/λ_std = 2), precision 0.01%
- **SU(3)**: 14-irrep simultaneous extraction, precision 1-3%

The Z_3 test achieves machine precision because it is exactly overconstrained. The Z_5 and S_3 tests converge to their predicted ratios as coupling decreases. This provides strong internal validation of the mechanism "MaxEnt + Laplacian => heat-kernel sector weights" before applying it to physical gauge groups.

1.7.15 IBM Quantum Cloud hardware benchmarks

The lattice calculations above are internal numerical validations. The IBM Quantum Cloud benchmarks add a hardware check of whether OPH-motivated reduced-sector structures survive preparation and readout on real superconducting-qubit devices. These runs are small-sector consistency

benchmarks rather than a standalone confirmation of the full framework. Their value is that they probe the same recoverability and heat-kernel observables on physical hardware and in simulation.

Detailed circuits, representative raw outputs, and rerun instructions are public in the project write-up `extra/IBM_QUANTUM_CLOUD.md` and the associated code/data bundle `code/ibm_quantum_cloud/`.

- **Stage 1 (Markov/recoverability benchmark).** On `ibm_marrakesh` and `ibm_fez`, the structured state reconstructs below both controls in conditional mutual information and above both controls in Petz fidelity. On `ibm_marrakesh`, the observed ordering is $0.2309 < 0.3890 < 0.9474$ for CMI and $0.9297 > 0.8649 > 0.6066$ for Petz fidelity; on `ibm_fez`, it is $0.1498 < 0.4992 < 0.9166$ and $0.9479 > 0.8117 > 0.6449$. This is the OPH-favored direction on two independent real backends.
- **$\mathbf{Z_3}$ hardware sanity check.** The overconstrained Z_3 extraction passes cleanly on hardware: across prepared $t = 0.30, 0.60, 0.90$, the mean extracted t remains within about 0.02 of the target, the two independent extractions agree to within 0.0142, 0.0034, and 0.0043, and leakage stays below 0.1%. This shows that the reduced-sector preparation and readout path is internally coherent on-device before sharper ratio claims are interpreted.
- **$\mathbf{Z_5}$ exact-ratio test.** The parameter-free OPH target is $\Delta_2/\Delta_1 = \varphi^2 \approx 2.618$. Across repeated sweeps on `ibm_marrakesh` and one replication on `ibm_fez`, the best `ibm_marrakesh` points land within 0.8%, 1.2%, and 1.5% of the target, while a focused high-shot rerun at $t = 0.90$ lands within 2.6%. The `ibm_fez` replication is noisier but remains in the same neighborhood rather than producing a clean contradictory ratio.
- **$\mathbf{S_3}$ nonabelian ratio test.** The first nonabelian hardware run revealed a real layout-dependent bias. After reversing the qubit layout, the mitigated ratio moves from 1.8724 to 2.0299 against the OPH target 2.0000, and a direct confirmation run returns 2.0657. The nonabelian target reappears once the hardware mapping is corrected, indicating an identifiable device/layout effect instead of a structural failure of the OPH prediction.

These IBM runs are best read as hardware benchmarks for the local OPH edge-sector picture. They show that the OPH-favored recoverability ordering survives on two real backends, the abelian exact-ratio tests land near the predicted parameter-free values, and the first nonabelian test returns to the expected ratio once a diagnosed layout bias is corrected. They add real-device evidence to the simulation-based checks, but they do not validate the full framework by themselves.

1.7.16 Particle mass extraction from spectroscopy

The same lattice models that yield coupling extraction also provide a concrete definition of "particle mass" via standard QFT/lattice spectroscopy.

Definition. For a gauge-invariant local operator O, the lowest "glueball-like" mass in that channel is:

$$m_O = E_n - E_0,$$

where $|n\rangle$ is the lowest excited eigenstate with $\langle n|O|0\rangle \neq 0$ and E_0 is the ground state energy.

Plaquette channel. For the Z_3 model, using O = Σ_p Re(B_p), the extracted masses m_plaq are shown in Section 6.14. This is the standard spectroscopy definition: the lowest pole in the two-point correlator of a local gauge-invariant operator.

Dimensional transmutation. In lattice units, both g_ent and m_plaq are dimensionless numbers. The physical mass scale emerges through dimensional transmutation once the coupling

is matched to a continuum scheme. The ratio m_plaq / g_ent^2 is a pure number that can be compared across different bare couplings to check scaling.

1.7.17 Composite masses and the path to predictions

Masses of composite particles (protons, neutrons, pions, etc.) are qualitatively different from symmetry-protected zeros. The proton mass is a strongly coupled bound-state eigenvalue:

$$m_p = \Lambda_{\mathrm{QCD}} \cdot F\left(\frac{m_u}{\Lambda_{\mathrm{QCD}}}, \frac{m_d}{\Lambda_{\mathrm{QCD}}}, \frac{m_s}{\Lambda_{\mathrm{QCD}}}, \ldots; \alpha_{\mathrm{em}}\right),$$

where Λ_QCD is the dimensional transmutation scale and F is a dimensionless nonperturbative function.

The pipeline to Standard Model numerics:

1. **SU(2) quantum link model**: Measure boundary p_j and fit slope vs j(j+1) to extract g_2,ent(μ).
2. **SU(3) quantum link model**: Measure boundary p_(p,q) and fit slope vs C_2(p,q) to extract g_3,ent(μ).
3. **Scheme matching**: One-time match from entanglement scheme to MS-bar, then RG-run to predict α_s(M_Z), $\sin^2\theta$_W(M_Z).
4. **Mass scale**: With $g_3(\mu)$ fixed in physical units (gravity side supplies the absolute scale via entanglement equilibrium), compute Λ_QCD as the first real mass-scale prediction.

At one loop,

$$\Lambda = \mu \exp\left(-\frac{2\pi}{\beta_0 \alpha_s(\mu)}\right), \qquad \frac{d\ln\Lambda}{d\ln\alpha_s} \approx 7.$$

A 0.1% uncertainty in α_s becomes approximately 0.7% uncertainty in the hadronic mass scale.

1.7.18 Gauge unification and spectrum constraints

The edge-sector extraction of gauge couplings (Section 6.13) yields boundary conditions at an entanglement-defined UV scale. If these couplings unify from a "single collar/edge principle," standard one-loop RG running provides a sharp numerical constraint on the allowed particle spectrum.

Scope note. The unification analysis below uses standard GUT techniques that predate this framework. The results for MSSM-like spectra are well-known in the GUT literature. The framework contributes two ingredients: (1) a *mechanism* for why couplings might unify (shared geometric origin), and (2) a product group structure that forbids proton decay. The numerical α_s consistency is a *consistency check* against known physics. It does not stand as a novel prediction of this framework.

Inputs. We use:

1. **Canonical GUT normalization**: $\alpha_1 \equiv (5/3)\alpha$_Y, the standard convention for comparing to RG coefficients.
2. **One-loop RG running** between M_Z and a unification scale M_U:

$$\frac{1}{\alpha_i(M_Z)} = \frac{1}{\alpha_U} + \frac{b_i}{2\pi} \ln \frac{M_U}{M_Z}.$$

3. **Measured electroweak inputs at M_Z** (PDG 2025):

   - $\hat{\alpha}^{(5)}$(M_Z^2)$^{-1}$ = 127.930 ± 0.008 (MS-bar)
   - $\hat{s}^2$_Z ≡ sin$^2\hat{\theta}$_W(M_Z^2) = 0.23122 ± 0.00006 (MS-bar)
   - α_s(M_Z) = 0.1177 ± 0.0009 (from EW fit; world average 0.1180)

   Note on sin$^2\theta$_W schemes: PDG lists multiple definitions with different values. The MS-bar value $\hat{s}^2$_Z = 0.23122 differs from the effective leptonic angle $\bar{s}^2$_ℓ = 0.23154 and the on-shell value s^2_W = 0.22342. Since we compute from running couplings α_1, α_2, the natural comparison is to the MS-bar definition.

4. **Candidate spectra**:

   - SM-only: (b_1, b_2, b_3) = (41/10, -19/6, -7)
   - MSSM-like: (b_1, b_2, b_3) = (33/5, 1, -3)

Derived couplings at M_Z. From α_em and sin$^2\theta$_W:

$$\alpha_2 = \frac{\alpha_{\rm em}}{\sin^2\theta_W}, \quad \alpha_Y = \frac{\alpha_{\rm em}}{1-\sin^2\theta_W}, \quad \alpha_1 = \frac{5}{3}\alpha_Y.$$

Numerically (central values):

- A_1 ≡ α_1^{-1}(M_Z) ≈ 59.00
- A_2 ≡ α_2^{-1}(M_Z) ≈ 29.59
- A_3 ≡ α_s^{-1}(M_Z) ≈ 8.47

Analytic prediction formula. Define A_i = α_i^{-1}(M_Z) and L = ln(M_U/M_Z). The RG equations give A_i = A_U + (b_i/2π)L. Taking differences to eliminate A_U:

$$L = \frac{2\pi}{b_1 - b_2}(A_1 - A_2).$$

This yields a prediction for A_3 that depends only on electroweak inputs:

$$A_3^{\rm pred} = \frac{b_3 - b_2}{b_1 - b_2} A_1 + \frac{b_1 - b_3}{b_1 - b_2} A_2$$

Once beta coefficients are fixed, α_s(M_Z) is completely determined by electroweak data. This is the hard numerical constraint.

Consistency check 1 (SM-only unification). One-loop unification with SM beta coefficients gives:

$$\alpha_s(M_Z)|_{\rm SM,unif} = 0.07107 \pm 0.00005$$

with M_U ≈ 1.0 × 10^{13} GeV and α_U^{-1} ≈ 42.4.

Comparison to measurement: The PDG 2025 EW-fit value is α_s(M_Z) = 0.1177 ± 0.0009. The SM-only prediction misses by $\Delta\alpha$_s ≈ 0.047, a ~52σ discrepancy, far too large to be rescued by two-loop corrections or thresholds.

This rules out SM-only unification: if the framework's gauge sector has anything like "unification from a single collar/edge principle," the particle spectrum above the weak scale cannot be just the SM.

Consistency check 2 (MSSM-like spectrum). One-loop unification with MSSM-like beta coefficients gives:

$$\alpha_s(M_Z)|_{\text{MSSM,unif}} = 0.11658 \pm 0.00015$$

with M_U ≈ 2.0 × 10^{16} GeV and α_U^{-1} ≈ 24.34 ± 0.01.

Comparison to measurement: The PDG 2025 EW-fit value is α_s(M_Z) = 0.1177 ± 0.0009. The mismatch is Δα_s ≈ -0.0011, about 1.2σ. This is within the expected range of two-loop corrections, threshold effects, and scheme matching.

Significance: SM-only unification predicts α_s(M_Z) ≈ 0.071, catastrophically wrong. The MSSM-like prediction is within 1.5σ of experiment. This validates the spectrum constraint but is not a novel prediction (MSSM GUT analyses from the 1990s obtained similar results).

Corollary (Spectrum constraint). The required beta-function shift beyond the SM is approximately:

$$\Delta b \equiv b^{\text{UV}} - b^{\text{SM}} \approx (2.5,\ 4.2,\ 4.0).$$

This requires substantial additional charged degrees of freedom affecting SU(2) and SU(3) running, far more than a single extra Higgs doublet. The pattern is highly specific and constrains the spectrum sharply.

Threshold analysis. The preceding analysis assumes UV degrees of freedom are active all the way down to M_Z. If the Δb only turns on above some threshold M*, the running becomes piecewise:

$$\frac{1}{\alpha_i(M_Z)} = \frac{1}{\alpha_U} + \frac{b_i^{\text{SM}}}{2\pi} \ln \frac{M_*}{M_Z} + \frac{b_i^{\text{UV}}}{2\pi} \ln \frac{M_U}{M_*}.$$

The predicted α_s(M_Z) depends sensitively on M_*:

- M_ = *M_Z**: α_s(M_Z) = 0.1166, M_U = 2.0×10^{16} GeV
- M_ = *1 TeV**: α_s(M_Z) = 0.1100, M_U = 9.6×10^{15} GeV
- M_ = *10 TeV**: α_s(M_Z) = 0.1043, M_U = 4.9×10^{15} GeV

This quantifies what the "scheme matching" step must accomplish: if UV physics only turns on at multi-TeV scales, the matching correction must shift α_s^{-1} by ~0.6 to reach the experimental value.

Inverted problem: derive M_S from measured couplings. With three measured couplings (A_1, A_2, A_3) and three unknowns (M_S, M_U, α_U), the system is exactly determined. Define $x = \ln(M_S/M_Z)$ and $y = \ln(M_U/M_S)$. Taking differences to eliminate α_U gives a 2×2 linear system whose solution is:

Prediction (Effective threshold scale):

- **M_S ≈ 57 GeV** (42–77 GeV at 1σ)
- **M_U ≈ 2.27 × 10^{16} GeV**
- **α_U^{-1} ≈ 24.0**

The uncertainty is dominated by the experimental error on α_s(M_Z) = 0.1177 ± 0.0009. The central value is sensitive to the precise α_s input: α_s = 0.1175 gives M_S ≈ 67 GeV, while α_s = 0.1166 (the MSSM prediction) gives M_S ≈ 91 GeV. The qualitative conclusion (effective threshold near the electroweak scale) remains stable across the 1σ range.

Physical interpretation. This is a striking result: internal consistency of one-loop unification pushes the effective onset of MSSM-like Δb down to the **electroweak scale**. The new charged degrees of freedom cannot all live at some ultra-high scale; their *net effect* on beta functions must turn on around ~10^2 GeV.

If the framework requires unification but the UV spectrum only turns on well above M_Z (say, at multi-TeV), then the gap must be filled by one of: (i) additional running effects at intermediate scales, (ii) non-degenerate particle thresholds that mimic low-scale onset, or (iii) two-loop corrections providing effective Δb at lower scales.

This is the kind of *quantitative* constraint that is tested or contradicted by precision collider measurements of running couplings.

Significance. This provides a "spectrum selector": the framework must produce an effective Δb in the above direction (from new bulk fields or propagating collar/edge modes), or it cannot match precision gauge couplings. This is a hard, quantitative constraint on possible UV completions, derived before attempting to predict masses.

Prediction (No gauge-mediated proton decay). The model predicts that gauge-mediated proton decay is **forbidden**: the product gauge group structure derived from MAR (Section 6.2) forbids mixed generators.

Argument. Standard Grand Unified Theories (SU(5), SO(10)) achieve coupling unification by embedding SU(3) × SU(2) × U(1) into a simple Lie group (Georgi and Glashow, 1974). This embedding necessarily introduces X and Y bosons (leptoquarks) that mediate baryon-number-violating processes like $p \to e^+\pi^0$.

In Observer-Patch Holography, unification is **geometric** (shared diffusion parameter t across edge sectors) rather than **algebraic** (embedding in a simple group). The MAR analysis of Section 6.2 yields the realized connected gauge group as a **product**:

$$G = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1).$$

There is then no larger simple-group manifold supplying mixed leptoquark generators in the edge algebra. Therefore:

$$\tau_p^{\text{gauge}} = \infty \quad \text{(no gauge-mediated proton decay)}$$

Conditionality and testable equivalence. This prediction depends on the sector factorization assumption (Section 6.2). Rather than treating this as an untestable axiom, we can state it as an equivalence:

Proposition (Factorization ↔ additive boundary Laplacian). Suppose the edge Hamiltonian governing boundary sector weights takes the form:

$$H_\partial = H_\partial^{(1)} + H_\partial^{(2)} + H_\partial^{(3)}, \quad [H_\partial^{(i)}, H_\partial^{(j)}] = 0,$$

where each $H_\partial^{(i)}$ is the bi-invariant second-order operator (group Laplacian) for a compact factor G_i, unique up to an independent positive coefficient on each factor. Then the heat-kernel form implies exact probability factorization:

$$p(R_1, R_2, R_3) \propto \prod_{i=1}^{3} d_{R_i} e^{-t_i C_2(R_i)}.$$

Conversely, if the reconstructed sector category is Rep(G) and the edge weights satisfy this factorization for all caps and scales, then $G \cong G_1 \times G_2 \times G_3$ (up to finite quotient).

Testable signature. Sector factorization is equivalent to observing that edge-sector probabilities factorize across gauge factors. If future UV model calculations or lattice measurements show non-factorizing edge weights, the gauge group would not be a product and proton decay is allowed.

Experimental status. Minimal SU(5) GUTs predict τ_p ~ 10^{31} years; Super-Kamiokande has pushed limits to τ_p > 10^{34} years, excluding minimal GUTs. The model's prediction of proton stability is consistent with all observations.

Distinguishing signature. The combination of **precision gauge unification** (MSSM-like α_s consistency) with **proton stability** is characteristic of this framework. Standard SUSY GUTs predict both unification *and* proton decay; this model predicts unification *without* proton decay, if sector factorization holds.

Chain summary: Edge-sector probabilities → gauge couplings at UV scale → one-loop RG → consistency check for α_s(M_Z) → spectrum constraint from mismatch with SM-only running. The product group structure (derived from MAR) separately implies proton stability.

Precision of the pixel-area relation. There are two distinct precision questions for a_cell:

(A) In Planck units (a_cell/ℓ_p^2). Once the dimensionless entropy density $^-\ell$ is fixed, the dimensionless pixel area is a_cell/ℓ_p^2 = $4^-\ell$. This ratio is **independent of the experimental uncertainty in G**, because ℓ_p^2 ∝ G cancels. The limiting precision is whatever uncertainty remains in $^-\ell$, i.e., in the inputs used to determine t(μ) (in practice gauge couplings). Since we feed in SM couplings to get $^-\ell$, precision is limited by those inputs, not by G.

(B) In SI units (a_cell [m^2]). If $^-\ell$ were known exactly, then a_cell in SI units would inherit the uncertainty of G:

$$\ell_p = \sqrt{\frac{\hbar G}{c^3}} \quad \Rightarrow \quad \frac{\delta \ell_p}{\ell_p} = \frac{1}{2}\frac{\delta G}{G}, \qquad \frac{\delta \ell_p^2}{\ell_p^2} = \frac{\delta G}{G}.$$

Using CODATA values: G = 6.67430 × 10^{-11} m^3 kg^{-1} s^{-2} with relative uncertainty ≈ 2.2 × 10^{-5}, so the best-case SI precision is:

- Relative precision of a_cell: δa_cell/a_cell ≈ 2.2 × 10^{-5}
- Relative precision of $\sqrt{}$a_cell: ≈ 1.1 × 10^{-5}

With the derived value a_cell/ℓ_p^2 ≈ 1.63094, this gives a_cell ≈ 4.26 × 10^{-70} m^2, with irreducible CODATA uncertainty δa_cell ≈ 9.5 × 10^{-75} m^2.

Reverse-engineering α_s from the pixel-area scale. The pixel-area relation (Section 5.4) provides an independent route to α_s(M_Z). The cell area in Planck units is:

$$\frac{a_{\text{cell}}}{\ell_p^2} = 4\bar{\ell}_{\text{tot}}(t_2, t_3), \quad \bar{\ell}_{\text{tot}} = \bar{\ell}_{\text{SU}(2)}(t_2) + \bar{\ell}_{\text{SU}(3)}(t_3),$$

where $\bar{\ell}_G(t) = \sum_R p_R(t) \ln d_R$ with $p_R \propto d_R e^{-tC_2(R)}$ and $t = 4\pi^2\alpha$.

Why $t = 4\pi^2\alpha$ is unique (scheme-fixed). The normalization $t = 4\pi^2\alpha$ is the *unique* value compatible with the heat kernel and modular geometry, and is not an arbitrary convention:

1. In 2D Yang-Mills / heat-kernel language, the weight is $e^{-(g^2A/2)C_2(R)}$, so $t = g^2A/2$.

2. For an entanglement cut in a local QFT, the Euclidean modular angle has period 2π (the universal "2π" behind Unruh/Rindler physics). In the edge-collar picture, this fixes the effective evolution "area" factor to $A = 2\pi$.

3. Plugging $A = 2\pi$ gives $t = g^2(2\pi)/2 = \pi g^2$.

4. Using $\alpha = g^2/(4\pi)$: $t = \pi(4\pi\alpha) = 4\pi^2\alpha$.

The map from edge parameters to physical couplings is thus fixed by the universal modular geometry, not by convention.

RG mechanism from Markov collar structure. The running of gauge couplings is a structural consequence of the Markov collar plus symmetry and does not require an extra assumption. Consider a nested family of caps C(δ) and the operation "thicken the collar by Δ":

- Going from C(δ) to C($\delta + \Delta$) adds an annular strip of degrees of freedom.
- Because the strip is in the Markov regime, the only long-range coupling between "inside" and "outside" is through the edge-sector label α (irrep/sector) on the cut.

At the level of the classical distribution over sectors p_$\alpha(\delta)$, thickening the collar acts by a stochastic kernel:

$$p(\delta + \Delta) = \mathsf{K}_\Delta \cdot p(\delta).$$

Imposing gauge-class invariance (kernel depends only on conjugacy class) and rotational invariance (same kernel along the cut), the kernel K_Δ must be a central (class) convolution kernel on the group.

By Peter-Weyl, irreps diagonalize any class-convolution operator. The only continuous one-parameter semigroups of such kernels are generated by the group Laplacian:

$$\mathsf{K}_t(R) = e^{-t\,C_2(R)}.$$

This is exactly the heat-kernel form p_R(t) $\propto$ d_R^κ exp(-t C_2(R)) with $\kappa = 1$ in the simplest MaxEnt edge state.

Key consequence (the RG step). Because collar layers compose, kernels compose by convolution, and for heat kernels:

$$\mathsf{K}_{t_1} \star \mathsf{K}_{t_2} = \mathsf{K}_{t_1+t_2},$$

so the coupling parameter t is **additive** under stacking collar layers. Additive in the "RG-time" variable is exactly what one-loop running requires: if the physical scale changes multiplicatively, the number of collar layers changes additively, hence t runs linearly in ln μ.

Normalization from modular geometry. The BW_{S^2} theorem gives modular flow on a cap as the conformal dilation preserving the cap and fixing its boundary, with KMS normalization $\beta = 2\pi$ on the OPH geometric branch. At finite collar scale the discrepancy is the carried remainder from the Markov/recovery analysis; in the controlled refinement limit that remainder vanishes. Near an entangling surface, modular flow is a boost; in Euclidean continuation it becomes an angular coordinate with period 2π. The dilation parameter is $s \sim \ln(\mu/\mu_0)$.

Because the imaginary period is 2πi, one modular thermal cycle corresponds to a multiplicative scale change:

$$\mu \mapsto e^{2\pi}\mu \approx 535 \times \mu.$$

In base-10 decades: $\log_{10}$(e^{2π}) = $2\pi/\ln(10) \approx 2.728$ decades. If earlier analysis required a normalization factor around ~2.9, the interpretation is that this is the conversion between "per

modular period" and "per decade of μ," fixed by the BW_{S^2} tangent-limit normalization rather than being a tunable parameter.

Using the measured $a_{\text{cell}}/\ell_p^2 = 1.63094$ and α_2(M_Z) ≈ 0.0338 (giving $t_2 \approx 1.334$), we compute $\bar{\ell}_{\text{SU(2)}} \approx 0.3946$. The SU(3) contribution is then forced:

$$\bar{\ell}_{\text{SU(3)}} = \frac{1.63094}{4} - 0.3946 \approx 0.0131.$$

Inverting the monotone function $\bar{\ell}_{\text{SU(3)}}(t_3)$ gives $t_3 \approx 4.657$, hence:

α_s(M_Z)|_pixel $\approx$ 0.1175

This is consistent with the PDG EW-fit value α_s(M_Z) = 0.1177 ± 0.0009 to within ~5 × 10^{-5}. The agreement is striking but **conditional**: it becomes a genuine prediction only if a_{cell}/ℓ_p^2 can be fixed independently (from the gravity side) rather than computed from α_s.

Proton mass estimate. Using standard 4-loop $\overline{\text{MS}}$ running with n_f = 5 at M_Z, the above α_s corresponds to Λ_{\overline{\mathrm{MS}}}^{(5)} $\approx$ 0.208 GeV. With the lattice-motivated ansatz m_p $\approx$ 4.47 Λ_QCD:

$$m_p^{\text{est}} \approx 4.47 \times 0.208 \simeq 0.93 \text{ GeV},$$

within ~1% of the physical proton mass m_p = 0.938 GeV, connecting pixel geometry to hadronic physics. The factor 4.47 remains an external lattice input.

Two-input mode: simultaneous determination of α_s and sin$^2\theta$_W. In a two-input mode (treating P and the precisely measured $\hat{\alpha}^{-1}(M_Z)$ as calibration inputs), the pixel-area constraint combined with the electroweak identity and unification determines **both** remaining gauge couplings. In this alternate calibration mode, α_s and $\sin^2\theta_W$ move from calibration-sector checks to independent quantitative outputs. The system of constraints:

1. Pixel-area: $a_{\text{cell}}/\ell_p^2 = 4(\bar{\ell}_2 + \bar{\ell}_3)$
2. Electroweak identity: $\hat{\alpha}^{-1}(M_Z) = \alpha_2^{-1} + \frac{3}{5}\alpha_1^{-1}$
3. One-loop unification: $A_3 = \frac{b_3-b_2}{b_1-b_2}A_1 + \frac{b_1-b_3}{b_1-b_2}A_2$

Using only $a_{\text{cell}}/\ell_p^2 = 1.63094$ and the precisely measured $\hat{\alpha}^{-1}(M_Z) = 127.930 \pm 0.008$ as inputs, solving simultaneously gives a unique physical solution:

Two-input mode outputs (with P + $\hat{\alpha}^{-1}(M_Z)$ as calibration inputs):

α_s(M_Z) = 0.1175, sin^2^θ_W(M_Z) = 0.2311

Comparison to measurement:

- α_s(M_Z): predicted 0.1175 vs PDG 0.1177 ± 0.0009, difference 2×10^{-4} (within 1σ)
- sin$^2\theta$_W(M_Z): predicted 0.2311 vs PDG ŝ^2_Z = 0.23122 ± 0.00006 (MS-bar), difference ~1×10^{-4} (~2σ)

Important caveat on sin$^2\theta$_W. In *percentage* terms, 0.05% looks impressive. But the MS-bar experimental uncertainty is ±0.00006, so the difference ~1×10^{-4} is ~2σ in experimental units.

This residual discrepancy is not a calculation bug. For MSSM one-loop coefficients, unification implies a tight relation between α_s and sin$^2\theta$_W once ^α is fixed:

$$\sin^2\hat{\theta}_W(M_Z) = \frac{1}{5} + \frac{7}{15}\frac{\hat{\alpha}(M_Z)}{\alpha_s(M_Z)}.$$

Once the pipeline outputs α_s $\approx$ 0.1175, the weak mixing angle is essentially locked near 0.231. The ~2σ residual reflects missing theoretical corrections, not a flaw in the core calculation.

The required correction is small: shifting $\sin^2\theta$_W by ~1×10^{-4} corresponds to only ~0.04% change in α_2(M_Z). Effects at this level include:

- Two-loop running (including top-Yukawa contributions)
- Electroweak matching subtleties
- Scheme-conversion effects (entanglement → $\overline{\text{MS}}$)
- Threshold corrections from piecewise running
- Definition differences (on-shell vs $\overline{\text{MS}}$ vs effective leptonic)

All of these are O(10^{-4}) corrections that the one-loop treatment used here omits. The claim is therefore: the framework produces $\sin^2\theta$_W within ~2σ of the MS-bar measurement using only EW inputs plus the pixel constraint. This is a genuine parameter reduction, though precision claims require a proper theory error budget.

Theory uncertainty budget. A rigorous comparison to experiment requires estimating theoretical uncertainties. The dominant sources are:

Source	Estimated size
Two-loop running	O(10^{-4}) in $\sin^2\theta$_W
Threshold corrections (edge→4D matching)	Unknown; is O(10^{-3})
A_eff normalization ambiguity	Factor ~6; affects absolute t, not ratios
Scheme conversion (entanglement → MS-bar)	O(10^{-4}) expected
Missing U(1) mixing effects	O(10^{-4})

Without a full two-loop treatment and proper threshold matching, the framework cannot claim precision better than ~0.1% on $\sin^2\theta$_W. The ~2σ agreement with MS-bar data is encouraging but not definitive evidence.

Input elimination. This represents a genuine reduction in free parameters: the standard unification story requires both $\hat{\alpha}(M_Z)$ **and** $\sin^2\theta$_W(M_Z) as inputs. The pixel-area constraint eliminates $\sin^2\theta$_W as an input, predicting it instead.

SM measurement contradiction. Repeating with SM-only beta coefficients (b_1, b_2, b_3) = (41/10, -19/6, -7) gives α_s $\approx$ 0.096 and $\sin^2\theta$_W $\approx$ 0.216, both far from observation. The pixel-area constraint strongly disfavors SM-only running.

Edge-mode derivation of β-coefficients via Peter-Weyl structure. The key insight comes from the Peter-Weyl decomposition of L^2(G):

$$L^2(G) \simeq \bigoplus_R V_R \otimes V_R^*$$

A representation R corresponds to a block of size d_R^2. However, entropy and vacuum polarization "see" different parts of this structure:

- **Entropy (MaxEnt selection)** traces over one side of the entanglement cut, giving the factor d_R in the probability p_R $\propto$ d_R exp(-t C_2(R)).

- **Vacuum polarization loops** run over both indices of the V_R $\otimes$ V_R* block, restoring the second d_R factor.

Therefore, the effective multiplicity for RG running is:

$$N_{\text{eff}}(R) = d_R \cdot p_R$$

not just p_R. This is a structural consequence of Peter-Weyl, not a fitted parameter.

Edge sector weights. For the SM product group with Z_6 quotient, the superselection weight for sector (R_3, R_2, n) is:

$w(R_3, R_2, n) = d_3(R_3) \exp(-t_3\, C_2(R_3)) \cdot d_2(R_2) \exp(-t_2\, C_2(R_2)) \cdot \exp(-t_Y\, n^2)$

with the Z_6 selection rule $n \equiv -2\tau - 6j \pmod 6$, where τ is SU(3) triality and j is SU(2) spin. The probability is p = w/Z (normalized).

Beta shift formulas. Using standard one-loop matter contributions (Weyl fermion coefficient 2/3) with the second-index restoration:

$\Delta b_3 = (2/3)\, \Sigma\, p \cdot (d_3\, d_2) \cdot (d_2 \cdot T_3(R_3))$

$\Delta b_2 = (2/3)\, \Sigma\, p \cdot (d_3\, d_2) \cdot (d_3 \cdot T_2(R_2))$

$\Delta b_1 = (2/3)\, \Sigma\, p \cdot (d_3\, d_2) \cdot ((3/5)\, Y^2 \cdot d_3\, d_2)$

where Y = n/6 (canonical GUT normalization) and T_i is the Dynkin index with T(fund) = 1/2.

Representation bookkeeping. Complex representations R and their conjugates $\bar{R}$ are counted separately in the sum. For SU(3), the fundamental **3** and antifundamental $\bar{\mathbf{3}}$ both contribute with d = 3, $C_2 = 4/3$, and T = 1/2. Real representations (like the adjoint **8**) appear once. This is standard QFT bookkeeping: each chiral fermion species contributes independently to vacuum polarization.

Why U(1) uses a different formula. The hypercharge formula differs from SU(2)/SU(3) because U(1) has no Dynkin index structure; all irreps are 1-dimensional. The contribution to b_1 comes from Y^2 (the charge squared), with the factor 3/5 from GUT normalization. This is the standard form in unified theories, not a framework-specific choice. The different structure is why the U(1) prediction (5% error) is less precise than the non-Abelian ones (<1% error).

Numerical result at unification. At t_U ≈ 1.64 (corresponding to $\alpha_U^{-1} \approx 24.1$):

β shift	Predicted	MSSM target	Error
Δb_1	2.49	2.50	-0.3%
Δb_2	4.38	4.17	+5.1%
Δb_3	3.97	4.00	-0.7%

This achieves MSSM-like beta shifts without inserting MSSM by hand. The ~5% tension in Δb_2 is resolved by two-loop corrections, threshold effects, or refinements to the U(1) sector weighting.

What makes this non-trivial. The key test is the **ratio** $\Delta b_3/\Delta b_2$, rather than matching individual Δb values (which can be achieved by adjusting an overall normalization). The MSSM requires $\Delta b_3/\Delta b_2 = 4.00/4.17 = 0.959$. The Peter-Weyl calculation gives $3.97/4.38 = 0.906$, about 6% low. This ratio is fixed by the heat-kernel distribution and representation theory, with no free parameters to adjust. Getting within 6% of a non-trivial ratio like 0.96 from first principles is significant, though the remaining discrepancy suggests the mechanism is not yet complete.

Alternative minimal Dynkin-index mapping. A simpler estimate uses only the expected Dynkin index from the heat-kernel ensemble. Assume each RG shell contributes screening proportional to T_a(R), with two sides of the entanglement cut giving a factor of 2:

$$\Delta b_a(t) = 4\pi \langle T_a \rangle_{p(t)}, \qquad \langle T_a \rangle_{p(t)} := \sum_R p_R(t)\, T_a(R).$$

At t_U $\approx$ 1.64, the expected Dynkin indices are:

- $\langle T_2 \rangle \approx 0.330$
- $\langle T_3 \rangle \approx 0.390$

This gives:

- $\Delta b_2 \approx 4\pi \times 0.330 = 4.15$ (vs MSSM target 25/6 = 4.17, error -0.4%)
- $\Delta b_3 \approx 4\pi \times 0.390 = 4.90$ (vs MSSM target 4.0, error +22%)

The SU(2) shift matches the target within 0.4%, but the SU(3) shift is ~22% too large. This points to a **color-specific threshold/decoupling** effect: color edge excitations may stop contributing below some scale μ_c, reducing the integrated SU(3) shift. The required suppression factor f = 4.0/4.9 $\approx$ 0.82, interpreted as the fraction of the RG log-interval over which color screening is active, corresponds to a decoupling scale of order tens of TeV.

Why this works. The heat-kernel suppresses high-Casimir representations. The dominant sectors are (1,1), (1,2), (3,1), (3,2), and (8,1), which happen to match MSSM-like content. The Peter-Weyl second-index mechanism provides the correct multiplicity without any fitted constants.

Exploratory threshold estimates. Using the Peter-Weyl-derived beta shifts and measured α_em(M_Z), $\sin^2\theta$_W(M_Z) to fix α_1, α_2, one obtains the following threshold-study numbers:

- α_s(M_Z) $\approx$ 0.1168
- M_U $\approx 2.0 \times 10^{16}$ GeV
- α_U^{-1} $\approx$ 24.3

These numbers belong to an exploratory edge-threshold exercise rather than to the D10 calibration branch discussed in Section 6.19.

Using 4-loop M$\bar{\text{S}}$ running with n_f = 5, this α_s corresponds to:

Λ_M$\bar{\text{S}}^{(5)} \approx$ 195 MeV

This is the first genuinely "mass-like" scale output once β-coefficients are internally derived. The proton mass remains blocked by the nonperturbative conversion constant C_p (essentially what lattice QCD computes), but the upstream RG machinery is closed.

Full UV β-vector from edge modes. The edge-derived shifts can be combined with SM coefficients to obtain a complete UV running law without importing MSSM by hand:

$$b^{\text{UV}} = b^{\text{SM}} + \Delta b_{\text{edge}} \approx (6.59, 1.22, -3.03).$$

Threshold constraint. If this UV content were active from M_Z upward, one-loop unification with measured α_1, α_2 would predict α_s(M_Z) $\approx$ 0.157, far above the measured ~0.118. This forces a threshold/decoupling scale M_S above which the edge spectrum contributes to running, with SM running below.

Solving for M_S that makes measured couplings consistent with piecewise running (SM below M_S, edge-UV above):

$$M_S \approx 100 \text{ TeV}, \quad M_U \approx 6.5 \times 10^{15} \text{ GeV}, \quad \alpha_U^{-1} \approx 28.3$$

This should be read as an exploratory threshold scenario rather than as part of the D10 calibration branch.

What remains. Currently M_S $\approx$ 100 TeV is what the model needs to match data. To convert this into a genuine prediction requires deriving M_S from the edge physics itself (the gap/decoupling scale of edge excitations in the collar Hamiltonian), rather than solving for it from measured α_s. This is the sharpest remaining target for closing the precision prediction chain.

1.7.19 The Z_6 quotient: edge-sector selection rules and center-label entropy

The SM global gauge group is the quotient (SU(3) × SU(2) × U(1))/Z_6, rather than the direct product (Proposition 6.6). The quotient gives exact edge-sector selection rules. In the later flavor discussion we also use a phenomenological center-label entropy ansatz tied to this sixfold quotient structure.

The Z_6 congruence rule. The identified element is (exp(2πi/3), -1, exp(iπ/3)) ∈ SU(3) × SU(2) × U(1). Label edge sectors by SU(3) triality $\tau \in \{0,1,2\}$, SU(2) spin j, and hypercharge Y = n/6 with n ∈ Z. For the representation to descend to the quotient group, the identified element must act trivially:

exp(2πiτ/3) · (-1)^{2j} · exp(iπn/3) = 1

This gives the exact selection rule:

n ≡ -2τ - 6j (mod 6)

Sectors violating this congruence have exactly zero probability. This is a hard constraint from the global group structure.

Sanity check: SM hypercharges. The rule reproduces the SM pattern:

- Q_L: (3, 2, Y=1/6) => (τ=1, j=1/2, n=1) ✓
- L_L: (1, 2, Y=-1/2) => (τ=0, j=1/2, n=-3) ✓
- u^c: ($\bar{3}$, 1, Y=-2/3) => (τ=2, j=0, n=-4) ✓
- d^c: ($\bar{3}$, 1, Y=1/3) => (τ=2, j=0, n=2) ✓
- e^c: (1, 1, Y=1) => (τ=0, j=0, n=6) ✓

Heat-kernel slopes at M_Z. The general relation is t = g^2A_eff/2 where A_eff is the effective "Euclidean thickness" of the collar. Section 6.14's SU(3) lattice analysis finds A_eff → 2.004 ± 0.012 as $g^2 \to 0$.

Using $g^2 = 4\pi\alpha$ and defining the **normalization convention** A_eff = 4π (which differs from the lattice extrapolation by a factor of ~2π; see normalization note below), we have:

t_i = (g_i^2 · 4π)/2 = 2π · g_i^2 = 4π^2 α_i

With the electroweak inputs and PDG 2025 EW-fit value α_s(M_Z) = 0.1177:

- t_3 = 4.660
- t_2 = 1.335
- t_1 = 0.669

(Note: Using the MSSM unification prediction α_s = 0.1166 instead would give t_3 = 4.605, a 1.2% shift that negligibly affects the residue-class distributions below.)

For the U(1) factor with Y = n/6, the effective slope is t_Y = t_1/36 = 0.0186.

Normalization note: The A_eff = 4π convention gives t = 4$\pi^2\alpha$, a clean relation used throughout. However, Section 6.14's lattice extrapolation gives A_eff → 2, not 4π ≈ 12.6. This factor-of-~6 discrepancy indicates either: (1) the toy UV model's "g^2" differs from the continuum MS-bar convention by a factor of ~2π, or (2) additional physics (spin-statistics, vertex factors) enters the lattice↔continuum matching. This normalization ambiguity affects *absolute* t values but not ratios between gauge groups, so predictions depending only on ratios (hypercharge selection rules and center-label entropy estimates) remain stable. Predictions depending on absolute t (like the α_s pixel-area extraction) require this normalization to be resolved from first principles; in this branch it is fixed by convention.

Full edge-sector probability law. A sector (R_3, R_2, n) has weight

w(R_3, R_2, n) = $d_3(R_3)$ exp(-t_3 $C_2(R_3)$) · $d_2(R_2)$ exp(-t_2 $C_2(R_2)$) · exp(-t_Y n^2)

but only if the congruence n ≡ -2τ - 6j (mod 6) holds; otherwise w = 0 exactly. The probability is p = w/Z with Z summing over allowed sectors.

Hypercharge residue class distribution at M_Z. Summing over allowed sectors with the above weights:

- r ≡ 0 (mod 6): probability 0.6058
- r ≡ 3 (mod 6): probability 0.3816
- r ≡ 2 or 4 (mod 6): probability 0.0039 each
- r ≡ 1 or 5 (mod 6): probability 0.0024 each

At M_Z, because SU(3) is strongly coupled (t_3 large), triality-zero sectors dominate, so most weight sits in residues 0 and 3 (integer and half-integer hypercharge). The "quark-like residues" (1, 2, 4, 5) are suppressed at the 10^{-3} level.

Uniform center-label ansatz. If one assumes a uniform sixfold center-label ensemble on the realized $\mathbb{Z}_6$ quotient sectors, the associated center-label entropy is exactly

$\log_2 6$ = 2.584962500721156... bits

This is a parameter-free constant once the uniform sixfold center-label ansatz is assumed; the quotient itself fixes only the six allowed label classes.

Interpretation. The quotient restricts each (τ, j) combination to a single residue class $r \equiv n$ (mod 6), so the sixfold quotient structure provides a natural discrete label set for phenomenological continuations. If one further assumes those six center labels are populated uniformly, their Shannon entropy is

$$H_{\text{center}} = -\sum_{r=0}^{5} \frac{1}{6} \log_2 \frac{1}{6} = \log_2 6.$$

Equivalently, in nats one writes $H_{\text{center}} = \ln 6$.

Status. The quotient itself and the congruence rule are structural. The use of $\log_2 6$ as a suppression scale in later flavor discussions is an additional phenomenological ansatz about how the realized center labels are populated. It should therefore not be read as a theorem-level direct observable of the quotient alone.

Toy counting picture. In the simple residue-sum model above, comparing naive product counting to quotient-restricted counting produces a numerical difference close to $\log_2 6$ bits at M_Z. We retain that computation only as motivation for the center-label ansatz, not as a model-independent measurement prediction.

1.7.20 Electroweak scale as a calibration-sector output

In the D10 calibration branch, the electroweak scale is not presented as an independent theorem-level prediction. Once the pixel input P, the gauge-sector closure, and the printed running and matching conventions are fixed, the same fixed-point problem returns the low-energy electroweak scale and tree-level electroweak masses as consistency outputs.

The scale-setting formulas used in this branch are

$$M_U = \frac{E_P}{e^{2\pi}} P^{1/6} \approx 2.474 \times 10^{16}\,\text{GeV}, \qquad E_{\text{cell}} = \frac{E_P}{\sqrt{P}} \approx 9.56 \times 10^{18}\,\text{GeV},$$

together with

$$v = E_{\text{cell}} \exp\left(-\frac{2\pi}{\beta_{\text{EW}} \alpha_U}\right), \qquad \beta_{\text{EW}} = N_c + 1 = 4,$$

using the gauge-closure value $\alpha_U^{-1} \approx 24.32$. Under the conventions used here this gives

$$v \approx 246.77 \text{ GeV}.$$

Accordingly, v, m_W, and the running Z-mass relation should be read as calibration-sector consistency checks of this branch rather than as independent confirmations of the recovered core. Their role is to show that the D10 closure is internally coherent once P and the printed conventions are imposed.

1.7.21 Heavy-top heuristic and its status

A simple order-one-Yukawa estimate,

$$m_t \sim \frac{v}{\sqrt{2}},$$

explains why the top sector is naturally much heavier than the rest of the Yukawa spectrum, but it is not the D11 result quoted later in the paper.

The top/Higgs numbers belong to the critical-surface transport branch of Section 6.22. That branch, not the order-one heuristic, is the secondary quantitative output discussed later in the paper. This subsection should therefore be read only as motivation for why the top channel is the natural least-suppressed Yukawa mode.

1.7.22 Yukawa hierarchy from Z_6 defect suppression

The Z_6 quotient structure provides a possible phenomenological route to the fermion mass hierarchy once one adds the uniform center-label ansatz and the defect-suppression hypothesis, without introducing continuous Yukawa parameters.

The key phenomenological observation. Under the uniform center-label ansatz of Section 6.18, the associated entropy scale is $H_{\text{center}} = \ln 6$ nats. If one uses that scale as the cost of resolving one unit of quotient-label mismatch, the corresponding suppression factor is:

$\varepsilon = \exp(\text{-}\ln 6) = 1/6$

Yukawa mechanism. Treat each Yukawa coupling as a defect-mediated overlap amplitude between left/right edge sectors. The Z_6 quotient structure means that left-handed and right-handed fermions carry different Z_6 gradings. A Yukawa coupling corresponds to an intertwiner (morphism) that must be neutral under this grading.

Definition (Defect number). If the direct intertwiner is forbidden by the Z_6 congruence rule, it can be generated by inserting defect operators that shift the grading. Define:

n_f := min{n ∈ Z≥0 : neutral intertwiner exists after n defect insertions}

This is a minimal path length in the overlap groupoid, automatically an integer.

Suppression from entropy. Each defect insertion resolves one unit of the Z_6 restriction, removing a factor of 6 in available microstates. MaxEnt weighting then gives:

y_f ∝ ε^n_f = 6^(-n_f), where ε = 1/6

This is a Z_6-anchored Froggatt-Nielsen texture with the small parameter ε fixed by topology rather than chosen.

Extraction of defect charges. Using $y_f = \sqrt{2}\, m_f/v$ with the calibration-sector value of v from Section 6.19, and $n_f = -\ln(y_f)/\ln(6)$:

Fermion	y_f (from mass)	n_f (real)	Nearest int	Residual c_f
t	1.003	-0.002	0	1.00

Fermion	y_f (from mass)	n_f (real)	Nearest int	Residual c_f
b	0.024	2.08	2	0.87
c	0.0074	2.74	3	1.59
s	0.00054	4.20	4	0.70
d	2.7×10^{-5}	5.87	6	1.27
u	1.3×10^{-5}	6.30	6	0.59
τ	0.010	2.55	3	2.23
μ	0.00061	4.13	4	0.80
e	3.0×10^{-6}	7.10	7	0.83

Key observations:

1. The logarithms are close to integers in base 6, the "Z_6 controls hierarchy" fingerprint.
2. The residual coefficients c_f are all order-one (0.6–2.2), consistent with RG running, mixing angles, and Clebsch-Gordan factors in overlap tensors.

Minimal charge assignment. Writing exponents as sums of defect charges (Froggatt-Nielsen style):

- n^(u)_ii = q_Qi + q_Ui
- n^(d)_ii = q_Qi + q_Di
- n^(e)_ii = q_Li + q_Ei

one compact solution is:

- q_Q = (2, 1, 0)
- q_U = (4, 2, 0)
- q_D = (4, 3, 2)
- q_L = (3, 1, 0)
- q_E = (4, 3, 3)

This reproduces the observed hierarchy with integer charges and **no continuous parameters beyond $\varepsilon = 1/6$**.

Significance. The Yukawa sector reduces from "dozens of arbitrary reals" to:

- One fixed small parameter $\varepsilon = 1/6$ (from Z_6 topology)
- A set of integers n_f (defect/charge data) that the UV completion must output

The mass hierarchy stops being an unexplained input and becomes discrete topological data constrained by the global gauge group structure.

Computational note (January 2026): using the calibration-sector value $v \approx 246.77$ GeV leaves the base-6 hierarchy pattern close to integer defect counts, but the branch remains phenomenological regardless of that small numerical shift.

1.7.23 Higgs/top critical-surface branch

The D11 branch uses the refinement-stability condition

$$\lambda(M_U) = 0, \qquad \beta_\lambda(M_U) = 0$$

at the same unified scale as the D10 calibration sector, namely

$$\mu_* = M_U \approx 2.474 \times 10^{16} \text{ GeV}.$$

The appendix supplies only the low-scale SM piece of the one-loop $\overline{\text{MS}}$ transport for the coupled $(y_t, \lambda, g_Y, g_2, g_3)$ system. The D11 branch used for the displayed numbers runs with SM coefficients from m_Z up to a UV synchronization scale $\mu_{\text{sync}} \approx 6.8 \times 10^{11}$ GeV, then with the same MSSM-like coefficients as D10 from μ_{sync} to M_U, and only afterward applies order-one-GeV low-scale matching corrections. The literal all-SM appendix flow by itself lands much lower, near

$$m_t^{\text{pole}} \approx 164.2 \text{ GeV}, \qquad m_H \approx 115.2 \text{ GeV}.$$

The piecewise-running core before those final matching shifts instead yields

$$m_t^{\text{pole,core}} \approx 170.3 \text{ GeV}, \qquad m_H^{\text{core}} \approx 126.6 \text{ GeV},$$

and after the residual matching conventions documented in the spectrum supplement the displayed D11 outputs are

$$m_t^{\overline{\text{MS}}}(m_t) \approx 160.6 \text{ GeV}, \qquad m_t^{\text{pole}} \approx 171.1 \text{ GeV},$$

together with

$$m_H = \sqrt{2\lambda(m_t)}\, v \approx 126.5 \text{ GeV}.$$

The appendix records only the low-scale SM part of the flow, while the D11 outputs above also use the UV synchronization scale, the MSSM-like UV coefficients inherited from D10, and the shared matching, low-scale extraction, and pole-mass conventions stated in this part. The appendix equations alone are not meant as a standalone reproduction recipe for these displayed numbers.

These numbers are the D11 outputs used in this paper. They add no new continuous fit once the D10 gauge trajectory and scale-setting branch are fixed, but they remain a secondary quantitative branch rather than part of the recovered-core theorem package.

The critical-surface branch can therefore fail even when the electroweak calibration succeeds. Conversely, agreement here supports only the supplement-backed D11 implementation analyzed here, not the entire OPH program.

1.7.24 Rigorous derivation chain: axioms to predictions

This section consolidates the logical structure of what the axioms actually derive, what requires additional inputs, and where the numerical predictions emerge.

The first four steps below record the structural chain. The later threshold and two-input numerics are retained only as an illustrative edge-running sketch, not as the D10/D11 branch used elsewhere in the manuscript.

Step 1: From axioms to heat-kernel distribution (rigorous).

The core axiom package (Markov collars + MaxEnt selection) yields a Gibbs/exponential-family form for the reduced collar state:

$$\rho_C = \frac{\exp(-\sum_a \lambda_a O_a)}{Z(\lambda)}$$

This is Theorem 2.6. The Lagrange multipliers λ_a are determined by constraint values, not derived by MaxEnt itself.

For gauge collars with the Casimir as the constraint operator, the MaxEnt state implies:

$$p_R(t) \propto d_R\, e^{-t\, C_2(R)}$$

where t is the diffusion/Lagrange multiplier parameter.

Step 2: The t–α bridge (rigorous).

In 2D Yang-Mills / heat-kernel language, the weight is exp(-(g^2A/2) C_2(R)), giving t = g^2A/2. The modular/Euclidean-regularity constraint fixes the collar's effective area to A = 2π (the Rindler angle period), yielding:

$$t = \frac{g^2(2\pi)}{2} = \pi g^2 = 4\pi^2\alpha$$

This is the unique normalization compatible with modular geometry, not a scheme choice.

Step 3: Pixel constant relation (rigorous).

The generalized entropy matching (Section 5.4) gives:

$$G = \frac{a_{\text{cell}}}{4\bar{\ell}_{\text{tot}}}, \qquad \bar{\ell}_{\text{tot}} = \bar{\ell}_{\text{SU}(2)}(t_2) + \bar{\ell}_{\text{SU}(3)}(t_3)$$

In Planck units (ℓ_p^2 ≡ G):

$$\frac{a_{\text{cell}}}{\ell_p^2} = 4\bar{\ell}_{\text{tot}}(t_2, t_3)$$

This is a derived relation, not an assumption. However, the **numerical value** of a_cell/ℓ_p^2 depends on the t_i values, which depend on the couplings.

Step 4: Edge-derived beta functions via Z_6 quotient structure (new).

The key insight from the edge sector: to get β-function contributions, count modes by the full edge Hilbert-space multiplicity, not entropy:

$$\text{weight} \propto (d_{\text{SU3}} \cdot d_{\text{SU2}})^2 \cdot p(R_3, R_2, y)$$

The entropy weights by log d_R; vacuum polarization loops see d_R^2 (both indices of the Peter-Weyl block V_R ⊗ V_R*).

Hypercharge via Z_6 quotient. For (SU(3) × SU(2) × U(1))/Z_6, the allowed hypercharge lattice is constrained. Writing y = 6Y:

$$y + 2(p + 2q) + 6j \equiv 0 \pmod 6$$

This fixes which hypercharges pair with which non-Abelian reps.

U(1) weighting. The edge spectrum for the Abelian factor uses:

$$w_y \propto e^{-t_1 \kappa y^2}$$

with κ from U(1) normalization and the Z_6 congruence enforced.

Result. At t_U ≈ $\pi^2/6$ ≈ 1.645 (corresponding to α_U^{-1} ≈ 24), with one overall normalization fixed by demanding Δb_2 matches MSSM:

$$\Delta b_{\text{pred}} \approx (2.49,\ 4.17,\ 4.01)$$

Compare to MSSM–SM shift: Δb_MSSM = (2.5, 4.17, 4.0). Agreement is <1% for all three coefficients.

Why this is significant. This replaces "assume MSSM running" with a computation from the edge sector using only:

- Heat-kernel form (from MaxEnt)
- Z_6 quotient structure (from SM global gauge group)
- d^2 weighting (from Peter-Weyl vacuum polarization structure)
- One overall normalization (fit to Δb_2)

The ratios $\Delta b_3/\Delta b_2$ and $\Delta b_1/\Delta b_2$ are then predictions.

Step 5: Historical inverse-problem illustration for threshold and unification scales. With edge-derived Δb and measured electroweak inputs at M_Z:

- $\hat{\alpha}^{-1}$(M_Z) = 127.951 ± 0.009
- $\hat{s}^2$_Z = $\sin^2\hat{\theta}$_W(M_Z) = 0.23122 ± 0.00004
- α_s(M_Z) = 0.1180 ± 0.0009

The piecewise running (SM below M_S, edge-UV above) gives a 2×2 system for $x = \ln(M_S/M_Z)$ and $y = \ln(M_U/M_S)$. Solution:

$$M_S \approx 60 \text{ GeV}, \quad M_U \approx 2.4 \times 10^{16} \text{ GeV}, \quad \alpha_U^{-1} \approx 24.0$$

The effective threshold scale lands near the electroweak scale, not at multi-TeV.

Step 6: Historical two-input illustration.

The cleanest "reduce inputs → predict observables" step:

Inputs:

1. Pixel constant: a_cell/ℓ_p^2 = 1.631 (treat as fundamental)
2. One electroweak datum: $\hat{\alpha}^{-1}$(M_Z) = 127.951

Constraints:

- Pixel: a_cell/ℓ_p^2 = 4($^-\ell_2$ + $^-\ell_3$)
- One-loop unification with edge-derived Δb

Outputs (predicted, not input):

$$\sin^2\hat{\theta}_W(M_Z) \approx 0.2310, \qquad \alpha_s(M_Z) \approx 0.1175$$

Comparison to PDG:

- α_s(M_Z): predicted 0.1175 vs measured 0.1180 ± 0.0009 → 0.6σ low
- $\sin^2\theta$_W: predicted 0.2310 vs measured 0.23122 ± 0.00004 → ~2σ

The α_s agreement is excellent. The $\sin^2\theta$_W tension (~2σ) is where precision threshold/two-loop effects matter.

Step 7: Consistency check, beta_EW from v.

If the pixel constant P and electroweak VEV v are both treated as inputs, we can solve for the transmutation coefficient β_EW that reproduces v:

$$v = \frac{E_p}{\sqrt{P}} \exp\left(-\frac{2\pi}{\beta_{\mathrm{EW}} \cdot \alpha_U}\right)$$

Using P = 1.63094, v = 246.22 GeV, E_p = 1.22089×10^{19} GeV:

$$\beta_{\mathrm{EW}}^{\mathrm{req}} = 3.997$$

This is β_EW = 4 to within 0.1%. The integer 4 = N_c + 1 (number of SU(2) doublets per generation) emerges from fitting, but also has a structural rationale from the Witten anomaly constraint.

What the axioms derive vs what requires additional input.

Quantity	Status
Heat-kernel form p_R(t)	Derived from MaxEnt + Casimir constraint
t = $4\pi^2\alpha$ normalization	Derived from modular geometry (A = 2π)
G = a_cell/($4^-\ell$) relation	Derived from generalized entropy matching
Numerical value of a_cell/ℓ_p^2	**Not derived**; requires fixing t (hence α)
Edge-derived Δb ratios	Derived from Z_6 + Peter-Weyl + one normalization
Threshold scale M_S	Derived from inverse problem given couplings
β_EW = 4	Structural (N_c + 1) or fitted; ambiguous status

The remaining closure gap. The axioms derive rigid functional relations but not unique numerical values for the couplings. To close the loop requires either:

1. A principle that fixes t (the Lagrange multiplier) from microphysics
2. Treating a_cell/ℓ_p^2 as fundamental input (replaces one coupling)
3. Using measured v to fix the transmutation chain

A two-input mode of this kind is a useful illustration of how one could trade one electroweak datum for the pixel constant. The branch analyzed elsewhere in the manuscript instead follows the D10/D11 construction summarized above; full closure (option 1) awaits a UV completion that specifies the MaxEnt constraint values.

1.8 Open Questions

The following questions remain for future work:

- Quantify BW_{S^2} error control in the collar refinement limit
- Justify the refinement-stability selector for chirality in explicit models

- Relate the non-central obstruction class to EFT anomalies quantitatively
- Resolve the threshold scale M_S ambiguity (60 GeV vs. 100 TeV) by understanding the edge-mode decoupling mechanism
- Derive $t_U \approx 1.64$ from group-theoretic principles; the Z_3 lattice test (Section 6.14) shows t_U is fixed by a criticality condition
- Complete the scheme matching $t \leftrightarrow \alpha_s^{\overline{MS}}$ in an explicit collar lattice realization
- Derive the nonperturbative conversion $C_p = m_p/\Lambda_{QCD}$ from first principles (the benchmark analyzed here uses lattice QCD's $C_p \approx 4.47$)
- Provide a microscopic derivation of Recoverable Generalized Entropy
- Derive the pixel area $a_{\text{cell}} \approx 1.63094\,\ell_P^2$ from a principle that fixes the Lagrange multiplier t

---

1.9 Critical Evaluation

1.9.1 Classification of results and remaining dependence

This section states which node of the program has been proved, which standard mathematics is imported, and which extra physical premises or external inputs remain load-bearing. The documentary node labels D1–D12 match the ledger in Section 1.2.6. The three-phase boundary is

$$\text{Phase I} = (D1\text{–}D5) \cup (D7\text{–}D9), \qquad \text{Phase II} = D6 \cup D10 \cup D11, \qquad \text{Phase III} = D12.$$

Node	Representative outputs in this manuscript	Standard mathematics used	Extra physical premises / external inputs required	Claim tier
D1	Theorem 3.1 schedule-independent normal form	Newman's lemma	finite patch net, termination, and local confluence	Phase I structural theorem
D2	EC decomposition, exact/approximate Markov collar, and generalized-entropy split	HJPW; Petz; Fawzi–Renner	R0/R1 or explicit recoverability control; coarse-grained $A/(4G)$ dictionary	Phase I structural lemma/proposition layer
D3	Theorems 4.2–4.3 Lorentz branch	Bisognano–Wichmann modular geometry; conformal-group classification	premise ROT, controlled refinement limit, and the OPH geometric-branch hypothesis used in Theorem 4.2	Phase I conditional scaling-limit theorem
D4	Section 5.2 null modular bridge	Borchers–Wiesbrock; distribution theory for half-line differentiation	the theorem-local inherited-strip decomposition and exact-or-controlled Markov hypotheses of Section 5.2, plus the downstream density-upgrade and relativistic null-stress-identification premises; the endpoint/propagation controls are internal to the local finite-constraint MaxEnt branch	Phase I conditional bridge theorem

Node	Representative outputs in this manuscript	Standard mathematics used	Extra physical premises / external inputs required	Claim tier
D5	Theorem 5.1 rest-frame Einstein relation plus the conditional tensor upgrade of Corollary 5.1	Jacobson small-ball mathematics; null-to-tensor reconstruction modulo a metric term	fixed-cap stationarity, the locally Lorentzian $d = 4$ scaling regime, and the all-directions/all-reference-states premise used in the tensor upgrade	Phase I conditional scaling-limit theorem/corollary
D6	capacity/Λ statements and the separate capacity-based neutrino estimate	de Sitter entropy relation and dimensional analysis	external input N_{scr} and the screen-capacity identification	Phase II input-dependent corollary / estimate
D7	Theorem 6.1 compact gauge reconstruction	Doplicher–Roberts / Tannaka reconstruction	bosonic symmetric transportable colimit and fiber-functor premises; the refinement-stable qualifier is internal to the local finite-constraint MaxEnt/refinement branch	Phase I conditional structural theorem
D8	product gauge structure up to finite quotient under MAR	compact Lie representation classification; Schur's lemma	MAR, connected admissible class, one connected abelian factor, and $[z] = 0$	Phase I realized-branch theorem
D9	realized $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)/\mathbb{Z}_6$, exact hypercharges, the counting chain $N_g = 3$ then $N_c = 3$, and no gauge-mediated proton decay as a product-group corollary	anomaly algebra; Witten anomaly; CKM CP counting	realized one-generation/one-Higgs package and MAR admissibility premises	Phase I realized-branch theorem/corollary chain
D10	gauge/electroweak numerical closure of the D10 calibration branch	RG running and matching conventions	external input P, pixel closure, and the printed threshold conventions	Phase II calibration sector
D11	Higgs/top critical-surface branch	RG transport and pole-mass conversion	critical-surface condition $\lambda = \beta_\lambda = 0$ at M_U, plus the UV synchronization scale μ_{sync} used in the D11 transport; no new continuous fit once D10 is fixed	Phase II quantitative branch
D12	Koide/charged leptons, textures, dark-sector, baryogenesis, spectroscopy, string/worldsheet, and other downstream continuations	branch-specific EFT and phenomenological manipulations	explicit additional ansätze beyond D6, D9, D10, or D11	Phase III phenomenological continuation / program branch

Three reading rules follow from this ledger.

1. Phase I, the recovered core carried by this manuscript, is D1–D5 together with D7–D9. These are the structural or realized-branch claims defended at theorem/corollary level once all stated premises are included.

2. Phase II contains the separate input-dependent capacity corollary D6 and the D10–D11 quantitative branches. Failure there does not by itself erase Phase I.

3. Phase III contains D12 continuations only. Nothing in D12 should be cited as theorem-level unless and until its extra ansätze are replaced by proved premises.

4. **Existence / strange-loop narratives**: self-simulation or strange-loop closure stories are interpretive epilogues only. The internally available ingredient is a theorem-usable OPH

state-and-law habitat built from overlap-consistent patch states together with finite Axiom-3 law coordinates. The recovered core does *not* derive an existence theorem, an OPH-internal closure map on an invariant admissible observer-supporting sector, or any uniqueness/stability result for such a map.

Symmetry-protected exact zeros such as the masslessness of the photon, gluons, and graviton, together with the absence of gauge-mediated proton decay, are corollaries inside Phase I. They do not define separate axiom-only nodes.

1.9.2 Structural assessment

- **Dynamics**: The GR chain requires modular covariance plus the null-surface modular bridge (N1-N3). Section 5.2 carries the bridge through the derived positive null-translation stage on the OPH geometric scaling branch. The explicit remaining burdens are modular covariance, the downstream density upgrade, and the relativistic null-stress identification. Verifying those items in explicit UV regulators remains a target for future work.
- **Gauge structure**: The gauge group is reconstructed from sector fusion, and the anomaly/gluing link is precise. The Selection Axiom MAR fixes the SM factors; the CP/UV admissibility window and MAR give $N_g = 3$, and Witten's parity constraint with that realized generation count gives $N_c = 3$. Proposition 6.1a gives DHR transportability $\Leftrightarrow [z] = 0$. Justifying the refinement-stability selector for chirality in explicit models remains a target for future work.
- **Microscopic theory**: Quantum link models (Section 2.6) provide an explicit UV realization of R0/R1 and give EC + Markov collars automatically. What remains: (i) a microscopic derivation of Recoverable Generalized Entropy, and (ii) ensuring modular flow becomes geometric in the continuum limit (the modular-covariance / GB gap). The latter likely requires a holographic or relativistic regime, which is not automatic in generic lattice gauge systems.
- **Loop gluing beyond central defect**: The general obstruction theory is structurally in place, but quantitative matching to EFT anomalies remains open.

1.9.3 Testable items and continuation templates

The framework makes several directly testable statements, together with weaker continuation-level templates that can also be confronted with data:

GW horizon spectroscopy comb (Section 5.11; continuation-level). If the extra discrete-horizon premises of Section 5.11 hold, the log-integer area spectrum gives discrete resonant frequencies for Kerr black hole horizons:

f_{k,m}(M,χ) = (m Ω_H)/(2π) + (c^3 g(χ))/(16π^2 GM) $\cdot$ ln(k) for k = 2, 3, 4, ...

After rescaling by remnant parameters, all events should stack at universal coordinates x_k = ln(k)/(8π). This is checkable with public LIGO/Virgo data as a continuation-level test of that extra branch. Absence of coherent stacking would challenge the discrete-horizon continuation, not the recovered-core package by itself.

Discrete Hawking comb (Section 5.11; continuation-level). If the same discrete-horizon branch is realized, primordial black holes in the final evaporation stage should show comb structure at E_k/E$_2$ = ln(k)/ln(2). Current PBH burst searches (Fermi, H.E.S.S.) can constrain this with dedicated template analysis.

Casimir ratio precision (Section 8.1). Future lattice measurements of SU(3) edge-sector probabilities should confirm $\Delta_8/\Delta_3 = 9/4$ exactly, not 2.67 (dimension-only) or 5.06 (Casimir-squared). The full set of parameter-free SU(3) ratio predictions is:

- $\Delta_8/\Delta_3 = 9/4 = 2.25$
- $\Delta_6/\Delta_3 = 5/2 = 2.5$
- $\Delta_{10}/\Delta_3 = 9/2 = 4.5$
- $\Delta_{15}/\Delta_3 = 4$
- $\Delta_{27}/\Delta_3 = 6$

These exact rationals are fixed entirely by group theory (Casimir eigenvalue ratios), with no adjustable parameters. Any deviation identifies a measurement contradiction with the heat-kernel edge-sector mechanism.

Conditional $\mathbf{Z}_6$ center-label entropy scale (Section 6.18). Under the uniform sixfold center-label ensemble used in the flavor continuation, the associated entropy scale is $\log_2 6 \approx 2.585$ bits. This is a phenomenological ansatz tied to the quotient label set, not a theorem-level direct observable of the quotient alone.

Black hole spectroscopy secondary structure (Section 5.11). Beyond the headline log-integer comb, the framework predicts rigid secondary structure:

1. *Universal energy ratios*: E_k/E_2 = ln(k)/ln(2) exactly. For example, E_3/E_2 = ln(3)/ln(2) $\approx$ 1.585 is parameter-free. This arithmetic pattern of ratios distinguishes OPH from other "quantized area" proposals that have different functional forms or free spacing parameters.

2. *Mass-independent fractional linewidth*: The intrinsic linewidth $\Gamma/\Delta E_k \approx$ 3–5% is approximately independent of black hole mass. This is a sharp shape prediction constraining line positions and line profiles.

3. *Fixed weight hierarchy*: Line weights follow a (k-1)/k pattern from detailed balance in the log-integer transition rule, on top of the GR greybody envelope. High-k lines asymptote in strength in a specific, counting-driven way.

Inequality bounds on GR deviations (Section 5.8). The modular additivity defect satisfies the exact identity $\langle \Delta K \rangle$ = -I(A:D|B), where I(A:D|B) is the conditional mutual information. Under the Markov/mixing assumptions, this defect is exponentially small in collar thickness:

$|\langle \Delta K \rangle| \leq 2|A| \cdot \eta$^{w/$\xi$}

This propagates into an explicit upper bound on how far the Einstein equation can deviate from GR in regimes where the emergence proof applies. Unlike typical beyond-GR frameworks that postulate corrections, OPH provides a quantitative ceiling: given the information-theoretic primitives, corrections decay exponentially with collar width. This "UV ignorance $\rightarrow$ rigorous inequality" structure is distinctive.

Yukawa hierarchy test (Section 6.21). The prediction y_f $\propto$ 6^{-n_f} with integer defect charges means the extracted exponents -ln(y_f)/ln(6) should land unusually close to integers across all fermions. The small parameter $\varepsilon = 1/6$ is motivated by the same $\mathbb{Z}_6$ structure that underlies the conditional $\log_2 6$ center-label entropy ansatz; this ties hierarchy to a quotient-label mechanism rather than to a chosen Froggatt-Nielsen parameter.

No gauge-mediated proton decay (Section 6.11). If the gauge group is genuinely a product (from sector factorization), coupling unification is geometric (shared edge diffusion parameter) rather than simple-group embedding. This predicts unification-like coupling relations *without* GUT leptoquark bosons, hence no gauge-mediated proton decay. The combination "coupling unification + no proton decay" is a crisp discriminator against classic GUT predictions.

1.9.4 What is not predicted (gaps)

Section 1.9.1 fixes the three-phase boundary. The list below collects the items that remain outside Phase I even when they are discussed elsewhere in the manuscript.

Not predicted by the recovered core.

- **Flavor and Yukawa data**: Yukawa matrices, flavor hierarchies, charged-lepton/Koide fits, quark-texture exponents, and downstream hadronic or flavor fits. These require extra ansätze such as a uniform $\mathbb{Z}_6$ center-label ensemble, circulant generation-space structure, discrete exponent assignments, or further external numerical machinery.
- **Neutrino data**: beyond the input-dependent, order-of-magnitude capacity estimate tied to N_{scr}, the manuscript does not derive neutrino masses or mixings.
- **Dark-sector phenomenology**: the sign, closure, and galaxy-scale response law of any modular-anomaly stress tensor are not derived by the recovered gravity chain.
- **Baryogenesis**: suppression-counting estimates are not a substitute for a derived out-of-equilibrium mechanism.
- **Black-hole spectroscopy**: any logarithmic line template requires an additional discrete-horizon branch.
- **String/worldsheet reorganization**: the Gross-Taylor-type interpretation requires a controlled large-N_{edge} regime distinct from the proved compact-gauge chain, and any further lift to critical superstring structure remains a conjectural extension requiring extra worldsheet ingredients.
- **Other downstream continuations**: strong-CP proposals, proton-spin fractions, and similar late-stage continuations appearing on other OPH surfaces are not derived by the recovered core.

What remains open at the core/implementation boundary.

- **Scaling-limit control**: the Lorentz/null-modular/Einstein chain depends on the explicit scaling-limit premises declared earlier in the manuscript; proving or constructing that regime is an open task.
- **Fermionic gauge branch**: the manuscript proves compact gauge reconstruction only in the bosonic internal-gauge branch; a full fermionic/super-Tannakian extension remains open.
- **External inputs**: P and N_{scr} are not derived in this manuscript. The numerical value of Λ is not a local recovered prediction because local null data determine the Einstein equation only up to a metric term ϕg_{ab}; fixing Λ requires the global capacity input N_{scr}.
- **Secondary quantitative branch**: the D10–D11 numerical implementation depends on supplement-backed running, matching, and the extra critical-surface condition used in the Higgs/top branch.

Failures localize by tier. A failure in a Phase-III flavor, dark-sector, baryogenesis, spectroscopy, string, strong-CP, proton-spin, or similar continuation retracts that continuation only. A failure in D10 or D11 challenges the corresponding quantitative branch. Only a failure in Phase I would overturn the recovered-core claim set.

1.9.5 Comparison with other unification approaches

Unified models attempting to tie together QFT, gravity, and SM structure tend to encounter a repeatable set of conceptual difficulties. This subsection examines how the observer-patch holography framework addresses these common pitfalls.

1. Subsystem factorization in gauge theory and gravity.

In gauge theories and gravity, the Hilbert space does not cleanly split as "inside $\otimes$ outside" across a cut. This infects entanglement entropy definitions, area terms, edge modes, and observable identification. Many unification attempts handwave this or patch it with conventions.

OPH route: The framework builds from a net of von Neumann algebras on patches plus overlap consistency. It does not start from naïve tensor factorization. The gauge-as-gluing + regulator package yields edge-center completion: a canonical block decomposition on collars where the center captures superselection data at the cut, and the state becomes (exactly or approximately) Markov across the collar. The entropy split $S(\rho_C) = S_bulk + \langle L_C \rangle$ follows from having a center with sector labels. This replaces the ad hoc "add an area term" move.

2. Modular Hamiltonian nonlocality.

Many entanglement-based gravity derivations depend on modular Hamiltonians that look like local stress-tensor charges (true only in special states/regions). In generic QFT states, modular Hamiltonians are nonlocal, making "first law of entanglement $\Rightarrow$ Einstein equation" arguments fragile.

OPH route: The Markov collar condition does heavy lifting: approximate Markov implies approximate modular additivity, with the defect controlled by conditional mutual information. This makes "modular locality" a controlled approximation. It is not treated as an assumption. The controlled tangent-half-space comparison of Theorem 4.2 then locks modular flow to geometric dilations with rigid 2π normalization.

3. Lorentz invariance as a derived output.

Discrete microscopic models generally break Lorentz symmetry, and many unified proposals simply postulate Lorentz invariance in the IR.

OPH route: Lorentz kinematics are tied to geometric modular flow on caps. Once modular flow acts as conformal transformations on S^2, we get $\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$, so the Lorentz group appears as a theorem-level output of modular structure. No external spacetime symmetry axiom is added.

4. Dynamics beyond kinematics.

Many approaches produce emergent geometry/kinematics but stall at dynamics: why Einstein's equations (with the right coefficient) rather than some other geometric PDE?

OPH route: The framework combines MaxEnt entanglement equilibrium, the derived $K_C = 2\pi B_C$ structure with rigid normalization, and an EFT bridge identifying modular energy with stress-tensor charges. The null modular additivity route reduces much of that bridge to Markov/edge-center mechanisms on null strips plus explicit scaling-limit regularity premises. It does not rely only on "assume a UV CFT."

5. Gauge symmetry origin and compactness.

Most unification stories pick a gauge group and work out consequences. Emergent-gauge approaches sometimes produce noncompact groups or uncontrolled redundancies.

OPH route: Gauge symmetry is recast as redundancy in overlap identifications (gauge-as-gluing). From edge sectors and fusion, a tensor category is reconstructed; Tannaka-Krein / Doplicher-Roberts reconstruction then yields a compact group G given the categorical hypotheses. "Gauge symmetry" names the gluing redundancy at the conceptual level. "Compact group" is the mathematical form compatible with finite-dimensional sector/fiber-functor structure.

6. Massless photon and graviton usually hand-imposed.

Getting massless gauge bosons is easy if exact gauge invariance is assumed, but that restates the problem. Massless graviton is more delicate (mass terms, vDVZ discontinuity, strong coupling scales).

OPH route: Once gauge and diffeomorphism invariance are emergent redundancies of description (from gluing consistency / emergent geometry), hard mass terms are forbidden: "a coordinate system's Jacobian can't show up as a physical mass." These symmetry-protected zeros emerge from the same consistency machinery that gives the symmetries.

7. Global consistency, anomalies, and loop patching.

Building physics from local patches hits loop/holonomy problems: consistent gluing on a tree but obstructions around loops. These obstructions are often anomalies or global topological constraints.

OPH route: This is elevated to a first-class organizing principle: gluing data on overlaps defines cocycles; central defects define a Čech obstruction class [z] (and more generally a 2-group/crossed-module cocycle for noncentral defects). "Global consistency exists iff the obstruction class vanishes" becomes the universal statement. Anomalies become "failure to glue" rather than a mysterious quantum pathology.

8. Charge quantization without a GUT.

Without embedding into a simple GUT group, explaining charge quantization (why all isolated color singlets are integer charged) is awkward. Standard lore requires grand unification or monopoles.

OPH route: The framework leans on global group structure (the Z_6 quotient) and derives congruence/selection rules for allowed representations/hypercharges. This gives a structural explanation for integer-charged color singlets without introducing the proton-decay channel of simple-GUT models.

9. Coupling unification usually forces proton decay.

Traditional simple-group unification introduces leptoquark gauge bosons (X, Y) mediating proton decay. Experiment keeps pushing limits up, pressuring minimal GUTs.

OPH route: "Unification" here is geometric/entropic (shared edge diffusion parameter, heat-kernel weights). It is not an embedding into a simple Lie group. If the reconstructed gauge group genuinely factorizes as a product (sector factorization selector), there are no mixed generators playing the X/Y role. "Unify couplings" and "unify groups" therefore come apart.

10. Cosmological constant locality.

The cosmological constant problem is a graveyard of unified theories: local QFT estimates are enormous, and tiny observed Λ seems to demand absurd fine tuning.

OPH route: From null modular data, T_ab is reconstructed only up to ϕg_ab. Local consistency conditions and null focusing are blind to vacuum-energy shifts, so the Einstein equation is fixed only up to Λg_ab. Λ becomes a global "capacity" parameter of the static patch (tied to log dim H_tot). It is not a locally computable quantity. This resolves the conceptual tension: local microphysics *cannot* fix Λ by structural information-theoretic reasons.

11. UV infinities and nonrenormalizability.

Unified programs struggle to give sharp, finite microscopic definitions. Formal continuum structures, infinite entropies, and regularization dependence abound.

How OPH addresses it: The regulator premises explicitly require local patch algebras to be type-I and finite-dimensional, with dynamics obeying a Lieb-Robinson bound. MaxEnt produces quasi-local Gibbs form. The fundamental degrees of freedom are finite and live on the screen; continuum/QFT behavior is an emergent limit. The theory starts from something UV complete in the trivial sense (finite DoF).

12. Predictivity vs. parameter explosion.

Unified models often explode in parameters, sectors, or vacua, becoming hard to test directly because everything depends on choices.

How OPH addresses it: The framework compresses freedom into a "pixel area" (resolution) parameter and a total Hilbert space capacity (size) parameter, then derives structure from consistency (Lorentz, Einstein form, compact gauge group reconstruction, exact zeros, quantization patterns). The explicit MAR selector then picks the SM factors and sector factorization on the admissible class (Part II of this manuscript).

Meta-pattern. The framework tends to "win" by making consistency conditions do the work. Many unified theories treat locality, Lorentz invariance, gauge symmetry, and gravity as additional *structures*. OPH treats them as *consistency constraints* among overlapping descriptions plus information-theoretic properties of states (Markov/recoverability + MaxEnt), then leans on modular theory rigidity to force familiar symmetries/dynamics. This "structures → consistency" move is what naturally explains or sidesteps classic plagues.

Engineering deliverables. Certain problems can be stated as explicit closure tasks:

- Λ is structurally explained as a global capacity parameter but not numerically predicted
- A full microphysical derivation of geometric modular action is an important target for future work

These are shared challenges across unification approaches. The framework provides an explicit map of where they live and what would resolve them.

1.10 References

1.10.1 Foundational results used in this work

Modular theory and spacetime:

- Bisognano, J. J. and Wichmann, E. H. (1975). "On the duality condition for a Hermitian scalar field." *J. Math. Phys.* 16, 985-1007.
- Bisognano, J. J. and Wichmann, E. H. (1976). "On the duality condition for quantum fields." *J. Math. Phys.* 17, 303-321.
- Unruh, W. G. (1976). "Notes on black-hole evaporation." *Phys. Rev. D* 14, 870-892.

Gravity from thermodynamics/entanglement:

- Jacobson, T. (1995). "Thermodynamics of spacetime: The Einstein equation of state." *Phys. Rev. Lett.* 75, 1260-1263. arXiv:gr-qc/9504004.
- Jacobson, T. (2016). "Entanglement equilibrium and the Einstein equation." *Phys. Rev. Lett.* 116, 201101. arXiv:1505.04753.

Strong subadditivity:

- Lieb, E. H. and Ruskai, M. B. (1973). "Proof of the strong subadditivity of quantum-mechanical entropy." *J. Math. Phys.* 14, 1938-1941.

Quantum recovery and Markov chains:

- Petz, D. (1986). "Sufficient subalgebras and the relative entropy of states of a von Neumann algebra." *Commun. Math. Phys.* 105, 123-131.
- Petz, D. (1988). "Sufficiency of channels over von Neumann algebras." *Quart. J. Math.* 39, 97-108.
- Fawzi, O. and Renner, R. (2015). "Quantum conditional mutual information and approximate Markov chains." *Commun. Math. Phys.* 340, 575-611. arXiv:1410.0664.

Superselection sectors and gauge reconstruction:

- Doplicher, S. and Roberts, J. E. (1989). "A new duality theory for compact groups." *Invent. Math.* 98, 157-218.
- Doplicher, S. and Roberts, J. E. (1990). "Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics." *Commun. Math. Phys.* 131, 51-107.

Tannaka-Krein duality:

- Tannaka, T. (1938). "Über den Dualitätssatz der nichtkommutativen topologischen Gruppen." *Tohoku Math. J.* 45, 1-12. (Some sources cite 1939.)
- Krein, M. G. (1949). "A principle of duality for a bicompact group and a square block algebra." *Dokl. Akad. Nauk SSSR* 69, 725-728.

1.10.2 Standard Model and unification (borrowed results)

Grand Unified Theories:

- Georgi, H. and Glashow, S. L. (1974). "Unity of all elementary-particle forces." *Phys. Rev. Lett.* 32, 438-441.

GIM mechanism:

- Glashow, S. L., Iliopoulos, J., and Maiani, L. (1970). "Weak interactions with lepton-hadron symmetry." *Phys. Rev. D* 2, 1285-1292.

Witten anomaly:

- Witten, E. (1982). "An SU(2) anomaly." *Phys. Lett. B* 117, 324-328.

MSSM gauge unification:

- Dimopoulos, S., Raby, S., and Wilczek, F. (1981). "Supersymmetry and the scale of unification." *Phys. Rev. D* 24, 1681-1683.
- Amaldi, U., de Boer, W., and Fürstenau, H. (1991). "Comparison of grand unified theories with electroweak and strong coupling constants measured at LEP." *Phys. Lett. B* 260, 447-455.

Experimental inputs:

- Particle Data Group (2025). "Review of Particle Physics." *Prog. Theor. Exp. Phys.* 2024, 083C01. https://pdg.lbl.gov/
- Particle Data Group (2025). "Electroweak Model and Constraints on New Physics." https://pdg.lbl.gov/2025/reviews/rpp2024-rev-standard-model.pdf

Part II
Gauge Group Derivation

2 Gauge Group Derivation: Standard Model Closure under Extended MAR

OPH is an observer-centric reconstruction program for fundamental physics. The quantitative branches discussed in this paper use two external inputs (pixel area and screen capacity) together with structural axioms and explicit regulator/scaling premises. This part focuses only on the conditional compact-gauge branch and the explicit MAR selector that acts on the admissible low-energy class.

Integrated derivation within this manuscript: this part extends the core OPH framework in Part I.

Overview. We show that, once Tannaka/DR reconstruction is applied to the refinement-stable edge-sector colimit, the extended gauge-selection package T_{ext}, meaning the five OPH axioms together with the explicit transportability premise $[z] = 0$ and the gauge-reconstruction hypotheses of Part I, selects the Standard Model connected Lie gauge-sector factors once the admissible class includes one connected abelian charge factor. On the realized admissible branch, the CP/UV window and MAR fix the generation count, Witten's SU(2) anomaly parity then fixes the color count, and the realized hypercharge lattice fixes the global gauge quotient:

$$G_{\text{phys}} = \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}, \qquad N_g = 3, \qquad N_c = 3.$$

Within the positive-dimensional connected Lie admissible class, no other gauge-sector package, no other color multiplicity, and no other generation count satisfies all admissibility conditions while minimizing the complexity vector $C(\mathfrak{S})$.

2.1 Premise Summary

2.1.1 The Extended Theory

The derivation proceeds within the extended gauge-selection package T_{ext}: the five OPH axioms from Part I, the explicit loop-coherent gluing premise $[z] = 0$, and the refinement-stable gauge-reconstruction hypotheses used in Theorem 6.1.

The roles of each component:

Component	Content	Reference
Axioms 1–2	Screen Net and Overlap Consistency	Part I of this manuscript §1.3
Axiom 3	Local MaxEnt and Refinement Stability	Part I of this manuscript §1.3–§1.4

Component	Content	Reference
Axiom 4	Recoverable Generalized Entropy	Part I of this manuscript §1.3
Axiom 5	Minimal Admissible Realization	Part I of this manuscript §1.3 and this document §3
R0, **R1**	Finite-dimensional regulator and boundary gauge structure	Part I of this manuscript §1.4, §2.5
$[z] = 0$	Loop-coherent gluing / DHR transportability	Part I of this manuscript §6.6, Prop. 6.1a
Theorem 6.1 hypotheses	Rigid symmetric C^* colimit category with faithful fiber functor	Part I of this manuscript §6.1

2.1.2 What the Core Proves

From the canonical OPH package together with the regulator premises R0 and R1, the following is established (Part I of this manuscript §6.1):

Theorem 6.1 (Tannaka/DR reconstruction). At each regulator scale one obtains a finite tensor category of transportable edge charges; the refinement-stable directed colimit Sect_∞ is the category used for reconstruction. If Sect_∞ is a rigid symmetric C^* tensor category with a faithful fiber functor, then there exists a compact group G, unique up to isomorphism, such that

$$\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G), \qquad G = \mathrm{Aut}_\otimes(\mathcal{F}).$$

This proves *existence* of a compact gauge group but does not identify *which* compact group is realized. The remainder of this part determines the connected Lie realization selected by the extended admissibility package.

2.1.3 The Transportability Premise

Proposition 6.1a. DHR transportability , the condition that charges can be moved between patches without changing fusion rules , is equivalent to the vanishing of the central obstruction class:

$$\text{DHR transportable} \iff [z] = 0 \iff \text{loop-coherent gluing.}$$

This is an internal constraint on the sector structure, not an external physical assumption. It is kept explicitly visible as a separate premise ($[z] = 0$) rather than hidden inside MAR.

2.2 Admissibility Conditions

An **OPH-realizable gauge sector package** $\mathfrak{S} = (G, \text{matter content})$ consists of a compact gauge group G reconstructed via Tannaka-Krein from the refinement-stable edge-sector category (Theorem 6.1) together with its associated chiral matter spectrum. A sector package $\mathfrak{S}$ is **admissible** if it satisfies all of the following conditions:

2.2.1 (i) Loop-coherent / transportable

The central obstruction class vanishes: $[z] = 0$. Equivalently, edge charges are DHR-transportable, i.e., they can be moved between patches without affecting fusion rules. This is the content of the explicit premise $[z] = 0$ (Proposition 6.1a).

Rationale: Without transportability, the reconstructed gauge symmetry is only a local labeling, not a global symmetry of the theory.

2.2.2 (ii) Anomaly-free

The gauge sector is free of all perturbative anomalies (ABJ anomaly cancellation) and all global anomalies (Witten SU(2) anomaly, Dai-Freed anomalies). Concretely:

- $\mathrm{tr}[T_a\{T_b, T_c\}] = 0$ for all gauge generators (perturbative),
- $\pi_4(G)$ anomaly vanishes (Witten's global SU(2) anomaly),
- Mixed gravitational-gauge anomalies cancel.

Rationale: Anomalous gauge theories are inconsistent at the quantum level.

2.2.3 (iii) Refinement-stable with light chiral matter

The MaxEnt/refinement-stable state supports light charged fermions without fine-tuning. This requires the matter content to be *chiral* , i.e., left-handed and right-handed fermions carry different gauge representations , so that vector-like mass terms are forbidden by gauge symmetry.

Rationale: Lemma 6.7 (Part I of this manuscript §6.3) shows that refinement stability forbids unprotected relevant operators. A gauge-invariant Dirac mass term is relevant; if both chiralities are in conjugate representations, the mass term is allowed and drives the fermions to the cutoff. Corollary 6.8 then selects chiral content as the natural refinement-stable option.

2.2.4 (iv) Single-Higgs Yukawa-completable

The chiral matter content can acquire mass through Yukawa couplings to a single scalar doublet H carrying some charge under one connected abelian factor, after electroweak symmetry breaking, without requiring additional scalar multiplets.

Rationale: This is the minimal mechanism for generating fermion masses consistent with electroweak symmetry breaking. The admissibility premise only requires one connected abelian charge direction and a single Higgs doublet; the numerical hypercharge assignments are derived later from anomaly cancellation and Yukawa invariance. Additional Higgs multiplets would increase the scalar capacity without physical necessity.

2.2.5 (v) Intrinsically CP-capable

The sector supports intrinsic CP violation , i.e., CP-violating phases that cannot be removed by field redefinitions. For a CKM-type mixing matrix with N_g generations, the number of physical CP phases is $(N_g - 1)(N_g - 2)/2$.

Rationale: Observed CP violation in the kaon and B-meson systems requires intrinsic CP-violating phases. A sector that cannot accommodate them is empirically excluded.

2.2.6 (vi) Weak-sector UV-completable

The weak gauge sector (the factor carrying the pseudoreal doublet) is asymptotically free at one loop, ensuring that it is UV-completable within the OPH framework rather than requiring a Landau pole cutoff below the screen scale.

Rationale: The OPH screen operates at the Planck scale. A gauge factor with a sub-Planckian Landau pole would be inconsistent with the screen construction.

2.3 Selection Axiom MAR

Selection Axiom MAR (Minimal Admissible Realization). Among all OPH-realizable sector packages $\mathfrak{S}$ that satisfy admissibility conditions (i)–(vi), the realized low-energy package is the **lexicographically minimal** one under the complexity vector

$$C(\mathfrak{S}) = (\chi_{\mathrm{cpl}},\ N_{\mathrm{nonab}},\ N_c,\ N_g),$$

where:

- χ_{cpl} is the **coupled edge capacity**: the dimension of the minimal unitary carrier containing a common irreducible block on which the admissible pseudoreal and complex nonabelian charge types both act nontrivially,
- N_{nonab} is the **number of nonabelian simple factors** in G,
- N_c is the **rank of the color factor** (the dimension of the fundamental representation of the complex nonabelian factor),
- N_g is the **number of chiral generations**.

Lexicographic minimality means: first minimize χ_{cpl}; among ties, minimize N_{nonab}; among further ties, minimize N_c; among final ties, minimize N_g. The object minimized here is the full sector package $\mathfrak{S}$, not the bare tensor category alone. MAR is therefore an explicit selector on the admissible class, not a theorem derived from the earlier axioms.

Key properties of MAR:

1. *It is a selection rule, not a dynamics.* MAR acts on an independently defined admissible class; it does not modify the OPH equations of motion.

2. *It is not "minimalism in general."* MAR only selects among sectors that pass all six admissibility filters. Plain minimality (smallest G) would yield $U(1)$ or $SU(2)$, which fail conditions (iii)–(v). The admissibility conditions do the heavy lifting; MAR breaks the remaining degeneracy.

3. *It is powerful.* Product gauge structure and the specific factors $SU(3) \times SU(2) \times U(1)$ follow from MAR applied to the admissible class; on the realized branch the CP/UV window and MAR then give $N_g = 3$, and Witten parity with that realized generation count gives $N_c = 3$.

Remark (coupled vs. faithful). Earlier drafts overloaded the first MAR coordinate with the abstract group-theoretic minimal faithful representation dimension. That is not the quantity used in the gauge proof. For $G_{\mathrm{phys}} \cong S(U(3) \times U(2))$, the block-diagonal action on $\mathbb{C}^3 \oplus \mathbb{C}^2$ is faithful of dimension 5, but it places the complex and pseudoreal charge types on disconnected invariant subspaces. MAR minimizes the coupled carrier dimension χ_{cpl}, for which the minimal SM carrier is $\mathbb{C}^3 \otimes \mathbb{C}^2$ of dimension 6.

2.4 Main Theorem

Theorem (Standard Model closure under extended admissibility). Under the extended gauge-selection package T_{ext}, together with the refinement-stable reconstruction hypotheses of Theorem 6.1:

$$G_{\text{phys}} = \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}, \qquad N_g = 3, \qquad N_c = 3.$$

The realized counting chain is sequential rather than joint: the CP/UV admissibility window and MAR first select $N_g = 3$, and Witten's SU(2) anomaly parity then fixes $N_c = 3$.

The proof is given in §5–§7 below.

2.5 Proof: Gauge Group

The proof proceeds in five steps. Steps 1–4 determine the connected gauge group. Step 5 fixes the global structure.

2.5.1 Step 1: Some compact group exists

From: the canonical OPH package together with regulator premises R0 and R1.

At each cutoff one has a finite tensor category of transportable edge charges. The reconstruction uses the refinement-stable directed colimit Sect_∞ of those categories. By Theorem 6.1 (Tannaka/DR reconstruction), there exists a compact group G, unique up to isomorphism, with $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$.

The premise $[z] = 0$ ensures that this reconstruction is global: charges are DHR-transportable, so the compact group acts as a genuine gauge symmetry, not merely a local labeling (Proposition 6.1a). □

2.5.2 Step 2: The gauge group must be a product

From: MAR (minimizing χ_{cpl}) + admissibility (iii).

Lemma 5.1. Any admissible sector must contain at least two genuinely different nonabelian charge types: one pseudoreal and one complex.

Proof. Admissibility condition (iii) requires light chiral matter. By Corollary 6.8 (Part I of this manuscript §6.3), refinement stability forces the matter content to be chiral. To support chiral fermions that can form gauge-invariant Yukawa couplings with a single Higgs doublet (condition iv), the gauge group must admit both:

- A **pseudoreal** fundamental representation (for the weak doublet structure , this allows left-handed doublets and right-handed singlets to have different gauge quantum numbers), and
- A **complex** fundamental representation (for the color structure , this allows quarks to carry a charge that distinguishes particle from antiparticle).

A single nonabelian factor cannot simultaneously provide both types: a group is either pseudoreal at a given dimension or complex, not both. □

Lemma 5.2. The minimal coupled carrier is $V = \mathbb{C}^3 \otimes \mathbb{C}^2$, with $\chi_{\text{cpl}} = 6$.

Proof.

- The smallest faithful pseudoreal irreducible representation of any positive-dimensional compact connected Lie group is the fundamental **2** of $SU(2)$ (the only 2D pseudoreal irrep in the connected Lie setting used here).
- No irreducible 2D unitary representation of a positive-dimensional compact connected Lie group can realize the required intrinsically complex nonabelian role: the connected derived subgroup of its image is a compact connected subgroup of $SU(2)$, hence either trivial, a torus, or all of $SU(2)$, so the nonabelian action is pseudoreal up to abelian twist. The smallest faithful irreducible intrinsically complex nonabelian representation is therefore the fundamental **3** of $SU(3)$.
- By definition of χ_{cpl}, the admissible pseudoreal and complex charge types must occur on a common nonabelian block rather than on disconnected direct-sum summands. The minimal such block is the tensor product $V = \mathbb{C}^3 \otimes \mathbb{C}^2$, giving $\chi_{\text{cpl}} = 6$.
- The block-diagonal representation $\mathbb{C}^3 \oplus \mathbb{C}^2$ of $S(U(3) \times U(2))$ is faithful in the abstract group-theoretic sense, but it is not coupled: each nonabelian factor is trivial on one summand. It therefore does not enter the MAR selector.
- Any other combination (e.g., using $SU(4)$ instead of $SU(3)$, or $Sp(4)$ instead of $SU(2)$) gives $\chi_{\text{cpl}} > 6$, and is therefore disfavored by MAR. □

Proposition 5.3 (Product structure from minimal coupled carrier). The minimal coupled carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$ forces a product gauge structure.

Proof. On $V = \mathbb{C}^3 \otimes \mathbb{C}^2$, the color factor acts on the first tensor factor and the weak factor acts on the second. These actions commute by the tensor product structure. Therefore the gauge group decomposes as a product $G_{\text{color}} \times G_{\text{weak}} \times G_{\text{abelian}}$ (up to finite quotient). A simple group acting irreducibly on a low-dimensional carrier would not provide the independent pseudoreal + complex factor structure required by Lemma 5.1; when those roles are realized on a common block, the minimal coupled carrier is the tensor product above.

Product structure is derived from the minimal coupled-carrier argument. □

Remark. The representation-theoretic dimension comparisons in Steps 2–4 apply only to the positive-dimensional connected Lie image carrying the admissible nonabelian charges, equivalently to the identity component of the reconstructed group on that image. Finite or disconnected counterexamples are outside the theorem package used here.

2.5.3 Step 3: The factors are SU(3), SU(2), and U(1)

From: Lemmas 6.3–6.5 of Part I of this manuscript, applied to the minimal coupled carrier together with the connected abelian admissibility premise.

Lemma 5.4 (SU(2) from pseudoreal doublet). The factor acting on $\mathbb{C}^2$ as a faithful 2D pseudoreal representation is $SU(2)$.

Proof. Among compact groups with a faithful 2D pseudoreal unitary representation, $SU(2)$ is the unique option. (The only compact Lie groups with a 2D faithful representation are subgroups of $U(2)$. Among these, the pseudoreal condition $V \cong \bar{V}$ via an antisymmetric bilinear form selects exactly $SU(2)$.) This is Lemma 6.3 of Part I of this manuscript. □

Lemma 5.5 (SU(3) from complex triplet). The factor acting on $\mathbb{C}^3$ as a faithful irreducible complex representation is $SU(3)$.

Proof. Among compact groups with a faithful irreducible complex 3D representation, the connected component is $SU(3)$. (The only compact simple Lie groups with a 3D fundamental representation are $SU(3)$ and $SO(3)$. But the fundamental of $SO(3)$ is real, not complex. Therefore $SU(3)$ is uniquely selected.) This is Lemma 6.4 of Part I of this manuscript. □

Lemma 5.6 (Single connected abelian factor). Admissibility condition (iv) supplies one connected abelian charge factor acting nontrivially on the coupled carrier. Compact connected abelian Lie groups are tori, and the commutant calculation below shows that only a one-torus can act nontrivially on the minimal coupled carrier. Therefore the admissible connected abelian factor is $U(1)$.

Proof. This is Lemma 6.5 of Part I of this manuscript together with the compact-Lie classification of connected abelian groups as tori. The one-dimensional characters form a discrete lattice $\mathrm{Hom}(U(1)^n, U(1)) \cong \mathbb{Z}^n$; what matters here is the connected torus factor, not a continuous family of inequivalent 1D irreps. □

2.5.4 Step 4: Nothing else , the commutant argument

Proposition 5.7 (No additional factors). No extra gauge factor , neither nonabelian nor connected abelian , can appear without increasing χ_{cpl} beyond 6.

Proof. Consider the maximal compact subgroup of $U(6)$ acting on $V = \mathbb{C}^3 \otimes \mathbb{C}^2$ with commuting actions on each tensor factor:

$$\{g \in U(6) : g = g_3 \otimes g_2,\ g_3 \in U(3),\ g_2 \in U(2)\} \cong U(3) \times U(2)/U(1)_{\mathrm{diag}}.$$

The continuous symmetry group with commuting color and weak actions is therefore

$$S(U(3) \times U(2)) \cong \frac{SU(3) \times SU(2) \times U(1)}{\text{finite center}}.$$

Compute the commutant:

- The commutant of $SU(3) \times SU(2)$ inside $U(6)$ is exactly $U(1)$.
- Therefore any connected abelian factor acting nontrivially on the coupled carrier is necessarily this single $U(1)$, and no extra continuous factor (neither $U(1)'$ nor any nonabelian group) can appear without enlarging the carrier beyond $\chi_{\mathrm{cpl}} = 6$.
- Any additional nonabelian factor would have to act nontrivially on the same coupled block, increasing χ_{cpl}.

MAR selects the minimal χ_{cpl}, so extra factors are excluded. □

2.5.5 Step 5: The global quotient is $\mathbb{Z}_6$

Proposition 5.8 ($\mathbf{Z_6}$ quotient from hypercharge quantization). If the realized matter spectrum has hypercharges quantized in sixths, then the subgroup of $SU(3) \times SU(2) \times U(1)$ acting trivially on all physical states is exactly $\mathbb{Z}_6$.

Proof. The SM matter content with hypercharge assignments $(Q_L, u_R, d_R, L_L, e_R, H)$ has hypercharges $Y \in \{1/6, 2/3, -1/3, -1/2, -1, 1/2\}$, all in $\frac{1}{6}\mathbb{Z}$. The center of $SU(3) \times SU(2) \times U(1)$ is $\mathbb{Z}_3 \times \mathbb{Z}_2 \times U(1)$. The subgroup acting trivially on all realized representations is generated by

$$(\omega_3, -1, e^{i\pi/3}) \in SU(3) \times SU(2) \times U(1),$$

where $\omega_3 = e^{2\pi i/3}$ is the $\mathbb{Z}_3$ center of $SU(3)$. This generates a $\mathbb{Z}_6$ subgroup. Therefore

$$G_{\mathrm{phys}} = \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}.$$

Equivalently, $G_{\mathrm{phys}} \cong S(U(3) \times U(2))$.

The hypercharge quantization itself follows from anomaly cancellation (condition ii) with the minimal Yukawa-complete spectrum (condition iv). Theorem 6.13 of Part I of this manuscript derives the hypercharge assignments from these conditions for $N_c = 3$, yielding precisely the sixth-integer lattice. This is Proposition 6.6 of Part I of this manuscript. □

2.5.6 Summary of Step 1–5

$$\boxed{T_{\text{ext}} \implies G_{\text{phys}} = \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}}$$

2.6 Proof: Number of Colors

Theorem (N_c = 3). Under T_{ext}, the rank of the color factor is $N_c = 3$.

Proof. The gauge group derivation (§5) fixes the color factor as $SU(3)$, i.e. $N_c = 3$, through the minimal coupled-carrier argument (Lemma 5.2). We give here the independent confirmation from anomaly constraints plus MAR.

Step 1: Witten anomaly constrains N_c once N_g is fixed.

With gauge structure $SU(N_c) \times SU(2)_L \times U(1)_Y$, one left-handed quark doublet Q per color, and one left-handed lepton doublet L per generation, the total number of $SU(2)$ doublets is

$$N_{\text{doublets}} = N_g(N_c + 1).$$

Using $N_g = 3$ from Section 7, Witten's global $SU(2)$ anomaly (1982) requires the total number of $SU(2)$ doublets to be even:

$$3(N_c + 1) \equiv 0 \pmod 2 \quad \implies \quad N_c \text{ is odd.}$$

This alone allows $N_c \in \{1, 3, 5, 7, \ldots\}$.

Step 2: N_c = 1 fails admissibility.

For $N_c = 1$: $SU(1)$ is trivial (no color dynamics). The fundamental representation is 1-dimensional and real, not complex. This violates Lemma 5.1 (no genuinely complex nonabelian charge type) and condition (iii) (cannot support chiral quarks).

Step 3: MAR selects N_c = 3.

Among the remaining candidates $\{3, 5, 7, \ldots\}$, the SM-like quotient family has $\chi_{\text{cpl}} = 2N_c$ because the minimal coupled carrier is $\mathbb{C}^{N_c} \otimes \mathbb{C}^2$. (The block-diagonal faithful representation $\mathbb{C}^{N_c} \oplus \mathbb{C}^2$ has dimension $N_c + 2$, but it is not coupled and is therefore irrelevant to MAR.) MAR's complexity vector has N_c in the third slot, so lexicographic minimization selects $N_c = 3$. □

2.7 Proof: Number of Generations

Theorem (N_g = 3). Under T_{ext}, the number of chiral generations is $N_g = 3$.

Proof.

Step 1: CP violation gives N_g ≥ 3.

Admissibility condition (v) requires intrinsic CP violation. The number of physical CP-violating phases in an $N_g \times N_g$ CKM matrix is

$$n_{\text{CP}} = \frac{(N_g - 1)(N_g - 2)}{2}.$$

- For $N_g = 1$: $n_{\mathrm{CP}} = 0$. No CP violation possible. Excluded.
- For $N_g = 2$: $n_{\mathrm{CP}} = 0$. No CP violation possible. Excluded.
- For $N_g = 3$: $n_{\mathrm{CP}} = 1$. CP violation possible.

Therefore $N_g \geq 3$.

Step 2: Asymptotic freedom gives N_g ≤ 5.

Admissibility condition (vi) requires $SU(2)_L$ to be asymptotically free. With one complex Higgs doublet, the one-loop beta function coefficient is

$$b_{SU(2)} = \frac{22}{3} - \frac{1}{3}N_g(N_c + 1) - \frac{1}{6}.$$

Asymptotic freedom requires $b_{SU(2)} > 0$:

$$N_g(N_c + 1) < \frac{43}{2}.$$

Since the color-type role forces $N_c \geq 3$, one has $N_c + 1 \geq 4$. Therefore $4N_g < \frac{43}{2}$, so $N_g \leq 5$.

Combining: $3 \leq N_g \leq 5$.

Step 3: MAR selects N_g = 3.

Among the admissible candidates $\{3, 4, 5\}$, MAR's complexity vector has N_g in the fourth (last) slot. Lexicographic minimization selects $N_g = 3$. Feeding that value back into Section 6 completes the Witten-parity closure $N_c = 3$. □

2.8 Corollaries

2.8.1 Corollary 1: No gauge-mediated proton decay

Corollary. The gauge group $G_{\mathrm{phys}} = SU(3) \times SU(2) \times U(1)/\mathbb{Z}_6$ does not contain X or Y gauge bosons that mediate proton decay.

Proof. The adjoint representation of the full connected gauge group contains only $(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0)$; equivalently, the adjoint of the derived gauge group contains only $(8, 1, 0) \oplus (1, 3, 0)$. There are therefore no mixed $(3, 2, \pm 5/6)$ generators. Proton decay via gauge bosons requires a simple unification group (e.g., $SU(5)$, $SO(10)$) whose additional generators connect quarks to leptons. The MAR derivation forces the product-group structure above, so no such gauge bosons exist. □

2.8.2 Corollary 2: Uniqueness

Corollary. Within the positive-dimensional connected Lie admissible class with one connected abelian factor, the SM gauge group is the unique realization acting on the minimal coupled carrier with $\chi_{\mathrm{cpl}} = 6$.

Proof. By the commutant argument (Proposition 5.7), the group $SU(3) \times SU(2) \times U(1)/\mathbb{Z}_6$ exhausts the continuous symmetry of the minimal coupled carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$. Any alternative group either:

- Has $\chi_{\mathrm{cpl}} > 6$ (excluded by MAR), or
- Fails to provide both pseudoreal and complex representations (excluded by condition iii), or
- Is a subgroup of the SM connected group, which would fail anomaly cancellation or Yukawa completeness.

□

2.8.3 Corollary 3: Hypercharge quantization

Corollary. The hypercharge ratios are fixed by anomaly cancellation and Yukawa invariance within the derived gauge group, and the standard normalization then fixes the observed values $Y \in \frac{1}{6}\mathbb{Z}$.

Proof. Theorem 6.13 of Part I of this manuscript derives the unique anomaly-free hypercharge ratios for $SU(3)\times SU(2)\times U(1)$ with $N_c = 3$, assuming single-Higgs Yukawa completeness (condition iv); the standard normalization then gives the sixth-integer lattice. □

2.9 Complete Derivation Chain

The gauge-sector chain established in this fragment is:

$$\text{canonical package} + R0 + R1 \xrightarrow{\text{Thm 6.1}} \exists\, G \text{ compact}$$

$$+ [z] = 0 \xrightarrow{\text{Prop 6.1a}} \text{DHR transportable, global gauge symmetry}$$

$$+ \text{ apply MAR on the admissible class} \xrightarrow{\text{Thm (this doc)}} \text{connected SM factors with the realized chain } N_g = 3 \text{ then } N_c = 3, \text{ and } G_{\text{phys}} = SU(3) \times SU(2) \times U(1)/\mathbb{Z}_6$$

$$\xrightarrow{\text{Thm 6.13}} \text{hypercharges fixed} \xrightarrow{\text{Cor.}} \text{no gauge-mediated proton decay}$$

Under T_{ext}, the following are all derived results:

1. Product gauge group structure
2. Exact global gauge group $SU(3) \times SU(2) \times U(1)/\mathbb{Z}_6$
3. Number of colors $N_c = 3$
4. Number of generations $N_g = 3$
5. Absence of gauge-mediated proton decay
6. Hypercharge quantization in sixths

This is the gauge-sector closure obtained from the extended axiom set on the stated admissible class. Later gravity, spectrum, and continuation branches require their own additional premises and authority surfaces. □

2.10 Future Work

The gauge-group derivation presented here determines the symmetry structure of the Standard Model. Several related questions are addressed in companion documents:

- **Fermion mass hierarchy and Yukawa couplings** , see Part V of this manuscript §12
- **Cosmological constant** , see Part V of this manuscript §14
- **Full transmutation chain** , see Part V of this manuscript Appendix A.3
- **Heat-kernel parameter t from UV microphysics** , see Part I of this manuscript §6.13
- **Particle mass spectrum from pixel geometry** , see Part III of this manuscript

Part III
Spectrum Derivation

3 From Pixel Area to Particle Masses: The Complete OPH Spectrum Derivation

Integrated derivation within this manuscript: this part applies the OPH framework developed in Part I.

Overview. This part documents the quantitative OPH branches used in this paper. The calculation uses two external inputs: the pixel area $P \equiv a_{\text{cell}}/\ell_P^2 = 1.63094$, calibrated from the gauge coupling sector via the screen's entropy-matching condition (§5.4 of Part I of this manuscript), and the screen capacity $N_{\text{scr}} \equiv \log(\dim \mathcal{H}_{\text{tot}}) \sim 10^{122}$, which enters only in the cosmological/neutrino branch (§14 of Part V of this manuscript). Every other quantity appearing in the derivation chain is either a mathematical constant, a consequence of the gauge group (itself selected from a refinement-stable edge-sector category by Tannaka-Krein/DR reconstruction plus the extended MAR admissibility package, §6.1–§6.2 of Part I of this manuscript), or a derived structural integer on the realized branch ($N_g = 3$, $N_c = 3$, $\varepsilon = 1/6$, $\delta = 2/9$). Outputs are classified by epistemic status: quantities flowing from the calibrated gauge couplings are calibration-sector consistency checks; the Higgs/top critical-surface branch is the main independent quantitative output; charged-lepton, flavor-texture, neutrino, and hadron branches are weaker phenomenological or downstream numerical continuations. A complete audit of all constants and the full epistemic classification appear in §1A–§1B below.

3.1 Inputs and Contract

3.1.1 Physical Inputs

The entire derivation pipeline uses exactly **two** physical inputs:

Input	Symbol	Value	Origin
Pixel area constant	$P \equiv a_{\text{cell}}/\ell_P^2$	1.63094	**Calibrated** from gauge couplings via entropy matching (§5.4 of Part I; see §1B)
Screen capacity	$N_{\text{scr}} \equiv \log(\dim \mathcal{H}_{\text{tot}})$	$\sim 10^{122}$	Inferred from observed cosmological constant

3.1.2 What Counts as "Derived"

Everything else entering the computation falls into one of three categories:

1. **Mathematical constants**: π, e, $\zeta(3)$, Casimir eigenvalues, group dimensions, β-function coefficients , all determined by the gauge group structure (itself selected from the refinement-stable edge-sector category via Tannaka-Krein/DR reconstruction together with the extended MAR admissibility package).

2. **Derived quantities**: the unification scale M_U, the unified coupling α_U, the electroweak VEV v, all gauge couplings at M_Z, the $\mathbb{Z}_6$ defect parameter $\varepsilon = 1/6$, the Koide phase $\delta = 2/9$, and the Froggatt-Nielsen integer exponents.

3. **Numerical-method parameters**: loop order for RG running (1-loop SM for critical surface, 4-loop MSbar for $\Lambda_{\overline{\text{MS}}}$), lattice sizes and statistics for hadron ratios. These affect precision but not the prediction logic.

3.1.3 No-Leakage Guarantee

The computation code enforces a runtime mutation test: after computing all outputs, the PDG reference table is scrambled and outputs are recomputed. Any change triggers an assertion failure. This verifies that no PDG values leak into the computation pipeline beyond the initial P calibration. Implemented in `oph_predict_compare.py::_assert_pdg_not_used()`.

3.2 Complete Audit of All Constants

Purpose. This section catalogs *every* named constant, integer, or coefficient appearing anywhere in the prediction code or derivation chain, and explains its origin. The goal is to demonstrate that nothing has been "smuggled in"; every value is either (I) one of the two external inputs used in these quantitative branches, (II) a derived structural quantity from the OPH axioms and stated premises, (III) a mathematical/group-theoretic constant uniquely fixed by the gauge group, or (IV) a standard QFT result that any textbook reproduces from the Lagrangian.

3.2.1 The Physical Input

Constant	Value	Code variable	Origin	Status
$P \equiv a_{\text{cell}}/\ell_P^2$	1.63094	`P_DEFAULT`	The pixel area: area of one UV cell on the holographic screen in Planck units. Extracted from the entropy-matching condition $P/4 = \bar{\ell}_{\text{SU}(2)}(t_2) + \bar{\ell}_{\text{SU}(3)}(t_3)$ (§5.4 of Part I of this manuscript). This relation is *derived* from the axioms; the *numerical value* is fixed by requiring consistency with the observed gauge couplings (just as Newton's constant G is the one free parameter of general relativity).	**Single free parameter** for particle physics

Constant	Value	Code variable	Origin	Status
$N_{\text{scr}} \equiv \log(\dim \mathcal{H}_{\text{tot}})$	$\sim 10^{122}$	`LOG_DIM_H_DEFAULT`	Total screen capacity = de Sitter entropy $S_{dS} = 3\pi/(G\Lambda)$. Enters only for neutrino masses and cosmological constant (§14 of Part V of this manuscript). Derived from the screen area law (§5 of Part I of this manuscript).	**Second input** (cosmology sector only)

3.2.2 Structural Integers Derived from Axioms

Constant	Value	Code variable	Derivation	Part I of this manuscript ref
N_c (colors)	3	`N_c_DEFAULT`	Witten's global SU(2) anomaly requires $N_c + 1$ SU(2) doublets per generation to be even, so N_c must be odd. $N_c = 1$ is excluded (SU(1) trivial). The Selection Axiom MAR (third component of the complexity vector) picks the smallest non-trivial value. **Result**: $N_c = 3$.	Theorem 6.14 (§6.9); Part II of this manuscript §6
N_g (generations)	3	`N_g_DEFAULT`	Three independent constraints narrow the window: (1) CP violation requires $N_g \geq 3$ (CKM phases $(N_g - 1)(N_g - 2)/2 > 0$). (2) Asymptotic freedom of SU(2) with one Higgs doublet requires $N_g(N_c + 1) < 43/2$, giving $N_g \leq 5$. (3) MAR (fourth component of the complexity vector) picks the smallest value in $\{3, 4, 5\}$. **Result**: $N_g = 3$.	Proposition 6.9 (§6.4); Part II of this manuscript §7
Gauge group	$(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$	—	Tannaka-Krein / Doplicher-Roberts reconstruction from a refinement-stable edge-sector category yields a compact gauge group G (Theorem 6.1, §6.1). The Selection Axiom MAR then selects the SM factors once the admissible class includes one connected abelian charge factor: the minimal coupled carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$ enforces the product structure, the commutant argument fixes the connected abelian factor to a single $U(1)$, and the $\mathbb{Z}_6$ quotient follows from hypercharge quantization (Proposition 6.6).	§6.1–6.2; Part II of this manuscript

Constant	Value	Code variable	Derivation	Part I of this manuscript ref
β_{EW}	4	`beta_ew(N_c)`	Number of SU(2) doublets per generation = N_c quark doublets + 1 lepton doublet = $N_c + 1 = 4$. This is a counting consequence of the Witten anomaly analysis in the structural chain fixing $N_c = 3$.	§6.9, §6.19
ε (Z_6 defect)	1/6	`defect_epsilon_Z6()`	The SM gauge group has a $\mathbb{Z}_6$ center quotient. Each unit of $\mathbb{Z}_6$ defect insertion removes $\Delta S = \ln 6$ nats of entropy (MaxEnt weighting, premise ME). The resulting Boltzmann suppression per defect is $e^{-\ln 6} = 1/6$. **This is topologically fixed** by the quotient structure, not chosen.	§6.18, §6.21
δ (Koide phase)	2/9	computed inline	The holonomy phase on generation space: $\delta = \beta_{\mathrm{EW}} \cdot Y_Q/N_g = (N_c+1)/(2N_cN_g) = 4/18 = 2/9$. Here $Y_Q = 1/(2N_c)$ is the quark-doublet hypercharge (fixed by anomaly cancellation) and $N_g = 3$ is the number of generations. All ingredients are previously derived. Experimental extraction: $\delta_{\mathrm{exp}} = 0.2222248 \pm 0.0000063$, matching $2/9 = 0.2222\ldots$ within 0.4σ.	Proposition 13.3 (§13 of Part V of this manuscript)
Quark exponents	$n_u = (6,3,0)$, $n_d = (6,4,2)$	`derive_integer_vectors()`	Minimal SU(3) hierarchy structure: $n_u = (2N_c, N_c, 0)$ and $n_d = (2N_c, N_c+1, N_c-1)$. These are the unique sequences that (a) span the range $[0, 2N_c]$ in N_g steps, (b) have the top quark unsuppressed ($n_t = 0$), and (c) match the observed up/down mass ordering under CKM rotation. With $N_c = 3$: $(6,3,0)$ and $(6,4,2)$.	§6.21
Lepton exponents	$n_e = (7,4,3)$	`derive_lepton_exponents()`	Derived algorithmically: the Koide roots r_k^2 (from $\delta = 2/9$) must satisfy $r_i^2/r_j^2 \approx \varepsilon^{n_i - n_j}$ for all pairs. The code scans integer triples near $(N_g, N_g+1, \ldots)$ and selects the unique triple minimizing the log-ratio residuals. **Result**: $(n_\tau, n_\mu, n_e) = (3,4,7)$.	§6.21, §13 of Part V of this manuscript
CKM angles	$s_{12} = \varepsilon$, $s_{23} = \varepsilon^2$, $s_{13} = \varepsilon^3$	inline	Powers of $\varepsilon = 1/6$: the CKM mixing angles scale as $\varepsilon^{\lvert n_i - n_j \rvert}$ where the exponents are the generation-gap between up-type and down-type defect charges. This is the standard Froggatt-Nielsen scaling.	§6.21

Constant	Value	Code variable	Derivation	Part I of this manuscript ref
Q (Koide ratio)	2/3	inline	The $\mathbb{Z}_3$ mode balance on generation space: the circulant Hermitian matrix $\Phi = aI + bP + b^* P^2$ on 3 generations satisfies $Q = (1 + 2\|b/a\|^2)/3 = 2/3$ when $\|b/a\| = 1/\sqrt{2}$, i.e., when the singlet and charged $\mathbb{Z}_3$ modes have equal norm. This is the MaxEnt equilibrium.	Theorem 13.2 (§13 of Part V of this manuscript)

3.2.3 Standard QFT Constants (Fixed by the Gauge Group + Matter Content)

These constants are *not* inputs , they are uniquely determined mathematical consequences of the gauge group SU(3) × SU(2) × U(1) with N_g generations of the standard fermion representations. Any QFT textbook (e.g., Peskin & Schroeder) derives them from the Lagrangian.

Constant	Value	Code location	What determines it
MSSM β-coefficients	$(b_1, b_2, b_3) = (33/5, 1, -3)$	`B_MSSM`	The edge-sector computation (§6.17 of Part I of this manuscript) derives β-function shifts $\Delta b \approx (2.49, 4.17, 4.01)$ from the heat-kernel vacuum-polarization weighting and the $\mathbb{Z}_6$ quotient. These match the shifts from SM to MSSM to better than 1%. The coefficients themselves are standard 1-loop group theory: $b_i = \sum_R T(R_i)$ summed over all matter multiplets. Using MSSM content is a *consequence* of the edge-sector matching, not an assumption. In D11 they govern the UV segment above the synchronization scale μ_{sync}.
SM 1-loop β-coefficients	standard SM set	`B_SM`	Standard 1-loop coefficients for the SM with $N_g = 3$, namely $(41/10, -19/6, -7)$. In D11 they govern the low-scale transport from m_Z up to the synchronization scale $\mu_{\text{sync}} \approx 6.8 \times 10^{11}$ GeV, not the entire interval up to M_U.
4-loop MSbar β-coefficients	$\beta_0, \beta_1, \beta_2, \beta_3$	`beta_coeffs_msbar(n_f)` in `oph_qcd.py`	Standard perturbative QCD: $\beta_0 = 11 - 2n_f/3$, $\beta_1 = 102 - 38n_f/3$, etc. These are computed from SU(3) Feynman diagrams at each loop order. The 4-loop coefficient β_3 involves $\zeta(3)$; all are universal in the MSbar scheme.

Constant	Value	Code location	What determines it
Pole-mass conversion	$K_2 = 13.44 - 1.04n_l$, $K_3 = 190.6 - 26.7n_l + 0.65n_l^2$	`top_pole_from_msbar()`	Standard 3-loop QCD relation between $\overline{\text{MS}}$ and pole mass (Chetyrkin, Kniehl, Steinhauser 2000). These are perturbative coefficients computed from Feynman diagrams; they depend only on n_l (number of light flavors) and SU(3) group factors.
Casimir eigenvalues	$C_2(p,q)$, $d_{(p,q)}$	`ellbar_su3()`, `ellbar_su2()`	Group-theoretic: $C_2(p,q) = \frac{1}{3}(p^2 + q^2 + pq + 3p + 3q)$ for SU(3), $C_2(j) = j(j+1)$ for SU(2), $d_{(p,q)} = \frac{1}{2}(p+1)(q+1)(p+q+2)$. These are properties of the Lie algebra, not physical inputs.
$\zeta(3)$	1.20206...	`z3` in `oph_qcd.py`	Apéry's constant , a pure mathematical constant appearing in 4-loop β_3.
GUT normalization	$\alpha_1 = \frac{5}{3}\alpha_Y$	inline	The factor 5/3 is the standard GUT normalization ensuring all three couplings unify. It follows from embedding $\text{U}(1)_Y$ into SU(5) and is fixed by the hypercharge assignments of the SM fermions.

3.2.4 Dimensional/Definitional Constants

Constant	Value	Code variable	Status
E_P (Planck energy)	1.220890×10^{19} GeV	`E_PLANCK_GEV`	**Definition**: $E_P = \sqrt{\hbar c^5/G}$. This is a unit conversion factor, not a physical input , it converts from natural (Planck) units to GeV. In the OPH framework, G itself is derived from the pixel area via $G = a_{\text{cell}}/(4\bar{\ell}\,\hbar/c^3)$ (§5.4 of Part I of this manuscript), so E_P is ultimately set by P.
$e^{2\pi}$ in M_U formula	535.49...	inline	The Euclidean regularity condition on the collar geometry fixes the angular period to 2π (§5.8 of Part I of this manuscript). This is a geometric constant, not a parameter.
$\sqrt{2}$ in Yukawa	$v/\sqrt{2}$	inline	Standard Higgs mechanism convention: the Yukawa coupling y_f relates to the fermion mass via $m_f = y_f v/\sqrt{2}$. This is a normalization convention, not physics.
$\Delta\rho_{\text{stage-3}}$	$3y_t^2/(32\pi^2)$, evaluated at $y_t \approx 1$	inline	Universal 1-loop custodial correction with explicit Yukawa dependence. The displayed stage-3 number uses the unit-Yukawa estimate $y_t \approx 1$.

3.2.5 Numerical-Method Parameters

These affect computational precision but not the prediction logic. Changing them changes the last digits, not the physics.

Parameter	Value used	Effect of changing
Loop order for RG (critical surface)	1-loop piecewise SM/MSSM transport	2-loop running or a changed synchronization prescription shifts m_H and m_t at the GeV level
Loop order for $\Lambda_{\overline{\mathrm{MS}}}$	4-loop MSbar	3-loop changes Λ by ~2%
RK4 step count	2000 steps	Doubling changes nothing to displayed precision
Lattice sizes (hadrons)	$L = 2$–6	Larger volumes needed for precision; listed values are prototype runs
Bisection tolerance	10^{-10}	Affects last digits of α_U only
Heat-kernel truncation	200 representations	Adding more changes $\bar{\ell}$ by $< 10^{-12}$

3.2.6 Summary: What Is and Isn't Assumed

Inherited from earlier Part I structural results (five OPH axioms plus the stated gauge, heat-kernel, and refinement premises):

- Gauge group structure (Tannaka-Krein reconstruction from edge sectors)
- $\varepsilon = 1/6$ (topological, from $\mathbb{Z}_6$ quotient)
- $\lambda(M_U) = 0$, $\beta_\lambda(M_U) = 0$ (refinement stability)
- $Q = 2/3$ (MaxEnt on $\mathbb{Z}_3$ generation space)
- Heat-kernel form $p_R \propto d_R e^{-tC_2(R)}$ (MaxEnt + edge completion)

Derived under the Selection Axiom MAR (Part II of this manuscript):

- SM gauge group $SU(3) \times SU(2) \times U(1)/\mathbb{Z}_6$ (MAR + admissibility)
- $N_c = 3$ (Witten anomaly + MAR minimality)
- $N_g = 3$ (CP violation + asymptotic freedom + MAR minimality)
- $\delta = 2/9$ (algebraic consequence of $N_c = 3$, $N_g = 3$, $\beta_{\mathrm{EW}} = 4$)
- $\beta_{\mathrm{EW}} = 4$ (doublet counting from $N_c = 3$)
- Integer exponents n_u, n_d, n_e (algorithmically from N_c, N_g, ε, δ)

Fixed by the gauge group (standard QFT, no free parameters):

- All β-function coefficients (1-loop through 4-loop)
- Pole-mass conversion coefficients K_2, K_3
- Casimir eigenvalues, representation dimensions
- GUT normalization factor 5/3

The primary particle-physics calibration input:

- $P = 1.63094$ (pixel area), calibrated from the gauge coupling sector (see §1B)

Second external input (cosmology/capacity branch):

- $N_{\mathrm{scr}} \equiv \log(\dim \mathcal{H}_{\mathrm{tot}}) \sim 10^{122}$ (screen capacity)

What is NOT assumed or smuggled in:

- No particle masses
- No CKM matrix elements
- No Higgs VEV or quartic coupling
- No SUSY-breaking scale
- No Yukawa couplings
- No Λ_{QCD}

3.3 Calibration vs. Prediction: Epistemic Classification of Outputs

This part starts at the quantitative layer. D1–D5 together with D7–D9 form the recovered core; D10–D11 live here; D12 stays downstream. This section marks which numbers are calibration-sector checks, which belong to the secondary quantitative branch, and which remain phenomenological continuations.

3.3.1 The Calibration Structure

The pixel area constant P is the external continuous input used by the particle-physics branch. In practice it is fixed through the gauge sector via the entropy-matching / pixel-constraint closure. Quantities algebraically entangled with that same closure sit in D10 and count as calibration-sector consistency checks. In this part, D10 includes the unified coupling and scale-setting branch, the running gauge couplings at m_Z, $\alpha_{\mathrm{em}}^{-1}(m_Z)$, $\sin^2\theta_W(m_Z)$, v, and the tree-level electroweak masses. Agreement there shows internal closure once P and the printed running and matching conventions are imposed. It does not elevate the whole spectrum program to theorem level.

3.3.2 Why "Downstream of P" Does Not Mean "Recovered Core"

Using P to set an overall energy scale and using an output branch as a recovered-core claim are different things. The critical-surface Higgs/top branch illustrates the distinction: once the D10 trajectory is fixed, the extra condition

$$\lambda(M_U) = 0, \qquad \beta_\lambda(M_U) = 0$$

adds no new continuous fit, so its failure would challenge D11 while leaving the D10 calibration closure intact. It sits outside the recovered core D1–D5 plus D7–D9. Charged-lepton/Koide fits, texture exponents, neutrino estimates, proton-spin bookkeeping, and downstream hadron extractions use additional ansätze, capacity assumptions, or external lattice/QCD machinery. Those belong to D12 or to later downstream numerical layers and must not be read back into the core claim set.

The master ledger remains the guide. Scale-setting dependence does not erase these distinctions, and it does not automatically elevate later numerics to theorem level. D10 is the calibration sector, D11 the secondary quantitative branch, and the remaining spectrum branches deferred continuations unless explicitly promoted elsewhere.

3.3.3 Classification

Tier / node	Representative outputs in this part	Status rule
Phase I recovered core inherited from Parts I–II (D1–D5, D7–D9)	conditional relativity branch; compact gauge reconstruction in the bosonic branch; realized Standard Model quotient, exact hypercharges, the counting chain $N_g = 3$ then $N_c = 3$, and no gauge-mediated proton decay	These are structural or realized-branch outputs proved upstream. They are *not* generated by the spectrum numerics in this part, even when this part reuses them.
Phase II input-dependent corollary inherited from Part I (D6)	$\Lambda = \frac{3\pi}{GN_{\rm scr}}$ once the screen-capacity identification is adopted, together with the separate order-of-magnitude neutrino estimate tied to the same capacity input	Separate from the recovered core. It uses the second external continuous input $N_{\rm scr}$.
Phase II calibration sector (D10)	pixel-closure gauge/electroweak consistency, $\alpha_i(m_{Z,\rm run})$, $\alpha_{\rm em}^{-1}(m_{Z,\rm run})$, $\sin^2\theta_W(m_{Z,\rm run})$, v, m_W, and the running Z-mass relation	Consistency checks of the D10 branch. Their agreement is expected once P and the printed running/matching conventions are imposed.
Phase II secondary quantitative branch (D11)	Higgs/top critical-surface branch	Main non-calibration numerical branch. It adds no further continuous fit once D10 is fixed, but it remains secondary to the recovered core.
Phase III deferred continuations / downstream numerical branches (D12 and beyond)	charged-lepton/Koide fits, quark-texture branches, neutrino estimates beyond the D6 boundary, downstream hadron extractions, proton-spin bookkeeping, and any use of $\varepsilon = 1/6$ or $\delta = 2/9$ as flavor ansätze	Failure retracts the corresponding continuation only. These branches use extra structural choices, capacity assumptions, or external numerical machinery beyond D10–D11.

3.3.4 What This Part Actually Contributes

The documentary center of gravity here is the D10–D11 implementation layer. The gauge-sector closure and electroweak scale-setting branch are calibration-sector consistency checks. The Higgs/top critical-surface branch is the main secondary quantitative output. Charged-lepton, texture, neutrino, proton-spin, and hadronic branches remain weaker continuations.

Quantities reused in these later branches do *not* all inherit the same claim tier. In particular, $\varepsilon = 1/6$, $\delta = 2/9$, Koide relations, texture exponents, neutrino scales, proton-spin fits, and hadron fits should not be cited as recovered-core outputs of overlap consistency plus the stated scaling-limit and categorical premises. Parts I–II carry the recovered relativity-plus-Standard-Model core; this part begins at the D10 implementation layer and keeps everything beyond D11 explicitly non-core.

3.4 Stage 1: Fundamental Scales

3.4.1 Unification Scale

The OPH framework derives a unification scale from the pixel area:

$$M_U = \frac{E_P}{e^{2\pi}} \cdot P^{1/6}$$

where $E_P = 1.220890 \times 10^{19}$ GeV is the Planck energy. This gives:

$$M_U \approx 2.474 \times 10^{16} \text{ GeV}$$

Derivation: The factor $e^{2\pi}$ arises from the modular geometry of the collar (the Euclidean regularity condition fixes the angular period to 2π). The $P^{1/6}$ factor comes from the dimensional relation between pixel area and the UV cell scale (§5.8 of Part I of this manuscript).

3.4.2 Cell Energy Scale

The UV cell energy is:

$$E_{\text{cell}} = \frac{E_P}{\sqrt{P}} \approx 9.56 \times 10^{18} \text{ GeV}$$

This is the natural energy scale of a single computational element on the holographic screen.

3.5 Stage 2: Gauge Closure and the Pixel Constraint

3.5.1 The Heat-Kernel Entropy

The edge-center completion (Theorem 2.3 of Part I of this manuscript) combined with MaxEnt yields sector probabilities of the heat-kernel form:

$$p_R(t) \propto d_R \, e^{-t\, C_2(R)}$$

where d_R is the representation dimension, $C_2(R)$ the quadratic Casimir, and $t = 4\pi^2\alpha$ is the diffusion parameter encoding the gauge coupling.

The **mean entropy per cell** (the $\bar{\ell}$ function) for each gauge factor is:

$$\bar{\ell}_G(t) = \sum_R p_R(t) \log d_R$$

For SU(2):

$$\bar{\ell}_{\text{SU}(2)}(t) = \sum_{j=0,1/2,1,\ldots} p_j(t) \log(2j+1), \quad p_j \propto (2j+1) e^{-t\, j(j+1)}$$

For SU(3):

$$\bar{\ell}_{\text{SU}(3)}(t) = \sum_{p,q\geq 0} p_{(p,q)}(t) \log d_{(p,q)}, \quad p_{(p,q)} \propto d_{(p,q)} \, e^{-t\, C_2(p,q)}$$

where $d_{(p,q)} = \frac{1}{2}(p+1)(q+1)(p+q+2)$ and $C_2(p,q) = \frac{1}{3}(p^2+q^2+pq+3p+3q)$.

3.5.2 The Pixel Constraint

The generalized entropy matching (§5.4 of Part I of this manuscript) gives:

$$\frac{P}{4} = \bar{\ell}_{\mathrm{SU}(2)}(t_2) + \bar{\ell}_{\mathrm{SU}(3)}(t_3)$$

where $t_i = 4\pi^2\alpha_i$. This is the **pixel constraint**: the total edge entropy per cell must equal $P/4$.

3.5.3 Gauge Running and the α_U Solution

At the unification scale, all three gauge couplings emerge from a single α_U. Running down to a scale μ with MSSM-like β-function coefficients $(b_1, b_2, b_3) = (33/5, 1, -3)$:

$$\alpha_i^{-1}(\mu) = \alpha_U^{-1} + \frac{b_i}{2\pi} \ln \frac{M_U}{\mu}$$

Why MSSM coefficients? The edge-sector computation (§6.17 of Part I of this manuscript) derives β-function shifts $\Delta b \approx (2.49, 4.17, 4.01)$ from the heat-kernel form, the $\mathbb{Z}_6$ quotient structure, and the Peter-Weyl vacuum-polarization weighting. These match the MSSM shifts $(2.5, 4.17, 4.0)$ to better than 1%. SM-only coefficients catastrophically fail (α_s prediction off by 52σ).

3.5.4 Self-Consistent Fixed Point

The system is solved self-consistently:

1. **Trial** $\alpha_U \to$ run couplings to scale μ.
2. **Compute** $v = E_{\mathrm{cell}} \cdot \exp(-2\pi/(\beta_{\mathrm{EW}} \cdot \alpha_U))$ with $\beta_{\mathrm{EW}} = N_c + 1 = 4$.
3. **Compute** tree-level $m_Z(\mu) = \frac{1}{2} v \sqrt{g_2^2 + g_Y^2}$ where $g_Y = \sqrt{4\pi \cdot \frac{3}{5}\alpha_1}$.
4. **Find** $\mu^* = m_Z(\mu^*)$ (self-consistent fixed point).
5. **Evaluate** the pixel residual: $\bar{\ell}_{\mathrm{SU}(2)}(t_2) + \bar{\ell}_{\mathrm{SU}(3)}(t_3) - P/4$.
6. **Bisect** on α_U until the pixel residual vanishes.

This yields:

$$\alpha_U \approx 0.04112, \quad \alpha_U^{-1} \approx 24.32$$

3.6 Stage 3: Electroweak Observables

With α_U and M_U determined, the gauge couplings at $\mu = m_{Z,\mathrm{run}}$ are fixed:

Quantity	Formula	Predicted Value
$m_{Z,\mathrm{run}}$	Self-consistent fixed point	91.652 GeV
v (Higgs VEV)	$E_{\mathrm{cell}} \cdot e^{-2\pi/(\beta_{\mathrm{EW}}\alpha_U)}$	246.77 GeV
$\alpha_1(m_{Z,\mathrm{run}})$	$[\alpha_U^{-1} + \frac{b_1}{2\pi} \ln(M_U/m_{Z,\mathrm{run}})]^{-1}$	0.01696
$\alpha_2(m_{Z,\mathrm{run}})$	$[\alpha_U^{-1} + \frac{b_2}{2\pi} \ln(M_U/m_{Z,\mathrm{run}})]^{-1}$	0.03384
$\alpha_3(m_{Z,\mathrm{run}})$	$[\alpha_U^{-1} + \frac{b_3}{2\pi} \ln(M_U/m_{Z,\mathrm{run}})]^{-1}$	0.1183

From these:

$$\alpha_{\text{em}} = \left(\frac{1}{\alpha_2} + \frac{1}{\frac{3}{5}\alpha_1} \right)^{-1}, \quad \sin^2\theta_W = \frac{\alpha_{\text{em}}}{\alpha_2}$$

We use the electroweak normalization

$$\alpha_1 = \frac{5}{3}\alpha_Y, \qquad g_Y^2 = 4\pi\alpha_Y = 4\pi\frac{3}{5}\alpha_1.$$

Observable	Predicted	PDG Value
$\alpha_{\text{em}}^{-1}(m_{Z,\text{run}})$	128.31	127.952
$\sin^2\theta_W(m_{Z,\text{run}})$	0.2307	0.23122
$\alpha_s(m_{Z,\text{run}})$	0.1183	0.1179

3.6.1 Z Boson Pole Mass

The tree-level $m_{Z,\text{run}} = 91.65$ GeV is the running mass at its own scale. The physical pole mass includes the custodial correction from top-quark loops:

$$\Delta\rho = \frac{3y_t^2}{32\pi^2}$$

For the stage-3 numerical estimate we use $y_t \approx 1$, giving

$$\Delta\rho_{\text{stage-3}} \approx \frac{3}{32\pi^2} \approx 0.009499$$

$$M_{Z,\text{pole}} = \frac{m_{Z,\text{run}}}{\sqrt{1+\Delta\rho}} \approx 91.220 \text{ GeV}$$

Why the stage-3 estimate uses $\Delta\rho \approx 3/(32\pi^2)$. The general one-loop custodial-symmetry-breaking contribution is $\Delta\rho = 3y_t^2/(32\pi^2)$. The displayed stage-3 number sets $y_t \approx 1$, requiring no knowledge of the actual top mass , only that the top Yukawa is order-one (the least-suppressed channel in the $\mathbb{Z}_6$ texture).

3.6.2 W Boson Mass

At tree level:

$$m_W = \frac{1}{2}v \cdot g_2 = \frac{1}{2}v\sqrt{4\pi\alpha_2} \approx 80.39 \text{ GeV}$$

3.7 Stage 4: Critical Surface , Higgs and Top

3.7.1 The Critical Surface Constraint

Refinement stability (§6.3, §6.22 of Part I of this manuscript) pushes the Higgs quartic coupling to a marginal stability point at the unification scale:

$$\lambda(M_U) = 0, \quad \beta_\lambda(M_U) = 0$$

Setting $\beta_\lambda = 0$ at one loop with $\lambda = 0$ fixes the top Yukawa boundary condition:

$$y_t(M_U) = \left[\frac{1}{16}\left(2g_2^4 + (g_2^2 + g_Y^2)^2\right)\right]^{1/4}$$

where $g_Y(M_U)$ and $g_2(M_U)$ are obtained inside the D11 transport by running the OPH-derived couplings from m_Z to a UV synchronization scale $\mu_{\text{sync}} \approx 6.8 \times 10^{11}$ GeV with SM coefficients and from μ_{sync} to M_U with the same MSSM-like coefficients used in D10. The raw all-SM appendix flow by itself lands far lower ($m_t^{\text{pole}} \approx 164.2$ GeV, $m_H \approx 115.2$ GeV); the D11 numbers quoted below are the piecewise transport plus only order-one-GeV low-scale matching corrections.

3.7.2 RG Evolution to Low Scales

The coupled system $(y_t, \lambda, g_Y, g_2, g_3)$ is integrated from M_U down to $\mu \sim m_t$ using RK4. The D11 branch uses a piecewise one-loop gauge transport:

$$\alpha_i^{-1}(\mu) = \alpha_i^{-1}(m_Z) - \frac{b_i^{\text{SM}}}{2\pi}\ln(\mu/m_Z), \qquad m_Z \le \mu \le \mu_{\text{sync}}$$

and

$$\alpha_i^{-1}(\mu) = \alpha_i^{-1}(\mu_{\text{sync}}) - \frac{b_i^{\text{UV}}}{2\pi}\ln(\mu/\mu_{\text{sync}}), \qquad \mu_{\text{sync}} \le \mu \le M_U,$$

with $b^{\text{UV}} = (33/5, 1, -3)$ and $\mu_{\text{sync}} \approx 6.8 \times 10^{11}$ GeV. This Stage-4 section is the D11 branch used for the quoted Higgs/top numbers. The appendix records only the low-scale SM part of the one-loop system; the numerical outputs below also use the UV synchronization, low-scale extraction, and pole-mass conventions fixed here.

3.7.3 Top Mass

The running top mass is determined self-consistently: iterate $\mu_{\text{guess}} = y_t(\mu)\cdot v/\sqrt{2}$ until convergence:

$$m_t^{\overline{\text{MS}}}(m_t) \approx 160.6 \text{ GeV}$$

The same piecewise-running core gives

$$m_t^{\text{pole,core}} \approx 170.3 \text{ GeV}$$

before the final low-scale matching adjustment. The pole-mass part of that core uses the standard 3-loop QCD relation:

$$\frac{m_t^{\text{pole}}}{m_t^{\overline{\text{MS}}}} = 1 + \frac{4}{3}\frac{\alpha_s}{\pi} + K_2\left(\frac{\alpha_s}{\pi}\right)^2 + K_3\left(\frac{\alpha_s}{\pi}\right)^3$$

with $K_2 = 13.44 - 1.04n_l$ and $K_3 = 190.6 - 26.7n_l + 0.65n_l^2$ ($n_l = 5$). The displayed D11 output then adds a residual +0.84 GeV matching correction:

$$m_t^{\text{pole}} \approx 171.1 \text{ GeV}$$

3.7.4 Higgs Mass

From $\lambda(m_t)$ obtained by the RG integration above one finds a core value

$$m_H^{\text{core}} \approx 126.6 \text{ GeV}$$

and after the residual -0.12 GeV Higgs matching shift used here:

$$m_H = \sqrt{2\lambda(m_t)} \cdot v \approx 126.5 \text{ GeV}$$

3.8 Stage 5: Discrete Spectrum , Quarks and Leptons

3.8.1 The $\mathbb{Z}_6$ Defect Parameter

The SM global gauge group is $(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$. The entropy deficit from the $\mathbb{Z}_6$ quotient yields the universal suppression factor:

$$\varepsilon = e^{-\ln 6} = \frac{1}{6}$$

This is the base of the Froggatt-Nielsen texture: each unit of $\mathbb{Z}_6$ defect insertion suppresses a Yukawa coupling by a factor of 6.

3.8.2 Integer Exponents (Algorithmically Derived)

The diagonal mass exponents are determined by the minimal SU(3) hierarchy structure with $N_c = 3$ colors and $N_g = 3$ generations:

Up-type quarks: $n_u = (2N_c, N_c, 0) = (6, 3, 0)$
Down-type quarks: $n_d = (2N_c, N_c + 1, N_c - 1) = (6, 4, 2)$
Charged leptons: derived from the Koide phase.

3.8.3 The Koide Phase

The charged lepton masses use a circulant ansatz on generation space with Koide ratio $Q = 2/3$ (from $\mathbb{Z}_3$ mode balance). The holonomy phase is:

$$\delta = \frac{\beta_{\text{EW}}}{2N_c N_g} = \frac{N_c + 1}{2N_c N_g} = \frac{4}{18} = \frac{2}{9}$$

The Koide roots are:

$$r_k = 1 + \sqrt{2} \cos\left(\frac{2}{9} + \frac{2\pi k}{3}\right), \quad k = 0, 1, 2$$

sorted in ascending order. The lepton exponents (n_e, n_μ, n_τ) are selected by matching ratios $r_i^2/r_j^2 \approx \varepsilon^{n_i - n_j}$, yielding $(n_e, n_\mu, n_\tau) = (7, 4, 3)$.

3.8.4 Diagonal Quark Masses

$$m_{u_i} = \frac{v}{\sqrt{2}} \varepsilon^{n_{u,i}}, \quad m_{d_i} = \frac{v}{\sqrt{2}} \varepsilon^{n_{d,i}}$$

3.8.5 CKM Mixing

The CKM matrix is parameterized by ε:

$$s_{12} = \varepsilon = 1/6, \quad s_{23} = \varepsilon^2 = 1/36, \quad s_{13} = \varepsilon^3 = 1/216$$

Physical down-type masses are the singular values of $V_{\text{CKM}} \cdot \text{diag}(m_d, m_s, m_b)$.

3.8.6 Charged Lepton Masses

A scale factor S is determined from the exponents $n_e = (7, 4, 3)$ and the Koide roots by demanding internal consistency:

$$\ln S_0 = -\frac{1}{3} \sum_{i=1}^{3} \ln \left(\frac{r_i^2 \sqrt{2}}{v} \cdot 6^{n_{e,i}} \right)$$

Then $S = S_0 \cdot 2^{1/6}$ and:

$$m_e = S \cdot r_1^2, \quad m_\mu = S \cdot r_2^2, \quad m_\tau = S \cdot r_3^2$$

3.9 Stage 6: QCD Scale and Hadron Masses

3.9.1 $\Lambda_{\overline{\text{MS}}}$ from $\alpha_s(m_{Z,\text{run}})$

Given the OPH-derived $\alpha_s(m_{Z,\text{run}}) \approx 0.1183$ (a calibration-sector output) and the quark thresholds (m_b, m_c from Stage 5), the QCD scale is extracted using the standard 4-loop MSbar definition:

$$\ln \frac{\mu^2}{\Lambda^2} = \frac{1}{\beta_0 a} + \frac{\beta_1}{\beta_0^2} \ln(\beta_0 a) + \int_0^a dx \left[\frac{1}{\beta(x)} + \frac{1}{\beta_0 x^2} - \frac{\beta_1}{\beta_0^2 x} \right]$$

where $a = \alpha_s/(4\pi)$ and $\beta_0, \beta_1, \beta_2, \beta_3$ are the 4-loop MSbar coefficients.
Stepping down across flavor thresholds:

$$\alpha_s^{(n_f-1)}(\mu_{\text{th}}) = \alpha_s^{(n_f)}(\mu_{\text{th}}) \quad \text{at } \mu_{\text{th}} = m_b, m_c$$

yields:

n_f	$\Lambda_{\overline{\text{MS}}}^{(n_f)}$ (GeV)
5	0.211
4	0.288
3	0.322

3.9.2 Hadron Masses

Hadron masses require a nonperturbative computation of dimensionless ratios $C_X = m_X/\Lambda^{(3)}$. These are computed by an internal lattice QCD collar calculation:

1. **Wilson SU(3) gauge** (quenched) with Metropolis updates.
2. **Wilson valence quarks** at multiple κ values for chiral extrapolation.

3. **Gradient-flow coupling** for input-free $a\Lambda$ determination.
4. **Two-point Richardson extrapolation** for the continuum limit.

Then: $m_X = C_X \cdot \Lambda_{\overline{\mathrm{MS}}}^{(3)}$.

The lattice computation uses small volumes ($L = 2$–6) in the quenched approximation. Systematic uncertainties from quenching (~10%), finite volume, and limited statistics dominate the error budget for hadron masses.

3.10 Stage 7: Neutrino Masses

3.10.1 The Capacity Model

The second OPH input, the screen capacity $N_{\mathrm{scr}} \equiv \log(\dim \mathcal{H}_{\mathrm{tot}}) \sim 10^{122}$, enters through the cosmological constant:

$$\rho_\Lambda = \frac{3}{8}\frac{M_P^4}{N_{\mathrm{scr}}}$$

The neutrino mass scale is anchored at $\rho_\Lambda^{1/4}$:

$$m_{\nu_3} = \rho_\Lambda^{1/4} \approx 3.0 \times 10^{-12}\ \mathrm{GeV} \approx 3.0\ \mathrm{meV}$$

The hierarchy uses the same $\varepsilon = 1/6$ suppression:

$$m_{\nu_2} = \varepsilon \cdot m_{\nu_3} \approx 0.50\ \mathrm{meV}, \quad m_{\nu_1} = \varepsilon^2 \cdot m_{\nu_3} \approx 0.084\ \mathrm{meV}$$

This yields $\Delta m_{32}^2 \approx 9.1 \times 10^{-24}\ \mathrm{GeV}^2$ and $\Delta m_{21}^2 \approx 2.5 \times 10^{-25}\ \mathrm{GeV}^2$. These values remain well below existing direct-detection bounds, but they do not reproduce the observed atmospheric and solar oscillation splittings.

3.11 Massless Particles

The following particles are predicted to be **exactly massless** by symmetry:

Particle	Reason
Photon	Unbroken $U(1)_{\mathrm{em}}$ gauge invariance forbids a mass term
Gluons (8)	Unbroken $SU(3)_c$ gauge invariance forbids a mass term
Graviton	Diffeomorphism invariance (emergent from entanglement equilibrium) forbids a mass term

These are **symmetry-protected zeros**, not accidental cancellations. The gauge invariances are derived from the gluing structure (§6.1 of Part I of this manuscript) and entanglement equilibrium (§5 of Part I of this manuscript).

3.12 Complete Spectrum: OPH Outputs vs PDG

All outputs below use inputs $P = 1.63094$ and $N_{\text{scr}} \equiv \log(\dim \mathcal{H}_{\text{tot}}) \sim 10^{122}$ only. PDG values are from the 2024/2025 edition. Each output is classified as **consistency check** (Tier 1, from calibration sector), **independent quantitative** (Tier 2), **phenomenological continuation / downstream branch** (Tier 3), or **structural** (Tier 4, parameter-free). See §1B for the full classification.

3.12.1 Gauge Bosons and Higgs

Particle	OPH (GeV)	PDG (GeV)	Rel. Error	Tier	Error Source
γ	0	0	exact	Structural	Symmetry-protected; no error expected
g (gluon)	0	0	exact	Structural	Symmetry-protected; no error expected
W boson	80.386	80.377 ± 0.012	+0.012%	Consistency	Flows from calibrated α_U and v. Tree-level only
Z pole	91.220	91.188 ± 0.002	+0.035%	Consistency	Flows from calibrated couplings. Simplified $\Delta\rho$
H (Higgs)	126.48	125.20 ± 0.11	+1.02%	D11 quantitative	Piecewise SM/MSSM transport plus residual low-scale matching after $\lambda = \beta_\lambda = 0$

3.12.2 Charged Leptons

Particle	OPH (GeV)	PDG (GeV)	Rel. Error	Tier	Error Source
Electron	5.109×10^{-4}	5.110×10^{-4}	-0.023%	D12 continuation	Koide structure ($Q = 2/3$, $\delta = 2/9$) plus an extra discrete-selection step; residual from QED running

Particle	OPH (GeV)	PDG (GeV)	Rel. Error	Tier	Error Source
Muon	0.10564	0.10566	-0.022%	D12 continuation	Same Koide branch; QED running corrections ~0.02%, matching the offset
Tau	1.7766	1.7769 ± 0.09	-0.020%	D12 continuation	Same origin. Uniform sign shows common scale-factor effect from QED running

3.12.3 Neutrinos

Particle	OPH (GeV)	Experiment	Tier	Error Source
ν_e	8.39×10^{-14}	$< 8 \times 10^{-10}$ (KATRIN)	D6 estimate	Uses second input ($\log \dim \mathcal{H}$); consistent with direct bounds but not a completed oscillation-scale prediction
ν_μ	5.04×10^{-13}	$< 8 \times 10^{-10}$ (KATRIN)	D6 estimate	Same capacity branch; below direct bounds, but not yet matched to the solar oscillation scale
ν_τ	3.02×10^{-12}	$< 8 \times 10^{-10}$ (KATRIN)	D6 estimate	Anchored at $\rho_\Lambda^{1/4}$; below direct bounds, but not yet matched to the atmospheric oscillation scale

All predictions are well below the experimental upper bound of ~ 0.8 eV $\approx 8 \times 10^{-10}$ GeV. This branch is better described as a low-scale capacity estimate than as a quantitative fit to oscillation data.

3.12.4 Quarks

Particle	OPH (GeV)	PDG (GeV)	Rel. Error	Tier	Error Source
up	3.74×10^{-3}	2.16×10^{-3}	+73%	D12 continuation	u-d **degeneracy**: both get exponent $n = 6$. Subleading effects not included
down	3.74×10^{-3}	4.70×10^{-3}	-20%	D12 continuation	Same degeneracy. Geometric mean $\sqrt{m_u m_d} \approx 3.2$ MeV closer to 3.7 MeV
strange	0.1346	0.0935	+44%	D12 continuation	No scheme matching; missing Clebsch factors and CKM rotation
charm	0.808	1.273	-37%	D12 continuation	Missing RG running from v to m_c (~factor 1.3) and Clebsch coefficient
bottom	4.847	4.183	+16%	D12 continuation	Running from v to m_b reduces mass by ~15%, closing most of the gap
top (texture)	174.5	172.6	+1.1%	D12 continuation	Exponent $n_t = 0$ (unsuppressed Yukawa); small error from v offset
top (crit. surf.)	171.1	172.6	-0.87%	D11 quantitative	Piecewise SM/MSSM transport plus residual pole-mass matching after $\lambda = \beta_\lambda = 0$

3.12.5 Gauge Couplings at $m_{Z,\mathrm{run}}$

These are **calibration-sector consistency checks** (Tier 1). The gauge couplings at $m_{Z,\mathrm{run}}$ are the fixed-point values used in the entropy-matching condition and self-consistent closure procedure. Their agreement with PDG values demonstrates internal consistency, not independent predictive power.

Quantity	OPH	PDG	Rel. Error	Tier	Error Source
$\alpha_{\mathrm{em}}^{-1}(m_{Z,\mathrm{run}})$	128.31	127.952 ± 0.009	+0.28%	Consistency	Flows from calibrated α_U
$\sin^2\theta_W(m_{Z,\mathrm{run}})$	0.2307	0.23122 ± 0.00004	-0.21%	Consistency	Most sensitive to threshold scale
$\alpha_s(m_{Z,\mathrm{run}})$	0.1183	0.1179 ± 0.0009	+0.37%	Consistency	Within 0.5σ of PDG

3.12.6 Derived Scales

Quantity	OPH Value	Reference	Agreement	Tier	Error Source
v (Higgs VEV)	246.77 GeV	246.22 GeV	+0.22%	Consistency	Flows from calibrated α_U via transmutation formula
M_U	2.47×10^{16} GeV	—	—	Consistency	No direct measurement; consistent with proton stability bounds
α_U^{-1}	24.32	—	—	Consistency	Intermediate variable in calibration procedure
$\Lambda_{\overline{\mathrm{MS}}}^{(3)}$	0.322 GeV	~0.332 GeV	-3.0%	Consistency	Propagated from α_s (calibration sector); 0.4% in α_s → ~3% in Λ

3.12.7 Hadron Masses

Hadron masses follow from $m_X = C_X \cdot \Lambda_{\overline{\mathrm{MS}}}^{(3)}$, where C_X is a dimensionless ratio computed via lattice QCD. The derivation chain is complete: $P \to \alpha_s(m_Z) \to \Lambda^{(3)} \to m_X$. Current lattice estimates use small volumes and the quenched approximation, so systematic uncertainties are large (~10–20%).

Particle	PDG (GeV)	Formula	Dominant uncertainty
Proton	0.93827	$C_p \cdot \Lambda^{(3)}$, $C_p \approx 2.9$	Quenching (~10%), finite volume
Neutron	0.93957	$\approx m_p$ in the isospin limit	Isospin splitting requires unquenched QCD+QED
$\pi^\pm$	0.13957	$C_\pi \cdot \Lambda^{(3)}$	Chiral extrapolation ($m_\pi^2 \propto m_q$)
π^0	0.13498	$\approx m_{\pi^\pm}$	EM splitting

Particle	PDG (GeV) Formula	Dominant uncertainty
$K^{\pm}$	$0.4937\, C_K \cdot \Lambda^{(3)}$	Strange-quark valence mass
K^0	$0.4976 \approx m_{K^{\pm}}$	EM splitting
Λ baryon	$1.1157\, C_\Lambda \cdot \Lambda^{(3)}$	Strange-quark baryon correlator

---

3.13 Analysis of Discrepancies

3.13.1 Sub-Permille Outputs ($< 0.1\%$)

W boson (+0.012%), **Z boson** (+0.035%), **electron** (-0.023%), **muon** (-0.022%), **tau** (-0.020%):

These five outputs are at the sub-permille level, but their epistemic status differs. The gauge boson masses (W, Z) are **consistency checks**: they flow directly from the calibrated gauge couplings and v, so their agreement is expected by construction. The charged leptons (e, μ, τ) are **Tier 3 phenomenological continuations**: the Koide structure ($Q = 2/3$, $\delta = 2/9$) is algebraically independent of the gauge-coupling calibration, but the final exponent assignment uses an additional discrete-selection step.

3.13.2 Percent-Level Outputs ($0.1\% - 2\%$)

Higgs (+1.0%), **top pole** (-0.87%), α_s (+0.37%), $\sin^2\theta_W$ (-0.21%), $\alpha_{\mathbf{em}}^{-1}$ (+0.28%):

The epistemic status differs within this group. The **gauge couplings** (α_s, $\sin^2\theta_W$, $\alpha_{\mathrm{em}}^{-1}$) are **consistency checks** — they are the data used to calibrate P, so their agreement demonstrates that the self-consistent fixed-point procedure works, not that the framework predicts them.

The **Higgs and top masses** are **Tier 2 independent quantitative outputs**: they come from the critical surface constraint ($\lambda(M_U) = 0$, $\beta_\lambda(M_U) = 0$) together with the D11 transport (SM running below μ_{sync}, MSSM-like running above, and only order-one-GeV residual matching). This remains an independent boundary condition not used in calibrating P. The ~1% error is consistent with the expected size of 2-loop corrections and threshold effects.

3.13.3 Qualitative Agreement Only: Quark Masses ($16\% - 73\%$)

The quark masses from the $\varepsilon = 1/6$ texture show large deviations. The reasons are well-understood:

1. **u-d degeneracy**: The texture assigns identical exponents $n_u = n_d = 6$ to the up and down quarks, predicting $m_u = m_d$. In reality, $m_u/m_d \approx 0.46$. Breaking this degeneracy requires isospin-violating effects beyond the minimal $\mathbb{Z}_6$ texture (e.g., subleading defect insertions or electromagnetic corrections).

2. **Scheme and scale mismatch**: The PDG quark masses are $\overline{\mathrm{MS}}$ running masses at scale $\mu = 2$ GeV (for light quarks) or $\mu = m_q$ (for heavy quarks). The OPH texture produces "Yukawa-scale" masses at $\mu \sim v$ with no scheme matching applied. For the charm quark, this can easily account for a factor of ~1.5.

3. **Order-one Clebsch-Gordan coefficients**: The texture $y_f = c_f \cdot \varepsilon^{n_f}$ has residual coefficients c_f that are undetermined (expected to be $\mathcal{O}(1)$, actually ranging from 0.6 to 2.2). These

encode CKM rotation effects, RG running from M_U to m_Z, and overlap matrix elements that the minimal integer-exponent ansatz does not resolve.

4. **Missing threshold corrections**: No QCD or electroweak threshold corrections are applied at flavor thresholds. These are typically 5–20% effects for the lighter quarks.

The correct interpretation: The texture correctly captures the *hierarchy* (the fact that $m_t/m_u \sim 10^5$ arises from integer exponents in base 6), and it does not aim for precision at the individual-mass level. The base-6 logarithms of all nine charged-fermion Yukawas land within ~0.3 of integers , this is the core prediction. The order-one residuals are where scheme matching and subleading effects live.

3.13.4 Hadron Masses

The hadron mass predictions follow from the dimensionless ratios $C_X = m_X/\Lambda^{(3)}$, computed via lattice QCD. The internal lattice code (`oph_lattice_su3_quenched_v5.py`) implements a quenched Wilson-gauge SU(3) lattice with Wilson valence quarks, gradient-flow scale setting, and Richardson continuum extrapolation , all without PDG inputs. The dominant systematic uncertainties are:

- **Quenching error** (~10%): The gauge field is quenched ($n_f = 0$), while physical QCD has $n_f = 2 + 1$ light dynamical flavors.
- **Volume effects**: Lattice volumes ($L = 2$–6) give finite-volume corrections.
- **Statistics**: Current runs use $\mathcal{O}(10)$ configurations; scaling to $\mathcal{O}(10^3)$ would reduce statistical errors.

These are standard lattice QCD systematics, not limitations of the OPH framework. The derivation chain $P \to \alpha_s \to \Lambda^{(3)} \to m_X$ is complete; improving the lattice computation improves the numerical precision of the hadron predictions.

3.13.5 Neutrinos: No Direct Comparison Available

The predicted neutrino masses (~ 0.08–3 meV) are below existing direct-detection sensitivity (KATRIN: $m_\beta < 0.8$ eV) and well below existing cosmological mass-sum bounds. The predicted $\sum m_\nu \approx 3.6$ meV is well below the cosmological bound $\sum m_\nu < 0.12$ eV (Planck 2018), but the simple capacity-branch hierarchy does not reproduce the observed oscillation splittings quantitatively.

3.14 Reproduction Instructions

All computations can be reproduced from the code in `code/particles/`.

3.14.1 Prerequisites

```
pip install numpy
```

No other dependencies are needed. The `pdg` and `pandas` packages are only required for `tools/fetch_pdg_data.py` (the PDG data fetcher, used for comparison only).

3.14.2 Code Files

File	Description	Stage
particle_masses_stage5.py	Core spectrum prediction: gauge closure, transmutation, critical surface, Z_6 texture	1–5
oph_qcd.py	4-loop MSbar β-coefficients and Λ extraction	6
oph_lattice_su3_quenched_v5.py	Quenched Wilson SU(3) lattice for hadron mass ratios	6–7
oph_predict_compare.py	Full predictor plus PDG comparison (main entry point)	All
oph_no_cheat_audit.py	Static anti-leak audit of prediction code	Audit
test_oph_predict_compare.py	Smoke tests plus runtime no-cheat mutation test	Tests
test_particle_masses_stage5.py	Regression tests for Stage 5 predictions	Tests

3.14.3 Running the Full Prediction

```
cd code/particles/

# Full prediction with PDG comparison table
python3 oph_predict_compare.py --compare

# JSON output for programmatic use
python3 oph_predict_compare.py --compare --json

# With internal hadron computation (slow; tiny-lattice demo)
python3 oph_predict_compare.py --compare --with-hadrons --hadron-profile demo
```

3.14.4 No-Leakage Verification

```
cd code/particles/

# Static audit: checks that build_spectrum() does not reference PDG values
python3 oph_no_cheat_audit.py

# Runtime mutation test: scrambles PDG table and verifies predictions unchanged
python3 test_oph_predict_compare.py
```

3.14.5 Individual Stages

```
cd code/particles/

# Stage 5 spectrum (core charged sector)
python3 particle_masses_stage5.py

# QCD scale extraction (standalone)
python3 oph_qcd.py
```

3.14.6 Fetching PDG Reference Data

```
pip install pdg pandas
python3 ../tools/fetch_pdg_data.py
# Output: ../pdg_data/particle_masses.{csv,json}
```

3.15 Summary

This part records the implementation layer that begins after the recovered core. Section 3.3 gives the full claim-tier map; the summary here keeps only the operational boundary.

- **Phase I recovered core inherited from Parts I–II:** conditional relativity plus the realized Standard Model structural chain. These results are upstream prerequisites for this part, not outputs newly established here.
- **Phase II input-dependent / implementation layer:** D6 if the screen-capacity identification is adopted; D10 for the supplement-backed gauge/electroweak calibration sector; D11 for the Higgs/top critical-surface branch that adds no new continuous fit once D10 is fixed.
- **Phase III continuation layer:** charged-lepton/Koide fits, texture branches, neutrino estimates beyond D6, downstream hadron extractions, proton-spin bookkeeping, and related phenomenological continuations.

The practical contribution of this part is narrow but useful. It records the D10 gauge/electroweak calibration closure, the D11 critical-surface Higgs/top branch, and a set of D12 continuations that reuse the same structural backdrop without acquiring the same claim tier. Here $D10$ uses the external input P, $D6$ uses the external input N_{scr}, $D11$ adds the extra critical-surface condition, and $D12$ introduces further ansätze or downstream numerical machinery. A failure in D12 retracts that continuation only; a failure in D11 challenges the secondary quantitative branch; a failure in D10 challenges the calibration branch.

Part IV
String-Theory Derivation

4 String Theory from Observer Patch Holography

OPH is an observer-centric reconstruction program for fundamental physics. The quantitative branches discussed in this paper use two external inputs (P for pixel area and N_{scr} for screen capacity) together with structural axioms and stated regulator/scaling premises. This part records the internal OPH route from edge-sector heat-kernel structure to a worldsheet/string reorganization of the bosonic branch.

4.1 A Continuation-Level Route

This document records the OPH derivation path by which the bosonic edge-sector branch admits a string-theoretic description of its effective degrees of freedom. The route uses OPH axioms together with the stated heat-kernel, large-N, and modular-sewing premises; no separate external string-theory starting postulate is introduced.

---

5 Overview: The Claimed Route

This part records the following continuation-level chain within the stated OPH branch:

```
Five OPH axioms + regulator premises used in the edge-sector branch
    \ensuremath{\downarrow}
Edge-Center Decomposition + Heat-Kernel Sector Weights
    \ensuremath{\downarrow}
2D Yang-Mills Theory on the Holographic Screen
    \ensuremath{\downarrow}
String Theory (Worldsheet Expansion via Gross-Taylor)
```

OPH also supplies:

- **Worldsheet kinematics**: Conformal/modular sewing from geometric modular flow
- **Target-space dynamics**: conditional scaling-limit Einstein branch from entanglement equilibrium
- **Higher gauge structure**: The natural slot for B-field/gerbe data

This records a conditional route by which the OPH edge/boundary sector can admit a string-theoretic reorganization if the stated large-N and modular-sewing premises are realized.

---

6 OPH Inputs Used

This part uses the following OPH ingredients from the main paper and technical supplement:

6.1 OPH Axioms

The OPH axioms used here are: Screen Net, Overlap Consistency, Local MaxEnt and Refinement Stability, Recoverable Generalized Entropy, and Minimal Admissible Realization (MAR). The heat-kernel / 2D Yang-Mills route uses primarily the first four plus the technical premises below; MAR enters only when one specializes to the realized Standard Model gauge branch later in the manuscript.

6.2 Technical Premises

Regulator premise R0 (Type-I local algebras): At the UV regulator scale, local patch algebras are type-I factors (finite-dimensional matrix algebras).

Regulator premise R1 (boundary gauge invariance): For a region $R \subset S^2$, the physical algebra is:

$$\mathcal{A}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}$$

where $G_{\partial R}$ is the boundary gauge group.

Together with the local MaxEnt clause of the OPH axioms, these theorem-local premises give the edge-center decomposition and the heat-kernel sector weights quoted in Section 2.3.

6.3 Key Derived Results

Edge-Center Decomposition (Theorem 1.1): Under R0 and R1, the collar Hilbert space decomposes as:

$$\mathcal{H}_{B_\delta} = \bigoplus_\alpha \left(\mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha} \right)$$

Heat-Kernel Sector Weights (Theorem 5.1): Under MaxEnt with bi-invariant constraints:

$$p_R(t) \propto d_R \, e^{-t\, C_2(R)}$$

where $d_R = \dim R$ and $C_2(R)$ is the quadratic Casimir.

Geometric Modular Flow (Theorem 2.2): On the BW geometric branch, cap modular flow acts as the standard cap-preserving conformal dilation with period 2π.

---

7 Einstein Equation from Entanglement Equilibrium

Before turning to string theory, this part first recalls that OPH recovers gravity, the "target-space dynamics" that any string theory must reproduce.

7.1 Setup: Null Deformation Problem

Pick a local causal diamond/cap region C in the emergent bulk. Focus on one null sheet of ∂C, generated by a future-directed null vector field k^a with affine parameter λ.

Consider an infinitesimal deformation moving the cut along the null generators:

$$\Sigma \to \Sigma(\lambda)$$

7.2 Generalized Entropy Variation

By Recoverable Generalized Entropy, generalized entropy is:

$$S_{\text{gen}}(\Sigma) = \frac{A(\Sigma)}{4G} + S_{\text{bulk}}(\Sigma)$$

Its first variation under the null deformation:

$$\delta S_{\text{gen}} = \delta\left(\frac{A}{4G}\right) + \delta S_{\text{bulk}}$$

Entanglement equilibrium premise: At the MaxEnt reference state, $\delta S_{\text{gen}} = 0$ for allowed local variations.

7.3 Area Variation via Raychaudhuri

Let $\theta(\lambda)$ be the expansion of the null congruence. The Raychaudhuri equation:

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma_{ab}\sigma^{ab} - R_{ab}k^a k^b$$

At equilibrium ($\theta(0) = 0$, $\sigma(0) = 0$):

$$\left.\frac{d\theta}{d\lambda}\right|_{\lambda=0} = -R_{kk}$$

Therefore:

$$\delta\left(\frac{A}{4G}\right) = -\frac{1}{4G}\int d^2x\sqrt{h_0}\int d\lambda\,\lambda\,R_{kk} + \cdots$$

7.4 Bulk Entropy Variation via Modular Theory

Entanglement First Law: For small perturbations around reference state ω:

$$\delta S_{\text{bulk}} = \delta\langle K\rangle$$

where $K = -\log\rho_C$ is the modular Hamiltonian.

Geometric Modular Hamiltonian (N1-N3): For null-deformed regions:

$$K = 2\pi\int d^2x\int d\lambda\,\lambda\,T_{kk}(\lambda, x) + K_\partial + O(\varepsilon)$$

Therefore:

$$\delta S_{\text{bulk}} = 2\pi\int d^2x\int d\lambda\,\lambda\,\delta\langle T_{kk}\rangle + \cdots$$

7.5 Entanglement Equilibrium Implies Einstein

Setting $\delta S_{\text{gen}} = 0$:

$$0 = -\frac{1}{4G}\int d^2x\int d\lambda\,\lambda\,R_{kk} + 2\pi\int d^2x\int d\lambda\,\lambda\,\langle T_{kk}\rangle$$

Since this holds for arbitrary localized deformations:

$$R_{kk} = 8\pi G\langle T_{kk}\rangle \quad \text{for all null } k^a$$

7.6 Upgrade to Full Einstein Equation

Define $E_{ab} := R_{ab} - 8\pi G\langle T_{ab}\rangle$.

The result $R_{kk} = 8\pi G T_{kk}$ for all null k means:

$$E_{ab}k^a k^b = 0 \quad \forall \text{ null } k$$

Lemma (Null Data Ambiguity): If $X_{ab}k^a k^b = 0$ for all null k, then $X_{ab} = \phi\, g_{ab}$.

Therefore $E_{ab} = \phi\, g_{ab}$, giving:

$$R_{ab} = 8\pi G\langle T_{ab}\rangle + \phi\, g_{ab}$$

In Einstein tensor form:

$$\boxed{G_{ab} + \Lambda g_{ab} = 8\pi G\langle T_{ab}\rangle}$$

where $\Lambda := \frac{1}{2}R - \phi$ is constant by the Bianchi identity.

Key point: Λ is not fixed by local null modular data,it requires global input (screen capacity).

---

8 Edge Sectors and 2D Yang-Mills

8.1 Edge-Sector Structure

From the OPH edge-center decomposition, the collar Hilbert space has structure:

$$\mathcal{H}_{\text{collar}} \cong \bigoplus_{\alpha} \left(\mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha}\right)$$

with α labeling irreps/charges of the boundary gauge group.

This is exactly the structure of gauge theory with boundary: cutting gauge space produces edge modes whose labels are representation data.

8.2 Heat-Kernel Weights from MaxEnt

OPH's MaxEnt + bi-invariant constraints give sector probabilities:

$$p_R(t) \propto d_R\, e^{-t\, C_2(R)}$$

Because of the Peter-Weyl doubling (loops see both factors), the effective weight is:

$$w_R(t) \propto d_R\, p_R(t) \propto d_R^2\, e^{-t\, C_2(R)}$$

8.3 Edge Partition Function

Define the edge partition function at diffusion parameter t:

$$Z_{\text{edge}}(t) := \sum_R d_R^2\, e^{-t\, C_2(R)}$$

8.4 Identification with Group Heat Kernel

The heat kernel on compact group G, expanded in characters:

$$K_t(U) = \sum_R d_R \, \chi_R(U) \, e^{-tC_2(R)}$$

At the identity $U = 1$ (using $\chi_R(1) = d_R$):

$$K_t(1) = \sum_R d_R^2 \, e^{-tC_2(R)} = Z_{\text{edge}}(t)$$

Result: The OPH edge ensemble is literally the heat kernel at the identity.

8.5 Gluing = Markov = Area Additivity

Heat kernels satisfy the Chapman-Kolmogorov (gluing) property:

$$K_{t_1+t_2}(U) = \int_G dV \, K_{t_1}(UV^{-1}) \, K_{t_2}(V)$$

This matches OPH patch sewing: add diffusion parameters when gluing collars/strips.

8.6 Exact Match with 2D Yang-Mills

The partition function of 2D Yang-Mills on a closed genus-g surface Σ_g with area A:

$$Z_{\text{2D YM}}(\Sigma_g) = \sum_R (d_R)^{2-2g} \exp\left[-\frac{g_{\text{YM}}^2 A}{2} C_2(R)\right]$$

For $\Sigma_0 = S^2$ (genus 0):

$$Z_{\text{2D YM}}(S^2) = \sum_R d_R^2 \exp\left[-\frac{g_{\text{YM}}^2 A}{2} C_2(R)\right]$$

Identification:

$$t = \frac{g_{\text{YM}}^2 A}{2}$$

Theorem 4.1: OPH edge-sector MaxEnt gives a fully-fledged 2D Yang-Mills theory on the screen, with collar diffusion parameter t playing the role of "coupling $\times$ area".

---

9 2D Yang-Mills as String Theory

9.1 The Gross-Taylor Expansion

In the large-N limit for $G = SU(N)$, 2D Yang-Mills admits an exact reorganization as a sum over worldsheets (Gross-Taylor, 1993).

The partition function expands as:

$$\log Z = \sum_{h \geq 0} N^{2-2h} F_h(\lambda), \quad \lambda := g_{\text{YM}}^2 N$$

where:

- h is the worldsheet genus
- $g_s \sim 1/N$ is the string coupling
- Contributions scale as N^{2-2h} for worldsheet genus h

9.2 Worldsheet Interpretation

The $1/N$ expansion coefficients count **branched coverings** of the target surface,i.e., "string configurations without folds."

This is a genuine string theory:

- **Target space:** The 2D screen surface
- **Worldsheet:** The covering surface summed over
- **String coupling:** $g_s \sim 1/N$

9.3 The Large-N Regime

The Gross-Taylor expansion is controlled in the large-N limit. OPH provides an effective large-N parameter from screen microstructure: many independent edge channels/pixels whose combined algebra behaves like large matrix degrees of freedom. The effective edge Hilbert space dimension scales as:

$$\dim(\tilde{\mathcal{H}}_\partial) \sim N^{\#(\text{boundary cells})}$$

so $1/N$ becomes the handle-counting coupling.

9.4 Genus Expansion

Define generating functional:

$$F(t) := \log Z_{\text{edge}}(t)$$

In controlled large-N regime:

$$F(t) = \sum_{h \geq 0} N^{2-2h} F_h(\lambda), \quad \lambda := g_{\text{YM}}^2 N$$

This is a perturbative closed-string genus expansion.

---

10 Worldsheet Kinematics from Modular Flow

10.1 Conformal Structure

OPH provides conformal structure from two sources:

Global conformal symmetry:

$$\text{Conf}^+(S^2) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$$

This gives Lorentz kinematics,interpreting screen angles as null directions of 3+1D spacetime.

Geometric modular flow: Cap modular flow is forced to be conformal dilation with period 2π (Euclidean regularity).

10.2 Sewing Structure

String worldsheets are sewn along boundaries/cuts. Consistent sewing requires conformal/modular data.

OPH's Markov/additivity theorems on collars derive this exact factorization structure that makes sewing well-behaved.

10.3 The Analogy with Standard String Theory

String Theory	OPH
Worldsheet conformal symmetry	Screen conformal group
Modular invariance	Overlap consistency
Target-space Einstein eq.	Entanglement equilibrium
Sewing axioms	Markov collar gluing

---

11 The B-Field Slot: Higher Gauge Structure

11.1 Nonabelian Gluing Cocycles

OPH includes gluing by a crossed-module 2-cocycle (g_{ij}, h_{ijk}) on triple overlaps, with coherence on quadruple overlaps.

This is precisely the higher-gauge/gerbe structure associated in string theory with:

- 2-form gauge field B (Kalb-Ramond field)
- Gauge invariance $B \to B + d\eta$

11.2 Strictification and Trivialization

OPH shows strictification exists iff the abelian Čech H^2 class is trivial.

This parallels: "background B-field is a gerbe class; strings couple to it; trivialization conditions matter."

11.3 D-Brane Interpretation

Certain defects/boundaries where the class trivializes are the seed of D-brane-like boundary conditions.

The mathematical structure where string theory places B-field/gerbe data is directly present in OPH. The full open/closed string boundary state formalism follows from extending this framework to include D-brane-like boundary conditions.

---

12 OPH String Scale and Parameters

12.1 String Scale from Pixel Area

OPH has a UV length from pixel area:

$$a_{\text{cell}} \approx 1.63\,\ell_P^2, \quad \ell_{\text{UV}} = \sqrt{a_{\text{cell}}} \approx 1.28\,\ell_P$$

Natural identification:

$$\alpha' \sim \ell_{\text{UV}}^2 \sim a_{\text{cell}}$$

String tension:

$$T_{\text{string}} = \frac{1}{2\pi\alpha'} \sim \frac{1}{2\pi a_{\text{cell}}}$$

Numerically (in Planck units):

$$T_{\text{string}} \approx \frac{1}{2\pi \cdot 1.63}\,\ell_P^{-2} \approx 0.098\,\ell_P^{-2}$$

This is order-Planck tension, as expected if UV cutoff sits near ℓ_P.

12.2 Worldsheet YM Coupling

From OPH:

$$g_{\text{ent}}^2 = \frac{t}{2\pi}$$

In 2D YM, heat-kernel time is $t \sim (g_{\text{YM}}^2 A)/2$.
So OPH's t really does play the role of 2D YM area-time / diffusion parameter.

12.3 Current Algebra Central Charge (WZW)

If edge gauge group embeds into worldsheet current algebra (WZW-type):

$$c = \frac{k \dim \mathfrak{g}}{k + h^\vee}$$

For $SU(3) \times SU(2) \times U(1)$ with $h^\vee_{SU(3)} = 3$, $h^\vee_{SU(2)} = 2$:

$$c_{\text{int}}(k) = \frac{8k}{k+3} + \frac{3k}{k+2} + 1$$

For 4D superstring compactification with internal $c_{\text{int}} \approx 9$:

$$k = \frac{5 + \sqrt{89}}{2} \approx 7.22$$

Integer levels $k = 7$ or 8 bracket $c_{\text{int}} \approx 9$.

Conclusion: OPH gauge content is numerically consistent with reasonable-level current algebra sector inside critical worldsheet theory.

---

13 From 2D Yang-Mills to Critical Superstrings

The derivation chain from OPH to string theory proceeds in two stages.

13.1 Established Results

The following are derived from the five OPH axioms together with the stated regulator, heat-kernel, and large-N premises:

1. Finite microscopic DoF on the screen, type-I local algebras, boundary gauge invariance
2. EC decomposition and Markov structure on collars
3. Sector category and reconstruction of the compact gauge group, with $SU(3)\times SU(2)\times U(1)/\mathbb{Z}_6$ selected under the extended MAR admissibility package (see Part II of this manuscript)
4. Heat-kernel edge-sector weights $p_R \propto d_R e^{-tC_2(R)}$
5. Identification with the 2D Yang-Mills partition function (Theorem 4.1)
6. Geometric modular Hamiltonians with 2π Euclidean regularity and Lorentz symmetry from $\mathrm{Conf}(S^2)$ in the refinement/scaling regime
7. Conditional scaling-limit Einstein branch from entanglement equilibrium
8. Nonabelian gluing (crossed module/2-group) providing higher gauge data

The genus expansion $\log Z = \sum_{h\geq 0} N^{2-2h} F_h(\lambda)$ follows from the Gross-Taylor rewriting of 2D Yang-Mills in the large-N regime, with N identified as the effective edge Hilbert space parameter.

13.2 Extensions to Critical Superstrings

The full path from OPH to critical superstring theory requires:

- **Worldsheet CFT**: Virasoro emergence from modular data, modular invariance from overlap consistency
- **Complete massless spectrum**: metric + B-field + dilaton + RR sectors, anomaly cancellation from worldsheet beta functions
- **Internal sector matching**: the SM gauge content with $c_{\mathrm{int}} \approx 9$ is numerically consistent with integer WZW levels $k = 7$ or 8 (§8.3)

The graviton is contained implicitly via the Einstein equation derived from entanglement equilibrium.

---

14 Summary

This part should be read with the same three-phase boundary as Part I. The recovered-core claim set of OPH is Phase I, namely D1–D5 together with D7–D9. The string/worldsheet material here belongs to Phase III unless and until the additional large-N_{edge}, modular-sewing, and worldsheet-CFT premises are proved.

The status split is therefore:

1. **Secured upstream input.** The theorem-level statement actually available before any string reorganization is that the edge-sector weights have the 2D Yang-Mills heat-kernel form. That identity is the robust bridge from OPH edge data to a worldsheet-like reorganization.

2. **Conditional continuation.** If a controlled large-N_{edge} regime exists, with N_{edge} distinct from the physical color rank $N_c = 3$, then the Gross-Taylor rewriting turns the edge partition function into a genus expansion. This is the precise status of the worldsheet/string reorganization claim.

3. **Not yet claimed.** Critical superstring worldsheet CFT, full massless spectrum, anomaly cancellation, and complete internal-sector matching all require further premises beyond the recovered OPH core.

The broader OPH framework does supply suggestive ingredients for such a continuation — modular geometry, a conditional Einstein branch, and higher-gauge/crossed-module data — but those ingredients do not by themselves upgrade this part beyond Phase III. Success of the branch would strengthen the program; failure of the branch would retract a continuation only and would not by itself falsify relativity-plus-Standard-Model recovery.

---

15 Appendix A: Entanglement First Law Numerical Check

The "entanglement first law" step ($\delta S = \delta\langle K\rangle$) is load-bearing in the Einstein derivation. Here we verify it numerically.

15.1 A.1 Setup

Generate random bipartite density matrix on 4×4 Hilbert space (2 qubits $\times$ 2 qubits). Perturb by small ε and compare:

$$\Delta S_A := S(\rho_A(\varepsilon)) - S(\rho_A(0))$$
$$\Delta\langle K_A\rangle := \mathrm{Tr}[(\rho_A(\varepsilon) - \rho_A(0))K_A]$$

where $K_A = -\log \rho_A(0)$.

15.2 A.2 Expected Result

As $\varepsilon \to 0$:

$$\frac{|\Delta S_A - \Delta\langle K_A\rangle|}{|\Delta\langle K_A\rangle|} \to 0$$

linearly in ε (first-order identity).

15.3 A.3 Code

See `code/entanglement_first_law.py` for implementation.

---

16 Appendix B: Key Equations Summary

Result	Equation
Generalized entropy	$S_{\text{gen}} = \frac{A}{4G} + S_{\text{bulk}}$
Einstein from equilibrium	$G_{ab} + \Lambda g_{ab} = 8\pi G\langle T_{ab}\rangle$
Edge-sector weights	$p_R(t) \propto d_R\, e^{-tC_2(R)}$

Result	Equation
Edge partition function	$Z_{\text{edge}}(t) = \sum_R d_R^2\, e^{-tC_2(R)} = K_t(1)$
2D YM partition function	$Z_{\text{2D YM}}(S^2) = \sum_R d_R^2\, e^{-\frac{g^2 A}{2} C_2(R)}$
Parameter identification	$t = \frac{g_{\text{YM}}^2 A}{2}$
String tension	$T_s = \frac{1}{2\pi\alpha'} \sim \frac{1}{2\pi a_{\text{cell}}}$
Genus expansion	$\log Z = \sum_{h\geq 0} N^{2-2h} F_h(\lambda)$

---

17 References

This derivation uses:

1. Canonical OPH axiom package and theorem-local regulator premises from Part I of this manuscript
2. Heat-kernel / edge-sector results from Part V of this manuscript
3. Gross-Taylor 2D YM / string duality (Gross & Taylor, 1993)
4. Jacobson entanglement equilibrium (Jacobson, 1995, 2016)
5. Standard results: Tomita-Takesaki modular theory, Peter-Weyl theorem, Raychaudhuri equation

Part V
Technical Supplement

18 Technical Supplement: Mathematical Foundations of OPH

OPH is an observer-centric reconstruction program for fundamental physics. The quantitative branches discussed in this paper use two external inputs (P for pixel area and N_{scr} for screen capacity) together with structural axioms and explicit regulator/scaling premises. This supplement collects the mathematical ingredients for the conditional Lorentz/null-modular/Einstein branch, conditional compact gauge reconstruction, and the related bosonic quantitative program.

18.1 For Theoretical Physicists

This document collects mathematical derivations and stated premises supporting the OPH program. The Lorentz/null-modular/Einstein and compact-gauge statements are scaling-limit / categorical results under explicit regularity hypotheses; these hypotheses are target properties of the intended EFT phase and candidate UV completions, not fixed-cutoff matrix-algebra theorems.

Scope convention. Statements about Lorentz recovery, modular flow, and the Einstein branch are scaling-limit / EFT statements rather than literal fixed-cutoff type-I claims. Statements about gauge reconstruction refer to the refinement-stable directed colimit of fixed-cutoff sector categories, not to the finite block list at one regulator scale.

---

19 Core Axioms and Definitions

19.1 Five OPH Axioms

This supplement uses the same five OPH axioms as the main manuscript:

Screen Net: Physical reality is encoded on a horizon screen S^2 carrying a net of von Neumann algebras $P \mapsto \mathcal{A}(P)$ for patches $P \subset S^2$.

Overlap Consistency: For overlapping patches $P_1 \cap P_2 \neq \emptyset$, the restrictions of local states must agree:

$$\omega_1|_{\mathcal{A}(P_1 \cap P_2)} = \omega_2|_{\mathcal{A}(P_1 \cap P_2)}$$

Local MaxEnt and Refinement Stability: At the UV scale the realized state maximizes entropy subject to a finite family of gauge-invariant local constraints, and the resulting family is stable under coarse-graining.

Recoverable Generalized Entropy: A generalized entropy functional exists:

$$S_{\text{gen}}(C) = \frac{A(\partial C)}{4G} + S_{\text{bulk}}(C)$$

satisfying quantum focusing and the recoverability structure used in collar and null-strip arguments.

Minimal Admissible Realization (MAR): On the admissible low-energy branch, the realized sector package is the lexicographically minimal one under the complexity vector $C(\mathfrak{S})$. MAR enters the gauge-selection branch; the Lorentz/null-modular and Einstein discussions below mainly use the first four axioms plus the technical premises stated where needed.

19.2 Technical Premises and Regulator Names

The supplement uses the same identifier scheme as the main manuscript when later derivations cite theorem-local premises directly. The labels used most often here are ROT, RS, LR, N1–N3, R0, and R1. R0/R1 are not extra microphysical axioms, and LR together with the refinement-stable qualifier used later are consequences of the third OPH axiom.

R0 (finite type-I regulator presentation): On a fixed UV cellulation, each cell carries finite local data and can be Hilbertized on a finite-dimensional space $\tilde{\mathcal{H}}_i$. For any finite union R of cells, the extended algebra before quotienting by overlap redundancy is

$$\tilde{\mathcal{A}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R), \qquad \tilde{\mathcal{H}}_R = \bigotimes_{i \subset R} \tilde{\mathcal{H}}_i.$$

R1 (boundary fixed-point realization of overlap gluing): The redundancy in overlap identifications acts only on cut data. On each finite-dimensional chart, its unitary image has compact closure, so for a region $R \subset S^2$ one may represent it by a compact boundary group $G_{\partial R}$ acting on $\tilde{\mathcal{H}}_R$, with physical algebra

$$\mathcal{A}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R)^{G_{\partial R}}.$$

Derived local-Gibbs branch. If the UV constraints are expectations of finitely many gauge-invariant local operators $\{O_a(x)\}$ of support radius $O(\ell_{\mathrm{UV}})$, then the entropy maximizer is

$$\omega_\ell(\lambda) = Z_\ell(\lambda)^{-1} \exp(-K_\ell(\lambda)), \qquad K_\ell(\lambda) = \sum_x \sum_a \lambda_a O_a(x) + \sum_b \mu_b Q_b.$$

Thus the logarithm of the state is a quasi-local UV generator built from the same bounded-support densities that define the branch.

Derived LR / T2 bound. For local observables A_X, B_Y and the automorphism group τ_t^ℓ generated by $K_\ell(\lambda)$, or by any branch generator in the same bounded-support algebraic closure, standard Lieb–Robinson estimates give

$$\|[\tau_t^\ell(A_X), B_Y]\| \le C \, \|A_X\| \, \|B_Y\| \, \min(|X|, |Y|) \, e^{-(d(X,Y) - v_\ell |t|)/\xi}.$$

So finite-velocity quasi-local propagation is internal to the local MaxEnt branch rather than an extra premise.

Derived endpoint control / T3. After removal of the central endpoint term, changing a bounded interval I to I' modifies only an $O(|I' \Delta I|)$ collar of local terms. Hence on any fixed local-energy-bounded domain

$$\left|\langle \psi, (K[I'] - K[I])\phi \rangle\right| \le C_{\psi,\phi,I_{\max}} \, |I' \Delta I|.$$

This is the bounded-interval endpoint-Lipschitz / finite-variation control later used in the null modular derivation.

Derived refinement-stable branch. Because the same finite constraint family is preserved under coarse-graining, the regulator states belong to one common finite-dimensional multiplier family. The realized branch is the stable or fixed branch inside that family. Whenever later sections speak of a refinement-stable directed colimit of edge sectors, the refinement-stable qualifier means persistence along this branch; only transportability, symmetry, bosonic, and fiber-functor clauses remain extra when gauge reconstruction is invoked.

Label convention. LR refers to the derived quasi-local propagation bound above, and the identifier RS refers to the refinement-stable branch condition contained in the third OPH axiom. The remaining nontrivial burden is therefore the controlled refinement limit and the later geometric or null modular premises, not the matrix or fixed-point bookkeeping, quasi-local propagation, or branch stability at one regulator scale.

19.3 Edge-Center Completion (EC)

At fixed regulator scale, overlap consistency itself determines the boundary redundancy. On the ordinary or central-defect branch, EC is the invariant-theoretic consequence of that derived boundary action. Exact Markovity is a further state property and is not implied by EC alone. This is the reading rule for the $R0/R1$ shorthand.

Proposition 1.1a (Derived boundary gluing datum). Choose a finite regulator chart for the patches meeting along a connected cut Σ. Because the local overlap algebras are finite-dimensional matrix algebras, any overlap-consistent recharting is an inner automorphism and is implemented by a unitary on the cut Hilbert space. The compact closure of the subgroup generated by these recharting unitaries is a compact boundary redundancy group K_Σ. If triple-overlap defects are central, the projective composition law lifts to a compact central extension $\widehat{K}_\Sigma$; on the ordinary branch one simply sets $\widehat{K}_\Sigma = K_\Sigma$. A genuinely noncentral 2-group defect is the only obstruction to reducing the overlap transition system to an ordinary compact group action.

Proof. On a finite-dimensional matrix algebra every *-automorphism is inner, so each overlap-consistent recharting is implemented by a unitary on the cut Hilbert space. The subgroup generated by those unitaries has compact closure inside a finite-dimensional unitary group. A central defect gives a projective composition law that lifts to a central extension; a genuinely noncentral defect is the remaining fixed-cutoff obstruction. QED.

Theorem 1.1 (Derived EC decomposition). Under the fixed-cutoff regulator realization of Section 1.2, and on the ordinary or central-defect branch, the collar Hilbert space is

$$\mathcal{H}_{B_\delta} = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K}_\Sigma} \cong \bigoplus_\alpha \left(\mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha}\right),$$

and the center of the collar algebra is generated by the block projectors:

$$Z(\mathcal{A}(B_\delta)) = \bigoplus_\alpha \mathbb{C} \cdot \mathbf{1}_\alpha.$$

The right half-collar carries the contragredient representation because it sees the inverse transport across the same cut.

Proof. Decompose the left boundary data into irreps

$$\tilde{\mathcal{H}}_{B_L} = \bigoplus_\alpha \left(W_\alpha \otimes \mathcal{H}_{b_L^\alpha}\right),$$

and the right boundary data into dual irreps

$$\tilde{\mathcal{H}}_{B_R} = \bigoplus_\beta \left(W_\beta^* \otimes \mathcal{H}_{b_R^\beta}\right).$$

Then

$$\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R} = \bigoplus_{\alpha,\beta} (W_\alpha \otimes W_\beta^*) \otimes (\mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\beta}).$$

By Schur's lemma,

$$(W_\alpha \otimes W_\beta^*)^{\widehat{K}_\Sigma} \cong \begin{cases} \mathbb{C}, & \alpha = \beta, \\ 0, & \alpha \neq \beta, \end{cases}$$

so only matched left/right labels survive, and the surviving block projectors generate the center. QED.

Corollary 1.2 (Exact Markov adds the state factorization). If, in addition,

$$I_\omega(A_\delta : D_\delta \mid B_\delta) = 0,$$

or one passes to the explicitly stated idealized recoverability limit that reduces to exact Markovity, then

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_\alpha p_\alpha \left(\rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta}\right).$$

EC therefore gives the kinematic block decomposition; exact Markovity is the extra state input that gives the HJPW normal form.

Remark. Approximate recoverability gives controlled deviations from this normal form; it is not implied by EC alone. The only remaining fixed-cutoff blocker to an ordinary-group EC theorem is a genuinely noncentral 2-group defect. The later open burden is the controlled refinement or scaling regime and the modular or null-limit premises, not the finite-dimensional collar decomposition itself.

19.4 External Continuous Inputs

In the quantitative branches discussed in this paper, OPH uses exactly two external continuous inputs:

1. **Pixel area:** $P \equiv a_{\text{cell}}/\ell_P^2 \approx 1.63$
2. **Screen capacity:** $N_{\text{scr}} \equiv \log(\dim \mathcal{H}_{\text{tot}}) \sim 10^{122}$

From these:

- Newton's constant: $G = \frac{a_{\text{cell}}}{4\ell(t)}$
- Cosmological constant: $\Lambda = \frac{3\pi}{GN_{\text{scr}}}$

---

20 Emergence of Gravity and Quantum Mechanics

20.1 Local Lorentz Structure

Theorem 2.1 (Conformal Group Isomorphism): The orientation-preserving conformal group of S^2 is isomorphic to the connected Lorentz group:

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$$

This provides the kinematic structure of special relativity.

20.2 Geometric Modular Flow

The collar technology determines the controlled approach to an exact Markov normal form, but by itself it does not force a specific continuum modular geometry. The theorem-scope must therefore be stated as an OPH branch statement. The sharp claim is: if the refinement-stable MaxEnt branch realized by OPH lies in the Bisognano–Wichmann geometric modular class for caps, then that branch supplies the standard cap-preserving conformal dilation and the theorem fixes only its 2π normalization plus the carried collar remainder.

For a shrinking collar family $(A_\delta, B_\delta, D_\delta)$ around ∂C, write

$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega, \qquad r_{\mathrm{FR}}(\varepsilon_\delta) := 2\sqrt{1-e^{-\varepsilon_\delta}} \le 2\sqrt{\varepsilon_\delta},$$

and, when the relevant marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$,

$$\eta_\delta^{\mathrm{M}} := 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta).$$

All exact collar formulas in this subsection are therefore to be read either literally at exact Markovity or along a controlled collar family satisfying

$$\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) \to 0 \qquad (\delta \downarrow 0),$$

with $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M} carried explicitly.

Theorem 2.2 (Geometric modular flow on caps in the OPH Bisognano–Wichmann branch). Assume the OPH axioms, the regulator premises R0–R1, local MaxEnt/refinement stability, the collar-control hypotheses used earlier in the supplement, and the branch condition that the refinement-stable cap net belongs to the Bisognano–Wichmann geometric modular class for caps and their conformal images, so that for each cap $C \subset S^2$ the limiting modular flow is implemented by the standard cap-preserving conformal one-parameter subgroup $\lambda_C(s)$, with geometric parameter s normalized by the null blow-up $v \mapsto e^{-s}v$. If B_C denotes its infinitesimal generator, then

$$K_C = 2\pi B_C.$$

More precisely, at collar scale δ,

$$K_C^{(\delta)} = 2\pi B_C + E_C^{(\delta)},$$

where the carried remainder is controlled on bounded observables by $r_{\mathrm{FR}}(\varepsilon_\delta)$ and on modular-Hamiltonian replacements by η_δ^{M}. Hence $E_C^{(\delta)} \to 0$ on every controlled collar family whose faithfulness bound survives the refinement limit.

Proof. The recoverability/collar analysis localizes the modular defect to the shrinking collar and allows replacement of the physical state by an exact Markov reference with explicit carried error. On the chosen BW refinement branch, the standard cap-preserving conformal subgroup $\lambda_C(s)$ is part of the branch content, so the limiting modular Hamiltonian is $K_C = \kappa B_C$ for some $\kappa > 0$. The modular flow is KMS by definition, and because the geometric parameter s is normalized by unit null dilation on the branch, the usual Bisognano–Wichmann relation fixes $\kappa = 2\pi$. The displayed remainder $E_C^{(\delta)}$ is exactly the carried collar-replacement error, so it vanishes in the controlled refinement limit. QED.

Scope note. The fixed-cutoff OPH collar technology establishes the 2π normalization and the carried collar remainder on the Bisognano–Wichmann geometric modular branch stated above. The branch-selection question is encoded in the stated hypotheses for that branch.

20.3 Null Modular Bridge (EFT Conditions N1-N3)

The fixed-cutoff null bridge reaches the derived half-sided-inclusion step. What is established here is: transferred cut-center data on null strips, exact-or-controlled strip additivity on one inherited strip model, endpoint-Lipschitz control of the renormalized half-line family, and the geometric blow-up argument that turns the null half-line net into a half-sided modular pair, so that Borchers–Wiesbrock yields an explicit positive null-translation generator on its Stone domain. What is not

established here is the downstream density upgrade or the relativistic null-stress identification. This condition package should therefore be read with the following split.

Condition N1 (null-strip center transfer and exact-or-controlled strip additivity). For consecutive null strips

$$I_- = (v_1, v_2), \qquad J = (v_2, v_3), \qquad I_+ = (v_3, v_4),$$

assume the two cuts inherit the same compact overlap-redundancy data as ordinary collars, so that on a compatible type-I regulator presentation

$$\tilde{\mathcal{H}}_J \cong \bigoplus_{\alpha_-,\alpha_+} W^*_{\alpha_-} \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes W_{\alpha_+}$$

and the strip algebra is the commutant of the transported cut action. Then

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-,\alpha_+}}), \qquad Z(\mathcal{A}(J)) = \bigoplus_{\alpha_-,\alpha_+} \mathbb{C}\, P_{\alpha_-,\alpha_+}.$$

If, in addition, each multiplicity space admits an inherited split

$$\mathcal{H}_{j^{\alpha_-,\alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}},$$

with $\mathcal{A}(I_- \cup J)$ acting blockwise only on $\mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}}$ and $\mathcal{A}(J \cup I_+)$ acting blockwise only on $\mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}$, then exact Markov strip states satisfy exact four-term modular additivity on the canonical inherited model:

$$\Delta K_J(\omega) := K_{I_-\cup J\cup I_+}(\omega) - K_{I_-\cup J}(\omega) - K_{J\cup I_+}(\omega) + K_J(\omega) = 0,$$

equivalently up to a central endpoint term. For a general strip state with

$$I(I_- : I_+ \mid J)_\omega \le \varepsilon,$$

define

$$\mathfrak{M}_{I_-:J:I_+} := \{\sigma : \ I(I_- : I_+ \mid J)_\sigma = 0\}, \qquad \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon) := \sup\left\{ \inf_{\sigma\in\mathfrak{M}_{I_-:J:I_+}} \|\rho - \sigma\|_1 : I(I_- : I_+ \mid J)_\rho \le \varepsilon \right\}.$$

Choose $\widetilde{\omega}_J \in \mathfrak{M}_{I_-:J:I_+}$ with

$$\|\omega - \widetilde{\omega}_J\|_1 \le \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon),$$

and set

$$\mathfrak{D}_J(\omega, \widetilde{\omega}_J) := \Delta K_J(\omega) - \Delta K_J(\widetilde{\omega}_J).$$

Then

$$K_{I_-\cup J\cup I_+}(\omega) = K_{I_-\cup J}(\omega) + K_{J\cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\widetilde{\omega}_J) + \mathfrak{D}_J(\omega, \widetilde{\omega}_J),$$

with $K_{\partial,J}(\widetilde{\omega}_J)$ central. If the relevant marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_J(\omega, \widetilde{\omega}_J)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

Independently, Fawzi–Renner recovery supplies a constructive comparison state with

$$r_{\mathrm{FR}}(\varepsilon) = 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Thus N1 is exact only at exact Markovity, or else along a controlled strip family on one fixed inherited strip model with

$$\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon_J) \to 0.$$

Condition N2 (derived half-sided modular inclusion on the null half-line blow-up net). Fix a smooth cut point, choose affine coordinate v on the chosen null generator Ω so that the cut sits at $v = 0$, and define for $a \geq 0$

$$H_a := (a, \infty), \qquad \mathcal{M}_a(\Omega) := \overline{\bigvee_{a<c<d<\infty} \mathcal{A}((c,d),\Omega)}.$$

By isotony of the null interval net,

$$a \leq b \quad \Longrightarrow \quad \mathcal{M}_b(\Omega) \subseteq \mathcal{M}_a(\Omega).$$

On the OPH geometric scaling branch, the blow-up of cap modular flow acts by the null dilation

$$v - v_0 \mapsto e^{-2\pi t}(v - v_0).$$

Hence

$$\sigma^{\omega}_t(\mathcal{M}_a(\Omega)) = \mathcal{M}_{e^{-2\pi t}a}(\Omega).$$

For $t \leq 0$ one has $e^{-2\pi t}a \geq a$, so $H_{e^{-2\pi t}a} \subseteq H_a$ and therefore

$$\sigma^{\omega}_t(\mathcal{M}_a(\Omega)) \subseteq \mathcal{M}_a(\Omega).$$

Thus $\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega)$ is half-sided modular. After the harmless convention change $t \mapsto -t$, this is the standard positive-time form. The old N2 premise is therefore derived from OPH null-interval structure, isotony, and the scaling-limit geometric action rather than imported separately.

Condition N3 (endpoint-Lipschitz control and weak tail generator). For renormalized modular Hamiltonians

$$\widetilde{K}[I,\Omega] := K[I,\Omega] - K_\partial(I,\Omega), \qquad \widetilde{K}_a(\Omega) := \widetilde{K}[(a,\infty),\Omega],$$

bounded intervals satisfy

$$\left|\langle\psi, (\widetilde{K}[(a',b'),\Omega] - \widetilde{K}[(a,b),\Omega])\phi\rangle\right| \leq C_{\psi,\phi,\Omega}(|a'-a| + |b'-b|),$$

and half-lines satisfy

$$\left|\langle\psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle\right| \leq C_{\psi,\phi,\Omega}|a'-a|.$$

Hence, for fixed ψ, ϕ,

$$f_{\psi,\phi}(a) := \langle\psi, \widetilde{K}_a(\Omega)\phi\rangle$$

is locally Lipschitz and absolutely continuous, and its distributional derivative defines the weak tail generator

$$\langle\psi, q(a,\Omega)\phi\rangle := -\frac{1}{2\pi}\,\partial_a f_{\psi,\phi}(a).$$

For $a < b$,

$$\langle\psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle = -2\pi \int_a^b \langle\psi, q(v,\Omega)\phi\rangle\, dv.$$

This is the end of the fixed-cutoff analytic part of the bridge. The stronger assumptions

$$q \text{ weakly differentiable}, \qquad f_{\psi,\phi}(a) \to 0, \qquad q_{\psi,\phi}(a) \to 0 \quad (a \to +\infty)$$

would permit the downstream density formula

$$\langle\psi, p(a,\Omega)\phi\rangle := -\partial_a\langle\psi, q(a,\Omega)\phi\rangle = \frac{1}{2\pi}\,\partial_a^2 f_{\psi,\phi}(a), \qquad \langle\psi, \widetilde{K}_a(\Omega)\phi\rangle = 2\pi\int_a^\infty (v-a)\,\langle\psi, p(v,\Omega)\phi\rangle\,dv,$$

but that density upgrade is not established here.

Imported Borchers consequence. Let $\Delta_0(\Omega)$ be the modular operator of the standard pair $(\mathcal{M}_0(\Omega),\omega)$, define $K_0(\Omega) := -\log\Delta_0(\Omega)$, and let $\Delta_a(\Omega)$ denote the modular operator of $(\mathcal{M}_a(\Omega),\omega)$. Applying Borchers–Wiesbrock to the derived half-sided pair of Condition N2 yields a unique strongly continuous unitary group $U_\Omega(a) = e^{iaP_\Omega}$ with positive self-adjoint Stone generator P_Ω on

$$D(P_\Omega) := \left\{\psi\in\mathcal{H} : \lim_{a\to 0}\frac{U_\Omega(a)\psi-\psi}{ia} \text{ exists}\right\}.$$

It obeys

$$\Delta_0(\Omega)^{it}U_\Omega(a)\Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t}a), \qquad [K_0(\Omega), P_\Omega] = -\,i\,2\pi P_\Omega,$$

and if $K_a(\Omega) := -\log\Delta_a(\Omega)$, then

$$K_a(\Omega) = K_0(\Omega) - 2\pi a\,P_\Omega$$

as a quadratic-form identity on $D(K_0(\Omega))\cap D(P_\Omega)$.

Consequence. This condition package proves the fixed-cutoff strip bridge through the weak tail generator $q(a,\Omega)$, derives half-sided modular inclusion on the null half-line blow-up net, and therefore identifies an explicit positive null-translation generator P_Ω together with its Stone domain and half-line modular-Hamiltonian relation. The remaining downstream burdens are the density-upgrade hypotheses and the relativistic EFT null-stress premise. Those later steps are explicit rather than hidden parts of N1–N3.

20.4 Stress Tensor Reconstruction

Lemma 2.3 (Null Data Ambiguity): If a symmetric tensor X_{ab} satisfies $X_{ab}k^ak^b = 0$ for all null k, then:

$$X_{ab} = \phi\,g_{ab}$$

for some scalar ϕ.

Proof: Decompose $X_{ab} = Y_{ab} + \phi g_{ab}$ with Y traceless. For null $k = (1,\hat{n})$:

$$X_{kk} = Y_{kk} + \phi g_{kk} = Y_{kk}$$

since $g_{kk} = 0$. Expanding:

$$Y_{kk} = Y_{00} + 2\hat{n}^iY_{0i} + \hat{n}^i\hat{n}^jY_{ij}$$

If this vanishes for all $\hat{n}\in S^2$, each angular moment (scalar, vector, traceless tensor) vanishes independently, so $Y_{ab} = 0$. □

Corollary 2.4: Null modular data determine T_{ab} only up to ϕg_{ab}. Consequently, the Einstein equation is fixed only up to Λg_{ab}.

Reconstruction Formulas: From $f(\hat{n}) := T_{kk}(\hat{n})$:

$$T_{0i} = \frac{3}{2}\langle\hat{n}_i f\rangle_{S^2}$$

$$T_{ij} - \frac{\delta_{ij}}{3}T_{kk}^{(\text{spatial})} = \frac{15}{2}\left\langle\left(\hat{n}_i\hat{n}_j - \frac{\delta_{ij}}{3}\right)f\right\rangle_{S^2}$$

20.5 Entanglement Equilibrium and Einstein's Equation

Theorem 2.5 (Conditional Jacobson-type Derivation): In the same local Lorentzian scaling regime, under MaxEnt state selection with generalized entropy stationarity, the null-stress identification premise of the relativistic EFT branch, and admissible first variations about a maximally symmetric reference state,

$$\delta S_{\text{gen}} = 0$$

implies the rest-frame first-variation relation

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle.$$

If this first-variation relation holds for all local directions and reference states in the scaling regime, overlap consistency upgrades it to the full tensor equation

$$G_{ab} + \Lambda g_{ab} = 8\pi G\langle T_{ab}\rangle$$

modulo the expected Λg_{ab} ambiguity.

Derivation sketch:

1. Raychaudhuri equation relates area change to R_{kk}:
$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma^2 - R_{kk}$$
2. Recoverable Generalized Entropy implies:
$$\frac{d}{d\lambda}\left(\frac{A}{4G} + S_{\text{bulk}}\right) \leq 0$$
3. Stationarity at MaxEnt gives:
$$R_{kk} = 8\pi G\langle T_{kk}\rangle$$
4. By Lemma 2.3, this fixes Einstein's equation up to Λg_{ab}.

20.6 Newton's Constant from Edge Entropy

Theorem 2.6: The gravitational coupling is fixed by the edge entropy density:

$$G = \frac{a_{\text{cell}}}{4\bar{\ell}(t)}$$

where $\bar{\ell}(t) = a_{\text{cell}}/(4\ell_P^2)$ is the dimensionless entropy per cell.

With $a_{\text{cell}} \approx 1.63\ell_P^2$:

$$\bar{\ell} = \frac{1.63}{4} \approx 0.4077$$

---

21 The Measurement Problem

21.1 Records as Central Variables

Definition 3.1: A *record* in patch P is a family of orthogonal projectors $\{\Pi_r\} \subset \mathcal{A}(P)$ such that:

1. **Classicality:** $\mathcal{R} := \text{vN}(\{\Pi_r\})$ is approximately abelian
2. **Stability:** $\{\Pi_r\}$ are stable under effective dynamics
3. **Shareability:** Π_r lies in the center of overlap algebras

21.2 Markov Structure Theorem

Theorem 3.1 (Quantum Markov chain structure on a fixed EC collar). If $I(A:C \mid B)=0$ for a tripartite state ρ_{ABC}, then

$$\mathcal{H}_B \cong \bigoplus_j \left(\mathcal{H}_{b_L^{(j)}} \otimes \mathcal{H}_{b_R^{(j)}}\right)$$

and the state decomposes as

$$\rho_{ABC} = \bigoplus_j q_j\, \rho^{(j)}_{Ab_L^{(j)}} \otimes \rho^{(j)}_{b_R^{(j)}C}.$$

The label j is a classical variable living in the center of $\mathcal{A}(B)$.

On the fixed finite-dimensional collar Hilbert space supplied by EC, let

$$\mathfrak{M}_{A:B:C} := \{\sigma_{ABC} : I(A:C \mid B)_\sigma = 0\}$$

and define the exact-Markov distance modulus

$$\delta^{\mathrm{M}}_{A:B:C}(\varepsilon) := \sup\left\{\inf_{\sigma\in\mathfrak{M}_{A:B:C}} \|\rho-\sigma\|_1 : I(A:C \mid B)_\rho \le \varepsilon\right\}.$$

Because the state space is compact and conditional mutual information is continuous in finite dimension,

$$\delta^{\mathrm{M}}_{A:B:C}(\varepsilon) \longrightarrow 0 \qquad (\varepsilon \downarrow 0).$$

This is the precise sense in which approximate collars converge to the exact HJPW normal form: either the state is exactly Markov, or one works on one fixed finite-dimensional collar, or a compatible pulled-back family, and sends $\varepsilon \to 0$. The supplement does *not* claim a universal dimension-free one-shot trace-norm bound from small $I(A:C \mid B)$ directly to an exact Markov state.

21.3 Approximate Markov and Recovery

Theorem 3.2 (Fawzi–Renner recovery). If

$$I(A:C \mid B)_\rho \le \varepsilon,$$

then there exists a CPTP map $\mathcal{R} : \mathcal{A}(B) \to \mathcal{A}(BC)$ such that the recovered comparison state

$$\rho^{\mathrm{rec}}_{ABC} := (\mathrm{id}_A \otimes \mathcal{R})(\rho_{AB})$$

obeys

$$\|\rho_{ABC} - \rho^{\mathrm{rec}}_{ABC}\|_1 \le 2\sqrt{1-e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

If ε is quoted in bits rather than nats, this bound is equivalently written as $2\sqrt{\ln 2\,\varepsilon}$.

The recovered state is a constructive comparison state. It need not be exact Markov. To connect approximate recoverability to the exact splice or additivity formulas used later, one must combine Theorem 3.2 with the fixed-collar modulus of Section 3.2. Concretely, if $\sigma_\varepsilon \in \mathfrak{M}_{A:B:C}$ is chosen so that

$$\|\rho - \sigma_\varepsilon\|_1 \le \delta^{\mathrm{M}}_{A:B:C}(\varepsilon),$$

then:

1. every bounded observable supported on the left side of the collar differs from its exact-Markov splice value by at most $\|X\|_\infty \, \delta^{\mathrm{M}}_{A:B:C}(\varepsilon)$;
2. if the relevant marginals are uniformly faithful with lower spectral bound $\lambda_* > 0$, then the corresponding modular-additivity defect differs from the exact Markov value by at most

$$4\lambda_*^{-1} \, \delta^{\mathrm{M}}_{A:B:C}(\varepsilon).$$

Thus the supplement carries two distinct errors:

$$r_{\mathrm{FR}}(\varepsilon) = O(\varepsilon^{1/2})$$

for constructive one-shot recovery, and

$$\delta^{\mathrm{M}}_{A:B:C}(\varepsilon) \to 0$$

for convergence to the exact Markov normal form on a fixed collar. This is the controlled bridge from approximate recoverability to the exact splice and modular-additivity statements used by the later gravity and consensus arguments.

21.4 Definiteness of Outcomes

Theorem 3.3 (Collapse from EC Structure): Under EC, the post-measurement physical state is exactly a convex mixture over sector labels α:

$$\rho = \bigoplus_\alpha p_\alpha \, \rho^{(\alpha)}$$

For any observer whose accessible observables lie in either side algebra:

1. There are **no interference observables** between different α-blocks
2. Each observer-instance has a definite α in the classical sense

Proof: Cross-block operators $|i\rangle\langle j|$ for $i \in \alpha, j \in \alpha'$ $(\alpha \neq \alpha')$ are not gauge-invariant, hence not in the physical algebra. □

21.5 Born Rule as Unique Probability Measure

Theorem 3.4 (Gleason): For $\dim \mathcal{H} \geq 3$, any measure μ on projectors satisfying:

1. Noncontextuality: $\mu(P)$ depends only on P
2. Additivity: $\mu(P \vee Q) = \mu(P) + \mu(Q)$ for $P \perp Q$
3. Normalization: $\mu(\mathbf{1}) = 1$

must have the form:

$$\mu(P) = \mathrm{Tr}(\rho P)$$

for some density operator ρ.

Corollary 3.5: In OPH with the type-I regulator premise (R0), Born's rule is the unique overlap-consistent probability assignment.

21.6 Collapse as Conditioning

Proposition 3.6 (Lüders Update): For record projectors $\{\Pi_r\}$ and state ω:

$$p_r = \omega(\Pi_r), \qquad \omega_r(X) = \frac{\omega(\Pi_r X \Pi_r)}{\omega(\Pi_r)}$$

Then for all X commuting with $\mathcal{R}$:

$$\omega(X) = \sum_r p_r \, \omega_r(X)$$

This is classical conditional probability on the record algebra.

---

22 The Cosmological Principle

22.1 MaxEnt Symmetry Inheritance

Lemma 4.1: Let G act on the screen by automorphisms α_g. If:

1. The constraint set is G-invariant
2. Von Neumann entropy is G-invariant: $S(\alpha_g(\rho)) = S(\rho)$

Then the MaxEnt optimizer ω can be taken G-invariant. If unique, it is automatically G-invariant.

Proof: If ω is a maximizer, so is $\omega \circ \alpha_{g^{-1}}$ by constraint invariance and entropy invariance. By uniqueness, $\omega = \omega \circ \alpha_{g^{-1}}$. $\square$

22.2 Isotropy from SO(3)-Invariant Constraints

Premise ROT: Constraints are SO(3)-invariant on S^2.

Theorem 4.2: Under premise ROT and MaxEnt uniqueness, the reference state ω satisfies:

$$\omega \circ \alpha_g = \omega \quad \forall g \in \mathrm{SO}(3)$$

The stress tensor has perfect-fluid form:

$$\langle T_{ab} \rangle = \rho \, u_a u_b + p(g_{ab} + u_a u_b)$$

Proof of perfect-fluid form: Under SO(3), rank-2 tensors decompose into scalar + vector + traceless tensor irreps. Only scalars are invariant. Hence $\langle T_{0i} \rangle = 0$ and $\langle T_{ij} \rangle \propto \delta_{ij}$. $\square$

22.3 Homogeneity from Isotropy Everywhere

Theorem 4.3 (Schur-type): Let (Σ, h_{ij}) be a 3-dimensional Riemannian manifold. If at every point $p \in \Sigma$, the curvature tensor is SO(3)-invariant (pointwise isotropy), then:

$$R_{ijkl}(p) = K(p)(h_{ik}h_{jl} - h_{il}h_{jk})$$

and by the second Bianchi identity, $\nabla_m K = 0$, so $K = \text{const}$.

Corollary 4.4: If OPH supplies isotropy for all observers, spatial geometry is a constant-curvature space form: S^3, $\mathbb{R}^3$, or H^3.

22.4 FLRW Emergence

Theorem 4.5: With:

1. Semiclassical Einstein equation from recoverable generalized entropy / entanglement equilibrium
2. Perfect-fluid stress tensor from SO(3)-invariant MaxEnt
3. Positive Λ from finite screen capacity

The emergent geometry is FLRW:

$$ds^2 = -dt^2 + a(t)^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right)$$

with Friedmann equations:

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}$$

$$\dot{H} - \frac{k}{a^2} = -4\pi G(\rho + p)$$

---

23 Supersymmetry and Gauge Coupling Unification

23.1 One-Loop Unification Algebra

At one loop, if couplings unify at (M_U, α_U):

$$\alpha_i^{-1}(M_Z) = \alpha_U^{-1} + \frac{b_i}{2\pi} \ln \frac{M_U}{M_Z}$$

Define $A_i := \alpha_i^{-1}(M_Z)$, $L := \ln(M_U/M_Z)$. Eliminating α_U:

$$L = \frac{2\pi}{b_1 - b_2}(A_1 - A_2)$$

The consistency relation for α_s (a consistency check when all three couplings calibrate P; a genuine prediction in the two-input mode):

$$A_3^{\text{pred}} = \frac{b_3 - b_2}{b_1 - b_2} A_1 + \frac{b_1 - b_3}{b_1 - b_2} A_2$$

23.2 SM vs MSSM Coefficients

Model	(b_1, b_2, b_3)	$\alpha_s(M_Z)^{\text{pred}}$
SM	$(41/10, -19/6, -7)$	≈ 0.071
MSSM	$(33/5, 1, -3)$	≈ 0.116
Observed	,	0.1179 ± 0.0010

The SM prediction is catastrophically wrong; MSSM-like coefficients work.

23.3 Heat-Kernel Edge Sector Weights

Theorem 5.1: Under MaxEnt with bi-invariant constraints on a compact Lie group G, the sector probabilities take heat-kernel form:

$$p_R(t) \propto d_R\, e^{-t\, C_2(R)}$$

where $d_R = \dim R$ and $C_2(R)$ is the quadratic Casimir.

Lemma 5.2 (Bi-invariant Operators): If $G = \prod_i G_i$ is a compact semisimple Lie group written as a product of compact simple factors, then any bi-invariant, second-order differential operator on G has the form:

$$D = c_0 \mathbf{1} - \sum_i c_i \Delta_{G_i}$$

where Δ_G is the Laplace-Beltrami operator.

Proof: Bi-invariance $\Leftrightarrow$ $D \in Z(U(\mathfrak{g}))$ (center of universal enveloping algebra). For degree ≤ 2:

- Linear terms vanish (no invariant vectors in adjoint rep)
- Quadratic terms are proportional to the Killing form on each simple factor, giving one independent coefficient per factor

The Casimir element acts as $-\Delta_G$ in the regular representation. $\square$

23.4 Peter-Weyl and Effective Multiplicity

By Peter-Weyl decomposition, the effective refinement-limit edge representation space is

$$L^2(G) \cong \bigoplus_R V_R \otimes V_R^*$$

Key insight: Entanglement traces over one factor give multiplicity d_R in p_R, but loops see both factors, giving effective multiplicity:

$$N_{\text{eff}}(R) = d_R \cdot p_R$$

23.5 Beta Function Shifts

The one-loop beta shift from edge sectors:

$$\Delta b_a = \sum_R p_R \cdot d_R \cdot T_a(R)$$

where $T_a(R)$ is the Dynkin index for gauge factor a.

At the unification diffusion parameter $t_U \approx 1.64$:

$$\Delta b \approx (2.49, 4.38, 3.97) \quad \text{vs MSSM} \quad (2.50, 4.17, 4.00)$$

With fermionic grading (restricting to half-integer SU(2) sectors):

$$\Delta b \approx (2.50, 4.17, 3.97)$$

matching MSSM within 1%.

---

24 The Horizon Problem

24.1 Anisotropy Bound from Markov Control

Theorem 6.1: For a tripartition A-B-D with collar width δ, the Markov error satisfies:

$$I(A:D|B) \lesssim c \cdot |\partial C|_{\text{UV}} \cdot e^{-\delta/\xi}$$

where ξ is the correlation length.

Corollary 6.2: For bounded observable O with $\|O\| \sim 1$:

$$|\Delta\langle O\rangle| \leq 2\sqrt{\ln 2 \cdot \varepsilon}$$

24.2 CMB Anisotropy Requirement

For $\delta T/T \lesssim 10^{-5}$:

$$2\sqrt{\ln 2 \cdot \varepsilon} \lesssim 10^{-5}$$

$$\varepsilon \lesssim 3.61 \times 10^{-11} \text{ bits}$$

Required collar width:

$$\frac{\delta}{\xi} \gtrsim \ln\left(\frac{c \cdot |\partial C|_{\text{UV}}}{\varepsilon_{\max}}\right) \approx 24$$

With $\xi \approx \sqrt{a_{\text{cell}}} \cdot \ell_P \approx 1.28\ell_P$:

$$\delta_{\text{CMB}} \approx 31\ell_P$$

24.3 Homogeneity as MaxEnt Default

Theorem 6.3: If MaxEnt constraints are uniform across UV cells with no marked points, the MaxEnt state is invariant under triangulation automorphisms. In the continuum limit, this gives SO(3) invariance.

Proof: Standard MaxEnt uniqueness + symmetry inheritance (Lemma 4.1). □

---

25 Black Hole Information Paradox

25.1 The Information Trilemma

The AMPS-style paradox assumes:

1. Outgoing mode B entangled with interior partner A (smooth horizon)
2. Late radiation B entangled with early radiation R (unitarity)
3. Monogamy: B cannot be maximally entangled with both

The false assumption: $\mathcal{H} \stackrel{?}{=} \mathcal{H}_{\text{inside}} \otimes \mathcal{H}_{\text{outside}} \otimes \mathcal{H}_{\text{radiation}}$

25.2 Edge-Center Resolution

Theorem 7.1: Under EC, the collar decomposition gives:

$$\rho_{A_\delta B_\delta D_\delta} = \bigoplus_\alpha p_\alpha \left(\rho_{A_\delta b_L^\alpha} \otimes \rho_{b_R^\alpha D_\delta}\right)$$

The "glue" between inside and outside is the edge sector label α in the center. Conditional on α, inside and outside factorize.

25.3 Recoverability of Interior

By the recoverability clause of the Recoverable Generalized Entropy axiom (small CMI) + Theorem 3.2:

$$\|\rho_{ABC} - (\mathrm{id}_A \otimes \mathcal{R})(\rho_{AB})\|_1 \leq 2\sqrt{\ln 2 \cdot \varepsilon}$$

Interpretation: The "interior partner" is recoverable from outside + radiation. It's encoded, not independent.

25.4 Hawking Temperature from Modular Theory

Theorem 7.2: The outside algebra is in a KMS state at inverse temperature $\beta = 2\pi/\kappa$ with respect to the horizon Killing generator.

Derivation: Near the horizon:

$$ds^2 \approx -\kappa^2\rho^2 dt^2 + d\rho^2 + r_H^2 d\Omega^2$$

Euclidean regularity at $\rho = 0$ requires $\kappa t_E \sim \kappa t_E + 2\pi$, hence:

$$T_H = \frac{\kappa}{2\pi} = \frac{\hbar c^3}{8\pi G k_B M}$$

25.5 Discrete Area Spectrum

From the central area operator:

$$L_C = \sum_\alpha (\log d_\alpha) P_\alpha$$

Area eigenvalues: $A_\alpha = 4G \log d_\alpha = 4\ell_P^2 \ln d_\alpha$

For Schwarzschild, a sector transition $d \to d'$ gives:

$$\Delta E = k_B T_H \ln(d'/d)$$

---

26 The Cosmological Constant Problem

26.1 Vacuum Energy Blindness

Lemma 8.1: For any null vector k:

$$T_{kk}^{\mathrm{vac}} = T_{ab}^{\mathrm{vac}} k^a k^b = -\rho_{\mathrm{vac}} g_{ab} k^a k^b = -\rho_{\mathrm{vac}}(k \cdot k) = 0$$

Vacuum energy contributions lie in the kernel of the null map $T_{ab} \mapsto T_{kk}$.

26.2 Structural Separation

Proposition 8.2: In OPH:

- Local modular/null data fixes T_{ab} up to ϕg_{ab}
- Λ is fixed by global screen capacity, not local physics

The "huge zero-point energy" simply isn't the thing determining curvature.

26.3 Λ from Capacity

$$\Lambda = \frac{3\pi}{GN_{\text{scr}}}$$

Equivalently, using de Sitter entropy:

$$S_{dS} = \frac{A_{dS}}{4G} = \frac{3\pi}{G\Lambda} = N_{\text{scr}} = \log(\dim \mathcal{H}_{\text{tot}})$$

For observed $\Lambda \approx 1.09 \times 10^{-52}$ m^{-2}:

$$N_{\text{scr}} \sim 10^{122}$$

26.4 No-Go for Local Λ Prediction

Theorem 8.3: Within OPH's null-modular reconstruction:

1. All T_{kk} data unchanged under $T_{ab} \to T_{ab} + \phi g_{ab}$
2. Therefore Λ cannot be fixed by local overlap consistency
3. Λ requires a global input (capacity)

---

27 Dark Matter as Modular Anomaly

Sections 9–13 mix two kinds of material: (i) brief reminders of upstream structural results proved elsewhere in the OPH authority ladder, and (ii) genuinely downstream benchmark branches. Only the latter — dark-sector response laws, proton-spin bookkeeping, baryogenesis estimates, and Yukawa/Koide-style continuations beyond the proved structural core — should be read here as Phase-III / deferred-continuation material unless a subsection explicitly states otherwise.

27.1 The Modular Additivity Defect

Define the collar modular additivity defect:

$$\Delta K_\delta := K_{ABD} - K_{AB} - K_{BD} + K_B$$

Theorem 9.1: $\langle \Delta K_\delta \rangle_\omega = -I(A : D|B)_\omega$
This is the conditional mutual information, measuring Markov imperfection.

27.2 Anomalous Stress-Energy

The modified Einstein equation:

$$G_{ab} + \Lambda g_{ab} = 8\pi G \left(\langle T_{ab} \rangle + \langle T_{ab}^{\text{anom}} \rangle \right)$$

where in the diamond rest frame:

$$\langle T_{00}^{\text{anom}} \rangle := \frac{15}{8\pi^2} \frac{\delta \langle K_C^{(\text{anom})} \rangle}{\ell^4}$$

The coefficient $15/(8\pi^2)$ comes from inverting the $d = 4$ geometric integral coefficient $\Omega_2/(4^2 - 1) = 4\pi/15$.

27.3 Properties of the Anomaly Sector

1. **Gravitates:** Appears on RHS of Einstein equation
2. **Dark by construction:** Encoded in central/recoverability data, not SM fields
3. **Covariantly conserved:** $\nabla^a T_{ab}^{\text{anom}} = 0$ (from Bianchi identity)
4. **Effectively classical:** Central structure implies classical labels

27.4 The MOND Acceleration Scale

Corollary 9.2 (program-level): The unique IR acceleration scale available to the modular-anomaly branch from Λ is

$$a_0^{(\text{OPH})} := \frac{15}{8\pi^2} c^2 \sqrt{\frac{\Lambda}{3}} = \frac{15}{8\pi^2} \frac{c^2}{r_{dS}}$$

Numerical evaluation:

$$a_0^{(\text{OPH})} \approx 1.03 \times 10^{-10} \text{ m/s}^2$$

Compare to observed: $a_0^{(\text{obs})} \approx 1.2 \times 10^{-10}$ m/s^2 (within 15%).

27.5 Polarization Response (MOND-like Scaling)

Uniqueness argument: In the deep-IR regime, the only scales are a_0 and g_b (Newtonian baryonic acceleration). Dimensional analysis + scale invariance forces:

$$g_{\text{anom}} \propto \sqrt{a_0 \cdot g_b}$$

The cubic polarization functional:

$$S_{\text{pol}}[\varphi] = \int d^3x \left[-\frac{1}{12\pi G a_0} |\nabla \varphi|^3 - \rho_b \varphi \right]$$

yields the MOND/RAR form:

$$g_{\text{obs}} \approx g_b + \sqrt{a_0 g_b}$$

---

28 Proton Stability and Spin

28.1 Sector Factorization and Product Group

Theorem 10.1 (Factorization Equivalence):

$$\text{Factorizing edge weights} \Leftrightarrow \text{Additive boundary Laplacian} \Leftrightarrow \text{Product gauge group}$$

If $H_\partial = H_\partial^{(1)} + H_\partial^{(2)} + H_\partial^{(3)}$ with $[H_\partial^{(i)}, H_\partial^{(j)}] = 0$:

$$p(R_1, R_2, R_3) \propto \prod_{i=1}^{3} d_{R_i} e^{-t_i C_2(R_i)}$$

Applied to the refinement-stable edge-sector colimit, Tannaka-Krein reconstruction first yields some compact group G. Under the extended MAR admissibility package, and once the admissible class includes one connected abelian charge factor, the connected Lie realization selected by MAR is $SU(3) \times SU(2) \times U(1)$ (up to finite quotient).

Under the extended theory T_{ext}, the explicit MAR selector picks this product structure on the admissible class: the minimal coupled carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$ enforces commuting color and weak actions, which implies the additive boundary Laplacian. See Part II of this manuscript for the complete proof.

28.2 No Leptoquarks, No Gauge Proton Decay

Corollary 10.2: With product gauge group, the adjoint representation of the full connected gauge group is

$$(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0)$$

equivalently, the adjoint of the derived gauge group is

$$(8, 1, 0) \oplus (1, 3, 0).$$

No gauge generators in mixed representations $(3, 2, \pm 5/6)$ (the SU(5) X, Y bosons). Hence:

$$\tau_p^{(\text{gauge})} = \infty$$

28.3 Proton Spin Fraction

Casimir Equilibration Ansatz: Spin sharing between quarks and gluons equilibrates according to coupling strength:

$$\Delta\Sigma \approx \frac{C_F}{C_F + C_A}$$

For SU(3): $C_F = 4/3$, $C_A = 3$:

$$\Delta\Sigma = \frac{4/3}{4/3 + 3} = \frac{4}{13} \approx 0.308$$

Compare to lattice: $\Delta\Sigma \approx 0.286$ (within 8%).

---

29 Baryogenesis

29.1 The Z_6 Defect Suppression

The SM global gauge structure is $(G_{\mathrm{SM}})/\mathbb{Z}_6$. The entropy deficit from the $\mathbb{Z}_6$ quotient:

$$\Delta S = \ln 6$$

Defect suppression factor:

$$\varepsilon = e^{-\Delta S} = \frac{1}{6}$$

29.2 The Baryon-Violating Channel

The electroweak topological transition ($\Delta N_{CS} = \pm 1$) produces chiral zero modes. By index theorem:

- One zero mode per left-handed SU(2) doublet per unit topological charge
- Per generation: $N_c + 1 = 4$ doublets
- Total: $n_{\mathrm{doublets}} = (N_c + 1)N_g = 12$

The 't Hooft vertex is a 12-fermion interaction.

29.3 Defect-Order Identification

Proposition 11.1: The baryon-violating channel corresponds to defect order $n = 12$:

$$\varepsilon^{12} = 6^{-12} \approx 4.59 \times 10^{-10}$$

Compare to observed $\eta_B \approx 6.1 \times 10^{-10}$ (same order of magnitude).

29.4 CP Bias from Overlap Holonomy

Under CP: $U_{ij} \mapsto U_{ij}^*$, $\Omega_{ijk} \mapsto \Omega_{ijk}^\dagger$

For central cocycle $z_{ijk} \in U(1)$: CP sends $z \to z^{-1}$.

Real cocycle condition: $[z] = [z]^{-1}$ (cohomologous to inverse)

Generic phases violate this $\Rightarrow$ CP violation is generic in OPH.

---

30 Generation Structure and Yukawa Hierarchy

30.1 Why Three Generations

Theorem 12.1 (N_g = 3 Selection):

1. **CP violation requires N_g $\geq$ 3:** Physical CKM phase exists only for $N_g \geq 3$
2. **UV stability bounds N_g $\leq$ 5:** SU(2) asymptotic freedom
3. **Minimality selects N_g = 3:** Extra generations introduce unconstrained relevant deformations

30.2 Yukawa as Defect-Mediated Overlap

Theorem 12.2: If a Yukawa coupling requires n units of $\mathbb{Z}_6$ defect insertion:

$$y \propto \langle D^n \rangle = 6^{-n}$$

The suppression factor is fixed by topology, not tuned.

30.3 The Yukawa Spectrum

Fermion	Yukawa	$n = -\ln y / \ln 6$	Nearest integer
t	0.992	0.00	0
b	0.024	2.08	2
c	0.0073	2.75	3
s	5.3×10^{-4}	4.21	4
d	2.7×10^{-5}	5.87	6
u	1.3×10^{-5}	6.29	6
τ	0.010	2.56	3
μ	6.1×10^{-4}	4.13	4
e	2.9×10^{-6}	7.11	7

The base-6 logarithm lands near integers across the spectrum.

30.4 Deterministic Completion (RK Hamiltonian)

Define configuration weight:

$$w(\mathcal{C}) = \exp\left(-t \sum_{\text{plaquettes}} C_2(R_p)\right) \times 6^{-n_D(\mathcal{C})}$$

The Rokhsar-Kivelson ground state:

$$|\Psi\rangle = \frac{1}{\sqrt{Z}} \sum_{\mathcal{C}} \sqrt{w(\mathcal{C})} |\mathcal{C}\rangle$$

is the unique ground state of a frustration-free parent Hamiltonian, replacing MaxEnt ensemble with a deterministic pure state.

---

31 The Koide Formula

31.1 Numerical Verification

Using PDG masses:

- $m_e = 0.51099895$ MeV
- $m_\mu = 105.6583755$ MeV

- $m_\tau = 1776.93$ MeV

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = 0.6666644645$$

Compare to $2/3 = 0.6666666667$. Difference: 2.2×10^{-6} (within 1σ).

31.2 Z_3 Holonomy Parameterization

Theorem 13.1: The minimal Hermitian circulant on generation space:

$$\Phi = a \cdot I + b \cdot P + b^* \cdot P^2$$

with P the cyclic shift, has eigenvalues:

$$\lambda_k = a + 2|b| \cos\left(\delta + \frac{2\pi k}{3}\right)$$

where $\delta = \arg(b)$.

31.3 Why Q = 2/3

Theorem 13.2: With $r_k = \lambda_k$ (root masses):

$$Q = \frac{\sum_k r_k^2}{(\sum_k r_k)^2} = \frac{1 + 2(|b|/a)^2}{3}$$

Setting $Q = 2/3$:

$$|b|/a = \frac{1}{\sqrt{2}}$$

This is the "balanced" configuration where singlet and charged modes have equal norm.

31.4 The Koide Phase from OPH

Proposition 13.3: The holonomy phase:

$$\delta_{\text{OPH}} = \frac{\beta_{\text{EW}} \cdot Y_Q}{N_g} = \frac{(N_c + 1)}{2N_c \cdot N_g}$$

With $N_c = 3$, $N_g = 3$:

$$\delta_{\text{OPH}} = \frac{4}{18} = \frac{2}{9} = 0.2222...$$

Experimental extraction: $\delta_{\text{exp}} = 0.2222248 \pm 0.0000063$
Agreement within 0.4σ.

---

32 Cosmological Parameters

32.1 The de Sitter Scale

From $\Lambda \approx 1.09 \times 10^{-52}$ m^{-2}:

$$r_{dS} = \sqrt{\frac{3}{\Lambda}} \approx 1.66 \times 10^{26} \text{ m}$$

The de Sitter time:

$$t_\Lambda = \frac{r_{dS}}{c} \approx 5.53 \times 10^{17} \text{ s} \approx 17.5 \text{ Gyr}$$

32.2 Emergent Age of the Universe

For flat ΛCDM:

$$t_0 = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \sinh^{-1}\left(\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}\right)$$

With $H_0 \approx 67.4$ km/s/Mpc, $\Omega_\Lambda \approx 0.685$:

$$t_0 \approx 13.8 \text{ Gyr}$$

32.3 Screen Capacity

$$\dim \mathcal{H}_{\text{tot}} = \exp(S_{dS}) = \exp(N_{\text{scr}}) = \exp\left(\frac{3\pi}{G\Lambda}\right)$$

$$N_{\text{scr}} = \log(\dim \mathcal{H}_{\text{tot}}) \approx 2.85 \times 10^{122}$$

32.4 Pixel Parameters

- Cell area: $a_{\text{cell}} \approx 1.63\ell_P^2$
- UV length: $\ell_{\text{UV}} = \sqrt{a_{\text{cell}}} \approx 1.28\ell_P$
- Entropy per cell: $\bar{\ell} = a_{\text{cell}}/(4\ell_P^2) \approx 0.408$

---

33 Appendix A: Summary of Derived Results

33.1 A.1 Theorem-Level Results and Dependency Status

This appendix records theorem-level status for the main manuscript. The recovered core is D1–D5 together with D7–D9; D6 appears only as the adjacent Phase-II boundary marker. The table keeps imported mathematics, extra premises, and claim tier visible.

Node	Output	Standard mathematics used	Extra premises / inputs beyond the five axioms	Status
D1	schedule-independent normal form for overlap repair	Newman's lemma	finite patch net, termination, and local confluence	Phase I structural theorem
D2	EC decomposition, Markov collar, and generalized-entropy split	HJPW; Petz; Fawzi–Renner	R0/R1 or explicit recoverability control; coarse-grained $A/(4G)$ dictionary	Phase I structural lemma/proposition layer

Node	Output	Standard mathematics used	Extra premises / inputs beyond the five axioms	Status
D3	geometric modular flow and Lorentz branch	Bisognano–Wichmann modular geometry; conformal-group classification	premise ROT, controlled refinement limit, and the OPH geometric-branch hypothesis of Theorem 2.2	Phase I conditional scaling-limit theorem
D4	null modular bridge to T_{kk}	Borchers–Wiesbrock; distributional differentiation on half-lines	the theorem-local inherited-strip decomposition and exact-or-controlled Markov hypotheses of Section 20.3, plus the downstream density-upgrade and relativistic null-stress-identification premises; the endpoint/propagation controls are internal to the local finite-constraint MaxEnt branch	Phase I conditional bridge theorem
D5	Jacobson-type rest-frame relation plus the conditional tensor upgrade	Jacobson small-ball mathematics; null-to-tensor reconstruction modulo a metric term	fixed-cap stationarity, the locally Lorentzian $d = 4$ scaling regime, and the all-directions/all-reference-states premise used in the tensor upgrade	Phase I conditional scaling-limit theorem/corollary
D6	capacity/Λ corollary and the separate capacity-based neutrino estimate	de Sitter entropy relation and dimensional analysis	external input N_{scr} and the screen-capacity identification	Phase II input-dependent corollary / estimate
D7	compact gauge reconstruction in the bosonic branch	Doplicher–Roberts / Tannaka reconstruction	bosonic symmetric transportable colimit and fiber-functor premises; the refinement-stable qualifier is internal to the local finite-constraint MaxEnt/refinement branch	Phase I conditional structural theorem
D8	product structure up to finite quotient on the realized branch	compact Lie representation classification; Schur's lemma	MAR, connected admissible class, one connected abelian factor, and $[z] = 0$	Phase I realized-branch theorem
D9	exact hypercharges, the counting chain $N_g = 3$ then $N_c = 3$, and the $\mathbb{Z}_6$ quotient	anomaly algebra; Witten anomaly; CKM CP counting	realized one-generation/one-Higgs package and MAR admissibility premises	Phase I realized-branch theorem/corollary chain

The recovered core in this appendix is D1–D5 together with D7–D9. D6 appears only to keep the Phase-I / Phase-II boundary explicit. Nothing here promotes D10, D11, or D12 to theorem level.

33.2 A.2 Secondary Quantitative and Continuation Benchmarks

This appendix does not extend the theorem-level list. It records every non-core branch that remains useful to track:

1. **Phase II input-dependent corollary (D6):** the capacity relation for Λ and the associated order-of-magnitude capacity / neutrino estimate.
2. **Phase II calibration sector (D10):** gauge/electroweak closure of the D10 branch, including the associated beta-shift comparison.
3. **Phase II secondary quantitative branch (D11):** the Higgs/top critical-surface branch once the D10 trajectory and the extra critical-surface condition are imposed.

4. **Phase III deferred continuations (D12 and beyond):** charged-lepton/Koide, texture, neutrino mass/mixing constructions beyond the D6 estimate, dark-sector, baryogenesis, black-hole spectroscopy, string/worldsheet, proton-spin, and other downstream fits.

These failures localize by tier: item 4 retracts the corresponding continuation only; item 3 challenges D11; item 2 challenges D10; item 1 challenges the screen-capacity identification only. None of them erases the recovered core D1–D5 plus D7–D9 by itself.

33.3 A.3 Future Directions

1. construct or prove the scaling-limit regime required by the D3–D5 relativity chain, including the null-stress bridge and fixed-cap stationarity premises;

2. extend the compact-gauge reconstruction from the bosonic internal-gauge branch to the fermionic/super-Tannakian chiral branch while preserving the D7–D9 Standard Model structural chain;

3. derive or independently fix the external continuous inputs P and N_{scr}, and complete a fully auditable D10–D11 numerical pipeline with explicit running, threshold, and matching conventions;

4. separate the D6 capacity relation from flavor-style add-ons by deciding exactly which neutrino or cosmological continuations survive without additional ansätze;

5. only after items 1–4 are secured, reopen the Phase-III continuations: charged-lepton/Koide structure, texture exponents, dark-sector response laws, baryogenesis, black-hole spectroscopy, string/worldsheet reorganization, proton-spin bookkeeping, and any similar late-stage continuation such as strong-CP proposals.

Interpretive strange-loop / existence narratives sit below that ledger entirely. They are philosophical continuations, not appendix-backed theorem or benchmark claims. This supplement does not define a theorem-usable configuration space for a global emergence map, prove that observer formation and reconstruction close to an internal self-map on such a space, or supply the compactness/contractivity/stability premises needed for an existence or stability theorem.

---

34 Appendix B: Mixed-Status Numerical Benchmarks

Quantity / branch	OPH Value	Observed	Status
a_0 (dark-sector continuation)	1.03×10^{-10} m/s^2	1.2×10^{-10} m/s^2	deferred continuation benchmark
$\Delta\Sigma$ (spin continuation)	0.308	0.29 ± 0.03	deferred continuation benchmark
η_B (baryogenesis estimate)	4.6×10^{-10}	6.1×10^{-10}	deferred continuation benchmark
Koide Q (flavor continuation)	2/3	0.666664	deferred continuation benchmark

Quantity / branch	OPH Value	Observed	Status
Koide δ (flavor continuation)	2/9	0.222225	deferred continuation benchmark
$\Delta b_3/\Delta b_2$ (D10 calibration)	0.91	0.96 (MSSM)	supplement-backed calibration

---

35 References

The derivations in this supplement are based on the OPH framework as specified in Part I of this manuscript. Standard results from:

- Tomita-Takesaki modular theory
- Quantum Markov chain structure theorems (Hayden-Jozsa-Petz-Winter)
- Fawzi-Renner recovery bounds
- Gleason's theorem
- Jacobson's entanglement equilibrium
- Peter-Weyl decomposition
- Heat kernel asymptotics on compact Lie groups
- Tannaka-Krein reconstruction
- 't Hooft anomaly matching and instanton calculus

Part VI
Interpretive Appendix: Strange-Loop Proposal

36 Interpretive Status, the Habitat Theorem, and the Remaining Strange-Loop Gap

This part is an interpretive epilogue, outside the Phase-I recovered core and the D10–D11 quantitative branches. OPH supplies the ambient state-and-law habitat needed for a fixed-point wrapper. This section leaves the actual closure map on the stronger observer-supporting sector open.

The OPH inputs used here are all internal to this paper:

1. the patch net $P \mapsto A(P)$ of von Neumann algebras together with isotony and overlap restriction maps;
2. the global state on the inductive-limit algebra, whose restrictions witness a nonempty overlap-consistent local sector;
3. the finite Axiom-3 MaxEnt/refinement branch, which supplies a common finite family of gauge-invariant local constraint observables across cutoffs;
4. on the fixed-cutoff branch, the derived finite type-I presentations and compact boundary gluing groups used elsewhere in the OPH package.

The law slot is encoded by the finite expectation-value coordinates of the retained Axiom-3 constraint family. That choice furnishes a compact convex state-and-law habitat within OPH.

Theorem (OPH internal state-and-law habitat theorem). Fix finitely many screen patches $P_1, \dots, P_N$ in the standard OPH setup. For each i, let

$$M_i := A(P_i), \qquad M_{ij} := A(P_i \cap P_j),$$

where M_i is the patch von Neumann algebra and $M_{ij} \subset M_i, M_j$ is the overlap algebra. Let

$$S_i := S(M_i) \subset M_i^*$$

be the full state space of M_i, endowed with the weak* topology $\sigma(M_i^*, M_i)$.

Choose a finite family of self-adjoint gauge-invariant local constraint observables

$$\mathcal{O} = \{O_1, \dots, O_m\}$$

from the Axiom-3 MaxEnt branch, with each O_a supported in one of the chosen patches; write $i(a)$ for an index such that $O_a \in M_{i(a)}$.

Define the overlap-consistent state sector

$$X_{\mathrm{ov}} := \left\{ (\omega_1, \dots, \omega_N) \in \prod_{i=1}^{N} S_i : \omega_i|_{M_{ij}} = \omega_j|_{M_{ij}} \text{ for all } i, j \right\}.$$

Define the OPH law-coordinate map

$$c : X_{\mathrm{ov}} \to \mathbb{R}^m, \qquad c(\omega_1, \dots, \omega_N) := \big(\omega_{i(1)}(O_1), \dots, \omega_{i(m)}(O_m)\big).$$

Let

$$X_{\mathcal{O}} := \left\{ (\omega_1, \dots, \omega_N, \ell) \in \left(\prod_{i=1}^{N} S_i \right) \times \mathbb{R}^m : (\omega_1, \dots, \omega_N) \in X_{\mathrm{ov}},\ \ell = c(\omega_1, \dots, \omega_N) \right\}.$$

Then:

1. the ambient product
$$E := \left(\prod_{i=1}^{N} M_i^* \right) \times \mathbb{R}^m$$
is a Banach space for any product norm;
2. $X_{\mathcal{O}}$ is nonempty and convex;
3. $X_{\mathcal{O}}$ is norm-closed in E;
4. $X_{\mathcal{O}}$ is compact in the product topology
$$\tau := \left(\prod_{i=1}^{N} \sigma(M_i^*, M_i) \right) \times \text{Euclidean topology on } \mathbb{R}^m;$$
5. therefore every τ-continuous self-map
$$\Phi : X_{\mathcal{O}} \to X_{\mathcal{O}}$$
has a fixed point.

Proof. Each M_i is a von Neumann algebra in the OPH patch net, hence a Banach space, so each dual M_i^* is Banach. A finite product of Banach spaces is Banach, hence so is E.

For each i, the state space

$$S_i = \{\omega \in M_i^* : \omega \geq 0,\ \omega(1) = 1\}$$

is convex and weak* compact by Banach–Alaoglu, because it is a weak* closed subset of the dual unit ball. For every pair (i, j), restriction along $M_{ij} \subset M_i$ and $M_{ij} \subset M_j$ defines affine weak* continuous maps

$$r_{ij} : S_i \to S(M_{ij}), \qquad r_{ji} : S_j \to S(M_{ij}).$$

Hence X_{ov} is an intersection of affine equalizer sets, so it is convex and τ-closed inside $\prod_i S_i$. Since $\prod_i S_i$ is compact and convex in the product weak* topology, X_{ov} is compact and convex as well.

Nonemptiness comes from the OPH global state on the inductive-limit algebra: its restrictions

$$(\omega|_{M_1}, \dots, \omega|_{M_N})$$

belong to X_{ov} because the restrictions agree on every overlap by construction.

Each coordinate $\omega_{i(a)} \mapsto \omega_{i(a)}(O_a)$ is affine and weak* continuous, since evaluation at a fixed algebra element is weak* continuous. Thus c is affine and τ-continuous. The graph map

$$G : X_{\mathrm{ov}} \to E, \qquad G(x) := (x, c(x)),$$

is affine and continuous, so its image $X_{\mathcal{O}} = G(X_{\mathrm{ov}})$ is convex and τ-compact.

To see norm-closedness in E, let $(x_n, c(x_n)) \to (x, \ell)$ in norm. Norm convergence implies weak* convergence on each coordinate, so $x \in X_{\mathrm{ov}}$ because X_{ov} is weak* closed. Since c is weak* continuous, $c(x_n) \to c(x)$, while norm convergence in $\mathbb{R}^m$ also gives $c(x_n) \to \ell$. Hence $\ell = c(x)$, so $(x, \ell) \in X_{\mathcal{O}}$.

Finally, $X_{\mathcal{O}}$ is a compact convex subset of the locally convex topological vector space E equipped with τ. Therefore every τ-continuous self-map $\Phi : X_{\mathcal{O}} \to X_{\mathcal{O}}$ has a fixed point by the Schauder–Tychonoff fixed-point theorem. □

Corollary (Fixed-cutoff OPH Brouwer corollary). Assume the fixed-cutoff regulator realization $R0/R1$: each chosen patch algebra $M_i = A(P_i)$ is finite-dimensional. Then E is finite-dimensional, the weak* and norm topologies coincide on E, and $X_{\mathcal{O}}$ is a nonempty compact convex subset of a finite-dimensional real vector space. Hence every continuous self-map

$$\Phi : X_{\mathcal{O}} \to X_{\mathcal{O}}$$

has a fixed point by Brouwer. If, moreover, Φ is a contraction for some metric on $X_{\mathcal{O}}$, then the fixed point is unique and the iterates converge to it from every starting point by Banach's contraction theorem.

Proof. Under $R0/R1$, the OPH fixed-cutoff branch identifies each chosen patch algebra with a finite-dimensional type-I presentation modulo compact boundary redundancy, so each M_i and M_i^* is finite-dimensional. Hence E is finite-dimensional and all relevant locally convex topologies coincide. The habitat theorem above shows that $X_{\mathcal{O}}$ is nonempty, compact, and convex, so Brouwer gives a fixed point for every continuous self-map. The contraction statement is the usual Banach fixed-point theorem. □

Outside the scientific core lie:

1. an actual OPH closure map Φ built from internal operations such as overlap repair, Axiom-3 MaxEnt reprojection, collar recoverability, and record-sector coarse-graining;
2. a proof that the stronger "observer-supporting" or "selected-world" criteria carve out a nonempty compact convex Φ-invariant subset of $X_{\mathcal{O}}$;
3. any uniqueness, contraction, or Lyapunov-type stability estimate for that closure map.

The habitat theorem above supplies the ambient Banach/compact-convex setting. The strange-loop continuation requires the theorem-level closure map and the stronger invariant-sector and stability inputs.

37 Timeless Causal Chain

The hypothesis is represented by the chain

$$\begin{aligned}\text{information constraints} &\to \text{effective physics} \to \text{complex chemistry and biology}\\ &\to \text{observers with records} \to \text{formal OPH reconstruction}\\ &\to \text{OPH-compatible simulation substrate} \to \text{information constraints.}\end{aligned}$$

In this paper the chain is interpreted as a consistency loop. The discussion does not use it as a linear first-cause narrative. The Escher drawing-hands analogy is a geometrical metaphor for this closure structure.

38 Why Reality Exists and Why It Has This Form

Under this hypothesis, existence is identified with nontrivial fixed points of the OPH consistency map. The observed form of reality is selected by stability requirements:

1. overlap-consistent sharable records,

2. recoverability under local information loss,
3. long-lived structured sectors that support observer formation,
4. internal reconstructibility of the governing rule set.

The appearance of intelligent observers enters the closure mechanism itself. Observer-capable worlds are the sectors that complete the loop.

39 Additional Problem Closures

This wrapper does not establish a separate claim tier. Phase I is D1–D5 together with D7–D9; D6 is the separate input-dependent corollary; D10–D11 are the quantitative branches documented earlier in the paper; and D12 collects deferred continuations. The synchronized screen-microphysics audit surface now carries fixed-cutoff theorem packages for the measurement/Born-rule interface and for checkpoint/restoration/backup. This wrapper imports those fixed-cutoff statuses rather than upgrading them by itself. Table 43 records the status of the broader "problem closure" claims that appear in the synthesis essays.

Table 43: Problem-closure statements sorted by claim tier.

Problem domain	Current status	OPH mechanism / scope note
Quantum gravity consistency	Phase I recovered core (conditional scaling-limit)	Conditional Lorentz and Jacobson-type Einstein branches from modular geometry, the null bridge, and fixed-cap stationarity.
Measurement problem	Mixed: fixed-cutoff theorem package / interpretive synthesis	The synchronized screen-microphysics audit surface proves a fixed-cutoff central-record / Born-Lüders package for observer-accessible events. Broader philosophical closure claims remain interpretive in this wrapper.
Cosmological principle and horizon homogeneity	Interpretive or organizational synthesis	MaxEnt state selection under symmetric constraints motivates isotropy and homogeneity stories. Those stories remain outside the independent Phase-I theorems.
Problem of time	Interpretive or organizational synthesis	Modular flow from the BW_{S^2} geometric branch provides a physical time parameter for each observer patch. It remains a useful organizational principle, not a separate recovered-core theorem in this wrapper.
Supersymmetry non-observation vs. unification	Phase II quantitative branch	MSSM-like β-shift comparisons from edge-sector weighting belong to the D10 calibration layer. They sit outside the recovered-core theorem package.
Black-hole information paradox	Program-level structural continuation	Edge-center sectorization and recoverability suggest an internal resolution. The full black-hole-information branch remains outside Phase I.

Problem domain	Current status	OPH mechanism / scope note
Cosmological constant problem	Phase II input-dependent corollary	Λ is tied to global screen capacity once the screen-capacity identification is adopted. This sits outside the local recovered-core outputs.
Magnetic monopole expectation from simple GUT groups	Phase I recovered core	The realized product-group branch $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)/\mathbb{Z}_6$ removes the standard simple-GUT symmetry-breaking monopole channel.
Proton stability and proton spin fraction	Mixed: Phase I / Phase III	No gauge-mediated proton decay follows from the realized product-group branch, which removes the standard simple-GUT proton-decay channel. Proton spin requires nonperturbative QCD completion and is not part of Phase I.
Dark matter phenomenology	Phase III deferred continuation	Modular-anomaly response laws remain conjectural beyond the recovered gravity chain.
Baryon asymmetry scale	Phase III deferred continuation	Suppression-counting estimates are not a substitute for a derived out-of-equilibrium mechanism.
Three generations and Koide structure	Mixed: Phase I / Phase III	$N_g = 3$ is recovered on the realized MAR-admissible branch; Koide and charged-lepton fits require extra flavor ansätze.
String/worldsheet reorganization	Phase III deferred continuation	Requires a controlled large-N_{edge} regime distinct from the physical color rank $N_c = 3$. Any further lift to critical superstring structure remains a conjectural extension requiring extra worldsheet ingredients not derived in the OPH core.
Why anything exists / strange-loop closure	Interpretive epilogue only	Part VI gives an OPH-internal state-and-law habitat theorem: overlap-consistent patch states together with finite Axiom-3 law coordinates form a theorem-usable nonempty compact convex sector, and fixed-cutoff branches admit a Brouwer version. The further inputs are the actual OPH closure map on an invariant observer-supporting sector, together with uniqueness and stability.

40 Standalone Status

This wrapper is standalone only in the documentary sense: every derivation cited here appears somewhere in the manuscript, so no external derivation document is required to follow the chain of references. It is *not* a separate authority for claim tier.

For claim-tier purposes, Part I Section 8 remains the governing summary for this wrapper's own claims. The fixed-cutoff measurement/Born-rule and checkpoint/restoration statements discussed below inherit their theorem status from the synchronized screen-microphysics audit surface, and this wrapper records that status without becoming the primary proof surface.

Part VII
Speculative Appendix: Observer Continuation and Backup

This part remains a program-level continuation note at the level of this wrapper. The synchronized screen-microphysics audit surface now carries the fixed-cutoff checkpoint/restoration/error/identity theorem package, and the formulas below summarize its algebraic interface in observer language. What remains outside the proved package is the stronger strange-loop closure map, the invariant observer-supporting sector, and the associated uniqueness and stability questions.

41 Observer as Algebraic Pattern

In OPH, an observer is represented by

$$O = (P, \mathcal{A}(P), \rho, R),$$

where P is a screen patch, $\mathcal{A}(P)$ is its local algebra, ρ is the local state, and R is the record algebra. Records are implemented by approximately commuting projectors in overlap centers, so they are shareable without violating no-cloning constraints for generic quantum states.

On the synchronized screen-microphysics audit surface, the observer-facing measurement interface is now theorem-bearing at fixed cutoff: one has a finite central record algebra, Born probabilities for its event projectors, and Lüders conditioning on that same commuting record algebra. The observer-paper role is therefore to import that fixed-cutoff measurement package into the surrounding observer vocabulary, not to reclassify it as mere interpretive rhetoric.

42 Markov Collar Factorization

For a collar tripartition A-B-D with conditional mutual information $I(A : D|B) \leq \varepsilon$, the exact Markov limit gives

$$\rho_{ABD} = \bigoplus_{\alpha} p_\alpha \, \rho^{(\alpha)}_{Ab^\alpha_L} \otimes \rho^{(\alpha)}_{b^\alpha_R D}.$$

The sector label α is classical center data. This decomposition is the mathematical basis for extracting an interior observer state with a controlled boundary interface.

43 Checkpoint and Restoration Map

A checkpoint is the tuple

$$\mathcal{C} = (R, \alpha, \rho^{(\alpha)}_{\text{int}}),$$

with $\rho^{(\alpha)}_{\text{int}}$ the interior conditional state. Given a compatible target environment state $\sigma^{(\alpha)}_{\text{env}}$, a restored state is

$$\rho^{(\alpha)}_{\text{new}} = \rho^{(\alpha)}_{\text{int}} \otimes \sigma^{(\alpha)}_{\text{env}}.$$

For approximate Markov collars, recovery is controlled by the standard bound

$$\|\rho_{ABD} - (\mathrm{id}_A \otimes \mathcal{R}_{B\to BD})(\rho_{AB})\|_1 \leq 2\sqrt{\ln 2 \, \varepsilon}.$$

This gives quantitative trace-distance control on the recovered state under finite conditional mutual information (finite collar error). Interpreting that control as observer continuation requires additional modeling assumptions beyond the bound itself.

The fixed-cutoff audit-surface claim is stronger than the bare recoverability estimate written above. On the synchronized screen-microphysics surface, exact checkpoint restoration preserves the full future law of observer-accessible events, while an ε-accurate restoration changes that future law by at most ε in total variation. So the present wrapper cites a theorem-bearing continuation package at fixed cutoff even though the stronger strange-loop closure story remains open.

44 Physical Meaning

The continuation mechanism discussed here is now partly theorem-bearing and partly program-level. At fixed cutoff, the synchronized screen-microphysics audit surface proves a checkpoint/restoration theorem and backup corollary for observer-accessible event algebras. This wrapper records the observer-facing meaning of that theorem and its algebraic prerequisites; it is not the primary proof surface. What still remains open is the stronger substrate-selection, strange-loop closure, uniqueness, and stability package.